

Topological Energy Condensate Theory (TECT): 23

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I. EXPLICIT HESSIAN-PROJECTOR EXTRACTION FOR THE REMAINING GRADED 2 + 2 COEFFICIENT POINT

II. GOAL

The only remaining microscopic coefficient problem is

$$\boxed{\mathcal{H}_{OO} \rightarrow \lambda_c, \quad \mathcal{H}_{TD}, \mathcal{H}_{DT} \rightarrow \Lambda_E, \Lambda_O,}$$

with possible subleading corrections from OD, DO, DD .

In the exact concrete Class II-compatible branch,

$$\Lambda_{\parallel}^{(0)} = 0,$$

so the longitudinal channel must be read from the Branch-B primitive coefficient

$$\lambda_c.$$

III. 1. FULL FIRST-ORDER TRANSPORT EXTRACTION

Write the background-dependent Hessian in block form

$$\Gamma^{(2)}[\bar{\Phi}_{\min}] = \begin{pmatrix} \mathcal{H}_{TT} & \mathcal{H}_{TO} & \mathcal{H}_{TD} \\ \mathcal{H}_{OT} & \mathcal{H}_{OO} & \mathcal{H}_{OD} \\ \mathcal{H}_{DT} & \mathcal{H}_{DO} & \mathcal{H}_{DD} \end{pmatrix},$$

and expand each block as

$$\mathcal{H}_{AB} = \mathcal{H}_{AB}^{(0)}(\xi) - i \mathcal{V}_{AB}^{\mu}(\xi) \partial_{y^{\mu}} + \mathcal{W}_{AB}^{\mu\nu}(\xi) \partial_{y^{\mu}} \partial_{y^{\nu}} + \cdots.$$

Only $\mathcal{V}_{AB}^{\mu}$ contributes to the principal graded transport symbol.

After restriction to the tangent bundle

$$Q_B(\xi) = (\mathbb{C}z_B(\xi))^{\perp}, \quad \dim_{\mathbb{C}} Q_B = 2,$$

define the Q_B -endomorphism-valued first-order block kernel

$$\mathcal{M}_{AB}^{\mu}(\xi) := \Pi_Q \mathcal{V}_{AB}^{\mu}(\xi) \Pi_Q.$$

IV. 2. CANONICAL PROJECTOR FUNCTIONALS

Let Σ_a denote the Pauli basis on Q_B , and let

$$(\hat{n}, e_1, e_2)$$

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be the shell-adapted orthonormal frame.

Define the three invariant projectors

$$\mathfrak{P}_{\parallel}[\mathcal{M}] := \frac{1}{2} \text{Tr}_{Q_B}(\Sigma_3 n_{\mu} \mathcal{M}^{\mu}), \quad (1)$$

$$\mathfrak{P}_E[\mathcal{M}] := \frac{1}{4} E^{\mu}{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_{\mu}), \quad (2)$$

$$\mathfrak{P}_O[\mathcal{M}] := \frac{1}{4} J^{\mu}{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_{\mu}). \quad (3)$$

These extract, respectively, the longitudinal singlet, the even transverse doublet, and the odd transverse doublet.

V. 3. OO-BLOCK LONGITUDINAL EXTRACTION

Define

$$V_{OO}^{\mu}(\xi) := \Pi_Q \mathcal{V}_{OO}^{\mu}(\xi) \Pi_Q. \quad (4)$$

Its longitudinal scalar is

$$a_{3;OO}(\xi) = \mathfrak{P}_{\parallel}[V_{OO}] = \frac{1}{2} \text{Tr}_{Q_B}(\Sigma_3 n_{\mu} V_{OO}^{\mu}(\xi)). \quad (5)$$

Because of the oddness filter, its principal branch-sensitive expansion has the form

$$a_{3;OO}(\xi) = \ell_{0;OO}^{(1)} \theta'(\xi) + \ell_{0;OO}^{(3)} \theta'(\xi)^3 + O(\theta'(\xi) \theta(\xi)^2, \theta'(\xi)^5). \quad (6)$$

Therefore the exact coefficient identification is

$$\Lambda_{\parallel;OO}^{(0)} = \ell_{0;OO}^{(1)}, \quad \frac{\lambda_{c;OO}}{2} = \ell_{0;OO}^{(3)}. \quad (7)$$

In the current concrete Class II-compatible branch,

$$\Lambda_{\parallel}^{(0)} = 0,$$

so the practical longitudinal coefficient is

$$\lambda_c = \lambda_{c;OO} = 2 \ell_{0;OO}^{(3)}. \quad (8)$$

Equivalently, one may write the executable limit formula

$$\lambda_c = 2 \lim_{\theta' \rightarrow 0} \frac{\mathfrak{P}_{\parallel}[V_{OO}]}{\theta'(\xi)^3}. \quad (9)$$

VI. 4. WHY THE PRIMITIVE LONGITUDINAL SEED STARTS AT θ^3

The microscopic longitudinal primitive channel is generated by

$$\mathcal{T}_{\xi} = i[P_B, \partial_{\xi} P_B].$$

Its direct Q_B -restriction vanishes,

$$\Pi_Q \mathcal{T}_\xi \Pi_Q = 0,$$

but its square is nontrivial:

$$\Pi_Q \mathcal{T}_\xi^2 \Pi_Q = \frac{\theta'(\xi)^2}{2} (\mathbf{1}_2 + \Sigma_3).$$

Taking the traceless part gives

$$S_\xi = \frac{\theta'(\xi)^2}{2} \Sigma_3.$$

Since the overlap $f_+ f_- \theta'(\xi)^2$ is parity-even and integrates to zero in the graded channel, one needs one additional odd factor. Hence the good longitudinal transport channel is

$$\boxed{V_{\parallel, \text{prim}}^\mu(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \Sigma_3.} \quad (10)$$

This is the Branch-B primitive longitudinal seed.

VII. 5. $TD + DT$ -BLOCK TRANSVERSE EXTRACTION

Define the mixed transverse carrier

$$\boxed{\mathcal{M}_{TD+DT}^\mu := \Pi_Q (\mathcal{V}_{TD}^\mu + \mathcal{V}_{DT}^\mu) \Pi_Q.} \quad (11)$$

Its leading even/odd transverse coefficients are extracted by

$$\boxed{e_0^{(1)} = \mathfrak{P}_E[\mathcal{M}_{TD+DT}^{\text{inv}}], \quad o_0^{(1)} = \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}].} \quad (12)$$

In the minimal mixed-block truncation, these are identified with the actual transverse coefficient point:

$$\boxed{\Lambda_E = e_0^{(1)}, \quad \Lambda_O = o_0^{(1)}.} \quad (13)$$

More broadly, beyond the minimal truncation one has

$$\Lambda_E = \sum_{AB \in \{TD, DT, OD, DO, DD\}} e_{0;AB}^{(1)}, \quad \Lambda_O = \sum_{AB \in \{TD, DT, OD, DO, DD\}} o_{0;AB}^{(1)}, \quad (14)$$

but the sharp current leading-order audit starts from $TD + DT$.

VIII. 6. REDUCED PDE COEFFICIENTS

Once $\lambda_c, \Lambda_E, \Lambda_O$ are extracted, the reduced coefficients follow from

$$\boxed{\lambda_\parallel = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O.} \quad (15)$$

Using

$$I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa, \quad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3,$$

one gets

$$\boxed{\lambda_\parallel = -\frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3,} \quad (16)$$

$$\boxed{\alpha = -\frac{\sqrt{3}\pi}{16} \Lambda_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} \Lambda_O \theta_0 \kappa.} \quad (17)$$

IX. 7. CURRENT WORKING-BRANCH SCHUR-COMPLEMENT REALIZATION

For the current refined locked + Class II working branch, integrating out the heavy mediator X_i^a gives the Schur correction

$$\Delta_X^{(2)} = -\frac{1}{M_X^2} (\alpha_X \delta J + \beta_X \delta K) (\alpha_X \delta J + \beta_X \delta K)^\dagger. \quad (18)$$

Hence

$$\Delta_X^{(2)} = -\frac{\alpha_X^2}{M_X^2} \delta J \delta J^\dagger - \frac{\alpha_X \beta_X}{M_X^2} (\delta J \delta K^\dagger + \delta K \delta J^\dagger) - \frac{\beta_X^2}{M_X^2} \delta K \delta K^\dagger. \quad (19)$$

The sharp working-branch identification is then:

1. the JJ -piece feeds the primitive longitudinal slot;
2. the mixed $JK + KJ$ -piece feeds the transverse E/O slots;
3. the pure K -sector is transverse at principal order and does not generate λ_c .

Accordingly, the branch-level coefficient point is

$$c_{\parallel} = \frac{\alpha_X^2}{M_X^2} j^2, \quad c_E = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}, \quad c_O = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)}. \quad (20)$$

Thus

$$\lambda_{\parallel} = \frac{1}{2} I_{\theta^3} \frac{\alpha_X^2}{M_X^2} j^2, \quad \alpha = I_{\theta} \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}, \quad \beta = I_{\theta} \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)}. \quad (21)$$

X. 8. HONEST CLOSURE STATEMENT

What is fully closed now is the extraction *protocol*:

$$\mathcal{H}_{OO} \rightarrow \lambda_c, \quad \mathcal{H}_{TD}, \mathcal{H}_{DT} \rightarrow \Lambda_E, \Lambda_O.$$

What is also closed on the current refined locked + Class II working branch is the Schur-reduced coefficient pattern (20)–(21).

What is still not yet unconditional is the final source-by-source differentiation of the *complete* microscopic TECT Hessian yielding the exact numerical functions

$$h_{\parallel}(\theta), \quad h_{11}(\theta), \quad h_{12}(\theta), \quad h_{21}(\theta), \quad h_{22}(\theta).$$

So the genuine remaining frontier is not structural. It is the explicit insertion of the full complete TECT background into

$$\mathcal{V}_{OO}^{\mu}(\xi), \quad \mathcal{V}_{TD}^{\mu}(\xi) + \mathcal{V}_{DT}^{\mu}(\xi).$$

XI. FINAL SUMMARY

The graded 2 + 2 bypass route is already structurally closed.

In the exact concrete Class II-compatible branch,

$$\Lambda_{\parallel}^{(0)} = 0,$$

so the remaining coefficient point is

$$\lambda_c, \quad \Lambda_E, \quad \Lambda_O.$$

The exact blockwise extraction protocol is

$$\mathcal{H}_{OO} \rightarrow V_{OO}^\mu \rightarrow a_{3;OO}(\xi) = \frac{1}{2} \text{Tr}_{Q_B}(\Sigma_3 n_\mu V_{OO}^\mu(\xi)) \rightarrow \lambda_c,$$

and

$$\mathcal{H}_{TD}, \mathcal{H}_{DT} \rightarrow \mathcal{M}_{TD+DT}^\mu \rightarrow e_0^{(1)}, o_0^{(1)} \rightarrow \Lambda_E, \Lambda_O.$$

After overlap insertion,

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O,$$

hence

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3, \quad \alpha = -\frac{\sqrt{3}\pi}{16} \Lambda_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} \Lambda_O \theta_0 \kappa.$$

Therefore the remaining frontier is not conceptual. It is the literal differentiation and projection of the complete background-dependent Hessian blocks

$$\mathcal{V}_{OO}^\mu(\xi), \quad \mathcal{V}_{TD}^\mu(\xi) + \mathcal{V}_{DT}^\mu(\xi).$$

XII. EXECUTABLE HESSIAN EXTRACTION OF THE REMAINING GRADED 2 + 2 COEFFICIENT POINT

XIII. GOAL

The remaining microscopic problem is not conceptual anymore. It is the explicit extraction of

$$\lambda_c, \quad \Lambda_E, \quad \Lambda_O$$

from the background-dependent first-order Hessian transport kernel.

At the current sharpest blockwise stage, this reduces to

$$\mathcal{H}_{OO} \rightarrow \lambda_c, \quad \mathcal{H}_{TD}, \mathcal{H}_{DT} \rightarrow \Lambda_E, \Lambda_O.$$

XIV. UNIVERSAL FIRST-ORDER HESSIAN TEMPLATE

Let the full fluctuation field be decomposed as

$$\varphi = (a, \nu, \zeta), \quad a \sim T, \quad \nu \sim O, \quad \zeta \sim D.$$

Around the shell-/defect-adapted background $\bar{\Phi}_{\min}$, the full Hessian has block form

$$\Gamma^{(2)}[\bar{\Phi}_{\min}] = \begin{pmatrix} \mathcal{H}_{TT} & \mathcal{H}_{TO} & \mathcal{H}_{TD} \\ \mathcal{H}_{OT} & \mathcal{H}_{OO} & \mathcal{H}_{OD} \\ \mathcal{H}_{DT} & \mathcal{H}_{DO} & \mathcal{H}_{DD} \end{pmatrix},$$

with blockwise expansion

$$\mathcal{H}_{AB} = \mathcal{H}_{AB}^{(0)}(\xi) - i \mathcal{V}_{AB}^\mu(\xi) \partial_{y^\mu} + \mathcal{W}_{AB}^{\mu\nu}(\xi) \partial_{y^\mu} \partial_{y^\nu} + \cdots. \quad (22)$$

Only the one-derivative kernel $\mathcal{V}_{AB}^\mu$ contributes to the principal graded symbol.

Project to the tangent bundle

$$Q_B(\xi) = (\mathbb{C} z_B(\xi))^\perp, \quad \dim_{\mathbb{C}} Q_B = 2,$$

and define the projected block kernels

$$\mathcal{M}_{AB}^\mu(\xi) := \Pi_Q \mathcal{V}_{AB}^\mu(\xi) \Pi_Q. \quad (23)$$

XV. CANONICAL PROJECTOR FUNCTIONALS

Let Σ_a ($a = 1, 2, 3$) denote the Pauli basis on Q_B , and let

$$(\hat{n}, E^\mu{}_a, J^\mu{}_a)$$

be the shell-adapted worldvolume frame.

Define the three invariant coefficient projectors

$$\mathfrak{P}_\parallel[\mathcal{M}] := \frac{1}{2} \text{Tr}_{Q_B}(\Sigma_3 n_\mu \mathcal{M}^\mu), \quad (24)$$

$$\mathfrak{P}_E[\mathcal{M}] := \frac{1}{4} E^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_\mu), \quad (25)$$

$$\mathfrak{P}_O[\mathcal{M}] := \frac{1}{4} J^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_\mu). \quad (26)$$

These extract, respectively,

$$n^\mu \Sigma_3, \quad E^\mu{}_a \Sigma_a, \quad J^\mu{}_a \Sigma_a.$$

XVI. OO BLOCK: EXACT LONGITUDINAL EXTRACTION

Define the OO -projected first-order transport block

$$V_{OO}^\mu(\xi) := \Pi_Q \mathcal{V}_{OO}^\mu(\xi) \Pi_Q. \quad (27)$$

Its longitudinal scalar is

$$a_{3;OO}(\xi) = \mathfrak{P}_\parallel[V_{OO}] = \frac{1}{2} \text{Tr}_{Q_B}(\Sigma_3 n_\mu V_{OO}^\mu(\xi)). \quad (28)$$

The branch-sensitive expansion is

$$a_{3;OO}(\xi) = \ell_{0;OO}^{(1)} \theta'(\xi) + \ell_{0;OO}^{(3)} \theta'(\xi)^3 + O(\theta'(\xi)^5, \theta'(\xi)\theta(\xi)^2). \quad (29)$$

Therefore

$$\Lambda_{\parallel;OO}^{(0)} = \ell_{0;OO}^{(1)}, \quad \frac{\lambda_{c;OO}}{2} = \ell_{0;OO}^{(3)}. \quad (30)$$

In the current concrete Class II-compatible branch,

$$\Lambda_{\parallel}^{(0)} = 0,$$

so the practical longitudinal coefficient is

$$\lambda_c = \lambda_{c;OO} = 2 \ell_{0;OO}^{(3)}. \quad (31)$$

Equivalently, the executable extraction rule is

$$\lambda_{c;OO} = \lim_{\theta \rightarrow 0} \frac{\text{Tr}_{Q_B}(\Sigma_3 n_\mu \Pi_Q \mathbb{H}_{1,OO}^\mu(\xi) \Pi_Q)}{\theta'(\xi)^3}. \quad (32)$$

XVII. SOURCE-BY-SOURCE DECOMPOSITION INSIDE THE OO AUDIT

At principal order, decompose the local transport kernel as

$$V^\mu(\xi) = V_{\text{prim}}^\mu(\xi) + V_K^\mu(\xi) + V_{EO}^\mu(\xi) + V_{\text{irr}}^\mu(\xi). \quad (33)$$

A. Primitive longitudinal source

The primitive commutator-square channel is

$$\mathcal{T}_\xi = i[P_B, \partial_\xi P_B], \quad S_\xi = (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}}.$$

For the explicit defect representative,

$$S_\xi = \frac{1}{2} \theta'(\xi)^2 \Sigma_3.$$

Hence the primitive transport block is

$$V_{\text{prim}}^\mu(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \Sigma_3. \quad (34)$$

Applying the projectors,

$$\mathfrak{P}_\parallel[V_{\text{prim}}] = \frac{\lambda_c}{2} \theta'(\xi)^3, \quad \mathfrak{P}_E[V_{\text{prim}}] = 0, \quad \mathfrak{P}_O[V_{\text{prim}}] = 0. \quad (35)$$

B. K -sector source

The K -sector tangent block has the universal form

$$V_K^\mu = \theta'(\xi) \left[\kappa_E(\xi) E^\mu{}_a \Sigma_a + \kappa_O(\xi) J^\mu{}_a \Sigma_a \right]. \quad (36)$$

Therefore

$$\mathfrak{P}_\parallel[V_K] = 0, \quad \mathfrak{P}_E[V_K] = \kappa_E(\xi) \theta'(\xi), \quad \mathfrak{P}_O[V_K] = \kappa_O(\xi) \theta'(\xi). \quad (37)$$

C. EO -slot source

The explicit E/O -slot contribution is

$$V_{EO}^\mu = \theta'(\xi) \cos \theta(\xi) \left[\tilde{\Lambda}_E E^\mu{}_a \Sigma_a + \tilde{\Lambda}_O J^\mu{}_a \Sigma_a \right]. \quad (38)$$

Thus

$$\mathfrak{P}_\parallel[V_{EO}] = 0, \quad \mathfrak{P}_E[V_{EO}] = \tilde{\Lambda}_E \theta'(\xi) \cos \theta(\xi), \quad \mathfrak{P}_O[V_{EO}] = \tilde{\Lambda}_O \theta'(\xi) \cos \theta(\xi). \quad (39)$$

D. Immediate principal-order consequence

Hence the principal source table is

source	$\mathfrak{P}_\parallel$	$\mathfrak{P}_E$	$\mathfrak{P}_O$
V_{prim}	$\frac{\lambda_c}{2} \theta'^3$	0	0
V_K	0	$\kappa_E \theta'$	$\kappa_O \theta'$
V_{EO}	0	$\tilde{\Lambda}_E \theta' \cos \theta$	$\tilde{\Lambda}_O \theta' \cos \theta$
V_{irr}	0 (at principal order)	0 (at principal order)	0 (at principal order)

(40)

Therefore the principal longitudinal source is unique:

$$\boxed{a_3^{(K)} = 0, \quad a_3^{(EO)} = 0, \quad a_3^{(\text{prim})} = \frac{\lambda_c}{2} \theta'(\xi)^3.} \quad (41)$$

This proves that, at the current solved leading stage, the OO longitudinal audit is exactly the extraction of the $\theta'(\xi)^3 n^\mu \Sigma_3$ coefficient.

XVIII. TD+DT BLOCK: EXACT TRANSVERSE EXTRACTION

Define the mixed transverse carrier

$$\boxed{\mathcal{M}_{TD+DT}^\mu := \Pi_Q (\mathcal{V}_{TD}^\mu + \mathcal{V}_{DT}^\mu) \Pi_Q.} \quad (42)$$

Its leading even/odd coefficients are extracted by

$$\boxed{e_0^{(1)} = \mathfrak{P}_E[\mathcal{M}_{TD+DT}^{\text{inv}}], \quad o_0^{(1)} = \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}].} \quad (43)$$

In the minimal mixed-block truncation,

$$\boxed{\Lambda_E = e_0^{(1)}, \quad \Lambda_O = o_0^{(1)}.} \quad (44)$$

Thus the leading transverse kernel is

$$\boxed{\mathcal{M}_{TD+DT}^\mu = \theta'(\xi) \left[\Lambda_E E^\mu{}_a \Sigma_a + \Lambda_O J^\mu{}_a \Sigma_a \right] + \mathcal{M}_{\text{corr}}^\mu,} \quad (45)$$

where $\mathcal{M}_{\text{corr}}^\mu$ starts at higher order and/or receives contributions from

$$OD, \quad DO, \quad DD.$$

XIX. HONEST CAVEAT ON THE TRANSVERSE SECTOR

At the reduced branch level, $TD + DT$ is the correct leading transverse summary. However, a naive local real Hermitian JK bilinear satisfies

$$\Sigma_3(\Lambda_E \Sigma_1 + \Lambda_O \Sigma_2) + (\Lambda_E \Sigma_1 + \Lambda_O \Sigma_2) \Sigma_3 = 0,$$

because

$$\{\Sigma_3, \Sigma_1\} = \{\Sigma_3, \Sigma_2\} = 0.$$

Therefore the statement

JK alone automatically gives the transverse pair

is too strong. A genuinely nonzero full-PDE transverse realization requires derivative-resolved and/or phase-resolved microscopic structure.

Thus the current statement is:

$$\boxed{TD + DT \rightarrow (\Lambda_E, \Lambda_O)} \quad (46)$$

is a correct branch-level reduced summary, but not yet the unconditional final microscopic closure of the complete TECT PDE.

XX. PROJECTION TO THE LOW MODES AND REDUCED PDE COEFFICIENTS

Let the even/odd low modes be

$$u_a^+(\xi) = f_+(\xi)e_a(\xi), \quad u_a^-(\xi) = f_-(\xi)e_a(\xi), \quad a = 1, 2.$$

Then the projected mixed symbol is

$$(M^\mu)_{ab} = \int d\xi f_+(\xi)f_-(\xi) e_a^\dagger(\xi)V^\mu(\xi)e_b(\xi). \quad (47)$$

Therefore

$$\lambda_{\parallel} = I_\theta \Lambda_{\parallel}^{(0)} + \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O. \quad (48)$$

Inside the exact concrete Class II branch,

$$\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O. \quad (49)$$

Using the already-evaluated overlap integrals

$$I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa, \quad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3,$$

the reduced PDE coefficients become

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3, \quad (50)$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} \Lambda_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} \Lambda_O \theta_0 \kappa. \quad (51)$$

XXI. EXACT HONEST CLOSURE STATEMENT

What is now fully closed is the executable extraction protocol:

$$\mathcal{H}_{OO} \rightarrow \lambda_c, \quad \mathcal{H}_{TD}, \mathcal{H}_{DT} \rightarrow \Lambda_E, \Lambda_O.$$

What is also closed at principal order is the source table: only the primitive commutator-square channel contributes to the longitudinal $n^\mu \Sigma_3$ slot, while the K -sector and EO -slot are purely transverse at this order.

What remains honestly open is the final source-by-source differentiation of the complete microscopic TECT Hessian yielding the explicit functions

$$\mathcal{V}_{OO}^\mu(\xi), \quad \mathcal{V}_{TD}^\mu(\xi) + \mathcal{V}_{DT}^\mu(\xi),$$

together with the derivative-resolved transverse realization beyond the naive local JK truncation.

XXII. FINAL SUMMARY

In the exact concrete Class II-compatible branch, the longitudinal channel is Branch B:

$$\Lambda_{\parallel}^{(0)} = 0.$$

Therefore the remaining coefficient point is

$$\lambda_c, \quad \Lambda_E, \quad \Lambda_O.$$

The exact executable extraction is

$$\mathcal{H}_{OO} \rightarrow a_{3;OO}(\xi) = \frac{1}{2} \text{Tr}_{Q_B}(\Sigma_3 n_\mu V_{OO}^\mu(\xi)) \rightarrow \lambda_{c;OO} \equiv \lambda_c,$$

and

$$\mathcal{H}_{TD}, \mathcal{H}_{DT} \rightarrow \mathcal{M}_{TD+DT}^\mu \rightarrow e_0^{(1)}, o_0^{(1)} \rightarrow \Lambda_E, \Lambda_O.$$

At principal order, the source table is:

$$a_3^{(\text{prim})} = \frac{\lambda_c}{2} \theta'(\xi)^3, \quad a_3^{(K)} = 0, \quad a_3^{(EO)} = 0,$$

so the longitudinal source is uniquely the primitive commutator-square channel, while the transverse sector is carried by the mixed transverse blocks.

After overlap insertion,

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O,$$

hence

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3, \quad \alpha = -\frac{\sqrt{3}\pi}{16} \Lambda_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} \Lambda_O \theta_0 \kappa.$$

Therefore the genuine remaining frontier is not structural. It is the literal differentiation and projection of the complete microscopic Hessian blocks

$$\mathcal{V}_{OO}^\mu(\xi), \quad \mathcal{V}_{TD}^\mu(\xi) + \mathcal{V}_{DT}^\mu(\xi).$$

XXIII. EXPLICIT WORKING-BRANCH EXTRACTION OF N_{ij}^a AND (g_c, g_E, g_O) FROM THE CLASS II TECT SECTOR

XXIV. SHORT EXPLANATION

The purpose of this section is not to claim the unconditional full microscopic closure of complete TECT. Rather, the goal is to isolate the strongest currently explicit *working microscopic branch* in which the transport coefficient point can be written algebraically and matched to the reduced graded $2+2$ PDE data.

The exact current mainline remains the graded $2+2$ Hessian extraction problem

$$OO \rightarrow \lambda_c, \quad TD + DT \rightarrow (\Lambda_E, \Lambda_O),$$

with

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O$$

in the present concrete Branch-B realization.

Thus the present section should be read as an explicit Class II *working-branch realization* of that coefficient point, not as the final unconditional microscopic completion of complete TECT.

XXV. 1. CLASS II WORKING MICROSCOPIC BRANCH

Introduce a heavy internal vector mediator X_i^a coupled to the refined locked TECT sector:

$$L_{\text{mic}} = L_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi]. \quad (52)$$

Since $M_X^2 \gg 1$, the mediator can be integrated out algebraically at quadratic order:

$$X_i^a = -\frac{1}{M_X^2} (\alpha_X J_i^a + \beta_X K_i^a). \quad (53)$$

Substituting back gives the exact quadratic Schur correction

$$\Delta_X^{(2)} = -\frac{1}{M_X^2} (\alpha_X J_i^a + \beta_X K_i^a) (\alpha_X J_i^a + \beta_X K_i^a). \quad (54)$$

Equivalently,

$$\Delta_X^{(2)} = -\frac{\alpha_X^2}{M_X^2} J J - \frac{\alpha_X \beta_X}{M_X^2} (J K + K J) - \frac{\beta_X^2}{M_X^2} K K. \quad (55)$$

Therefore, if one wishes to keep only the mixed current-current piece, one must say explicitly that one is working in the reduced mixed-channel truncation:

$$\Delta_X^{(2)} \rightsquigarrow -\frac{\alpha_X \beta_X}{M_X^2} (J K + K J). \quad (56)$$

Writing only

$$-\frac{\alpha_X \beta_X}{M_X^2} J K$$

without this caveat is too strong.

XXVI. 2. WORKING-BRANCH TENSOR EXTRACTION

Linearizing the Class II working branch around the locked 12-mode BCC background, one obtains a shell-enriched gradient-linear transport coupling of the form

$$\delta L_{\text{enr}} = \mathbf{N}_{ij}^a (\partial_i \phi_0) (-i \partial_j) \otimes s^a. \quad (57)$$

At the working-branch level, the induced tensor can be parametrized as

$$\delta \mathbf{N}_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a \mathcal{C}_{ij}^{ab}, \quad (58)$$

provided one makes explicit that this is a *branch-level reduced tensor identification*, not yet the final complete-TEXT Hessian-derived universal formula.

In the shell-adapted orthonormal frame $(\hat{n}, e_1, e_2)$, write the irreducible decomposition

$$\mathbf{N}_{ij}^3 = g_c \hat{n}_i \hat{n}_j, \quad (59)$$

$$\mathbf{N}_{ij}^1 = g_E \hat{n}_i e_{1j} - g_O \hat{n}_i e_{2j}, \quad (60)$$

$$\mathbf{N}_{ij}^2 = g_E \hat{n}_i e_{2j} + g_O \hat{n}_i e_{1j}. \quad (61)$$

Here the notation (g_c, g_E, g_O) is chosen to match directly the graded Branch-B transport point.

XXVII. 3. RELATION TO THE ACTUAL GRADED $2+2$ MAINLINE

The true current mainline is not the tensor ansatz by itself. It is the background-dependent Hessian extraction:

$$\boxed{\mathcal{H}_{OO} \rightarrow \lambda_c, \quad \mathcal{H}_{TD}, \mathcal{H}_{DT} \rightarrow \Lambda_E, \Lambda_O.} \quad (62)$$

Inside the exact concrete Class II-compatible branch, the present lowest-order solved result is

$$\boxed{\Lambda_{\parallel}^{(0)} = 0,} \quad (63)$$

so the longitudinal channel is Branch B, not Branch A. Accordingly,

$$\boxed{\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O.} \quad (64)$$

Therefore the working-branch tensor coefficients must be understood as a branch-level realization of the identifications

$$\boxed{g_c \leftrightarrow \lambda_c, \quad g_E \leftrightarrow \Lambda_E, \quad g_O \leftrightarrow \Lambda_O,} \quad (65)$$

up to the already-isolated overlap normalization layer.

XXVIII. 4. EXACT NORMALIZATION LAYER

The overlap-completed reduced coefficients are

$$\boxed{\lambda_{\parallel} = C_c g_c, \quad \alpha = C_E g_E, \quad \beta = C_O g_O.} \quad (66)$$

Equivalently,

$$\boxed{g_c = \frac{\lambda_{\parallel}}{C_c}, \quad g_E = \frac{\alpha}{C_E}, \quad g_O = \frac{\beta}{C_O}.} \quad (67)$$

Only after explicit overlap evaluation may one specialize to

$$\boxed{C_c = \frac{1}{2} I_{\theta'^3}, \quad C_E = I_{\theta}^{(E)}, \quad C_O = I_{\theta}^{(O)}.} \quad (68)$$

If the locked branch symmetry enforces equal transverse overlap, one may then set

$$I_{\theta}^{(E)} = I_{\theta}^{(O)} =: I_{\theta}.$$

Thus the reduced Branch-B coefficients become

$$\boxed{\lambda_{\parallel} = \frac{g_c}{2} I_{\theta'^3}, \quad \alpha = g_E I_{\theta}, \quad \beta = g_O I_{\theta}.} \quad (69)$$

XXIX. 5. EXPLICIT WORKING-BRANCH COEFFICIENT POINT

At the branch level, the Class II reduction yields the coefficient point

$$\boxed{c_{\parallel} = \frac{\alpha_X^2}{M_X^2} j^2, \quad c_E = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}, \quad c_O = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)}.} \quad (70)$$

Therefore a natural working-branch identification is

$$\boxed{g_c \sim c_{\parallel}, \quad g_E \sim c_E, \quad g_O \sim c_O.} \quad (71)$$

In the canonical normalization where the proportionality constants are absorbed into the overlap layer, this may be written as

$$\boxed{g_c = \frac{\alpha_X^2}{M_X^2} j^2, \quad g_E = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}, \quad g_O = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)}}. \quad (72)$$

However, this must be labeled explicitly as a *working-branch canonical normalization*, not as the unconditional exact coefficient formula of the complete microscopic TECT action.

XXX. 6. FINITE PROJECTED EXTRACTION PIPELINE

The finite extraction protocol may be written as

$$\boxed{L_{\text{mic}} \rightarrow \Gamma^{(2)}[\bar{\Phi}] \rightarrow \mathcal{H}_{\text{Bloch}}(k) \rightarrow H_{\text{eff}}(k) \rightarrow K_i = \partial_{k_i} H_{\text{eff}}(k) \big|_{k_*} \rightarrow P_* K_i P_* \rightarrow (\lambda_{\parallel}, \alpha, \beta)}. \quad (73)$$

At the projected level,

$$\lambda_{\parallel} = \frac{1}{2} \text{Tr}(\sigma_3 P_* K_{\parallel} P_*), \quad (74)$$

$$\alpha = \frac{1}{4} \sum_{a=1}^2 \text{Tr}(\sigma_a P_* (e_{ai} K_i) P_*), \quad (75)$$

$$\beta = \frac{1}{4} \left[\text{Tr}(\sigma_2 P_* (e_{1i} K_i) P_*) - \text{Tr}(\sigma_1 P_* (e_{2i} K_i) P_*) \right]. \quad (76)$$

This is the exact finite projected extraction layer. What remains open is the literal computation of the projected derivative blocks from the final complete microscopic operator.

XXXI. 7. HONEST NONDEGENERACY STATEMENT

The most accurate nondegeneracy criterion is not an informal determinant statement for an unspecified matrix M , but the positive-definite symbol condition

$$\boxed{G_{ij} > 0 \iff \lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0)}. \quad (77)$$

Equivalently,

$$\boxed{G_{ij} > 0 \iff g_c \neq 0, \quad (g_E, g_O) \neq (0, 0)}. \quad (78)$$

Thus the precise finite working-branch test is

$$\boxed{g_c \neq 0, \quad (g_E, g_O) \neq (0, 0)}. \quad (79)$$

XXXII. 8. HONEST STATUS

What is already closed:

1. the graded $2 + 2$ bypass is the correct mainline;
2. the exact concrete Class II-compatible lowest-order branch is Branch B:

$$\Lambda_{\parallel}^{(0)} = 0;$$

3. the overlap-completed reduced PDE dictionary is explicit:

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O;$$

4. the finite projected extraction protocol

$$\Gamma^{(2)} \rightarrow K_i \rightarrow P_* K_i P_* \rightarrow (\lambda_{\parallel}, \alpha, \beta)$$

is explicit;

5. the explicit Class II working branch provides nontrivial witness seeds for

$$g_c, \quad g_E, \quad g_O.$$

What remains honestly open:

1. derive the actual full microscopic transport kernel from the complete TECT Hessian rather than from a reduced working ansatz;
2. compute the actual microscopic values of

$$\lambda_c, \quad \Lambda_E, \quad \Lambda_O;$$

3. verify directly that the complete TECT defect background realizes

$$\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).$$

Therefore the correct conclusion is not that the full microscopic TECT Dirac problem is completely closed. The correct conclusion is that the problem has been reduced to a finite and explicit coefficient-extraction computation, and that the concrete Class II working branch already realizes a nontrivial witness sector.

XXXIII. FINAL SUMMARY

The proposed section is mathematically useful, but only after a strict honesty correction. The heavy Class II mediator does not generate only a single mixed term

$$-\frac{\alpha_X \beta_X}{M_X^2} JK;$$

rather, the exact quadratic Schur correction is

$$-\frac{\alpha_X^2}{M_X^2} JJ - \frac{\alpha_X \beta_X}{M_X^2} (JK + KJ) - \frac{\beta_X^2}{M_X^2} KK.$$

The current exact mainline is the graded 2 + 2 Branch-B extraction:

$$OO \rightarrow \lambda_c, \quad TD + DT \rightarrow (\Lambda_E, \Lambda_O), \quad \Lambda_{\parallel}^{(0)} = 0.$$

Hence

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O.$$

The transport-side coefficients satisfy the exact normalization layer

$$\lambda_{\parallel} = C_c g_c, \quad \alpha = C_E g_E, \quad \beta = C_O g_O,$$

so only after overlap specialization may one write

$$\lambda_{\parallel} = \frac{g_c}{2} I_{\theta'^3}, \quad \alpha = g_E I_{\theta}, \quad \beta = g_O I_{\theta}.$$

Inside the explicit Class II working branch, one may use the canonical branch-level coefficient point

$$g_c = \frac{\alpha_X^2}{M_X^2} j^2, \quad g_E = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}, \quad g_O = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)},$$

but this must be labeled as a working-branch realization, not yet the unconditional exact formula of complete TECT. The precise finite nondegeneracy criterion is

$$G_{ij} > 0 \iff \lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0),$$

equivalently

$$G_{ij} > 0 \iff g_c \neq 0, \quad (g_E, g_O) \neq (0, 0).$$

Therefore the honest current status is:

the finite extraction protocol is explicit,

the concrete Class II working branch provides a nontrivial witness sector,

but

the full microscopic closure of complete TECT still requires the explicit Hessian extraction of $\lambda_c, \Lambda_E, \Lambda_O$.

XXXIV. MINIMAL WORKING MICROSCOPIC TECT PDE CONSISTENT WITH THE CLOSED CONSTRAINTS

XXXV. SHORT EXPLANATION

At the present stage, we do *not* claim that the full complete TECT action has already been derived from absolute first principles. What is currently justified is weaker and more precise:

1. the reduced infrared Dirac-type PDE form is already fixed;
2. the plain scalar Bragg route is insufficient inside the currently safe scalar library;
3. enriched microscopic carrier structure is therefore necessary for the current full-PDE program;
4. the strongest explicit currently available microscopic realization is the refined locked TECT branch coupled to a heavy Class II mediator;
5. the remaining frontier is the explicit coefficient extraction

$$L(k) \rightarrow K_i \rightarrow P_* K_i P_* \rightarrow (\lambda_{\parallel}, \alpha, \beta),$$

equivalently

$$(\Lambda_{\parallel}^{(0)}, \lambda_c, \Lambda_E, \Lambda_O),$$

from the full background-dependent TECT Hessian.

Therefore the correct goal of this section is not to declare the final complete microscopic TECT PDE already finished, but to record the minimal *working microscopic PDE* and the minimal *final-complete action ansatz* that are consistent with all already-closed constraints.

XXXVI. 1. WHAT IS ALREADY CLOSED AND WHAT IS NOT

The following points are already structurally closed:

$$\boxed{A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2).} \quad (80)$$

In the current exact concrete Class II-compatible realization, the sharp reduced form is

$$\boxed{A_1^{(B)}(p) = \frac{\lambda_c}{2} I_{\theta^3} p_{\parallel} \sigma_3 + I_{\theta} \Lambda_E (p_1 \sigma_1 + p_2 \sigma_2) + I_{\theta} \Lambda_O (-p_2 \sigma_1 + p_1 \sigma_2).} \quad (81)$$

What is *not* yet fully closed is the derivation of the actual coefficient point of the complete microscopic TECT theory:

$$\boxed{(\Lambda_{\parallel}^{(0)}, \lambda_c, \Lambda_E, \Lambda_O)} \quad (82)$$

or equivalently

$$\boxed{(\lambda_{\parallel}, \alpha, \beta)} \quad (83)$$

from the full background-dependent TECT Hessian.

Therefore the exact current statement is:

$$\boxed{\text{the reduced infrared PDE form is closed,}} \quad (84)$$

$$\boxed{\text{the strongest explicit currently available microscopic realization is the refined locked + Class II working branch,}} \quad (85)$$

but

$$\boxed{\text{the final complete microscopic TECT action is not yet fully derived unconditionally.}} \quad (86)$$

XXXVII. 2. MINIMAL MICROSCOPIC WORKING BRANCH

The strongest explicit microscopic branch currently available is

$$\boxed{L_{\text{mic}} = L_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \Psi].} \quad (87)$$

Here:

- $\Psi(x, t) \in \mathbb{C}^3$ is the condensate triplet field;
- $P(x, t) = z(x, t)z(x, t)^{\dagger}$, $z^{\dagger}z = 1$, encodes the projector/orientational sector;
- $X_i^a(x, t)$ is a heavy Class II internal carrier field;
- $L_{\Psi}^{\text{ref}}[\Psi, P]$ denotes the refined locked microscopic TECT sector.

We do *not* claim that (87) is already the final complete TECT action. It is the strongest explicit working microscopic branch in which the full-PDE derivation can currently be pushed concretely.

XXXVIII. 3. MINIMAL CONDENSATE CORE

Inside the minimal local isotropic shell-forming truncation, the condensate core contains

$$S_0[\Psi] = \int dt d^3x \left[\frac{1}{2} \partial_t \Psi^\dagger \partial_t \Psi - \frac{\mu^2}{2} \Psi^\dagger \Psi + \frac{Z}{2} \partial_i \Psi^\dagger \partial_i \Psi - \frac{Y}{2} (\nabla^2 \Psi)^\dagger (\nabla^2 \Psi) - \frac{g}{4} (\Psi^\dagger \Psi)^2 \right]. \quad (88)$$

Its Euler–Lagrange equation is

$$\partial_t^2 \Psi = -\mu^2 \Psi + Z \nabla^2 \Psi - Y \nabla^4 \Psi - g(\Psi^\dagger \Psi) \Psi. \quad (89)$$

This is the already-fixed minimal shell-forming condensate PDE. What the current full-PDE program adds is the enriched carrier structure required to generate the shell-relevant transport coefficients.

XXXIX. 4. HEAVY CLASS II MEDIATOR ELIMINATION

The heavy mediator satisfies the algebraic equation

$$X_i^a = -\frac{1}{M_X^2} \left(\alpha_X J_i^a[\Psi] + \beta_X K_i^a[P, \Psi] \right). \quad (90)$$

Substituting back gives the exact quadratic Schur correction

$$\Delta_X^{(2)} = -\frac{1}{M_X^2} (\alpha_X J_i^a + \beta_X K_i^a) (\alpha_X J_i^a + \beta_X K_i^a). \quad (91)$$

Equivalently,

$$\Delta_X^{(2)} = -\frac{\alpha_X^2}{M_X^2} J J - \frac{\alpha_X \beta_X}{M_X^2} (J K + K J) - \frac{\beta_X^2}{M_X^2} K K. \quad (92)$$

Thus it is not correct to retain only a single mixed term

$$-\frac{\alpha_X \beta_X}{M_X^2} J_i^a K_i^a$$

unless one explicitly states that one is working in a reduced mixed-channel truncation.

XL. 5. WORKING MICROSCOPIC CONDENSATE EQUATION

Varying the working branch with respect to $\Psi^\dagger$ yields the condensate equation

$$\partial_t^2 \Psi = -\mu^2 \Psi + Z \nabla^2 \Psi - Y \nabla^4 \Psi - g(\Psi^\dagger \Psi) \Psi + \mathcal{F}_\Psi^{\text{ref}} - \frac{\delta \Delta_X^{(2)}}{\delta \Psi^\dagger}. \quad (93)$$

After inserting (91), this may be written more compactly as

$$\partial_t^2 \Psi = -\mu^2 \Psi + Z \nabla^2 \Psi - Y \nabla^4 \Psi - g(\Psi^\dagger \Psi) \Psi + \mathcal{F}_\Psi^{\text{ref}} - \frac{1}{M_X^2} \frac{\delta}{\delta \Psi^\dagger} \left[(\alpha_X J_i^a + \beta_X K_i^a) (\alpha_X J_i^a + \beta_X K_i^a) \right]. \quad (94)$$

If one defines

$$\mathcal{J}_i^a[\Psi, P] := \alpha_X J_i^a[\Psi] + \beta_X K_i^a[P, \Psi], \quad (95)$$

then

$$\partial_t^2 \Psi = -\mu^2 \Psi + Z \nabla^2 \Psi - Y \nabla^4 \Psi - g(\Psi^\dagger \Psi) \Psi + \mathcal{F}_\Psi^{\text{ref}} - \frac{1}{M_X^2} \frac{\delta}{\delta \Psi^\dagger} (\mathcal{J}_i^a \mathcal{J}_i^a). \quad (96)$$

This is a legitimate compact working-branch PDE. However, it should be called the *working microscopic TECT PDE with a heavy Class II carrier*, not the final complete microscopic TECT PDE.

XLI. 6. WHY THE CLASS II BRANCH IS JUSTIFIED BUT NOT YET UNIQUE

What is currently established is:

1. the plain scalar Bragg route is insufficient in the currently safe scalar library;
2. enriched internal carrier structure must therefore be retained;
3. the refined locked + Class II branch is the strongest explicit currently available working microscopic branch.

What is *not* yet established is that the Class II field is the unique logically possible UV completion. Indeed, the minimal UV classification still allows multiple effective (i, a) -carrier classes, of which Class II is the most explicit and currently most useful one.

Therefore the correct language is:

$$\boxed{\text{Class II is the strongest explicit currently available working branch,}} \quad (97)$$

not

$$\boxed{\text{Class II is already proved to be uniquely and absolutely necessitated.}} \quad (98)$$

XLII. 7. RELATION TO THE TRUE CURRENT FRONTIER

The actual remaining frontier is not to postulate a deeper ontology, but to derive the actual transport coefficients from the full background-dependent operator.

The exact finite extraction target is

$$\boxed{L(k) \rightarrow K_i \rightarrow P_* K_i P_* \rightarrow (\lambda_{\parallel}, \alpha, \beta),} \quad (99)$$

equivalently

$$\boxed{\mathcal{H}_{OO} \rightarrow \lambda_c, \quad \mathcal{H}_{TD}, \mathcal{H}_{DT} \rightarrow \Lambda_E, \Lambda_O.} \quad (100)$$

At the current sharpest lowest-order realization, the longitudinal sector is on Branch B:

$$\boxed{\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0,} \quad (101)$$

and therefore

$$\boxed{\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O.} \quad (102)$$

Thus the correct conclusion of the present section is not “Phase I closed from absolute first principles”, but rather:

$$\boxed{\text{the minimal working microscopic PDE and the finite extraction problem are now explicit.}} \quad (103)$$

XLIII. 8. HONEST STATUS

What is already closed:

1. the reduced infrared Dirac-type PDE form;
2. the need for enriched carrier structure beyond the plain scalar route;
3. the strongest explicit current working branch (87);
4. the exact quadratic Schur correction (91)–(92);

5. the finite extraction target (99)–(100).

What remains honestly open:

1. the actual final complete microscopic TECT action;
2. the full background-dependent Hessian extraction of

$$(\Lambda_{\parallel}^{(0)}, \lambda_c, \Lambda_E, \Lambda_O);$$

3. the direct verification that the complete microscopic TECT theory realizes

$$\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0)$$

on the physically relevant defect background.

XLIV. FINAL SUMMARY

The proposed section becomes mathematically correct only after a strict honesty correction.

It is not yet justified to say that the full complete microscopic TECT action has been derived from absolute first principles, nor that the heavy Class II mediator has been proved to be the unique possible completion.

What is currently justified is the following:

$$\boxed{\text{The reduced infrared Dirac-type PDE form is already fixed.}} \quad (104)$$

$$\boxed{\text{The refined locked + heavy Class II sector is the strongest explicit currently available microscopic working branch.}} \quad (105)$$

$$\boxed{\text{After eliminating the heavy mediator, the exact quadratic correction is } -\frac{1}{M_X^2}(\alpha_X J + \beta_X K)(\alpha_X J + \beta_X K), \text{ containing } JJ, J} \quad (106)$$

$$\boxed{\text{The true remaining frontier is the explicit coefficient extraction } L(k) \rightarrow K_i \rightarrow P_* K_i P_* \rightarrow (\lambda_{\parallel}, \alpha, \beta),} \quad (107)$$

equivalently

$$\mathcal{H}_{OO} \rightarrow \lambda_c, \quad \mathcal{H}_{TD}, \mathcal{H}_{DT} \rightarrow \Lambda_E, \Lambda_O.$$

In the current sharpest lowest-order realization,

$$\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0,$$

so the reduced Branch-B dictionary is

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O.$$

Therefore the correct present conclusion is:

$$\boxed{\text{the minimal working microscopic PDE is explicit,}}$$

$$\boxed{\text{the finite extraction problem is explicit,}}$$

but

$$\boxed{\text{the final complete microscopic closure of TECT is still open.}}$$

XLV. TERM-BY-TERM *OO*-BLOCK LONGITUDINAL AUDIT

XLVI. GOAL

The current sharpest blockwise longitudinal problem is to determine the primitive *OO*-block coefficient

$$\ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2},$$

equivalently to decide whether

$$\lambda_{c;OO} \neq 0.$$

At the present stage, this is the only remaining transport-level longitudinal question in the current explicit library.

XLVII. 1. BACKGROUND HESSIAN AND FIRST-ORDER TRANSPORT LAYER

Let the fluctuation field be decomposed as

$$\varphi = (a, \nu, \zeta),$$

with

$$a \sim T, \quad \nu \sim O, \quad \zeta \sim D.$$

Then the background Hessian has the exact block form

$$\Gamma^{(2)}[\bar{\Phi}_{\min}] = \begin{pmatrix} \mathcal{H}_{TT} & \mathcal{H}_{TO} & \mathcal{H}_{TD} \\ \mathcal{H}_{OT} & \mathcal{H}_{OO} & \mathcal{H}_{OD} \\ \mathcal{H}_{DT} & \mathcal{H}_{DO} & \mathcal{H}_{DD} \end{pmatrix}. \quad (108)$$

Each block admits the worldvolume derivative expansion

$$\mathcal{H}_{AB} = \mathcal{H}_{AB}^{(0)}(\xi) - i \mathcal{V}_{AB}^{\mu}(\xi) \partial_{y^{\mu}} + \mathcal{W}_{AB}^{\mu\nu}(\xi) \partial_{y^{\mu}} \partial_{y^{\nu}} + \cdots. \quad (109)$$

The shell-relevant first-order transport content is therefore carried by

$$\mathcal{V}_{AB}^{\mu}(\xi).$$

XLVIII. 2. CANONICAL *OO*-BLOCK DECOMPOSITION

In the current explicit library, the orientational diagonal block is recalled in the form

$$\boxed{\mathcal{H}_{OO} = -Z_{O,k} \partial^2 + \mathcal{M}_{OO}^{\text{pot}}(\xi) + Y_k \mathcal{K}_{OO}^{\nabla \text{ani}}(\xi, \partial) + \mathcal{T}_{OO}(\xi, \partial_y)}. \quad (110)$$

This decomposition is to be read term-by-term as follows:

1. the kinetic piece

$$\mathcal{H}_{OO}^{\text{kin}} := -Z_{O,k} \partial^2;$$

2. the local potential/background-mass piece

$$\mathcal{H}_{OO}^{\text{pot}} := \mathcal{M}_{OO}^{\text{pot}}(\xi);$$

3. the derivative-anisotropy piece

$$\mathcal{H}_{OO}^{\nabla \text{ani}} := Y_k \mathcal{K}_{OO}^{\nabla \text{ani}}(\xi, \partial);$$

4. the explicit background-induced first-order transport piece

$$\mathcal{H}_{OO}^{\text{tr}} := \mathcal{T}_{OO}(\xi, \partial_y).$$

We now audit these four terms against the longitudinal target channel.

XLIX. 3. LONGITUDINAL PROJECTOR AND COEFFICIENT EXPANSION

Project to the tangent doublet bundle

$$Q_B(\xi) = (\mathbb{C}z_B(\xi))^\perp, \quad \dim_{\mathbb{C}} Q_B = 2,$$

and let Σ_a be the Pauli basis on Q_B .

Define the OO -block longitudinal projector

$$a_{3;OO}(\xi) := \frac{1}{2} \text{Tr}_{Q_B}(\Sigma_3 n_\mu \Pi_Q \mathcal{V}_{OO}^\mu(\xi) \Pi_Q). \quad (111)$$

Its branch-sensitive expansion is

$$a_{3;OO}(\xi) = \ell_{0;OO}^{(1)} \theta'(\xi) + \ell_{0;OO}^{(3)} \theta'(\xi)^3 + a_{3;OO}^{\text{higher}}(\xi). \quad (112)$$

The coefficient identification is

$$\Lambda_{\parallel;OO}^{(0)} = \ell_{0;OO}^{(1)}, \quad \frac{\lambda_{c;OO}}{2} = \ell_{0;OO}^{(3)}. \quad (113)$$

L. 4. PRIMITIVE LONGITUDINAL SEED

For the explicit real CP^2 defect representative, define

$$\mathcal{T}_\xi := i[P_B, \partial_\xi P_B].$$

Then the established tangent identities are

$$\Pi_Q \mathcal{T}_\xi \Pi_Q = 0, \quad (114)$$

but

$$S_\xi := (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'(\xi)^2 \Sigma_3. \quad (115)$$

Therefore the primitive longitudinal transport seed is

$$\boxed{V_{(\parallel),\text{prim}}^\mu(\xi) = \theta'(\xi) S_\xi n^\mu = \frac{1}{2} \theta'(\xi)^3 n^\mu \Sigma_3.} \quad (116)$$

This is exactly the Branch-B longitudinal carrier.

LI. 5. TERM-BY-TERM ELIMINATION INSIDE THE OO BLOCK

We now inspect each term in (110).

A. 5.1 Kinetic term

The kinetic piece

$$-Z_{O,k} \partial^2$$

is second order in derivatives and does not by itself produce an odd first-order worldvolume transport tensor of the target form

$$\theta'(\xi)^m n^\mu \Sigma_3.$$

Hence it does not contribute to the principal OO -block longitudinal coefficient:

$$a_{3;OO}^{\text{kin}}(\xi) = 0 \quad \text{at the principal first-order transport level.} \quad (117)$$

B. 5.2 Potential term

The background potential term

$$\mathcal{M}_{OO}^{\text{pot}}(\xi)$$

is local and contains no worldvolume derivative. Therefore it contributes only to zeroth-order local mass/background structure and not to the first-order transport kernel:

$$a_{3;OO}^{\text{pot}}(\xi) = 0 \quad \text{at the principal transport level.} \quad (118)$$

C. 5.3 Derivative-anisotropy term

The anisotropy piece

$$Y_k \mathcal{K}_{OO}^{\nabla \text{ani}}(\xi, \partial)$$

must be checked more carefully because it contains derivatives.

However, in the solved lowest-order realization, the reflection-symmetry and projector-commutator audit shows that the linear Branch-A longitudinal monomial

$$\theta'(\xi) n^\mu \Sigma_3$$

is forbidden, while the first allowed longitudinal monomial is

$$\theta'(\xi)^3 n^\mu \Sigma_3.$$

Therefore

$$\boxed{\ell_0^{(1)} = 0.} \quad (119)$$

In particular, the derivative-anisotropy term does not generate a nonzero Branch-A coefficient in the current solved lowest-order class:

$$a_{3;OO}^{\nabla \text{ani}}(\xi) \text{ does not contribute a nonzero } \ell_{0;OO}^{(1)}. \quad (120)$$

D. 5.4 Transport term

After eliminating the kinetic, potential, and Branch-A-forbidden contributions, the only surviving leading longitudinal carrier inside the OO block is

$$\boxed{\mathcal{T}_{OO}.} \quad (121)$$

Its shell-relevant traceless longitudinal projection is precisely the primitive channel (116). Hence

$$\boxed{\mathcal{T}_{OO} \Rightarrow \ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2}.} \quad (122)$$

LII. 6. FINAL OO -BLOCK VERDICT

Combining the previous steps, the global OO -block longitudinal result is

$$\boxed{\ell_0^{(1)} = 0, \quad \ell_0^{(3)} = \ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2}.} \quad (123)$$

Equivalently,

$$\boxed{a_{3;OO}(\xi) = \frac{\lambda_{c;OO}}{2} \theta'(\xi)^3 + a_{3;OO}^{\text{higher}}(\xi).} \quad (124)$$

Thus, in the current solved lowest-order class, the OO -block does not realize Branch A and carries only the primitive Branch-B longitudinal seed.

LIII. 7. REDUCED LONGITUDINAL COEFFICIENT

After shell localization,

$$\lambda_{\parallel} = I_{\theta'^3} \ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2} I_{\theta'^3}. \quad (125)$$

Using

$$I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3, \quad (126)$$

we obtain

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \lambda_{c;OO} \theta_0^3 \kappa^3. \quad (127)$$

LIV. 8. PRACTICAL FINITE TEST

Therefore the entire Gate-2 longitudinal problem in the current explicit library reduces to the single finite extraction

$$\lambda_{c;OO} \neq 0 \text{ ?} \quad (128)$$

If yes, then

$$\lambda_{\parallel} \neq 0$$

immediately follows because $I_{\theta'^3} \neq 0$.

LV. 9. HONEST STATUS

What is proved at the present stage is:

1. the OO -block is the natural primitive longitudinal candidate block;
2. the linear Branch-A coefficient vanishes in the solved lowest-order realization:

$$\ell_0^{(1)} = 0;$$

3. the only surviving leading OO -block longitudinal carrier is $\mathcal{T}_{OO}$;
4. therefore

$$\ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2}$$

is the sharp current transport-level target.

What is still not yet fully inserted is the literal term-by-term evaluation of the complete final microscopic TECT operator that yields the numerical/symbolic value of $\lambda_{c;OO}$. So the frontier is no longer structural. It is the explicit finite extraction $\lambda_{c;OO} \neq 0$.

LVI. FINAL SUMMARY

The OO -block longitudinal audit is now reduced to a single primitive transport coefficient. The exact orientational diagonal block is organized as

$$\mathcal{H}_{OO} = -Z_{O,k}\partial^2 + \mathcal{M}_{OO}^{\text{pot}}(\xi) + Y_k \mathcal{K}_{OO}^{\nabla\text{ani}}(\xi, \partial) + \mathcal{T}_{OO}(\xi, \partial_y).$$

In the solved lowest-order realization,

$$\Lambda_{\parallel}^{(0)} = 0, \quad \ell_0^{(1)} = 0,$$

because the linear longitudinal monomial

$$\theta'(\xi) n^\mu \Sigma_3$$

is forbidden, while the first allowed longitudinal traceless seed is

$$V_{(\parallel),\text{prim}}^\mu(\xi) = \frac{1}{2} \theta'(\xi)^3 n^\mu \Sigma_3.$$

Hence the kinetic term, the local potential term, and the Branch-A-forbidden linear anisotropy sector do not provide the leading longitudinal carrier. The only surviving leading OO -block source is

$$\mathcal{T}_{OO},$$

and therefore

$$\ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2}.$$

After shell localization,

$$\lambda_{\parallel} = \frac{\lambda_{c;OO}}{2} I_{\theta^3} = -\frac{3\sqrt{3}\pi}{512} \lambda_{c;OO} \theta_0^3 \kappa^3.$$

Therefore the entire current Gate-2 longitudinal problem is reduced to the single finite question

$$\boxed{\lambda_{c;OO} \neq 0 ?}.$$

What remains open is not the structure of the argument. It is the explicit insertion of the final complete microscopic TECT operator into the OO -block transport extraction.

LVII. SCHUR-CORRECTED MICROSCOPIC SOURCE SHEET FOR THE OO -BLOCK LONGITUDINAL CARRIER

LVIII. SHORT EXPLANATION

At the present stage, the longitudinal carrier problem is no longer a broad structural classification problem. It has reduced to the explicit source audit of the OO -block transport operator.

What is already structurally fixed is:

1. the heavy Class II correction decomposes into

$$JJ, \quad JK, \quad KK;$$

2. the shell-adapted leading branch has

$$J \text{ purely longitudinal,} \quad K \text{ purely transverse;}$$

3. therefore the primitive longitudinal Σ_3 -singlet can survive only through the JJ -sector at principal shell order.

What is not yet fully frozen is the exact convention-level coupling prefactor of the resulting longitudinal coefficient, because different working notes currently use two notationally inequivalent normalizations.

LIX. 1. STARTING POINT: EXACT SCHUR DECOMPOSITION

The heavy-mediated correction to the effective microscopic operator is

$$\Delta_X = -\frac{1}{M_X^2}(\alpha_X \mathcal{J} + \beta_X \mathcal{K})(\alpha_X \mathcal{J} + \beta_X \mathcal{K})^\dagger. \quad (129)$$

Hence the transport contribution decomposes channelwise as

$$\Delta_X = \Delta_X^{JJ} + \Delta_X^{JK} + \Delta_X^{KK}, \quad (130)$$

with

$$\Delta_X^{JJ} = -\frac{\alpha_X^2}{M_X^2} \mathcal{J} \mathcal{J}^\dagger, \quad (131)$$

$$\Delta_X^{JK} = -\frac{\alpha_X \beta_X}{M_X^2} (\mathcal{J} \mathcal{K}^\dagger + \mathcal{K} \mathcal{J}^\dagger), \quad (132)$$

$$\Delta_X^{KK} = -\frac{\beta_X^2}{M_X^2} \mathcal{K} \mathcal{K}^\dagger. \quad (133)$$

Thus the OO -block first-order transport term may be written as

$$\mathcal{T}_{OO} = \mathcal{T}_{OO}^{(\text{ref})} + \mathcal{T}_{OO}^{(JJ)} + \mathcal{T}_{OO}^{(JK)} + \mathcal{T}_{OO}^{(KK)}. \quad (134)$$

LX. 2. SHELL-ADAPTED LEADING BRANCH FORMS

At the current explicit leading shell level, the intervalley tensors are taken in the form

$$\mathbf{J}_{i,r}^a(G_*; 0) = \hat{n}_i j^a(\Sigma_3)_r + \mathbf{J}_{i,r}^{a,\text{null}}, \quad (135)$$

$$\mathbf{K}_{i,r}^a(G_*; 0) = \hat{n}_i k^a \left[\Lambda_E^{(0)}(\Sigma_1)_r + \Lambda_O^{(0)}(\Sigma_2)_r \right] + \mathbf{K}_{i,r}^{a,\text{null}}. \quad (136)$$

Therefore, at principal shell order,

$$J \sim \Sigma_3, \quad K \sim \Lambda_E^{(0)} \Sigma_1 + \Lambda_O^{(0)} \Sigma_2.$$

LXI. 3. LONGITUDINAL PROJECTOR

Project to the tangent doublet space Q_B and define the OO -block longitudinal scalar

$$a_{3;OO}(\xi) = \frac{1}{2} \text{Tr}_{Q_B}(\Sigma_3 n_\mu V_{OO}^\mu(\xi)), \quad V_{OO}^\mu := \Pi_Q \mathbb{H}_{1,OO}^\mu \Pi_Q. \quad (137)$$

Its branch-sensitive expansion is

$$a_{3;OO}(\xi) = \ell_{0;OO}^{(1)} \theta'(\xi) + \ell_{0;OO}^{(3)} \theta'(\xi)^3 + \dots, \quad \ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2}. \quad (138)$$

In the solved current lowest-order realization,

$$\ell_{0;OO}^{(1)} = 0, \quad (139)$$

so the actual longitudinal target is the primitive Branch-B coefficient

$$\ell_{0;OO}^{(3)}.$$

LXII. 4. SOURCE-BY-SOURCE LONGITUDINAL AUDIT

Decompose

$$a_{3;OO} = a_{3;OO}^{(\text{ref})} + a_{3;OO}^{(JJ)} + a_{3;OO}^{(JK)} + a_{3;OO}^{(KK)}. \quad (140)$$

A. 4.1 Refined locked core

The refined locked core may in principle contribute background-dependent transport terms, but in the current solved library it does not provide an allowed principal Branch-A longitudinal seed. Thus, at the sharpest present stage, it is not the distinguished carrier of the primitive odd longitudinal singlet.

So the current longitudinal frontier is not a general “refined core” question. It is the explicit primitive source audit of the Schur-generated carrier channels.

B. 4.2 JJ -sector

Because J is purely shell-normal and aligned with Σ_3 , the JJ -sector is axial only at principal shell order. Equivalently,

$$\boxed{\lambda_{\parallel}^{(JJ)} \propto m_3, \quad \alpha^{(JJ)} = 0, \quad \beta^{(JJ)} = 0.} \quad (141)$$

Hence

$$\boxed{a_{3;OO}^{(JJ)}(\xi) = \frac{\lambda_c^{(JJ)}}{2} \theta'(\xi)^3 + \dots} \quad (142)$$

is allowed and is the unique principal longitudinal channel.

C. 4.3 JK -sector

At the naive local Hermitian level,

$$\Sigma_3(\Lambda_E \Sigma_1 + \Lambda_O \Sigma_2) + (\Lambda_E \Sigma_1 + \Lambda_O \Sigma_2) \Sigma_3 = 0$$

by Pauli anticommutation. At the derivative-resolved shell level, the mixed sector rotates into transverse Pauli directions, since

$$\Sigma_3 \Sigma_1 = i \Sigma_2, \quad \Sigma_3 \Sigma_2 = -i \Sigma_1.$$

Therefore the mixed channel feeds the transverse coefficients only:

$$\boxed{c_{\parallel}^{(JK)} = 0 \quad \text{at principal shell order.}} \quad (143)$$

Hence

$$\boxed{a_{3;OO}^{(JK)}(\xi) = 0 \quad \text{at principal longitudinal shell order.}} \quad (144)$$

D. 4.4 KK -sector

Since the K -sector itself carries no leading Σ_3 -singlet component, its quadratic self-channel remains in the transverse family at the shell-relevant leading order. Therefore

$$\boxed{c_{\parallel}^{(KK)} = 0 \quad \text{at the primitive shell-relevant longitudinal level,}} \quad (145)$$

and hence

$$\boxed{a_{3;OO}^{(KK)}(\xi) = 0 \quad \text{at principal longitudinal shell order.}} \quad (146)$$

LXIII. 5. PRINCIPAL CLASSIFICATION THEOREM

Collecting the previous results,

$$\boxed{a_{3;OO}(\xi) = a_{3;OO}^{(JJ)}(\xi) + \text{higher/nonleading terms},} \quad (147)$$

with

$$\boxed{a_{3;OO}^{(JJ)}(\xi) = \frac{\lambda_c^{(JJ)}}{2} \theta'(\xi)^3 + \dots, \quad a_{3;OO}^{(JK)} = 0, \quad a_{3;OO}^{(KK)} = 0} \quad (148)$$

at principal longitudinal shell order.

Therefore the unique principal OO -block longitudinal source is

$$\boxed{\mathcal{T}_{OO}^{(JJ)}}. \quad (149)$$

LXIV. 6. REDUCED LONGITUDINAL COEFFICIENT

After shell localization,

$$\boxed{\lambda_{\parallel} = \frac{\lambda_{c;OO}}{2} I_{\theta'^3}, \quad \lambda_{c;OO} \equiv \lambda_c^{(JJ)} \quad \text{at principal shell order.}} \quad (150)$$

Thus the entire current longitudinal carrier problem reduces to the single finite question

$$\boxed{\lambda_c^{(JJ)} \neq 0 ?}. \quad (151)$$

LXV. 7. HONEST CONVENTION MISMATCH

Here one must distinguish two layers.

a. Layer A: structural source classification. This layer is already closed:

JJ is the unique principal longitudinal source, JK , KK do not contribute to $c_{\parallel}$

at principal shell order.

b. Layer B: explicit coupling-prefactor normalization. At this layer, the current note package contains two different working normalizations:

1. a channel-decomposed Bloch-renormalization form

$$\Delta\lambda_{\parallel} = -\frac{\alpha_X^2}{M_X^2} \Lambda_c^{JJ} - \frac{\alpha_X \beta_X}{M_X^2} \Lambda_c^{JK} - \frac{\beta_X^2}{M_X^2} \Lambda_c^{KK},$$

which reduces structurally to a JJ -only longitudinal channel;

2. a one-mediator canonical shell-projection form

$$\lambda_{c;OO} = -\frac{\alpha_X \beta_X}{M_X^2} C_{\parallel}.$$

These two expressions are not yet manifestly identical at the notation level. Therefore the exact coupling prefactor of $\lambda_{c;OO}$ is not yet fully frozen across all working-note conventions.

What is robustly true is only the convention-free statement:

$$\boxed{\text{the principal longitudinal source is unique and localized to the allowed longitudinal channel.}} \quad (152)$$

LXVI. 8. PRACTICAL EXECUTION TARGET

The correct next direct execution is therefore not another structural theorem. It is to evaluate the actual projected Bloch normalization entering the unique longitudinal source, for example through either of the equivalent routes:

$$\boxed{V_{OO}^\mu \rightarrow a_{3;OO}(\xi) \rightarrow \lambda_{c;OO},} \quad (153)$$

or

$$\boxed{M_{\parallel} = P_* \partial_{k_{\parallel}} H_{\text{eff}}(k_*) P_* \rightarrow \lambda_{\parallel} = \frac{1}{2} \text{Tr}(\sigma_3 M_{\parallel}).} \quad (154)$$

So the remaining frontier is a finite microscopic insertion problem, not a missing conceptual structure.

LXVII. FINAL SUMMARY

The OO -block longitudinal transport operator admits the Schur-corrected source decomposition

$$\mathcal{T}_{OO} = \mathcal{T}_{OO}^{(\text{ref})} + \mathcal{T}_{OO}^{(JJ)} + \mathcal{T}_{OO}^{(JK)} + \mathcal{T}_{OO}^{(KK)}.$$

At the current shell-adapted leading branch,

$$J \sim \Sigma_3 \quad (\text{longitudinal}), \quad K \sim \Lambda_E^{(0)} \Sigma_1 + \Lambda_O^{(0)} \Sigma_2 \quad (\text{transverse}).$$

Therefore the principal longitudinal source classification is:

$$\boxed{a_{3;OO}^{(JJ)} \neq 0 \text{ is allowed,} \quad a_{3;OO}^{(JK)} = 0, \quad a_{3;OO}^{(KK)} = 0 \quad \text{at principal shell order.}}$$

Equivalently, the primitive OO -block longitudinal coefficient localizes to the JJ -sector:

$$\boxed{\lambda_{c;OO} \equiv \lambda_c^{(JJ)} \quad \text{at principal shell order.}}$$

Hence the reduced longitudinal coefficient is

$$\boxed{\lambda_{\parallel} = \frac{\lambda_{c;OO}}{2} I_{\theta'^3},}$$

and the remaining microscopic longitudinal problem is the single finite test

$$\boxed{\lambda_c^{(JJ)} \neq 0 ? .}$$

What is fully closed:

$$\boxed{\text{the principal source classification } (JJ \text{ only}) \text{ is closed.}}$$

What is not yet fully frozen:

$$\boxed{\text{the exact coupling-prefactor normalization of } \lambda_{c;OO} \text{ across all current note conventions.}}$$

Thus the true next step is not another structural theorem, but the explicit projected Bloch evaluation of the unique longitudinal source.

LXVIII. CONVENTION AUDIT FOR Λ_c^{JJ} VERSUS $C_\parallel$

LXIX. SHORT EXPLANATION

The current note package contains two different longitudinal normalizations.

One normalization is the exact Class-II renormalization of the reduced locked coefficients:

$$\Delta\lambda_\parallel = -\frac{\alpha_X^2}{M_X^2}\Lambda_c^{JJ} - \frac{\alpha_X\beta_X}{M_X^2}\Lambda_c^{JK} - \frac{\beta_X^2}{M_X^2}\Lambda_c^{KK}.$$

The other normalization is the Branch-B one-mediator Bloch calibration:

$$\lambda_\parallel = -\frac{\alpha_X\beta_X}{2M_X^2}C_\parallel I_3.$$

These are not manifestly the same definition. Therefore the correct task is not to assert equality immediately, but to separate what is convention-free from what still depends on the chosen normalization layer.

LXX. 1. EXACT CHANNEL-RENORMALIZATION LAYER

In the exact Class-II renormalization notes, the reduced locked coefficients are split as

$$\lambda_\parallel = \lambda_\parallel^{\text{ref}} + \Delta\lambda_\parallel, \quad \alpha = \alpha^{\text{ref}} + \Delta\alpha, \quad \beta = \beta^{\text{ref}} + \Delta\beta,$$

with

$$\Delta\lambda_\parallel = -\frac{\alpha_X^2}{M_X^2}\Lambda_c^{JJ} - \frac{\alpha_X\beta_X}{M_X^2}\Lambda_c^{JK} - \frac{\beta_X^2}{M_X^2}\Lambda_c^{KK}. \quad (155)$$

The channel overlaps are defined by projected Pauli traces:

$$\Lambda_c^{UV} := \frac{1}{2} \text{Tr}(\sigma_3 P_*(\hat{n}_\ell \mathfrak{D}_\ell^{UV}(k_*))P_*), \quad (U, V) \in \{(J, J), (J, K), (K, K)\}. \quad (156)$$

So Λ_c^{JJ} is a transport-side projected Bloch-derivative invariant attached specifically to the JJ channel.

LXXI. 2. PRINCIPAL SOURCE CLASSIFICATION

Independently of normalization, the current source-classification notes imply:

$$JJ \text{ alone probes only the longitudinal-type shell singlet,} \quad (157)$$

while

$$JJ \text{ alone does not generate } (g_E, g_O), \quad JK/KK \text{ are required for the transverse pair.} \quad (158)$$

Equivalently, at principal longitudinal shell order,

$$\Lambda_c^{JK} = 0, \quad \Lambda_c^{KK} = 0 \quad (\text{or, more precisely, they do not contribute to the principal allowed longitudinal slot}). \quad (159)$$

Hence the exact channel formula reduces structurally to

$$\Delta\lambda_\parallel^{(\text{med}, \parallel)} = -\frac{\alpha_X^2}{M_X^2}\Lambda_c^{JJ} \quad (160)$$

at principal longitudinal shell order.

This part is convention-free and already closed.

LXXII. 3. BRANCH-B OVERLAP-COMPLETED LAYER

A different family of notes uses the Branch-B reduced dictionary

$$\lambda_{\parallel} = \frac{\lambda_{c;OO}}{2} I_3, \quad I_3 = I_{\theta'^3}. \quad (161)$$

Then, in the one-mediator shell-projected calibration,

$$\lambda_{c;OO} = -\frac{\alpha_X \beta_X}{M_X^2} C_{\parallel}, \quad (162)$$

so that

$$\lambda_{\parallel} = -\frac{\alpha_X \beta_X}{2M_X^2} C_{\parallel} I_3. \quad (163)$$

Moreover,

$$C_{\parallel} = -\frac{M_X^2}{\alpha_X \beta_X I_3} \left[(\hat{n} \cdot \vec{W}_+)_{11} - (\hat{n} \cdot \vec{W}_+)_{22} \right], \quad (164)$$

hence $C_{\parallel}$ is a Bloch-calibration constant extracted from the diagonal difference of a projected shell matrix.

Thus $C_{\parallel}$ lives at a different explicit normalization layer from Λ_c^{JJ} .

LXXIII. 4. WHY THE TWO FORMULAS CANNOT YET BE IDENTIFIED UNCONDITIONALLY

Comparing (160) and (163), one is tempted to write

$$-\frac{\alpha_X^2}{M_X^2} \Lambda_c^{JJ} \stackrel{?}{=} -\frac{\alpha_X \beta_X}{2M_X^2} C_{\parallel} I_3. \quad (165)$$

If this were justified, it would imply

$$C_{\parallel} = \frac{2\alpha_X}{\beta_X I_3} \Lambda_c^{JJ}. \quad (166)$$

However, this step is not yet licensed by the current notes alone. To make (166) rigorous, one would need all of the following:

1. the $\lambda_{\parallel}$ appearing in (160) and (163) must be exactly the same mediated contribution, not one total coefficient and one branch-reduced coefficient;
2. the reference/core contribution must either vanish or be subtracted in the same convention:

$$\lambda_{\parallel}^{\text{ref}} = 0 \quad \text{or} \quad \lambda_{\parallel} \rightarrow \Delta \lambda_{\parallel};$$

3. the JJ -trace object

$$P_*(\hat{n}_{\ell} \mathfrak{D}_{\ell}^{JJ}(k_*)) P_*$$

must be shown explicitly to equal, after overlap completion and basis normalization, the diagonal-difference object

$$(\hat{n} \cdot \vec{W}_+)_{11} - (\hat{n} \cdot \vec{W}_+)_{22};$$

4. the same shell overlap normalization I_3 and the same sign conventions must be used on both sides.

The currently uploaded notes do not yet contain this explicit identification chain. Therefore (166) is at present only a *candidate conversion formula*, not a proved identity.

LXXIV. 5. WHAT IS ACTUALLY PROVED RIGHT NOW

The following statement is rigorous and convention-free:

$$\boxed{\text{At principal longitudinal shell order, the unique allowed source is the } JJ \text{ channel.}} \quad (167)$$

Hence the mediated longitudinal coefficient is controlled by a single scalar, which may be represented in different conventions as:

1. Λ_c^{JJ} : projected Pauli-trace of the JJ derivative block;
2. $\lambda_c^{(JJ)}$: primitive longitudinal OO -block coefficient;
3. $C_{\parallel}$: one-mediator Bloch calibration constant of the Branch-B shell reduction.

So the correct current statement is not

$$\Lambda_c^{JJ} = C_{\parallel}$$

but rather

$$\boxed{\Lambda_c^{JJ}, \lambda_c^{(JJ)}, C_{\parallel} \text{ are three longitudinal calibration representatives belonging to different note conventions.}} \quad (168)$$

LXXV. 6. SAFE CANONICAL CHOICE GOING FORWARD

To eliminate the ambiguity, the safest master definition is to define the mediated longitudinal coefficient directly by the projected Bloch derivative:

$$\boxed{\lambda_{\parallel}^{(\text{med})} := \frac{1}{2} \text{Tr} \left(\sigma_3 P_* (\hat{n}_i \partial_{k_i} H_{\text{eff}}^{(\text{med})}(k_*) P_*) \right).} \quad (169)$$

Then define the conversion quantities only *from* this master coefficient:

$$\boxed{\Lambda_c^{JJ} := -\frac{M_X^2}{\alpha_X^2} \lambda_{\parallel}^{(\text{med}, JJ)},} \quad (170)$$

$$\boxed{C_{\parallel} := -\frac{2M_X^2}{\alpha_X \beta_X I_3} \lambda_{\parallel}^{(\text{med})}.} \quad (171)$$

With this convention, any future equality between Λ_c^{JJ} and $C_{\parallel}$ must emerge as a derived conversion identity, not as an assumption.

LXXVI. 7. HONEST FINAL VERDICT

Therefore the convention audit yields the following precise verdict:

$$\boxed{\text{Source classification: closed.}} \quad (172)$$

$$\boxed{JJ \text{ is the unique principal longitudinal source.}} \quad (173)$$

$$\boxed{\text{Normalization equivalence between } \Lambda_c^{JJ} \text{ and } C_{\parallel}: \text{ not yet proved.}} \quad (174)$$

$$\boxed{\text{The remaining task is an explicit same-projector same-basis same-overlap conversion computation.}} \quad (175)$$

LXXVII. FINAL SUMMARY

The uploaded notes contain two different longitudinal normalizations.
The first is the exact Class-II channel-renormalization form

$$\Delta\lambda_{\parallel} = -\frac{\alpha_X^2}{M_X^2}\Lambda_c^{JJ} - \frac{\alpha_X\beta_X}{M_X^2}\Lambda_c^{JK} - \frac{\beta_X^2}{M_X^2}\Lambda_c^{KK},$$

where Λ_c^{UV} is defined by projected Pauli traces of derivative blocks.

The second is the Branch-B one-mediator shell-calibration form

$$\lambda_{\parallel} = -\frac{\alpha_X\beta_X}{2M_X^2}C_{\parallel}I_3, \quad \lambda_{c;OO} = -\frac{\alpha_X\beta_X}{M_X^2}C_{\parallel},$$

where $C_{\parallel}$ is extracted from the diagonal difference of the projected shell matrix.

What is rigorously closed is the source classification:

JJ is the unique principal longitudinal source, JK, KK do not contribute to the principal allowed longitudinal slot.

Therefore the exact channel formula reduces structurally to

$$\Delta\lambda_{\parallel}^{(\text{med}, \parallel)} = -\frac{\alpha_X^2}{M_X^2}\Lambda_c^{JJ}.$$

However, the present notes do not yet prove that

$$\Lambda_c^{JJ} = \frac{\beta_X I_3}{2\alpha_X}C_{\parallel}$$

or any equivalent relation, because the two objects are introduced at different normalization layers and have not yet been written in the same projector/basis/overlap convention.

Thus the correct present verdict is:

source classification is closed,

but

normalization equivalence between Λ_c^{JJ} and $C_{\parallel}$ is still open.

The safest canonical next step is to define

$$\lambda_{\parallel}^{(\text{med})} = \frac{1}{2}\text{Tr}\left(\sigma_3 P_*(\hat{n}_i\partial_{k_i}H_{\text{eff}}^{(\text{med})}(k_*))P_*\right)$$

as the master longitudinal scalar and derive both Λ_c^{JJ} and $C_{\parallel}$ from it.

LXXVIII. MASTER-OBJECT UNIFICATION OF Λ_c^{JJ} , $\lambda_{c;OO}$, AND $C_{\parallel}$

LXXIX. SHORT EXPLANATION

The safest way to remove the current notation drift is to avoid identifying

$$\Lambda_c^{JJ}, \quad \lambda_{c;OO}, \quad C_{\parallel}$$

directly.

Instead, one first defines a single master projected longitudinal scalar from the mediated Bloch operator, and only afterwards introduces the three note-dependent calibration variables as derived quantities.

This removes the ambiguity between the channel-renormalization notes and the Branch-B one-mediator shell-calibration notes.

LXXX. 1. MASTER PROJECTED BLOCH LONGITUDINAL SCALAR

Let

$$K_i^{(\text{med})} := \partial_{k_i} H_{\text{eff}}^{(\text{med})}(k) \Big|_{k=k_*}, \quad K_{\parallel}^{(\text{med})} := \hat{n}_i K_i^{(\text{med})}.$$

Define the exact mediated longitudinal master scalar by

$$\lambda_{\parallel}^{(\text{med})} := \frac{1}{2} \text{Tr} \left(\sigma_3 P_* K_{\parallel}^{(\text{med})} P_* \right). \quad (176)$$

This is the common projector-trace object underlying all current longitudinal notations.

LXXXI. 2. CHANNEL-RENORMALIZATION CONVENTION

In the exact Class-II channel-renormalization notes, one writes

$$K_{\ell} = \partial_{k_{\ell}} H_{\Psi\Psi}(k) \Big|_{k=k_*} - \frac{\alpha_X^2}{M_X^2} \mathfrak{D}_{\ell}^{JJ}(k_*) - \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{D}_{\ell}^{JK}(k_*) - \frac{\beta_X^2}{M_X^2} \mathfrak{D}_{\ell}^{KK}(k_*),$$

and hence

$$\lambda_{\parallel} = \lambda_{\parallel}^{\text{ref}} + \Delta\lambda_{\parallel},$$

with

$$\Delta\lambda_{\parallel} = -\frac{\alpha_X^2}{M_X^2} \Lambda_c^{JJ} - \frac{\alpha_X \beta_X}{M_X^2} \Lambda_c^{JK} - \frac{\beta_X^2}{M_X^2} \Lambda_c^{KK}. \quad (177)$$

The channel overlap is defined by

$$\Lambda_c^{UV} := \frac{1}{2} \text{Tr} \left(\sigma_3 P_* (\hat{n}_{\ell} \mathfrak{D}_{\ell}^{UV}(k_*)) P_* \right), \quad (U, V) \in \{(J, J), (J, K), (K, K)\}. \quad (178)$$

At principal longitudinal shell order, the source classification is already fixed:

JJ is the unique allowed longitudinal source, JK , KK do not contribute to the principal allowed longitudinal slot.

Therefore the structural reduction is

$$\lambda_{\parallel}^{(\text{med})} = \Delta\lambda_{\parallel}^{(\parallel)} = -\frac{\alpha_X^2}{M_X^2} \Lambda_c^{JJ}. \quad (179)$$

Thus Λ_c^{JJ} is the channel-renormalization representative of the master longitudinal scalar.

LXXXII. 3. BRANCH-B TRANSPORT CONVENTION

In the Branch-B transport notes, the primitive longitudinal seed is

$$V_{(\parallel),\text{tr}}^{\mu}(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^{\mu} \sigma_3,$$

so the reduced longitudinal coefficient is

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}. \quad (180)$$

Under the current OO -candidate interpretation,

$$\ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2},$$

hence

$$\boxed{\lambda_{\parallel} = \frac{\lambda_{c;OO}}{2} I_3, \quad I_3 := I_{\theta'^3}.} \quad (181)$$

Thus $\lambda_{c;OO}$ is the transport-side primitive longitudinal representative of the same Branch-B longitudinal slot.

LXXXIII. 4. ONE-MEDIATOR SHELL-CALIBRATION CONVENTION

In the one-mediator shell-projected calibration notes,

$$\boxed{\lambda_{c;OO} = -\frac{\alpha_X \beta_X}{M_X^2} C_{\parallel},} \quad (182)$$

and equivalently

$$\boxed{\lambda_{\parallel} = -\frac{\alpha_X \beta_X}{2M_X^2} C_{\parallel} I_3.} \quad (183)$$

Therefore $C_{\parallel}$ is the shell-calibration representative of the same longitudinal Branch-B coefficient, but after a different normalization choice.

LXXXIV. 5. UNIFIED DICTIONARY

Equations (179), (181), and (183) show that the clean unified hierarchy is

$$\boxed{\lambda_{\parallel}^{(\text{med})} \longleftrightarrow \Lambda_c^{JJ} \quad (\text{channel-renormalization convention}),} \quad (184)$$

$$\boxed{\lambda_{\parallel}^{(\text{med})} \longleftrightarrow \lambda_{c;OO} \quad (\text{primitive Branch-B transport convention}),} \quad (185)$$

$$\boxed{\lambda_{\parallel}^{(\text{med})} \longleftrightarrow C_{\parallel} \quad (\text{one-mediator shell-calibration convention}).} \quad (186)$$

So the three variables should not be treated as primary objects of equal status. The primary object is the master Bloch scalar $\lambda_{\parallel}^{(\text{med})}$.

LXXXV. 6. CONDITIONAL CONVERSION FORMULAS

If one assumes that all three note families refer to the same pure mediated longitudinal contribution, with the same projector, the same valley basis, and the same overlap normalization, then one may combine the formulas to obtain

$$\boxed{\lambda_{c;OO} = \frac{2\lambda_{\parallel}^{(\text{med})}}{I_3},} \quad (187)$$

$$\Lambda_c^{JJ} = -\frac{M_X^2}{\alpha_X^2} \lambda_{\parallel}^{(\text{med})}, \quad (188)$$

$$C_{\parallel} = -\frac{2M_X^2}{\alpha_X \beta_X I_3} \lambda_{\parallel}^{(\text{med})}. \quad (189)$$

Under these same assumptions, one gets the candidate conversion laws

$$\lambda_{c;OO} = -\frac{\alpha_X \beta_X}{M_X^2} C_{\parallel}, \quad (190)$$

$$C_{\parallel} = \frac{2\alpha_X}{\beta_X I_3} \Lambda_c^{JJ}, \quad (191)$$

and

$$\lambda_{c;OO} = -\frac{2\alpha_X^2}{M_X^2 I_3} \Lambda_c^{JJ}. \quad (192)$$

However, (191)–(192) must be labeled explicitly as *conditional conversion formulas*, because the uploaded notes do not yet provide the fully explicit same-projector same-basis same-overlap derivation linking the three conventions.

LXXXVI. 7. FINITE-SUM REALIZATION OF THE MASTER SCALAR

The carrier-side longitudinal audit notes already provide an explicit finite harmonic representative:

$$V_{\parallel}^{(JJ)} = -\Gamma_J M, \quad m_3 = \frac{1}{2}(M_{11} - M_{22}),$$

with the finite sum

$$m_3 = -\frac{|\Omega_{\text{cell}}|}{2\Gamma_J} \sum_{G, G' \in \mathcal{S}_N} \left(\overline{c_{1,G}^{(+)}} c_{1,G'}^{(-)} - \overline{c_{2,G}^{(+)}} c_{2,G'}^{(-)} \right) \widehat{V}_{\parallel}^{(JJ)}(G_* + G - G'). \quad (193)$$

This is the explicit computational realization of the longitudinal master slot. If

$$m_3 \neq 0,$$

then

$$\lambda_c^{(JJ)} \neq 0 \implies \lambda_{\parallel} \neq 0.$$

Hence the remaining microscopic longitudinal frontier is no longer conceptual. It is the literal computation of this projected finite harmonic quantity from the refined locked Bloch data.

LXXXVII. 8. SAFE CANONICAL CONVENTION GOING FORWARD

To avoid further notation drift, the safest convention is:

1. define the master longitudinal scalar by (176);
2. define Λ_c^{JJ} only through (188);
3. define $\lambda_{c;OO}$ only through (187);
4. define $C_{\parallel}$ only through (189).

Then any future equality among Λ_c^{JJ} , $\lambda_{c;OO}$, and $C_{\parallel}$ becomes a derived conversion identity rather than an assumed identification.

LXXXVIII. 9. HONEST FINAL VERDICT

The current notation audit yields the following exact conclusion:

$$\boxed{\text{Source classification is closed: } JJ \text{ is the unique principal longitudinal source.}} \quad (194)$$

$$\boxed{\Lambda_c^{JJ}, \lambda_{c;OO}, C_{\parallel} \text{ are three convention-dependent representatives of one longitudinal master slot.}} \quad (195)$$

$$\boxed{\text{Their full explicit conversion is not yet unconditional, because the same-projector same-basis same-overlap derivation is still un}} \quad (196)$$

$$\boxed{\text{The remaining task is a finite projected Bloch evaluation of the master scalar } \lambda_{\parallel}^{(\text{med})}.} \quad (197)$$

LXXXIX. FINAL SUMMARY

The safest unification of the current longitudinal notations is to introduce the master projected Bloch scalar

$$\boxed{\lambda_{\parallel}^{(\text{med})} := \frac{1}{2} \text{Tr} \left(\sigma_3 P_* (\hat{n}_i \partial_{k_i} H_{\text{eff}}^{(\text{med})}(k_*)) P_* \right).}$$

Then the three currently used longitudinal quantities are interpreted as follows:

$$\boxed{\Lambda_c^{JJ} \text{ is the channel-renormalization representative of } \lambda_{\parallel}^{(\text{med})},}$$

through

$$\lambda_{\parallel}^{(\text{med})} = -\frac{\alpha_X^2}{M_X^2} \Lambda_c^{JJ} \quad \text{at principal longitudinal shell order,}$$

because JJ is the unique principal longitudinal source.

$$\boxed{\lambda_{c;OO} \text{ is the primitive Branch-B transport representative of } \lambda_{\parallel}^{(\text{med})},}$$

through

$$\lambda_{\parallel} = \frac{\lambda_{c;OO}}{2} I_3.$$

$$\boxed{C_{\parallel} \text{ is the one-mediator shell-calibration representative of } \lambda_{\parallel}^{(\text{med})},}$$

through

$$\lambda_{\parallel} = -\frac{\alpha_X \beta_X}{2M_X^2} C_{\parallel} I_3.$$

Therefore the exact current verdict is:

$\Lambda_c^{JJ}, \lambda_{c;OO}, C_{\parallel}$ are not three independent physical coefficients.

They are three convention-dependent representatives of the same longitudinal master slot.

What is fully closed:

 the source classification JJ only is closed.

What is not yet fully closed:

 the unconditional explicit conversion between Λ_c^{JJ} and $C_{\parallel}$,

because the same-projector same-basis same-overlap derivation is still unwritten.

Hence the true remaining longitudinal task is:

 compute $\lambda_{\parallel}^{(\text{med})}$ directly from the projected refined locked Bloch operator.

XC. HARMONIC-DATA UNIFICATION OF THE LONGITUDINAL JJ SLOT

XCI. SHORT EXPLANATION

The safest way to remove the remaining notation drift is to define a single longitudinal master scalar directly from the harmonic finite-sum realization of the pure- JJ Class-II transport contribution.

The key point is that the source classification is already closed: the principal longitudinal slot is carried only by the axial JJ -sector, while JK and KK remain transverse-family at principal longitudinal shell order.

Hence the true primary object is not

$$\Lambda_c^{JJ}, \quad \lambda_{c;OO}, \quad C_{\parallel}$$

separately, but one unique longitudinal master scalar extracted from the same harmonic data.

XCII. 1. EXACT CLASS-II LONGITUDINAL MASTER SCALAR

The pure- JJ longitudinal Class-II contribution is already reduced to

$$v_{\parallel}^{(II)} = -\Gamma_J m_3. \tag{198}$$

At the same time, the finite harmonic extraction gives

$$m_3 = -\frac{|\Omega_{\text{cell}}|}{2\Gamma_J} \left(\Delta_{00} \widehat{V}_{\parallel}^{(JJ)}(G_*) + R_N \right). \tag{199}$$

Therefore

$$v_{\parallel}^{(II)} = \frac{|\Omega_{\text{cell}}|}{2} \left(\Delta_{00} \widehat{V}_{\parallel}^{(JJ)}(G_*) + R_N \right). \tag{200}$$

We now define the pure longitudinal harmonic master slot by

$$\mathfrak{L}_{\parallel}^{(JJ)} := v_{\parallel}^{(II)}. \tag{201}$$

This is the cleanest convention-free representative of the pure JJ longitudinal channel.

XCIII. 2. CANONICAL BRANCH REDUCTION OF THE HARMONIC INPUT

In the current canonical branch,

$$\widehat{V}_{\parallel}^{(JJ)}(G_*) \propto \alpha_X L_* A_{G_*}, \quad L_* = m_{\parallel} u_{\parallel}. \quad (202)$$

Moreover, the valley-mixing scalar and the shell-normal J -bias are given by

$$\Delta_{00} = \eta_+^{\dagger} \Sigma_3 \eta_-, \quad (203)$$

$$u_{\parallel} = \frac{1}{2} \text{Tr}(\Sigma_3 U_{\parallel}) = \frac{1}{2}(u_{11} - u_{22}), \quad L_* = m_{\parallel} u_{\parallel}. \quad (204)$$

Hence the dominant one-mode approximation to the longitudinal master scalar is

$$\mathfrak{L}_{\parallel,0}^{(JJ)} = \frac{|\Omega_{\text{cell}}|}{2} \Delta_{00} \widehat{V}_{\parallel}^{(JJ)}(G_*) \propto \frac{|\Omega_{\text{cell}}|}{2} \Delta_{00} \alpha_X L_* A_{G_*}. \quad (205)$$

Thus the pure longitudinal slot is controlled by the finite harmonic data

$$(E_+(0), E_-(0), U_{\parallel}, m_{\parallel}, A_{G_*}, R_N). \quad (206)$$

XCIV. 3. TOTAL LONGITUDINAL COLUMN VERSUS PURE CLASS-II SLOT

The total longitudinal column is not identical to the pure Class-II contribution. In the current Bloch-operator working truncation one has

$$v_{\parallel} = \widetilde{\mathcal{N}}_{\parallel}^3 + v_{\parallel}^{(II)} + v_{\parallel}^{(\text{sp})}. \quad (207)$$

In the one-step scalar spectator truncation,

$$v_{\parallel}^{(\text{sp})} = 0, \quad (208)$$

so

$$v_{\parallel} = \widetilde{\mathcal{N}}_{\parallel}^3 + \mathfrak{L}_{\parallel}^{(JJ)}. \quad (209)$$

Therefore it is crucial to distinguish:

1. the *pure mediated/Class-II longitudinal slot*

$$\mathfrak{L}_{\parallel}^{(JJ)};$$

2. the *total longitudinal coefficient*

$$v_{\parallel}.$$

Any exact conversion formula among Λ_c^{JJ} , $\lambda_{c;OO}$, and $C_{\parallel}$ must specify clearly which of these two layers is being represented.

XCV. 4. CHANNEL-RENORMALIZATION REPRESENTATIVE

In the exact channel-renormalization convention, the mediated longitudinal correction is written as

$$\Delta\lambda_{\parallel}^{(\parallel)} = -\frac{\alpha_X^2}{M_X^2}\Lambda_c^{JJ}, \quad (210)$$

because the JJ -channel is the unique principal longitudinal source.

Therefore the channel-renormalization representative of the master slot is defined by

$$\Lambda_c^{JJ} := -\frac{M_X^2}{\alpha_X^2}\mathfrak{L}_{\parallel}^{(JJ)}. \quad (211)$$

This is the clean way to interpret Λ_c^{JJ} : it is not an independent primitive object, but the JJ -channel normalization of the same master scalar.

XCVI. 5. PRIMITIVE BRANCH-B TRANSPORT REPRESENTATIVE

In the Branch-B reduced transport convention,

$$\lambda_{\parallel} = \frac{\lambda_c}{2}I_3, \quad I_3 := I_{\theta'^3}. \quad (212)$$

For the OO -candidate primitive slot, this is written as

$$\lambda_{\parallel} = \frac{\lambda_{c;OO}}{2}I_3. \quad (213)$$

Hence, if one wishes to use the pure Class-II master slot as the primitive longitudinal representative, the natural definition is

$$\lambda_{c;OO}^{(JJ)} := \frac{2}{I_3}\mathfrak{L}_{\parallel}^{(JJ)}. \quad (214)$$

So $\lambda_{c;OO}^{(JJ)}$ is the primitive Branch-B transport normalization of the same harmonic master scalar.

XCVII. 6. ONE-MEDIATOR SHELL-CALIBRATION REPRESENTATIVE

The one-mediator shell-calibration notes write

$$\lambda_{c;OO} = -\frac{\alpha_X\beta_X}{M_X^2}C_{\parallel}, \quad (215)$$

equivalently

$$\lambda_{\parallel} = -\frac{\alpha_X\beta_X}{2M_X^2}C_{\parallel}I_3. \quad (216)$$

Therefore the shell-calibration representative of the same master slot is

$$C_{\parallel}^{(JJ)} := -\frac{2M_X^2}{\alpha_X\beta_X I_3}\mathfrak{L}_{\parallel}^{(JJ)}. \quad (217)$$

Again, this should be understood as a derived definition from the master scalar, not as an independent primary coefficient.

XCVIII. 7. UNIFIED DICTIONARY

The correct unification is therefore

$$\boxed{\mathfrak{L}_{\parallel}^{(JJ)} \longleftrightarrow \Lambda_c^{JJ} \quad (\text{channel-renormalization representative}),} \quad (218)$$

$$\boxed{\mathfrak{L}_{\parallel}^{(JJ)} \longleftrightarrow \lambda_{c;OO}^{(JJ)} \quad (\text{primitive Branch-B transport representative}),} \quad (219)$$

$$\boxed{\mathfrak{L}_{\parallel}^{(JJ)} \longleftrightarrow C_{\parallel}^{(JJ)} \quad (\text{one-mediator shell-calibration representative}).} \quad (220)$$

Hence Λ_c^{JJ} , $\lambda_{c;OO}$, and $C_{\parallel}$ should not be treated as three separate physical quantities. They are three convention-dependent representatives of one and the same harmonic longitudinal slot.

XCIX. 8. CONDITIONAL CONVERSION FORMULAS

Once the master definition is adopted, the formal conversions are immediate:

$$\boxed{\lambda_{c;OO}^{(JJ)} = \frac{2}{I_3} \mathfrak{L}_{\parallel}^{(JJ)}, \quad \Lambda_c^{JJ} = -\frac{M_X^2}{\alpha_X^2} \mathfrak{L}_{\parallel}^{(JJ)}, \quad C_{\parallel}^{(JJ)} = -\frac{2M_X^2}{\alpha_X \beta_X I_3} \mathfrak{L}_{\parallel}^{(JJ)}. \quad (221)}$$

Combining them gives the candidate pairwise conversion laws

$$\boxed{C_{\parallel}^{(JJ)} = \frac{2\alpha_X}{\beta_X I_3} \Lambda_c^{JJ}, \quad (222)}$$

$$\boxed{\lambda_{c;OO}^{(JJ)} = -\frac{2\alpha_X^2}{M_X^2 I_3} \Lambda_c^{JJ}, \quad (223)}$$

$$\boxed{\lambda_{c;OO}^{(JJ)} = -\frac{\alpha_X \beta_X}{M_X^2} C_{\parallel}^{(JJ)}. \quad (224)}$$

However, these must still be labeled as *master-convention conversion formulas*. They become note-by-note identities only after one verifies that the compared notes are indeed referring to the same pure mediated contribution and not to different total/reduced/truncated coefficient layers.

C. 9. PRACTICAL FINITE SUFFICIENT CONDITION

The master scalar formulation now makes the actual longitudinal survival test transparent.

Since

$$\mathfrak{L}_{\parallel}^{(JJ)} = \frac{|\Omega_{\text{cell}}|}{2} \left(\Delta_{00} \hat{V}_{\parallel}^{(JJ)}(G_*) + R_N \right),$$

and

$$\hat{V}_{\parallel}^{(JJ)}(G_*) \propto \alpha_X L_* A_{G_*},$$

a practical sufficient condition is

$$\boxed{|\Delta_{00} \alpha_X L_* A_{G_*}| > |R_N|. \quad (225)}$$

In the ultra-minimal truncation $R_N = 0$, it is enough to verify

$$\boxed{\Delta_{00} \neq 0, \quad L_* \neq 0, \quad \alpha_X \neq 0, \quad A_{G_*} \neq 0. \quad (226)}$$

Thus the longitudinal frontier is now reduced to a finite matrix-element insertion problem in the harmonic master data.

CI. 10. HONEST FINAL VERDICT

The genuinely primary longitudinal quantity is

$$\mathfrak{L}_{\parallel}^{(JJ)} = \frac{|\Omega_{\text{cell}}|}{2} \left(\Delta_{00} \widehat{V}_{\parallel}^{(JJ)}(G_*) + R_N \right). \quad (227)$$

What is fully closed:

1. JJ is the unique principal longitudinal source;
2. m_3 , $v_{\parallel}^{(II)}$, Δ_{00} , and L_* all belong to one finite harmonic-data pipeline;
3. Λ_c^{JJ} , $\lambda_{c;OO}$, and $C_{\parallel}$ can all be defined consistently from this same master scalar.

What is still honest-open:

1. the actual microscopic numerical/symbolic insertion of

$$a_{\pm}, b_{\pm}, c_{\pm}, u_{11}, u_{22}, m_{\parallel}, A_{G_*}, R_N;$$

2. the exact note-by-note identification of every historical formula with this unified master convention.

CII. FINAL SUMMARY

The cleanest resolution of the remaining longitudinal notation drift is to introduce the pure- JJ harmonic master scalar

$$\mathfrak{L}_{\parallel}^{(JJ)} := v_{\parallel}^{(II)} = \frac{|\Omega_{\text{cell}}|}{2} \left(\Delta_{00} \widehat{V}_{\parallel}^{(JJ)}(G_*) + R_N \right).$$

This scalar is the unique principal Class-II longitudinal slot because the source classification is already closed:

$$JJ \text{ is the unique principal longitudinal source, } JK, KK \text{ do not contribute to the principal longitudinal slot.}$$

In the current canonical branch,

$$\widehat{V}_{\parallel}^{(JJ)}(G_*) \propto \alpha_X L_* A_{G_*}, \quad L_* = m_{\parallel} u_{\parallel}, \quad u_{\parallel} = \frac{1}{2} \text{Tr}(\Sigma_3 U_{\parallel}),$$

and

$$\Delta_{00} = \eta_+^{\dagger} \Sigma_3 \eta_-.$$

Therefore the longitudinal frontier is completely reduced to finite harmonic data.

The three previously competing notations are then understood as derived representatives of the same master scalar:

$$\Lambda_c^{JJ} = -\frac{M_X^2}{\alpha_X^2} \mathfrak{L}_{\parallel}^{(JJ)},$$

$$\lambda_{c;OO}^{(JJ)} = \frac{2}{I_3} \mathfrak{L}_{\parallel}^{(JJ)},$$

$$C_{\parallel}^{(JJ)} = -\frac{2M_X^2}{\alpha_X \beta_X I_3} \mathfrak{L}_{\parallel}^{(JJ)}.$$

Hence

$$\boxed{\Lambda_c^{JJ}, \lambda_{c;OO}, C_{\parallel} \text{ are not three independent physical coefficients.}}$$

They are three convention-dependent representatives of one and the same longitudinal harmonic slot. The practical sufficient condition for longitudinal survival is

$$\boxed{|\Delta_{00} \alpha_X L_* A_{G_*}| > |R_N|}.$$

In the ultra-minimal truncation $R_N = 0$, it is enough to verify

$$\boxed{\Delta_{00} \neq 0, \quad L_* \neq 0, \quad \alpha_X \neq 0, \quad A_{G_*} \neq 0.}$$

Therefore the remaining open part is no longer structural. It is the explicit microscopic insertion of the finite harmonic data

$$a_{\pm}, b_{\pm}, c_{\pm}, u_{11}, u_{22}, m_{\parallel}, A_{G_*}, R_N.$$

CIII. DIRECT FINITE EXTRACTION OF Δ_{00} AND L_* IN THE CANONICAL LOCKED BASIS

CIV. SHORT EXPLANATION

The remaining carrier-side longitudinal problem has been reduced to a literal finite 2×2 matrix audit. The only nontrivial longitudinal scalars are

$$\Delta_{00} = \eta_+^\dagger \Sigma_3 \eta_-, \quad L_* = m_{\parallel} u_{\parallel},$$

where $\eta_{\pm}$ are the normalized low eigenvectors of the valley blocks $E_{\pm}(0)$, and

$$u_{\parallel} = \frac{1}{2} \text{Tr}(\Sigma_3 U_{\parallel}), \quad U_{\parallel} = \hat{n}^j \mathcal{U}_j(0).$$

CV. 1. EXPLICIT VALLEY BLOCKS

Write the two valley-diagonal projected blocks as general Hermitian matrices:

$$E_+(0) = \begin{pmatrix} a_+ & c_+ \\ c_+^* & b_+ \end{pmatrix}, \quad E_-(0) = \begin{pmatrix} a_- & c_- \\ c_-^* & b_- \end{pmatrix}, \quad (228)$$

with

$$a_{\pm}, b_{\pm} \in \mathbb{R}, \quad c_{\pm} \in \mathbb{C}.$$

Define

$$t_{\pm} := \frac{a_{\pm} + b_{\pm}}{2}, \quad d_{\pm} := \frac{a_{\pm} - b_{\pm}}{2}, \quad \rho_{\pm} := |c_{\pm}|. \quad (229)$$

Then

$$E_{\pm}(0) = t_{\pm} \mathbf{1}_2 + d_{\pm} \Sigma_3 + (\Re c_{\pm}) \Sigma_1 - (\Im c_{\pm}) \Sigma_2. \quad (230)$$

Hence the nontrivial Bloch vector of each valley block is

$$\mathbf{h}_{\pm} = (\Re c_{\pm}, -\Im c_{\pm}, d_{\pm}).$$

CVI. 2. LOW EIGENVALUES AND NORMALIZED LOW EIGENVECTORS

The two eigenvalues of $E_{\pm}(0)$ are

$$\varepsilon_{\pm}^{(\text{low/high})} = t_{\pm} \mp R_{\pm}, \quad R_{\pm} := \sqrt{d_{\pm}^2 + \rho_{\pm}^2}. \quad (231)$$

Write

$$c_{\pm} = \rho_{\pm} e^{i\phi_{\pm}}, \quad \rho_{\pm} \geq 0, \quad (232)$$

and define the polar angle by

$$\cos \theta_{\pm} = \frac{d_{\pm}}{R_{\pm}}, \quad \sin \theta_{\pm} = \frac{\rho_{\pm}}{R_{\pm}}. \quad (233)$$

A convenient normalized low eigenvector is

$$\eta_{\pm} = \begin{pmatrix} -e^{-i\phi_{\pm}} \sin(\theta_{\pm}/2) \\ \cos(\theta_{\pm}/2) \end{pmatrix}, \quad (234)$$

up to an irrelevant overall phase.

Equivalently, in algebraic ratio form, define

$$r_{\pm} := -\frac{c_{\pm}}{a_{\pm} - \varepsilon_{\pm}^{(\text{low})}} = -\frac{c_{\pm}}{d_{\pm} + R_{\pm}}, \quad (235)$$

whenever $d_{\pm} + R_{\pm} \neq 0$, and then

$$\eta_{\pm} = \frac{1}{\sqrt{1 + |r_{\pm}|^2}} \begin{pmatrix} 1 \\ r_{\pm} \end{pmatrix}. \quad (236)$$

CVII. 3. EXACT FORMULA FOR Δ_{00}

By definition,

$$\Delta_{00} = \eta_+^\dagger \Sigma_3 \eta_- = \overline{A_1^{(+)}} A_1^{(-)} - \overline{A_2^{(+)}} A_2^{(-)}, \quad (237)$$

where

$$\eta_{\pm} = \begin{pmatrix} A_1^{(\pm)} \\ A_2^{(\pm)} \end{pmatrix}.$$

Using (234), one gets

$$\Delta_{00} = e^{i(\phi_+ - \phi_-)} \sin \frac{\theta_+}{2} \sin \frac{\theta_-}{2} - \cos \frac{\theta_+}{2} \cos \frac{\theta_-}{2}. \quad (238)$$

Using (236), one obtains the fully algebraic expression

$$\Delta_{00} = \frac{1 - \overline{r_+} r_-}{\sqrt{(1 + |r_+|^2)(1 + |r_-|^2)}}. \quad (239)$$

Therefore the exact vanishing condition is

$$\Delta_{00} = 0 \iff \overline{r_+} r_- = 1. \quad (240)$$

Hence Δ_{00} is nonzero generically. Its vanishing requires a tuned matching relation between the two low-valley eigenvectors.

CVIII. 4. EXACT FORMULA FOR $u_{\parallel}$

Define the shell-normal intervalley J -matrix

$$U_{\parallel} := \hat{n}^j \mathcal{U}_j(0) = \begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix}. \quad (241)$$

Then

$$u_{\parallel} = \frac{1}{2} \text{Tr}(\Sigma_3 U_{\parallel}) = \frac{1}{2}(u_{11} - u_{22}). \quad (242)$$

Equivalently, if

$$U_{\parallel} = u_0 \mathbf{1}_2 + u_1 \Sigma_1 + u_2 \Sigma_2 + u_3 \Sigma_3, \quad (243)$$

then

$$u_{\parallel} = u_3. \quad (244)$$

So the shell-normal longitudinal bias depends only on the Σ_3 -component of the intervalley J -block.

CIX. 5. EXACT FORMULA FOR L_*

By definition,

$$L_* = m_{\parallel} u_{\parallel}. \quad (245)$$

Therefore

$$L_* = \frac{m_{\parallel}}{2}(u_{11} - u_{22}) = m_{\parallel} u_3. \quad (246)$$

Hence

$$L_* = 0 \iff m_{\parallel} = 0 \quad \text{or} \quad u_{11} = u_{22} \iff m_{\parallel} = 0 \quad \text{or} \quad u_3 = 0. \quad (247)$$

So once $m_{\parallel} \neq 0$, the only way the longitudinal slot dies is the exact diagonal cancellation

$$u_{11} = u_{22}.$$

CX. 6. PRACTICAL SUFFICIENT CONDITION FOR LONGITUDINAL SURVIVAL

In the current canonical branch, the longitudinal witness is controlled by the dominant product

$$\Delta_{00} \alpha_X L_* A_{G_*}, \quad (248)$$

up to the explicit remainder R_N . Therefore a practical sufficient condition is

$$|\Delta_{00} \alpha_X L_* A_{G_*}| > |R_N|. \quad (249)$$

In the ultra-minimal truncation $R_N = 0$, it is enough to verify

$$\Delta_{00} \neq 0, \quad L_* \neq 0, \quad \alpha_X \neq 0, \quad A_{G_*} \neq 0. \quad (250)$$

Substituting the explicit formulas, this becomes

$$\left| \frac{1 - \bar{r}_+ r_-}{\sqrt{(1 + |r_+|^2)(1 + |r_-|^2)}} \right| \cdot |\alpha_X| \cdot \left| \frac{m_{\parallel}}{2}(u_{11} - u_{22}) \right| \cdot |A_{G_*}| > |R_N|. \quad (251)$$

CXI. 7. REDUCED LONGITUDINAL COEFFICIENT IN THE CURRENT CANONICAL BRANCH

The canonical witness formula gives

$$\boxed{\lambda_{\parallel} = \frac{1}{2} \alpha_X L_* I_{\theta^3}.} \quad (252)$$

Thus, if $\Delta_{00} \neq 0$, $L_* \neq 0$, $\alpha_X \neq 0$, and $A_{G_*} \neq 0$, then the first-shell dominant compression yields

$$m_3 \neq 0 \implies \lambda_c^{(JJ)} \neq 0 \implies \lambda_{\parallel} \neq 0.$$

CXII. 8. GENERIC NONVANISHING STATEMENT

The two potential failure mechanisms are now explicit:

1. $\Delta_{00} = 0$ only on the tuned locus

$$\overline{r_+} r_- = 1;$$

2. $L_* = 0$ only if

$$m_{\parallel} = 0 \quad \text{or} \quad u_{11} = u_{22}.$$

Therefore, absent an exact symmetry enforcing either condition, the longitudinal slot is generically nonzero once

$$\alpha_X \neq 0, \quad A_{G_*} \neq 0.$$

CXIII. 9. HONEST STATUS

What is closed is the exact finite audit map

$$(E_+(0), E_-(0), U_{\parallel}, m_{\parallel}) \longmapsto (\Delta_{00}, L_*).$$

What is still open is only the final microscopic insertion of the entries

$$a_{\pm}, b_{\pm}, c_{\pm}, u_{11}, u_{22}, m_{\parallel}.$$

So the remaining frontier is not structural. It is a finite symbolic or numerical matrix-element computation.

CXIV. FINAL SUMMARY

The carrier-side longitudinal frontier is now reduced to the finite extraction of two scalars:

$$\Delta_{00}, \quad L_* = m_{\parallel} u_{\parallel}.$$

Write

$$E_{\pm}(0) = \begin{pmatrix} a_{\pm} & c_{\pm} \\ \overline{c_{\pm}} & b_{\pm} \end{pmatrix}, \quad U_{\parallel} = \hat{n}^j \mathcal{U}_j(0) = \begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix}.$$

Then the normalized low eigenvectors of $E_{\pm}(0)$ determine

$$\boxed{\Delta_{00} = \frac{1 - \overline{r_+} r_-}{\sqrt{(1 + |r_+|^2)(1 + |r_-|^2)}}, \quad r_{\pm} = -\frac{c_{\pm}}{a_{\pm} - \varepsilon_{\pm}^{(\text{low})}}.}$$

Hence

$$\boxed{\Delta_{00} = 0 \iff \overline{r_+} r_- = 1.}$$

Next,

$$u_{\parallel} = \frac{1}{2} \text{Tr}(\Sigma_3 U_{\parallel}) = \frac{1}{2}(u_{11} - u_{22}),$$

and therefore

$$L_* = m_{\parallel} u_{\parallel} = \frac{m_{\parallel}}{2}(u_{11} - u_{22}).$$

Hence

$$L_* = 0 \iff m_{\parallel} = 0 \quad \text{or} \quad u_{11} = u_{22}.$$

In the current canonical branch, the practical sufficient condition for longitudinal survival is

$$|\Delta_{00} \alpha_X L_* A_{G_*}| > |R_N|.$$

In the ultra-minimal truncation $R_N = 0$, it is enough to verify

$$\Delta_{00} \neq 0, \quad L_* \neq 0, \quad \alpha_X \neq 0, \quad A_{G_*} \neq 0.$$

Therefore the longitudinal frontier is no longer structural. It is reduced to the explicit microscopic insertion of the finite matrix entries

$$a_{\pm}, b_{\pm}, c_{\pm}, u_{11}, u_{22}, m_{\parallel}.$$

CXV. EXACT SHELL-HARMONIC INSERTION SHEET FOR $E_{\pm}(0)$, $U_{\parallel}$, AND $m_{\parallel}$ ON THE LOCKED 12-MODE BCC BACKGROUND

CXVI. SHORT EXPLANATION

The remaining longitudinal audit is now a literal finite matrix-element computation. Its exact finite audit map is

$$(E_+(0), E_-(0), U_{\parallel}, m_{\parallel}) \mapsto (\Delta_{00}, L_*),$$

with

$$\Delta_{00} = \eta_+^{\dagger} \Sigma_3 \eta_-, \quad L_* = m_{\parallel} u_{\parallel}, \quad u_{\parallel} = \frac{1}{2} \text{Tr}(\Sigma_3 U_{\parallel}).$$

Thus the correct next step is to write the actual matrix entries

$$a_{\pm}, b_{\pm}, c_{\pm}, u_{11}, u_{22}, m_{\parallel}$$

directly from the shell-harmonic Bloch operator of the locked 12-mode BCC background.

CXVII. 1. FOLDED BRAGG CONVENTION AND LOCKED 12-MODE BCC BACKGROUND

Fix the folded Bragg convention

$$k_* = \frac{G_*}{2}, \quad k = k_* + p, \quad |p| \ll |G_*|. \quad (253)$$

Choose the first BCC reciprocal shell

$$\mathcal{S}_{12} = \frac{q_*}{\sqrt{2}} \{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\}, \quad |\mathcal{S}_{12}| = 12. \quad (254)$$

The locked periodic background is

$$\phi_0(x) = A \sum_{Q \in \mathcal{S}_{12}} e^{iQ \cdot x}, \quad \Psi_0(x) = z_0 \phi_0(x), \quad P_0 = z_0 z_0^{\dagger}. \quad (255)$$

Choose an internal basis adapted to the locked splitting

$$u_0 = z_0, \quad u_1, u_2 \in Q_{z_0}, \quad u_r^{\dagger} u_s = \delta_{rs}, \quad \mathbb{C}^3 = \mathbb{C} z_0 \oplus Q_{z_0}. \quad (256)$$

CXVIII. 2. BLOCH-HARMONIC EXPANSION OF THE MICROSCOPIC FLUCTUATIONS

Expand the triplet fluctuation as

$$\delta\Psi_k(x) = e^{ik \cdot x} \sum_{G \in \Lambda^*} \sum_{r=0}^2 c_{G,r} u_r e^{iG \cdot x}, \quad (257)$$

where Λ^* is the reciprocal lattice generated by the periodic BCC background.

The effective Bloch operator has the convolution form

$$(H_{\text{eff}}(k))_{(G,r),(G',s)} = h_{rs}^{\text{diag}}(k+G) \delta_{G,G'} + V_{rs}^{\text{eff}}(G'-G; k+G), \quad (258)$$

with

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - \frac{1}{M_X^2} (\alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k)) (\alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k))^\dagger. \quad (259)$$

Thus every matrix entry needed below is a finite matrix element of the actual refined locked Bloch operator.

CXIX. 3. SHELL-RELEVANT DOUBLED LOCAL BLOCK

At the selected shell pair $\pm G_*$, define the local doubled shell basis

$$|+, a\rangle := |G=0, u_a\rangle, \quad |-, a\rangle := |G=-G_*, u_a\rangle, \quad a=1, 2. \quad (260)$$

Then the shell-relevant doubled space is

$$\mathcal{H}_D = \text{span}\{|+, 1\rangle, |+, 2\rangle, |-, 1\rangle, |-, 2\rangle\} \cong \mathbb{C}_{\text{valley}}^2 \otimes \mathbb{C}_{\text{int}}^2. \quad (261)$$

The shell block at $p=0$ is therefore

$$H_D(0) = \begin{pmatrix} E_+(0) & M(0) \\ M(0)^\dagger & E_-(0) \end{pmatrix}, \quad (262)$$

where $E_\pm(0)$ are the 2×2 valley-diagonal blocks and $M(0)$ is the Bragg mixing block.

CXX. 4. EXACT SHELL-HARMONIC FORMULAS FOR $E_\pm(0)$

By definition,

$$(E_+(0))_{ab} = \langle +, a | H_{\text{eff}}(k_*) | +, b \rangle, \quad (E_-(0))_{ab} = \langle -, a | H_{\text{eff}}(k_*) | -, b \rangle. \quad (263)$$

Using (258) and (260), this becomes

$$(E_+(0))_{ab} = h_{ab}^{\text{diag}}(k_*) + V_{ab}^{\text{eff}}(0; k_*), \quad (264)$$

$$(E_-(0))_{ab} = h_{ab}^{\text{diag}}(-k_*) + V_{ab}^{\text{eff}}(0; -k_*). \quad (265)$$

Therefore the finite matrix entries are exactly

$$a_+ = (E_+(0))_{11}, \quad b_+ = (E_+(0))_{22}, \quad c_+ = (E_+(0))_{12}, \quad (266)$$

$$a_- = (E_-(0))_{11}, \quad b_- = (E_-(0))_{22}, \quad c_- = (E_-(0))_{12}. \quad (267)$$

So once the refined locked core h^{diag} and the Schur-corrected periodic coefficient V^{eff} are fixed, the valley matrices $E_\pm(0)$ are explicit finite 2×2 Hermitian matrices.

CXXI. 5. BRAGG MIXING BLOCK AND THE ROLE OF A_{G_*}

The intervalley Bragg block is

$$M_{ab}(0) = \langle +, a | H_{\text{eff}}(k_*) | -, b \rangle. \quad (268)$$

Using the convolution rule,

$$M_{ab}(0) = V_{ab}^{\text{eff}}(-G_*, k_*). \quad (269)$$

Equivalently, since the leading background harmonics satisfy $Q \in \mathcal{S}_{12}$, the first-shell dominant intervalley mixing is controlled by the single shell Fourier amplitude A_{G_*} .

Thus the finite shell problem separates naturally into:

1. the valley-diagonal 2×2 Hermitian blocks $E_{\pm}(0)$;
2. the intervalley shell amplitude A_{G_*} through $M(0)$.

CXXII. 6. LOW VALLEY EIGENVECTORS AND Δ_{00}

Let

$$E_{\pm}(0) = \begin{pmatrix} a_{\pm} & c_{\pm} \\ \bar{c}_{\pm} & b_{\pm} \end{pmatrix}. \quad (270)$$

Diagonalize $E_{\pm}(0)$ and let $\eta_{\pm}$ be the normalized low eigenvectors. Then

$$\Delta_{00} = \eta_+^{\dagger} \Sigma_3 \eta_-. \quad (271)$$

Equivalently, if

$$r_{\pm} = -\frac{c_{\pm}}{a_{\pm} - \varepsilon_{\pm}^{(\text{low})}}, \quad (272)$$

then

$$\Delta_{00} = \frac{1 - \bar{r}_+ r_-}{\sqrt{(1 + |r_+|^2)(1 + |r_-|^2)}}. \quad (273)$$

So Δ_{00} is already an explicit rational function of the six finite entries

$$a_{\pm}, b_{\pm}, c_{\pm}.$$

CXXIII. 7. EXACT SHELL-HARMONIC FORMULA FOR $U_{\parallel}$

Define the shell-relevant reduced J -matrix by

$$\mathcal{U}_j(0) = \left(\mathcal{U}_{j,ab}(0) \right)_{a,b=1}^2, \quad \mathcal{U}_{j,ab}(0) = \langle -, a | J_j(G_*; 0) | +, b \rangle. \quad (274)$$

Then the shell-normal reduced matrix is

$$U_{\parallel} := \hat{n}^j \mathcal{U}_j(0) = \begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix}. \quad (275)$$

By direct shell-harmonic insertion,

$$u_{ab} = \hat{n}^j \sum_{G, G' \in \Lambda^*} \overline{c_{a,G}^{(-)}} J_j^{\text{eff}}(G' - G; k_* + G) c_{b,G'}^{(+)}.$$
 (276)

This is the exact finite-harmonic matrix element formula. In the first-shell dominant truncation, only the leading intervalley harmonic survives, so operationally

$$u_{ab} \approx \langle -, a | \hat{n}^j J_j(G_*; 0) | +, b \rangle.$$
 (277)

CXXIV. 8. CANONICAL SHELL FORM OF THE J -KERNEL AND EXTRACTION OF $u_{\parallel}$

In the canonical locked basis, the shell-relevant J -harmonic coefficient has the form

$$J_{i,r}^a(G_*; 0) = m_{\parallel} \delta^{a3} i(G_*)_i A_{G_*} \left[u_{\parallel} \hat{n}_j \Sigma_3 + u_E (e_{1j} \Sigma_1 + e_{2j} \Sigma_2) + u_O (-e_{2j} \Sigma_1 + e_{1j} \Sigma_2) \right]_r^b.$$
 (278)

Hence the shell-normal singlet coefficient is extracted by

$$u_{\parallel} = \frac{1}{2} \text{Tr}(\Sigma_3 U_{\parallel}) = \frac{1}{2} [(U_{\parallel})_{11} - (U_{\parallel})_{22}].$$
 (279)

So the only longitudinally relevant part of the J -kernel is the diagonal shell-normal asymmetry of the reduced 2×2 intervalley matrix.

CXXV. 9. EXACT FORMULA FOR $m_{\parallel}$ AND L_*

In the canonical locked frame, the background selector is aligned as

$$m^a(P_0) = m_{\parallel} \delta^{a3}.$$
 (280)

Therefore

$$m_{\parallel} = m^3(P_0).$$
 (281)

No further reduction is needed.

Thus the final scalar invariant is

$$L_* = m_{\parallel} u_{\parallel} = \frac{m_{\parallel}}{2} [(U_{\parallel})_{11} - (U_{\parallel})_{22}].$$
 (282)

CXXVI. 10. DIRECT LONGITUDINAL WITNESS RECONSTRUCTION

The first-shell dominant pure- JJ longitudinal contribution is controlled by

$$\mathfrak{L}_{\parallel}^{(JJ)} = \frac{|\Omega_{\text{cell}}|}{2} \left(\Delta_{00} \hat{V}_{\parallel}^{(JJ)}(G_*) + R_N \right),$$
 (283)

and in the current canonical branch

$$\hat{V}_{\parallel}^{(JJ)}(G_*) \propto \alpha_X L_* A_{G_*}.$$
 (284)

Therefore the practical sufficient condition is

$$|\Delta_{00} \alpha_X L_* A_{G_*}| > |R_N|.$$
 (285)

In the ultra-minimal truncation $R_N = 0$, it is enough to require

$$\Delta_{00} \neq 0, \quad L_* \neq 0, \quad \alpha_X \neq 0, \quad A_{G_*} \neq 0.$$
 (286)

CXXVII. 11. FINAL REDUCED COEFFICIENT

Once the primitive longitudinal slot survives, the reduced Branch-B coefficient is

$$\boxed{\lambda_{\parallel} = \frac{1}{2} \alpha_X L_* I_{\theta'^3}} \quad (287)$$

in the current canonical witness normalization.

Thus the complete longitudinal audit is now reduced to the explicit finite insertion of

$$a_{\pm}, \quad b_{\pm}, \quad c_{\pm}, \quad u_{11}, \quad u_{22}, \quad m_{\parallel}, \quad A_{G_*}, \quad R_N.$$

CXXVIII. 12. HONEST STATUS

What is fully reduced is:

1. the shell-harmonic formulas for $E_{\pm}(0)$;
2. the shell-harmonic formula for $U_{\parallel}$;
3. the exact extraction formulas for Δ_{00} and L_* ;
4. the practical sufficient condition (285).

What remains open is only the explicit symbolic or numerical insertion of the finite harmonic matrix elements for the final chosen refined locked microscopic operator.

CXXIX. FINAL SUMMARY

On the locked 12-mode BCC background

$$\phi_0(x) = A \sum_{Q \in \mathcal{S}_{12}} e^{iQ \cdot x}, \quad \Psi_0(x) = z_0 \phi_0(x), \quad P_0 = z_0 z_0^{\dagger},$$

the remaining longitudinal audit is a finite shell-harmonic matrix-element problem.

The valley-diagonal 2×2 blocks are

$$(E_+(0))_{ab} = \langle +, a | H_{\text{eff}}(k_*) | +, b \rangle, \quad (E_-(0))_{ab} = \langle -, a | H_{\text{eff}}(k_*) | -, b \rangle,$$

so

$$a_{\pm} = (E_{\pm}(0))_{11}, \quad b_{\pm} = (E_{\pm}(0))_{22}, \quad c_{\pm} = (E_{\pm}(0))_{12}.$$

Their low eigenvectors $\eta_{\pm}$ determine

$$\Delta_{00} = \eta_+^{\dagger} \Sigma_3 \eta_-,$$

equivalently

$$\Delta_{00} = \frac{1 - \bar{r}_+ r_-}{\sqrt{(1 + |r_+|^2)(1 + |r_-|^2)}}, \quad r_{\pm} = -\frac{c_{\pm}}{a_{\pm} - \varepsilon_{\pm}^{(\text{low})}}.$$

The reduced shell-normal J -matrix is

$$U_{\parallel} = \hat{n}^j \mathcal{U}_j(0) = \left(\langle -, a | \hat{n}^j J_j(G_*; 0) | +, b \rangle \right)_{a,b=1}^2,$$

and the shell-normal singlet coefficient is

$$u_{\parallel} = \frac{1}{2} \text{Tr}(\Sigma_3 U_{\parallel}) = \frac{1}{2} [(U_{\parallel})_{11} - (U_{\parallel})_{22}].$$

With

$$m_{\parallel} = m^3(P_0),$$

the final scalar invariant is

$$L_* = m_{\parallel} u_{\parallel} = \frac{m_{\parallel}}{2} [(U_{\parallel})_{11} - (U_{\parallel})_{22}].$$

In the canonical first-shell dominant branch,

$$\widehat{V}_{\parallel}^{(JJ)}(G_*) \propto \alpha_X L_* A_{G_*},$$

so the practical sufficient condition is

$$|\Delta_{00} \alpha_X L_* A_{G_*}| > |R_N|.$$

In the ultra-minimal truncation $R_N = 0$, it is enough that

$$\Delta_{00} \neq 0, \quad L_* \neq 0, \quad \alpha_X \neq 0, \quad A_{G_*} \neq 0.$$

Therefore the remaining longitudinal frontier is no longer structural. It is the explicit finite insertion of the shell-harmonic data

$$a_{\pm}, b_{\pm}, c_{\pm}, u_{11}, u_{22}, m_{\parallel}, A_{G_*}, R_N.$$

CXXX. DIRECT SHELL-MATRIX INSERTION OF $H_{\Psi\Psi}(k_*)$, $\mathcal{J}(k_*)$, AND $\mathcal{K}(k_*)$

CXXXI. SHORT EXPLANATION

At the present stage, the longitudinal audit is no longer a structural theorem problem. It is the explicit shell-matrix insertion problem

$$H_{\Psi\Psi}(k_*), \quad \mathcal{J}(k_*), \quad \mathcal{K}(k_*) \implies E_{\pm}(0), M(0), U_{\parallel}, m_{\parallel}.$$

The honest status is:

1. the J -sector is explicit at the microscopic level;
2. the K -sector is only fixed in shell-relevant template form;
3. therefore the J -induced longitudinal block $U_{\parallel}$ can already be written explicitly, whereas the final K -dependent Schur-corrected entries remain partly parametric.

CXXXII. 1. WORKING BRANCH AND FOLDED BRAGG CONVENTION

We work in the strongest explicit currently available microscopic branch

$$L_{\text{mic}} = L_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi]. \quad (288)$$

We use the folded Bragg convention

$$k_* = \frac{G_*}{2}, \quad k = k_* + p, \quad |p| \ll |G_*|. \quad (289)$$

The locked 12-mode BCC background is

$$\phi_0(x) = A \sum_{Q \in \mathcal{S}_{12}} e^{iQ \cdot x}, \quad \Psi_0(x) = z_0 \phi_0(x), \quad P_0 = z_0 z_0^{\dagger}. \quad (290)$$

Choose the canonical locked internal basis

$$u_0 = z_0, \quad u_1, u_2 \in Q_{z_0}, \quad u_r^{\dagger} u_s = \delta_{rs}, \quad \mathbb{C}^3 = \mathbb{C} z_0 \oplus Q_{z_0}. \quad (291)$$

CXXXIII. 2. BLOCH-HARMONIC BASIS AND SHELL CARRIER BLOCK

Expand the triplet fluctuation in reciprocal harmonics:

$$\delta\Psi_k(x) = e^{ik \cdot x} \sum_{G \in \Lambda^*} \sum_{r=0}^2 c_{G,r} u_r e^{iG \cdot x}. \quad (292)$$

Fix the shell carrier basis

$$|+, a\rangle := |G = 0, u_a\rangle, \quad |-, a\rangle := |G = -G_*, u_a\rangle, \quad a = 1, 2. \quad (293)$$

Then

$$\mathcal{H}_D = \text{span}\{|+, 1\rangle, |+, 2\rangle, |-, 1\rangle, |-, 2\rangle\} \cong \mathbb{C}_{\text{valley}}^2 \otimes \mathbb{C}_{\text{int}}^2. \quad (294)$$

The shell-restricted effective operator is

$$H_D(0) = \begin{pmatrix} E_+(0) & M(0) \\ M(0)^\dagger & E_-(0) \end{pmatrix}, \quad (295)$$

with 2×2 internal blocks $E_\pm(0)$ and intervalley Bragg block $M(0)$.

CXXXIV. 3. EXACT SCHUR-CORRECTED SHELL ENTRIES

The effective triplet Bloch operator is

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - \mathcal{C}(k)H_{XX}(k)^{-1}\mathcal{C}(k)^\dagger, \quad \mathcal{C}(k) := \alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k). \quad (296)$$

At leading heavy-mass order,

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - \frac{1}{M_X^2} \mathcal{C}(k)\mathcal{C}(k)^\dagger + O(M_X^{-4}). \quad (297)$$

Therefore the exact shell entries are

$$(E_+(0))_{ab} = \langle +, a | H_{\Psi\Psi}(k_*) | +, b \rangle - \frac{1}{M_X^2} \langle \mathcal{C}_+^{(b)}, \mathcal{C}_+^{(a)} \rangle_X + O(M_X^{-4}), \quad (298)$$

$$(E_-(0))_{ab} = \langle -, a | H_{\Psi\Psi}(k_*) | -, b \rangle - \frac{1}{M_X^2} \langle \mathcal{C}_-^{(b)}, \mathcal{C}_-^{(a)} \rangle_X + O(M_X^{-4}), \quad (299)$$

where

$$\mathcal{C}_\pm^{(a)} := \mathcal{C}(k_*) | \pm, a \rangle,$$

and $\langle \cdot, \cdot \rangle_X$ is the internal mediator-fiber inner product.

Thus the six finite Hermitian valley entries are

$$a_+ = (E_+(0))_{11}, \quad b_+ = (E_+(0))_{22}, \quad c_+ = (E_+(0))_{12}, \quad (300)$$

$$a_- = (E_-(0))_{11}, \quad b_- = (E_-(0))_{22}, \quad c_- = (E_-(0))_{12}. \quad (301)$$

Similarly, the intervalley Bragg block is

$$M_{ab}(0) = \langle +, a | H_{\Psi\Psi}(k_*) | -, b \rangle - \frac{1}{M_X^2} \langle \mathcal{C}_-^{(b)}, \mathcal{C}_+^{(a)} \rangle_X + O(M_X^{-4}). \quad (302)$$

CXXXV. 4. WHAT CAN BE WRITTEN EXPLICITLY FOR $H_{\Psi\Psi}$

Because the background is periodic, $H_{\Psi\Psi}(k)$ has reciprocal-convolution form:

$$\left(H_{\Psi\Psi}(k) \right)_{(G,r),(G',s)} = h_{rs}^{\Psi\Psi}(k+G)\delta_{G,G'} + V_{rs}^{\Psi\Psi}(G'-G;k). \quad (303)$$

Hence

$$\left(E_+(0) \right)_{ab}^{(\Psi\Psi)} = h_{ab}^{\Psi\Psi}(k_*) + V_{ab}^{\Psi\Psi}(0;k_*), \quad (304)$$

$$\left(E_-(0) \right)_{ab}^{(\Psi\Psi)} = h_{ab}^{\Psi\Psi}(-k_*) + V_{ab}^{\Psi\Psi}(0;-k_*), \quad (305)$$

and

$$M_{ab}^{(\Psi\Psi)}(0) = V_{ab}^{\Psi\Psi}(-G_*;k_*). \quad (306)$$

At the level of the locked Brazovskii core, the diagonal symbol contains at least the scalar shell-forming piece. The exact refined internal matrix part is encoded in $h_{ab}^{\Psi\Psi}$ and remains dependent on the specific choice of L_{Ψ}^{ref} . Therefore the insertion formulas (304)–(306) are exact, but not yet numerically closed until L_{Ψ}^{ref} is fixed completely.

CXXXVI. 5. EXPLICIT LINEARIZATION OF $J_i^a[\Psi]$

The microscopic current is explicitly

$$J_i^a[\Psi] = (\partial_i \Psi^\dagger) T^a \Psi + \Psi^\dagger T^a (\partial_i \Psi). \quad (307)$$

Linearizing around Ψ_0 ,

$$\delta J_i^a = (\partial_i \delta \Psi^\dagger) T^a \Psi_0 + (\partial_i \Psi_0^\dagger) T^a \delta \Psi + \delta \Psi^\dagger T^a (\partial_i \Psi_0) + \Psi_0^\dagger T^a (\partial_i \delta \Psi). \quad (308)$$

Using

$$\Psi_0(x) = z_0 \sum_{Q \in S_{12}} A_Q e^{iQ \cdot x}, \quad \delta \Psi_k(x) = e^{ik \cdot x} \sum_{G,r} c_{G,r} u_r e^{iG \cdot x},$$

one gets the reciprocal-harmonic expansion

$$\delta J_i^a(x) = e^{ik \cdot x} \sum_{G,r} \sum_{Q \in S_{12}} c_{G,r} e^{i(G+Q) \cdot x} J_{i,r}^a(Q; k+G) + \text{h.c.}, \quad (309)$$

with explicit coefficient

$$J_{i,r}^a(Q; q) = i q_i A_Q u_r^\dagger T^a z_0 + i Q_i A_Q z_0^\dagger T^a u_r, \quad q := k+G. \quad (310)$$

Thus the $Q = G_*$ harmonic needed for the shell-pair intervalley channel is already explicit:

$$J_{i,r}^a(G_*; q) = i q_i A_{G_*} u_r^\dagger T^a z_0 + i (G_*)_i A_{G_*} z_0^\dagger T^a u_r. \quad (311)$$

CXXXVII. 6. REDUCED J -MATRIX AND EXTRACTION OF $U_{\parallel}$

Define the reduced intervalley J -matrix by

$$\mathcal{U}_{j,ab}(0) = \langle -, a | \widehat{J}_j(G_*; 0) | +, b \rangle. \quad (312)$$

Then the shell-normal reduced matrix is

$$U_{\parallel} := \hat{n}^j \mathcal{U}_j(0) = \begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix}. \quad (313)$$

Its exact finite-harmonic matrix-element form is

$$u_{ab} = \hat{n}^j \sum_{G, G' \in \Lambda^*} \overline{c_{a,G}^{(-)}} J_j^{\text{eff}}(G' - G; k_* + G) c_{b,G'}^{(+)}. \quad (314)$$

In the first-shell dominant truncation,

$$u_{ab} \approx \langle -, a | \hat{n}^j \hat{J}_j(G_*; 0) | +, b \rangle. \quad (315)$$

Therefore

$$u_{\parallel} = \frac{1}{2} \text{Tr}(\Sigma_3 U_{\parallel}) = \frac{1}{2} (u_{11} - u_{22}), \quad (316)$$

and

$$L_* = m_{\parallel} u_{\parallel} = \frac{m_{\parallel}}{2} (u_{11} - u_{22}), \quad m_{\parallel} = m^3(P_0). \quad (317)$$

CXXXVIII. 7. WHAT REMAINS ONLY TEMPLATE-LEVEL FOR K_i^a

The current notes fix only the shell-relevant template

$$K_i^a[P, \psi] = \text{Tr}(T^a \partial_i P) \left(\Lambda_E^{(0)} E_j^b + \Lambda_O^{(0)} J_j^b \right) D_j \psi_b. \quad (318)$$

On the defect background,

$$K_i^a[P_B, \psi] = \delta_{i\xi} \theta'(\xi) k_a(\theta) \left(\Lambda_E^{(0)} E_j^b + \Lambda_O^{(0)} J_j^b \right) D_j \psi_b. \quad (319)$$

Therefore the K -sector is sufficient to parameterize the transverse E/O channels, but it is not yet specified enough to give a fully explicit microscopic numerical matrix for the K -dependent Schur correction in (298)–(302).

So the honest current split is:

1. J -sector and $U_{\parallel}$: explicit;
2. K -sector: shell-template explicit, but not yet fully microscopic numeric.

CXXXIX. 8. FINAL EXECUTABLE INSERTION FORMULAS

The actual remaining finite entries are therefore computed by the following chain:

$$a_{\pm}, b_{\pm}, c_{\pm} \text{ from (298), (299),} \quad (320)$$

$$u_{11}, u_{22} \text{ from (314) or (315),} \quad (321)$$

$$\Delta_{00} = \frac{1 - \bar{r}_+ r_-}{\sqrt{(1 + |r_+|^2)(1 + |r_-|^2)}}, \quad r_{\pm} = -\frac{c_{\pm}}{a_{\pm} - \varepsilon_{\pm}^{(\text{low})}}, \quad (322)$$

$$\boxed{L_* = \frac{m_{\parallel}}{2}(u_{11} - u_{22}),} \quad (323)$$

and finally

$$\boxed{|\Delta_{00} \alpha_X L_* A_{G_*}| > |R_N| \implies \lambda_{\parallel} \neq 0} \quad (324)$$

in the first-shell dominant canonical branch.

CXL. 9. HONEST STATUS

What is already explicit:

1. the shell basis;
2. the Schur-corrected shell-entry formulas;
3. the microscopic J -current and its $Q = G_*$ harmonic coefficient;
4. the reduced J -matrix $U_{\parallel}$ and the scalar $L_* = m_{\parallel} u_{\parallel}$.

What is not yet fully explicit:

1. the complete refined internal symbol inside $H_{\Psi\Psi}(k)$, because L_{Ψ}^{ref} is not yet fixed uniquely;
2. the full microscopic K -matrix beyond its shell-relevant template form.

Thus the remaining frontier is no longer structural. It is the explicit insertion of the final chosen refined locked microscopic constitutive data into the finite formulas above.

CXLI. FINAL SUMMARY

The remaining longitudinal shell audit is now an explicit finite matrix-element insertion problem. The shell carrier basis is

$$|+, a\rangle := |G = 0, u_a\rangle, \quad |-, a\rangle := |G = -G_*, u_a\rangle, \quad a = 1, 2,$$

and the shell-restricted block is

$$H_D(0) = \begin{pmatrix} E_+(0) & M(0) \\ M(0)^{\dagger} & E_-(0) \end{pmatrix}.$$

Its exact Schur-corrected entries are

$$(E_{\pm}(0))_{ab} = \langle \pm, a | H_{\Psi\Psi}(k_*) | \pm, b \rangle - \frac{1}{M_X^2} \langle \mathcal{C}_{\pm}^{(b)}, \mathcal{C}_{\pm}^{(a)} \rangle_X + O(M_X^{-4}),$$

with

$$\mathcal{C}(k) = \alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k).$$

The J -sector is already explicit:

$$J_i^a[\Psi] = (\partial_i \Psi^{\dagger}) T^a \Psi + \Psi^{\dagger} T^a (\partial_i \Psi),$$

so after linearization around

$$\Psi_0(x) = z_0 \sum_{Q \in S_{12}} A_Q e^{iQ \cdot x},$$

the $Q = G_*$ harmonic coefficient is

$$J_{i,r}^a(G_*; q) = i q_i A_{G_*} u_r^\dagger T^a z_0 + i (G_*)_i A_{G_*} z_0^\dagger T^a u_r.$$

Therefore the reduced intervalley shell-normal J -matrix

$$U_{\parallel} = \hat{n}^j \mathcal{U}_j(0), \quad \mathcal{U}_{j,ab}(0) = \langle -, a | \hat{\mathcal{J}}_j(G_*; 0) | +, b \rangle,$$

is explicit, and

$$u_{\parallel} = \frac{1}{2} \text{Tr}(\Sigma_3 U_{\parallel}) = \frac{1}{2}(u_{11} - u_{22}), \quad L_* = m_{\parallel} u_{\parallel} = \frac{m_{\parallel}}{2}(u_{11} - u_{22}).$$

By contrast, the K -sector is currently fixed only at shell-template level:

$$K_i^a[P, \psi] = \text{Tr}(T^a \partial_i P) (\Lambda_E^{(0)} E_j^b + \Lambda_O^{(0)} J_j^b) D_j \psi_b,$$

so the full K -dependent Schur correction is not yet microscopically numerically closed.

Hence the exact present status is:

J -sector and $U_{\parallel}$: explicit,

K -sector: template explicit only,

$a_{\pm}, b_{\pm}, c_{\pm}, u_{11}, u_{22}$ are now reduced to a finite constitutive-data insertion problem.

CXLII. CORE-ONLY EXPLICIT SHELL MATRIX FOR $H_{\Psi\Psi}(k)$ ON THE LOCKED 12-MODE BCC BACKGROUND

CXLIII. SHORT EXPLANATION

At the present stage, the cleanest nonredundant move is to separate the full shell matrix into:

1. the *core/refined triplet block* $H_{\Psi\Psi}(k)$, which can already be written explicitly up to the still-unspecified refined internal terms;
2. the *Class II Schur correction*, whose J -sector is explicit but whose full K -sector remains only template-level.

This immediately yields an important partial result:

the pure locked Brazovskii core contributes an internal scalar shell block,

so any nontrivial Σ_3 -resolved longitudinal carrier data must come from refined internal structure and/or the mediated J -sector.

CXLIV. 1. LOCKED 12-MODE BCC BACKGROUND AND SHELL BASIS

Fix the equal-amplitude locked background

$$\boxed{\phi_0(x) = A \sum_{Q \in \mathcal{S}_{12}} e^{iQ \cdot x}, \quad \Psi_0(x) = z_0 \phi_0(x), \quad P_0 = z_0 z_0^\dagger,} \quad (325)$$

with

$$\mathcal{S}_{12} = \frac{q_*}{\sqrt{2}} \{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\}, \quad k_* = \frac{G_*}{2}. \quad (326)$$

Choose the locked internal basis

$$u_0 = z_0, \quad u_1, u_2 \in Q_{z_0}, \quad u_r^\dagger u_s = \delta_{rs}. \quad (327)$$

The shell carrier basis is

$$|+, a\rangle := |G = 0, u_a\rangle, \quad |-, a\rangle := |G = -G_*, u_a\rangle, \quad a = 1, 2. \quad (328)$$

CXLV. 2. CORE CONDENSATE OPERATOR IN THE TANGENT SECTOR

The core/refined condensate Lagrangian is

$$\mathcal{L}_\Psi^{\text{core}} = |\partial_t \Psi|^2 - \mu^2 \Psi^\dagger \Psi + Z \partial_i \Psi^\dagger \partial_i \Psi - Y (\nabla^2 \Psi)^\dagger (\nabla^2 \Psi) - u (\Psi^\dagger \Psi)^2 + \mathcal{L}_\Psi^{\text{ref}}. \quad (329)$$

Linearize around

$$\Psi_0 = z_0 \phi_0$$

using tangent fluctuations

$$\delta \Psi_k(x) = e^{ik \cdot x} \sum_{G \in \Lambda^*} \sum_{a=1}^2 c_{G,a} u_a e^{iG \cdot x}. \quad (330)$$

Because $u_a^\dagger z_0 = 0$, the quartic density term contributes only a scalar periodic density potential in the tangent sector. Therefore the tangent-sector Bloch operator has the form

$$(H_{\Psi\Psi}^{\text{core}}(k))_{(G,a),(G',b)} = \delta_{ab} [\omega_B(k+G) \delta_{G,G'} + 2u \hat{\rho}_0(G' - G)] + R_{ab}^{\text{ref}}(G' - G; k), \quad (331)$$

where

$$\omega_B(q) := -\mu^2 + Z|q|^2 - Y|q|^4, \quad (332)$$

and

$$\rho_0(x) := |\phi_0(x)|^2, \quad \hat{\rho}_0(Q) = \int_{\text{cell}} d^3x e^{-iQ \cdot x} \rho_0(x). \quad (333)$$

The matrix $R_{ab}^{\text{ref}}(Q; k)$ collects all still-unspecified refined internal derivative terms coming from $\mathcal{L}_\Psi^{\text{ref}}$.

CXLVI. 3. EXPLICIT FOURIER COEFFICIENTS OF THE LOCKED BCC DENSITY

From

$$\phi_0(x) = A \sum_{Q \in \mathcal{S}_{12}} e^{iQ \cdot x},$$

one gets

$$\rho_0(x) = |\phi_0(x)|^2 = A^2 \sum_{Q, Q' \in \mathcal{S}_{12}} e^{i(Q-Q') \cdot x}. \quad (334)$$

Hence

$$\widehat{\rho}_0(0) = 12 A^2 |\Omega_{\text{cell}}|, \quad (335)$$

and for the selected Bragg vector G_* ,

$$\widehat{\rho}_0(G_*) = N_{G_*} A^2 |\Omega_{\text{cell}}|, \quad (336)$$

where

$$N_{G_*} := \# \{ (Q, Q') \in \mathcal{S}_{12}^2 : Q - Q' = G_* \}. \quad (337)$$

So the core Bragg amplitude is an explicit finite shell-combinatorial quantity.

CXLVII. 4. EXACT CORE FORMULAS FOR $E_{\pm}(0)$ AND $M(0)$

By restriction to the shell basis,

$$(E_+^{\text{core}}(0))_{ab} = \langle +, a | H_{\Psi\Psi}^{\text{core}}(k_*) | +, b \rangle, \quad (338)$$

$$(E_-^{\text{core}}(0))_{ab} = \langle -, a | H_{\Psi\Psi}^{\text{core}}(k_*) | -, b \rangle. \quad (339)$$

Using (331),

$$(E_+^{\text{core}}(0))_{ab} = \delta_{ab} [\omega_B(k_*) + 2u \widehat{\rho}_0(0)] + R_{ab}^{\text{ref}}(0; k_*), \quad (340)$$

$$(E_-^{\text{core}}(0))_{ab} = \delta_{ab} [\omega_B(-k_*) + 2u \widehat{\rho}_0(0)] + R_{ab}^{\text{ref}}(0; -k_*). \quad (341)$$

Similarly, the core intervalley Bragg block is

$$M_{ab}^{\text{core}}(0) = \langle +, a | H_{\Psi\Psi}^{\text{core}}(k_*) | -, b \rangle = \delta_{ab} 2u \widehat{\rho}_0(G_*) + R_{ab}^{\text{ref}}(-G_*; k_*). \quad (342)$$

Therefore the finite valley entries are

$$a_+^{\text{core}} = \omega_B(k_*) + 2u \widehat{\rho}_0(0) + R_{11}^{\text{ref}}(0; k_*), \quad b_+^{\text{core}} = \omega_B(k_*) + 2u \widehat{\rho}_0(0) + R_{22}^{\text{ref}}(0; k_*), \quad (343)$$

$$c_+^{\text{core}} = R_{12}^{\text{ref}}(0; k_*), \quad (344)$$

and

$$a_-^{\text{core}} = \omega_B(-k_*) + 2u \widehat{\rho}_0(0) + R_{11}^{\text{ref}}(0; -k_*), \quad b_-^{\text{core}} = \omega_B(-k_*) + 2u \widehat{\rho}_0(0) + R_{22}^{\text{ref}}(0; -k_*), \quad (345)$$

$$c_-^{\text{core}} = R_{12}^{\text{ref}}(0; -k_*). \quad (346)$$

CXLVIII. 5. IMPORTANT IMMEDIATE CONSEQUENCE

If the refined internal terms are absent or internally scalar in the locked basis, i.e.

$$R_{ab}^{\text{ref}}(Q; k) = r^{\text{ref}}(Q; k) \delta_{ab},$$

then

$$\boxed{a_+^{\text{core}} = b_+^{\text{core}}, \quad a_-^{\text{core}} = b_-^{\text{core}}, \quad c_+^{\text{core}} = c_-^{\text{core}} = 0.} \quad (347)$$

Thus the pure locked Brazovskii core produces an internal identity block on the tangent doublet:

$$\boxed{E_{\pm}^{\text{core}}(0) \propto \mathbf{1}_2, \quad M^{\text{core}}(0) \propto \mathbf{1}_2.} \quad (348)$$

This is exactly the dangerous scalar Bragg route: it produces the constant scalar Bragg block but no internal Σ_3 -resolved longitudinal selector.

Therefore any genuinely nontrivial longitudinal $2 + 2$ carrier data must come from:

1. refined non-scalar internal structure inside $\mathcal{L}_{\Psi}^{\text{ref}}$, and/or
2. the Class II mediated J -sector.

CXLIX. 6. EXPLICIT J -SECTOR HARMONIC INPUT FOR $U_{\parallel}$

The microscopic current is explicitly

$$\boxed{J_i^a[\Psi] = (\partial_i \Psi^\dagger) T^a \Psi + \Psi^\dagger T^a (\partial_i \Psi).} \quad (349)$$

Linearizing around $\Psi_0 = z_0 \phi_0$, one obtains the shell-relevant harmonic coefficient

$$\boxed{J_{i,r}^a(G_*; q) = i q_i A_{G_*} u_r^\dagger T^a z_0 + i (G_*)_i A_{G_*} z_0^\dagger T^a u_r.} \quad (350)$$

Hence the reduced intervalley J -matrix is

$$\boxed{(\mathcal{U}_j(0))_{ab} = \langle -, a | \hat{J}_j(G_*; 0) | +, b \rangle, \quad U_{\parallel} := \hat{n}^j \mathcal{U}_j(0).} \quad (351)$$

Its shell-normal Σ_3 -component is

$$\boxed{u_{\parallel} = \frac{1}{2} \text{Tr}(\Sigma_3 U_{\parallel}) = \frac{1}{2} [(U_{\parallel})_{11} - (U_{\parallel})_{22}].} \quad (352)$$

Thus

$$\boxed{L_* = m_{\parallel} u_{\parallel} = \frac{m_{\parallel}}{2} [(U_{\parallel})_{11} - (U_{\parallel})_{22}], \quad m_{\parallel} = m^3(P_0).} \quad (353)$$

So $U_{\parallel}$ is already explicit once the canonical locked internal basis (z_0, u_1, u_2) and the chosen generator T^a are fixed.

CL. 7. HONEST SPLIT OF WHAT IS EXPLICIT AND WHAT IS STILL PARAMETRIC

At the present stage, the following are already explicit:

1. the shell basis $|\pm, a\rangle$;
2. the core shell matrix formulas (340)–(342);
3. the density Fourier coefficients $\hat{\rho}_0(0)$ and $\hat{\rho}_0(G_*)$;

4. the microscopic J -current and its $Q = G_*$ harmonic coefficient (350);
5. the reduced J -matrix $U_{\parallel}$ and $L_* = m_{\parallel} u_{\parallel}$.

What is still parametric:

1. the non-scalar refined internal block $R_{ab}^{\text{ref}}(Q; k)$, because $\mathcal{L}_{\Psi}^{\text{ref}}$ has not yet been fixed uniquely;
2. the full microscopic K -matrix beyond its shell-template form.

Thus the remaining frontier is not structural. It is the insertion of the final constitutive data into the already explicit finite shell formulas.

CLI. FINAL SUMMARY

For the locked 12-mode BCC background, the tangent-sector core Bloch operator is

$$(H_{\Psi\Psi}^{\text{core}}(k))_{(G,a),(G',b)} = \delta_{ab} \left[\omega_B(k+G) \delta_{G,G'} + 2u \widehat{\rho}_0(G'-G) \right] + R_{ab}^{\text{ref}}(G'-G; k),$$

with

$$\omega_B(q) = -\mu^2 + Z|q|^2 - Y|q|^4, \quad \rho_0(x) = |\phi_0(x)|^2.$$

Therefore the shell entries are

$$(E_+^{\text{core}}(0))_{ab} = \delta_{ab} \left[\omega_B(k_*) + 2u \widehat{\rho}_0(0) \right] + R_{ab}^{\text{ref}}(0; k_*),$$

$$(E_-^{\text{core}}(0))_{ab} = \delta_{ab} \left[\omega_B(-k_*) + 2u \widehat{\rho}_0(0) \right] + R_{ab}^{\text{ref}}(0; -k_*),$$

$$M_{ab}^{\text{core}}(0) = \delta_{ab} 2u \widehat{\rho}_0(G_*) + R_{ab}^{\text{ref}}(-G_*; k_*).$$

If the refined internal terms are absent or internally scalar in the locked basis, then

$$E_{\pm}^{\text{core}}(0) \propto \mathbf{1}_2, \quad M^{\text{core}}(0) \propto \mathbf{1}_2.$$

Hence the pure locked Brazovskii core reproduces the dangerous scalar Bragg block but does not produce a nontrivial Σ_3 -resolved longitudinal selector.

The microscopic J -sector is already explicit:

$$J_i^a[\Psi] = (\partial_i \Psi^\dagger) T^a \Psi + \Psi^\dagger T^a (\partial_i \Psi),$$

and its shell-relevant harmonic coefficient is

$$J_{i,r}^a(G_*; q) = i q_i A_{G_*} u_r^\dagger T^a z_0 + i (G_*)_i A_{G_*} z_0^\dagger T^a u_r.$$

Therefore the reduced intervalley shell-normal matrix

$$U_{\parallel} = \hat{n}^j \mathcal{U}_j(0), \quad (\mathcal{U}_j(0))_{ab} = \langle -, a | \widehat{J}_j(G_*; 0) | +, b \rangle,$$

is explicit, and

$$u_{\parallel} = \frac{1}{2} \text{Tr}(\Sigma_3 U_{\parallel}), \quad L_* = m_{\parallel} u_{\parallel}.$$

So the exact present verdict is:

core shell matrix: explicit up to R_{ab}^{ref} ,

J -sector longitudinal selector: explicit,

K -sector full microscopic closure: still template-level.

Thus the remaining frontier is only the insertion of the final constitutive data into an already explicit finite shell matrix problem.

CLII. EXPLICIT EVALUATION OF R_{ab}^{ref} FOR THE MINIMAL SAFE REFINED ALIGNMENT TERM

CLIII. SHORT EXPLANATION

The previous core-only shell matrix formulas contained the still-unspecified refined internal block

$$R_{ab}^{\text{ref}}(Q; k).$$

To avoid leaving this completely abstract, we now choose the only refined term that is both explicitly available in the current notes and rigorously safe on the locked sector: the projector-stiffness/alignment term

$$\boxed{\mathcal{F}_{\text{align}} = \frac{\chi}{4} \int d^3x \rho^2 \text{Tr}(\partial_i P \partial_i P).} \quad (354)$$

We show that this term gives

$$R_{ab}^{\text{ref,align}}(0; \pm k_*) = 0, \quad R_{ab}^{\text{ref,align}}(-G_*; k_*) = 0$$

at tree level on the locked constant-projector background. Its effect appears only indirectly after integrating out the heavy projector tangent mode.

CLIV. 1. LOCKED BACKGROUND AND FLUCTUATION SPLIT

Write

$$\Psi = \rho z, \quad P = z z^\dagger, \quad z^\dagger z = 1.$$

On the locked 12-mode BCC background,

$$\boxed{\Psi_0(x) = z_0 \phi_0(x), \quad P_0 = z_0 z_0^\dagger, \quad \partial_i P_0 = 0.} \quad (355)$$

Introduce tangent fluctuations of the projector by

$$\boxed{z = z_0 + \eta - \frac{1}{2}(\eta^\dagger \eta) z_0 + \dots, \quad z_0^\dagger \eta = 0, \quad \eta \in Q_{z_0}.} \quad (356)$$

Then

$$\boxed{\delta P = \eta z_0^\dagger + z_0 \eta^\dagger, \quad \partial_i \delta P = (\partial_i \eta) z_0^\dagger + z_0 (\partial_i \eta^\dagger).} \quad (357)$$

CLV. 2. EXACT VANISHING ON THE LOCKED SECTOR

Because P_0 is constant,

$$\boxed{\mathcal{F}_{\text{align}}[\Psi_0, P_0] = 0.} \quad (358)$$

Therefore this term does not modify:

1. the shell radius q_* ;
2. the scalar locked BCC selection at the background level;
3. the background shell Fourier amplitudes at tree level.

Equivalently, in the shell-matrix language, it contributes no background constant Bragg block at zeroth order.

CLVI. 3. QUADRATIC EXPANSION OF THE ALIGNMENT TERM

Expand (354) around (355). Since $\partial_i P_0 = 0$, the first nonzero term is quadratic:

$$\mathcal{F}_{\text{align}}^{(2)} = \frac{\chi}{4} \int d^3x \bar{\rho}^2 \text{Tr}(\partial_i \delta P \partial_i \delta P) + O(\delta \rho \eta^2, \eta^3), \quad (359)$$

where $\bar{\rho} := |\phi_0|$ on the locked background.

Using (357),

$$\begin{aligned} \text{Tr}(\partial_i \delta P \partial_i \delta P) &= \text{Tr} \left[(\partial_i \eta) z_0^\dagger + z_0 (\partial_i \eta^\dagger) \right] \left[(\partial_i \eta) z_0^\dagger + z_0 (\partial_i \eta^\dagger) \right] \\ &= 2 (\partial_i \eta^\dagger) (\partial_i \eta), \end{aligned} \quad (360)$$

so

$$\mathcal{F}_{\text{align}}^{(2)} = \frac{\chi}{2} \int d^3x \bar{\rho}^2 (\partial_i \eta^\dagger) (\partial_i \eta). \quad (361)$$

Thus the alignment term contributes a positive quadratic projector-tangent operator

$$H_{\eta\eta}^{\text{align}} = \chi \bar{\rho}^2 (-\nabla^2) \quad (362)$$

on the heavy projector mode.

CLVII. 4. TREE-LEVEL SHELL-MATRIX CONTRIBUTION TO THE TANGENT TRIPLET BLOCK

Now compare (361) with the tangent triplet shell block

$$H_{\Psi\Psi}(k).$$

The key point is that $\mathcal{F}_{\text{align}}^{(2)}$ is quadratic in the projector tangent mode η , not in the shell-carrying tangent triplet coefficients $c_{G,a}$ directly.

Therefore, in the shell basis

$$|+, a\rangle, \quad |-, a\rangle, \quad a = 1, 2,$$

the direct tree-level contribution of the alignment term to the triplet shell block vanishes:

$$R_{ab}^{\text{ref,align}}(Q; k) = 0 \quad \text{at tree level on the locked constant-projector background.} \quad (363)$$

In particular,

$$R_{ab}^{\text{ref,align}}(0; k_*) = 0, \quad R_{ab}^{\text{ref,align}}(0; -k_*) = 0, \quad R_{ab}^{\text{ref,align}}(-G_*; k_*) = 0. \quad (364)$$

Hence the core shell entries remain

$$(E_+^{\text{core+align,tree}}(0))_{ab} = \delta_{ab} \left[\omega_B(k_*) + 2u \hat{\rho}_0(0) \right], \quad (365)$$

$$(E_-^{\text{core+align,tree}}(0))_{ab} = \delta_{ab} \left[\omega_B(-k_*) + 2u \hat{\rho}_0(0) \right], \quad (366)$$

and

$$M_{ab}^{\text{core+align,tree}}(0) = \delta_{ab} 2u \hat{\rho}_0(G_*). \quad (367)$$

So the direct shell block remains proportional to the internal identity:

$$E_{\pm}^{\text{core+align,tree}}(0) \propto \mathbf{1}_2, \quad M^{\text{core+align,tree}}(0) \propto \mathbf{1}_2. \quad (368)$$

CLVIII. 5. INTERPRETATION: NO DIRECT BRANCH-A SEED FROM THE EXPLICIT SAFE REFINED TERM

Equation (368) implies immediately that the explicit safe refined term $\mathcal{F}_{\text{align}}$ does *not* generate a nontrivial internal Σ_3 -resolved valley selector at tree level. Therefore it does not provide a direct constant-leading Branch-A coefficient:

$$\boxed{\Lambda_{\parallel}^{(0)}|_{\text{align,tree}} = 0.} \quad (369)$$

So the projector-stiffness refinement is structurally safe but not a direct longitudinal source.

CLIX. 6. INDUCED EFFECT AFTER HEAVY-MODE ELIMINATION

Although the direct tree-level contribution vanishes, $\mathcal{F}_{\text{align}}$ makes the projector tangent mode heavy. Hence, if an admissible shell-projector mixing source

$$\mathcal{J}_{\Psi\eta}$$

exists, then integrating out η induces a second-order correction of Schur type:

$$\boxed{\delta H_{\Psi\Psi}^{(\text{align,ind})} = -H_{\Psi\eta} (H_{\eta\eta}^{\text{align}})^{-1} H_{\eta\Psi}.} \quad (370)$$

Using (362),

$$\boxed{(H_{\eta\eta}^{\text{align}})^{-1} \sim \frac{1}{\chi \bar{\rho}^2 q_*^2}} \quad (371)$$

at shell scale. Therefore the induced shell corrections satisfy the parametric bound

$$\boxed{\delta E_{\pm}^{(\text{align,ind})}, \quad \delta M^{(\text{align,ind})}, \quad \delta W_{\parallel}^{(\text{align,ind})}, \quad \delta W_I^{(\text{align,ind})}, \quad \delta W_J^{(\text{align,ind})} = O\left(\frac{1}{\chi \bar{\rho}^2 q_*^2}\right),} \quad (372)$$

provided the microscopic shell sector couples to the projector tangent mode with the admissible tensor structure.

CLX. 7. HONEST ENDPOINT

Thus the explicit refined alignment term gives the following exact status:

1. it is zero on the locked background and preserves the scalar locked reduction;
2. it gives a positive heavy quadratic operator to the projector tangent mode;
3. it does not generate a direct tree-level shell-enriched tensor;
4. consequently

$$R_{ab}^{\text{ref,align}}(Q; k) = 0 \quad \text{for the shell triplet block at tree level;}$$

5. any effect on the actual witness coefficients arises only indirectly, after heavy-mode elimination, and depends on the still-open explicit shell-projector mixing source.

Therefore the explicit safe refined term confirms, rather than changes, the current mainline: the direct explicit longitudinal selector is still supplied by the Branch-B/Class-II route, not by the tree-level projector stiffness term.

CLXI. FINAL SUMMARY

Choose the explicit safe refined term

$$\mathcal{F}_{\text{align}} = \frac{\chi}{4} \int d^3x \rho^2 \text{Tr}(\partial_i P \partial_i P).$$

On the locked 12-mode BCC background

$$\Psi_0 = z_0 \phi_0, \quad P_0 = z_0 z_0^\dagger, \quad \partial_i P_0 = 0,$$

this term vanishes exactly at the background level:

$$\mathcal{F}_{\text{align}}[\Psi_0, P_0] = 0.$$

Its first nonzero contribution is quadratic in the projector tangent fluctuation η :

$$\mathcal{F}_{\text{align}}^{(2)} = \frac{\chi}{2} \int d^3x \bar{\rho}^2 (\partial_i \eta^\dagger)(\partial_i \eta),$$

so it defines the heavy projector operator

$$H_{\eta\eta}^{\text{align}} = \chi \bar{\rho}^2 (-\nabla^2).$$

Therefore it contributes no direct tree-level correction to the tangent-triplet shell block:

$$\boxed{R_{ab}^{\text{ref,align}}(0; \pm k_*) = 0, \quad R_{ab}^{\text{ref,align}}(-G_*; k_*) = 0.}$$

Hence

$$\boxed{E_{\pm}^{\text{core+align,tree}}(0) \propto \mathbf{1}_2, \quad M^{\text{core+align,tree}}(0) \propto \mathbf{1}_2.}$$

So the explicit safe refined term does not generate a direct internal Σ_3 -resolved selector and does not provide a tree-level Branch-A longitudinal seed:

$$\boxed{\Lambda_{\parallel}^{(0)}|_{\text{align,tree}} = 0.}$$

Its effect appears only indirectly after integrating out the heavy projector tangent mode:

$$\delta H_{\Psi\Psi}^{(\text{align,ind})} = -H_{\Psi\eta} (H_{\eta\eta}^{\text{align}})^{-1} H_{\eta\Psi},$$

which yields induced shell corrections of order

$$\boxed{O\left(\frac{1}{\chi \bar{\rho}^2 q_*^2}\right),}$$

provided an admissible shell-projector mixing source exists.

Therefore the explicit safe refined term confirms the current mainline:

$$\boxed{\text{the direct explicit longitudinal selector is still the Branch-B/Class-II route,}}$$

not the tree-level projector-stiffness sector.

CLXII. DIRECT CANONICAL EXTRACTION OF $U_{\parallel}$ AND $u_{\parallel}$

CLXIII. SHORT EXPLANATION

At the present stage, the carrier-side longitudinal frontier is no longer a tensor-classification problem. It is the extraction of the single scalar

$$u_{\parallel} = \frac{1}{2} \text{Tr}(\Sigma_3 \hat{n}^j \mathcal{U}_j(0)),$$

in the canonical locked basis.

Once this scalar is known, the longitudinal invariant is

$$L_* = m_{\parallel} u_{\parallel},$$

and in the minimal explicit canonical split,

$$\lambda_{\parallel} = \frac{1}{2} \alpha_X m_{\parallel} u_{\parallel} I_{\theta'^3}.$$

CLXIV. 1. CANONICAL LOCKED BASIS

Fix the canonical locked basis

$$\boxed{z_0 = (1, 0, 0)^T, \quad u_1 = (0, 1, 0)^T, \quad u_2 = (0, 0, 1)^T.} \quad (373)$$

Then

$$Q_{z_0} = \text{span}_{\mathbb{C}}\{u_1, u_2\},$$

and the tangent Pauli matrices are

$$\boxed{\Sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \Sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \Sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.} \quad (374)$$

Fix also the shell frame

$$\boxed{\{e_1, e_2, \hat{n}\}, \quad \hat{n} = \frac{G_*}{|G_*|}.} \quad (375)$$

CLXV. 2. EXPLICIT MICROSCOPIC J -CRITERION IN THIS BASIS

The J -block is explicitly known once z_0 and (u_1, u_2) are fixed, and the relevant tangent-shell couplings are controlled by

$$\boxed{t_{\alpha 0}^a := u_{\alpha}^{\dagger} T^a z_0, \quad \alpha = 1, 2.} \quad (376)$$

In the canonical basis (373),

$$\boxed{t_{10}^a = (T^a)_{21}, \quad t_{20}^a = (T^a)_{31}.} \quad (377)$$

Therefore the exact microscopic coupling criterion becomes

$$\boxed{\mathcal{J}(k) \text{ couples the shell-tangent doublet to the mediator} \iff (T^a)_{21} \neq 0 \text{ or } (T^a)_{31} \neq 0 \text{ for some active } a.} \quad (378)$$

So in the canonical locked basis, the J -sector coupling question is a literal finite internal-matrix test.

CLXVI. 3. BRAGG-MATCHED SHELL RESTRICTION

At the folded Bragg point, the shell basis is

$$|+, \alpha\rangle \equiv |G = 0, \alpha\rangle, \quad |-, \alpha\rangle \equiv |G = -G_*, \alpha\rangle, \quad \alpha = 1, 2. \quad (379)$$

The exact Bragg matching condition is

$$0 = (-G_*) + Q \iff Q = G_*, \quad (380)$$

so only the $Q = G_*$ harmonic contributes directly to the intervalley shell block.

Thus the reduced J -matrix is defined by

$$(\mathcal{U}_j(0))_{\alpha\beta} := \langle -, \alpha | \hat{J}_j(G_*; 0) | +, \beta \rangle. \quad (381)$$

CLXVII. 4. CANONICAL DECOMPOSITION OF THE J -SECTOR TENSOR

The shell-relevant J -tensor decomposes as

$$\mathcal{U}_{jr}{}^b(0) = u_{\parallel} \hat{n}_j (\Sigma_3)_r{}^b + u_E (e_{1j} \Sigma_1 + e_{2j} \Sigma_2)_r{}^b + u_O (-e_{2j} \Sigma_1 + e_{1j} \Sigma_2)_r{}^b + \mathcal{U}_{jr}^{\text{irr}b}. \quad (382)$$

Projecting along $\hat{n}^j$, one gets

$$\hat{n}^j \mathcal{U}_j(0) = u_{\parallel} \Sigma_3 + \hat{n}^j \mathcal{U}_j^{\text{irr}}(0), \quad (383)$$

because

$$\hat{n} \cdot e_1 = \hat{n} \cdot e_2 = 0.$$

Hence the exact projector formula is

$$u_{\parallel} = \frac{1}{2} \text{Tr}(\Sigma_3 \hat{n}^j \mathcal{U}_j(0)). \quad (384)$$

If the irreducible remainder has no shell-normal Σ_3 -singlet component in the chosen truncation, then

$$U_{\parallel} := \hat{n}^j \mathcal{U}_j(0) = u_{\parallel} \Sigma_3. \quad (385)$$

CLXVIII. 5. EXPLICIT MATRIX FORM OF $U_{\parallel}$

Under the minimal explicit canonical split,

$$U_{\parallel} = u_{\parallel} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (386)$$

Therefore

$$(U_{\parallel})_{11} = u_{\parallel}, \quad (U_{\parallel})_{22} = -u_{\parallel}, \quad (U_{\parallel})_{12} = (U_{\parallel})_{21} = 0. \quad (387)$$

Equivalently,

$$u_{\parallel} = \frac{1}{2} [(U_{\parallel})_{11} - (U_{\parallel})_{22}]. \quad (388)$$

Thus the longitudinal shell-normal J -singlet survives if and only if the two carrier basis states are split in the Σ_3 direction:

$$u_{\parallel} \neq 0 \iff (U_{\parallel})_{11} \neq (U_{\parallel})_{22}. \quad (389)$$

CLXIX. 6. BACKGROUND SELECTOR AND L_*

In the canonical locked reduction, the background selector aligns with the singlet axis:

$$\boxed{m^a(P_0) = m_{\parallel} \delta^{a3}.} \quad (390)$$

Therefore the longitudinal invariant is

$$\boxed{L_* = m_{\parallel} u_{\parallel}.} \quad (391)$$

Using (388),

$$\boxed{L_* = \frac{m_{\parallel}}{2} [(U_{\parallel})_{11} - (U_{\parallel})_{22}].} \quad (392)$$

So the actual remaining longitudinal scalar check is simply

$$\boxed{m_{\parallel} \neq 0 \quad \text{and} \quad (U_{\parallel})_{11} \neq (U_{\parallel})_{22}.} \quad (393)$$

CLXX. 7. REDUCED WITNESS COEFFICIENT

In the current canonical split, the reduced witness coefficient is

$$\boxed{\lambda_{\parallel} = \frac{1}{2} \alpha_X m_{\parallel} u_{\parallel} I_{\theta'^3}.} \quad (394)$$

Hence

$$\boxed{\alpha_X \neq 0, \quad m_{\parallel} \neq 0, \quad u_{\parallel} \neq 0 \implies \lambda_{\parallel} \neq 0.} \quad (395)$$

CLXXI. 8. OPTIONAL EXPLICIT INTERNAL-ALGEBRA CHECK

In the canonical basis, the shell-tangent couplings are the first-column off-diagonal entries

$$t_{10}^a = (T^a)_{21}, \quad t_{20}^a = (T^a)_{31}.$$

So the microscopic origin of a nonzero $u_{\parallel}$ is an actual J -induced asymmetry between the two tangent channels (u_1, u_2) .

Therefore the last honest fork is exactly:

$$\boxed{\text{either verify an actual internal } J\text{-split between } u_1, u_2, \quad \text{or compute } (U_{\parallel})_{11} - (U_{\parallel})_{22} \text{ directly.}} \quad (396)$$

CLXXII. 9. HONEST STATUS

What is fully closed:

1. the canonical locked basis;
2. the exact projector formula for $u_{\parallel}$;
3. the identification

$$L_* = m_{\parallel} u_{\parallel};$$

4. the reduced witness formula

$$\lambda_{\parallel} = \frac{1}{2} \alpha_X m_{\parallel} u_{\parallel} I_{\theta'^3}.$$

What remains open is only the explicit microscopic value of $u_{\parallel}$, equivalently the direct evaluation of

$$(U_{\parallel})_{11} - (U_{\parallel})_{22}$$

for the final chosen refined locked operator.

CLXXIII. FINAL SUMMARY

In the canonical locked basis

$$z_0 = (1, 0, 0)^T, \quad u_1 = (0, 1, 0)^T, \quad u_2 = (0, 0, 1)^T,$$

the shell-relevant J -tensor decomposes as

$$\mathcal{U}_{jr}{}^b(0) = u_{\parallel} \hat{n}_j (\Sigma_3)_r{}^b + u_E (e_{1j}\Sigma_1 + e_{2j}\Sigma_2)_r{}^b + u_O (-e_{2j}\Sigma_1 + e_{1j}\Sigma_2)_r{}^b + \mathcal{U}_{jr}^{\text{irr}b}.$$

Therefore the shell-normal reduced matrix is

$$U_{\parallel} := \hat{n}^j \mathcal{U}_j(0),$$

and the longitudinal scalar is extracted by the exact projector formula

$$u_{\parallel} = \frac{1}{2} \text{Tr}(\Sigma_3 U_{\parallel}) = \frac{1}{2} [(U_{\parallel})_{11} - (U_{\parallel})_{22}].$$

Under the minimal explicit canonical split, one has

$$U_{\parallel} = u_{\parallel} \Sigma_3,$$

so explicitly

$$(U_{\parallel})_{11} = u_{\parallel}, \quad (U_{\parallel})_{22} = -u_{\parallel}.$$

The background selector aligns as

$$m^a(P_0) = m_{\parallel} \delta^{a3},$$

hence the longitudinal invariant is

$$L_* = m_{\parallel} u_{\parallel} = \frac{m_{\parallel}}{2} [(U_{\parallel})_{11} - (U_{\parallel})_{22}].$$

In the current canonical reduced witness normalization,

$$\lambda_{\parallel} = \frac{1}{2} \alpha_X m_{\parallel} u_{\parallel} I_{\theta'^3}.$$

So

$$\alpha_X \neq 0, \quad m_{\parallel} \neq 0, \quad (U_{\parallel})_{11} \neq (U_{\parallel})_{22} \implies \lambda_{\parallel} \neq 0.$$

Finally, the microscopic J -coupling criterion is finite and explicit:

$$t_{10}^a = (T^a)_{21}, \quad t_{20}^a = (T^a)_{31},$$

so the shell-tangent doublet couples to the mediator iff some active generator has a nonzero first-column off-diagonal entry.

Therefore the longitudinal carrier frontier is no longer structural. It is the explicit finite evaluation of the diagonal asymmetry

$$(U_{\parallel})_{11} - (U_{\parallel})_{22}.$$

CLXXIV. ACTUAL INTERNAL-GENERATOR CALCULATION OF THE LONGITUDINAL J -SPLIT

Fix the canonical locked basis

$$z_0 = e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad u_1 = e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad u_2 = e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

Let the internal generators be the standard Hermitian $SU(3)$ basis

$$T^a = \frac{\lambda^a}{2}.$$

Then the only entries relevant for the shell-tangent coupling are

$$t_{10}^a := u_1^\dagger T^a z_0 = (T^a)_{21}, \quad t_{20}^a := u_2^\dagger T^a z_0 = (T^a)_{31}.$$

For the standard Gell-Mann basis one gets explicitly

$$T^1 = \frac{1}{2} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad T^2 = \frac{1}{2} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

$$T^3 = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad T^4 = \frac{1}{2} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix},$$

$$T^5 = \frac{1}{2} \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, \quad T^6 = \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix},$$

$$T^7 = \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad T^8 = \frac{1}{2\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}.$$

Therefore

$$(t_{10}^a, t_{20}^a) = \begin{cases} (\frac{1}{2}, 0), & a = 1, \\ (\frac{i}{2}, 0), & a = 2, \\ (0, 0), & a = 3, \\ (0, \frac{1}{2}), & a = 4, \\ (0, \frac{i}{2}), & a = 5, \\ (0, 0), & a = 6, 7, 8. \end{cases}$$

So only the four generators

$$T^1, T^2, T^4, T^5$$

couple the locked direction z_0 to the tangent doublet (u_1, u_2) . Moreover,

$$T^{1,2} : z_0 \leftrightarrow u_1, \quad T^{4,5} : z_0 \leftrightarrow u_2,$$

while

$$T^3, T^6, T^7, T^8$$

do *not* contribute to the tangent shell-coupling at all.

Now take a general active mediator direction

$$T_\star = \sum_{a=1}^8 c_a T^a, \quad c_a \in \mathbb{R}.$$

Then

$$t_{10}(T_\star) = \frac{1}{2}(c_1 + ic_2), \quad t_{20}(T_\star) = \frac{1}{2}(c_4 + ic_5).$$

In the minimal shell-diagonal truncation, the shell-normal reduced J -matrix has the form

$$U_\parallel = \begin{pmatrix} \kappa_J t_{10} + \overline{\kappa_J t_{10}} & 0 \\ 0 & \kappa_J t_{20} + \overline{\kappa_J t_{20}} \end{pmatrix} = \begin{pmatrix} 2\Re(\kappa_J t_{10}) & 0 \\ 0 & 2\Re(\kappa_J t_{20}) \end{pmatrix},$$

where $\kappa_J \neq 0$ is the shell-normal Bragg prefactor coming from the derivative/harmonic projection.

Hence

$$u_\parallel = \frac{1}{2}[(U_\parallel)_{11} - (U_\parallel)_{22}] = \Re[\kappa_J(t_{10} - t_{20})].$$

Substituting the explicit canonical couplings,

$$u_\parallel = \frac{1}{2} \Re[\kappa_J((c_1 + ic_2) - (c_4 + ic_5))].$$

Therefore the exact canonical vanishing condition is

$$u_\parallel = 0 \iff \Re[\kappa_J((c_1 + ic_2) - (c_4 + ic_5))] = 0.$$

Equivalently, the longitudinal slot survives whenever the effective mediator direction has unequal projected weight on the two coupling sectors

$$\{T^1, T^2\} \quad \text{and} \quad \{T^4, T^5\},$$

after the shell-normal phase κ_J is included.

Finally,

$$L_\star = m_\parallel u_\parallel, \quad \lambda_\parallel = \frac{1}{2} \alpha_X m_\parallel u_\parallel I_{\theta^3},$$

so the explicit canonical sufficient condition is

$$m_\parallel \neq 0, \quad \alpha_X \neq 0, \quad I_{\theta^3} \neq 0, \quad \Re[\kappa_J((c_1 + ic_2) - (c_4 + ic_5))] \neq 0.$$

CLXXV. ACTUAL GENERATOR-LEVEL COMPUTATION OF $u_\parallel$

CLXXVI. STEP 1: CANONICAL BASIS AND ACTIVE COUPLINGS

Fix the canonical locked basis

$$z_0 = e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad u_1 = e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad u_2 = e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

Let

$$T^a = \frac{\lambda^a}{2}$$

be the standard Hermitian $SU(3)$ generators. Then the shell-tangent couplings are

$$t_{10}^a := u_1^\dagger T^a z_0 = (T^a)_{21}, \quad t_{20}^a := u_2^\dagger T^a z_0 = (T^a)_{31}.$$

For the Gell-Mann basis one obtains:

$$T^1 = \frac{1}{2} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow t_{10}^1 = \frac{1}{2}, \quad t_{20}^1 = 0,$$

$$T^2 = \frac{1}{2} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow t_{10}^2 = \frac{i}{2}, \quad t_{20}^2 = 0,$$

$$T^3 = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow t_{10}^3 = 0, \quad t_{20}^3 = 0,$$

$$T^4 = \frac{1}{2} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \Rightarrow t_{10}^4 = 0, \quad t_{20}^4 = \frac{1}{2},$$

$$T^5 = \frac{1}{2} \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix} \Rightarrow t_{10}^5 = 0, \quad t_{20}^5 = \frac{i}{2},$$

$$T^6, T^7, T^8 \Rightarrow t_{10}^a = t_{20}^a = 0.$$

So only

$$T^1, T^2, T^4, T^5$$

couple the locked direction z_0 to the tangent doublet.

CLXXVII. STEP 2: SHELL-NORMAL REDUCED MATRIX

In the minimal shell-diagonal truncation, define the shell-normal complex prefactor

$$\kappa_J \neq 0$$

coming from the actual Bragg/harmonic/derivative projection.

Then the reduced shell-normal J -matrix is

$$U_{\parallel} = \begin{pmatrix} 2\Re(\kappa_J t_{10}) & 0 \\ 0 & 2\Re(\kappa_J t_{20}) \end{pmatrix}.$$

Therefore

$$u_{\parallel} = \frac{1}{2} \left[(U_{\parallel})_{11} - (U_{\parallel})_{22} \right] = \Re[\kappa_J (t_{10} - t_{20})].$$

CLXXVIII. STEP 3: EXPLICIT SINGLE-GENERATOR VALUES

Case A: $T_\star = T^1$

$$t_{10} = \frac{1}{2}, \quad t_{20} = 0,$$

hence

$$u_\parallel(T^1) = \frac{1}{2} \Re(\kappa_J).$$

Therefore

$$L_\ast(T^1) = \frac{m_\parallel}{2} \Re(\kappa_J),$$

and

$$\lambda_\parallel(T^1) = \frac{1}{4} \alpha_X m_\parallel \Re(\kappa_J) I_{\theta'^3}.$$

Case B: $T_\star = T^2$

$$t_{10} = \frac{i}{2}, \quad t_{20} = 0,$$

so

$$u_\parallel(T^2) = \frac{1}{2} \Re(i\kappa_J) = -\frac{1}{2} \Im(\kappa_J).$$

Hence

$$L_\ast(T^2) = -\frac{m_\parallel}{2} \Im(\kappa_J),$$

and

$$\lambda_\parallel(T^2) = -\frac{1}{4} \alpha_X m_\parallel \Im(\kappa_J) I_{\theta'^3}.$$

Case C: $T_\star = T^4$

$$t_{10} = 0, \quad t_{20} = \frac{1}{2},$$

thus

$$u_\parallel(T^4) = -\frac{1}{2} \Re(\kappa_J).$$

Hence

$$L_\ast(T^4) = -\frac{m_\parallel}{2} \Re(\kappa_J),$$

and

$$\lambda_\parallel(T^4) = -\frac{1}{4} \alpha_X m_\parallel \Re(\kappa_J) I_{\theta'^3}.$$

Case D: $T_\star = T^5$

$$t_{10} = 0, \quad t_{20} = \frac{i}{2},$$

so

$$u_{\parallel}(T^5) = -\Re\left(\kappa_J \frac{i}{2}\right) = \frac{1}{2} \Im(\kappa_J).$$

Hence

$$L_\star(T^5) = \frac{m_{\parallel}}{2} \Im(\kappa_J),$$

and

$$\lambda_{\parallel}(T^5) = \frac{1}{4} \alpha_X m_{\parallel} \Im(\kappa_J) I_{\theta'^3}.$$

Case E: $T_\star = T^3, T^6, T^7, T^8$

Since

$$t_{10} = t_{20} = 0,$$

we get

$$u_{\parallel}(T^3) = u_{\parallel}(T^6) = u_{\parallel}(T^7) = u_{\parallel}(T^8) = 0.$$

Therefore these directions do not produce a longitudinal J -split in the canonical locked basis.

CLXXIX. STEP 4: GENERIC ACTIVE DIRECTION

Take a general active mediator direction

$$T_\star = c_1 T^1 + c_2 T^2 + c_4 T^4 + c_5 T^5, \quad c_i \in \mathbb{R}.$$

Then

$$t_{10}(T_\star) = \frac{1}{2}(c_1 + ic_2), \quad t_{20}(T_\star) = \frac{1}{2}(c_4 + ic_5),$$

so

$$u_{\parallel}(T_\star) = \frac{1}{2} \Re\left[\kappa_J((c_1 + ic_2) - (c_4 + ic_5))\right].$$

Hence the exact canonical vanishing condition is

$$u_{\parallel} = 0 \iff \Re\left[\kappa_J((c_1 + ic_2) - (c_4 + ic_5))\right] = 0.$$

CLXXX. STEP 5: EXPLICIT EXAMPLES OF CANCELLATION AND NONCANCELLATION

Noncancelling example 1

$$T_\star = T^1 \quad \Rightarrow \quad u_{\parallel} = \frac{1}{2} \Re(\kappa_J).$$

So if $\Re(\kappa_J) \neq 0$, longitudinal survival is immediate.

Noncancelling example 2

$$T_\star = \frac{1}{\sqrt{2}}(T^1 - T^4)$$

gives

$$u_\parallel = \frac{1}{\sqrt{2}}\Re(\kappa_J).$$

Cancelling example

$$T_\star = \frac{1}{\sqrt{2}}(T^1 + T^4)$$

gives

$$u_\parallel = \frac{1}{2}\Re\left[\kappa_J\left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}\right)\right] = 0.$$

So exact equal shell-normal weight on the u_1 - and u_2 -coupling sectors produces longitudinal cancellation.

CLXXXI. STEP 6: FINAL WITNESS CONDITION

Since

$$L_\star = m_\parallel u_\parallel, \quad \lambda_\parallel = \frac{1}{2}\alpha_X L_\star I_{\theta'^3},$$

the explicit canonical sufficient condition is

$$\boxed{m_\parallel \neq 0, \quad \alpha_X \neq 0, \quad I_{\theta'^3} \neq 0, \quad \Re\left[\kappa_J((c_1 + ic_2) - (c_4 + ic_5))\right] \neq 0.}$$

Under these conditions,

$$\boxed{\lambda_\parallel \neq 0.}$$

CLXXXII. FINAL SUMMARY

In the canonical locked basis

$$z_0 = e_1, \quad u_1 = e_2, \quad u_2 = e_3,$$

the shell-tangent couplings are

$$t_{10}^a = (T^a)_{21}, \quad t_{20}^a = (T^a)_{31}.$$

For the standard $SU(3)$ generators:

$$T^1, T^2 \text{ couple only to } u_1, \quad T^4, T^5 \text{ couple only to } u_2,$$

while

$$T^3, T^6, T^7, T^8$$

do not couple the locked direction to the tangent doublet at all.

With shell-normal prefactor κ_J , the reduced matrix is

$$U_{\parallel} = \begin{pmatrix} 2\Re(\kappa_J t_{10}) & 0 \\ 0 & 2\Re(\kappa_J t_{20}) \end{pmatrix},$$

hence

$$u_{\parallel} = \Re[\kappa_J(t_{10} - t_{20})].$$

For a general active direction

$$T_{\star} = c_1 T^1 + c_2 T^2 + c_4 T^4 + c_5 T^5,$$

this becomes

$$u_{\parallel} = \frac{1}{2} \Re[\kappa_J((c_1 + ic_2) - (c_4 + ic_5))].$$

Therefore the exact canonical cancellation condition is

$$u_{\parallel} = 0 \iff \Re[\kappa_J((c_1 + ic_2) - (c_4 + ic_5))] = 0.$$

So the longitudinal slot survives generically unless the mediator direction is tuned to give equal effective shell-normal weight on the two coupling sectors

$$\{T^1, T^2\} \quad \text{and} \quad \{T^4, T^5\}.$$

Finally,

$$L_{\star} = m_{\parallel} u_{\parallel}, \quad \lambda_{\parallel} = \frac{1}{2} \alpha_X L_{\star} I_{\theta'^3},$$

so the explicit canonical sufficient condition for longitudinal survival is

$$m_{\parallel} \neq 0, \quad \alpha_X \neq 0, \quad I_{\theta'^3} \neq 0, \quad \Re[\kappa_J((c_1 + ic_2) - (c_4 + ic_5))] \neq 0.$$

CLXXXIII. ACTUAL SINGLE-GENERATOR COMPUTATION OF $U_{\parallel}$, $u_{\parallel}$, AND $\lambda_{\parallel}$

CLXXXIV. STEP 1: CANONICAL BASIS

Use the canonical locked basis

$$z_0 = e_1, \quad u_1 = e_2, \quad u_2 = e_3.$$

Then the shell-normal reduced J -matrix satisfies

$$u_{\parallel} = \frac{1}{2} [(U_{\parallel})_{11} - (U_{\parallel})_{22}].$$

Let the shell-normal projection prefactor be a complex scalar κ_J . For a single active generator $T_{\star}$, the reduced diagonal entries are

$$(U_{\parallel})_{11} = 2\Re(\kappa_J t_{10}), \quad (U_{\parallel})_{22} = 2\Re(\kappa_J t_{20}),$$

where

$$t_{10} = u_1^{\dagger} T_{\star} z_0, \quad t_{20} = u_2^{\dagger} T_{\star} z_0.$$

CLXXXV. STEP 2: EXPLICIT $SU(3)$ COUPLINGS

For $T^a = \lambda^a/2$, the only nonzero canonical couplings are

$$T^1 : \quad t_{10} = \frac{1}{2}, \quad t_{20} = 0,$$

$$T^2 : \quad t_{10} = \frac{i}{2}, \quad t_{20} = 0,$$

$$T^4 : \quad t_{10} = 0, \quad t_{20} = \frac{1}{2},$$

$$T^5 : \quad t_{10} = 0, \quad t_{20} = \frac{i}{2},$$

while

$$T^3, T^6, T^7, T^8$$

have

$$t_{10} = t_{20} = 0.$$

CLXXXVI. STEP 3: ACTUAL $U_{\parallel}$ MATRICES

Case A: $T_{\star} = T^1$

$$(U_{\parallel})_{11} = 2\Re\left(\kappa_J \frac{1}{2}\right) = \Re(\kappa_J), \quad (U_{\parallel})_{22} = 0.$$

Hence

$$U_{\parallel}(T^1) = \begin{pmatrix} \Re(\kappa_J) & 0 \\ 0 & 0 \end{pmatrix} = \frac{\Re(\kappa_J)}{2} \mathbf{1}_2 + \frac{\Re(\kappa_J)}{2} \Sigma_3.$$

Therefore

$$u_{\parallel}(T^1) = \frac{1}{2} \Re(\kappa_J).$$

Case B: $T_{\star} = T^2$

$$(U_{\parallel})_{11} = 2\Re\left(\kappa_J \frac{i}{2}\right) = \Re(i\kappa_J) = -\Im(\kappa_J), \quad (U_{\parallel})_{22} = 0.$$

Hence

$$U_{\parallel}(T^2) = \begin{pmatrix} -\Im(\kappa_J) & 0 \\ 0 & 0 \end{pmatrix} = -\frac{\Im(\kappa_J)}{2} \mathbf{1}_2 - \frac{\Im(\kappa_J)}{2} \Sigma_3,$$

so

$$u_{\parallel}(T^2) = -\frac{1}{2} \Im(\kappa_J).$$

Case C: $T_\star = T^4$

$$(U_\parallel)_{11} = 0, \quad (U_\parallel)_{22} = 2\Re\left(\kappa_J \frac{1}{2}\right) = \Re(\kappa_J).$$

Hence

$$U_\parallel(T^4) = \begin{pmatrix} 0 & 0 \\ 0 & \Re(\kappa_J) \end{pmatrix} = \frac{\Re(\kappa_J)}{2} \mathbf{1}_2 - \frac{\Re(\kappa_J)}{2} \Sigma_3,$$

so

$$u_\parallel(T^4) = -\frac{1}{2} \Re(\kappa_J).$$

Case D: $T_\star = T^5$

$$(U_\parallel)_{11} = 0, \quad (U_\parallel)_{22} = 2\Re\left(\kappa_J \frac{i}{2}\right) = -\Im(\kappa_J).$$

Hence

$$U_\parallel(T^5) = \begin{pmatrix} 0 & 0 \\ 0 & -\Im(\kappa_J) \end{pmatrix} = -\frac{\Im(\kappa_J)}{2} \mathbf{1}_2 + \frac{\Im(\kappa_J)}{2} \Sigma_3,$$

so

$$u_\parallel(T^5) = \frac{1}{2} \Im(\kappa_J).$$

Case E: $T_\star = T^3, T^6, T^7, T^8$

Since

$$t_{10} = t_{20} = 0,$$

one gets

$$U_\parallel = 0, \quad u_\parallel = 0.$$

CLXXXVII. STEP 4: ACTUAL $L_\star$ AND $\lambda_\parallel$

The canonical longitudinal invariant is

$$L_\star = m_\parallel u_\parallel,$$

and under the minimal explicit specialization the reduced longitudinal coefficient is

$$\lambda_\parallel = \frac{1}{2} \alpha_X m_\parallel u_\parallel I_{\theta'^3}.$$

Since

$$I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3,$$

we get:

$$T_{\star} = T^1$$

$$L_{\star}(T^1) = \frac{m_{\parallel}}{2} \Re(\kappa_J),$$

$$\lambda_{\parallel}(T^1) = \frac{1}{4} \alpha_X m_{\parallel} \Re(\kappa_J) I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{1024} \alpha_X m_{\parallel} \Re(\kappa_J) \theta_0^3 \kappa^3.$$

$$T_{\star} = T^2$$

$$L_{\star}(T^2) = -\frac{m_{\parallel}}{2} \Im(\kappa_J),$$

$$\lambda_{\parallel}(T^2) = -\frac{1}{4} \alpha_X m_{\parallel} \Im(\kappa_J) I_{\theta'^3} = +\frac{3\sqrt{3}\pi}{1024} \alpha_X m_{\parallel} \Im(\kappa_J) \theta_0^3 \kappa^3.$$

$$T_{\star} = T^4$$

$$L_{\star}(T^4) = -\frac{m_{\parallel}}{2} \Re(\kappa_J),$$

$$\lambda_{\parallel}(T^4) = -\frac{1}{4} \alpha_X m_{\parallel} \Re(\kappa_J) I_{\theta'^3} = +\frac{3\sqrt{3}\pi}{1024} \alpha_X m_{\parallel} \Re(\kappa_J) \theta_0^3 \kappa^3.$$

$$T_{\star} = T^5$$

$$L_{\star}(T^5) = \frac{m_{\parallel}}{2} \Im(\kappa_J),$$

$$\lambda_{\parallel}(T^5) = \frac{1}{4} \alpha_X m_{\parallel} \Im(\kappa_J) I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{1024} \alpha_X m_{\parallel} \Im(\kappa_J) \theta_0^3 \kappa^3.$$

CLXXXVIII. STEP 5: MIXED ACTIVE DIRECTION

For a general active direction

$$T_{\star} = c_1 T^1 + c_2 T^2 + c_4 T^4 + c_5 T^5, \quad c_i \in \mathbb{R},$$

one has

$$t_{10} = \frac{1}{2}(c_1 + ic_2), \quad t_{20} = \frac{1}{2}(c_4 + ic_5),$$

hence

$$u_{\parallel}(T_{\star}) = \frac{1}{2} \Re \left[\kappa_J ((c_1 + ic_2) - (c_4 + ic_5)) \right].$$

Therefore

$$\lambda_{\parallel}(T_{\star}) = \frac{1}{4} \alpha_X m_{\parallel} \Re \left[\kappa_J ((c_1 + ic_2) - (c_4 + ic_5)) \right] I_{\theta'^3}.$$

The exact cancellation locus is

$$\Re \left[\kappa_J ((c_1 + ic_2) - (c_4 + ic_5)) \right] = 0.$$

CLXXXIX. ACTUAL EXPLICIT NONDEGENERATE WITNESS BRANCH

CXC. 1. FIX THE LONGITUDINAL GENERATOR

Take the active mediator direction to be

$$T_{\star} = T^1.$$

In the canonical locked basis

$$z_0 = e_1, \quad u_1 = e_2, \quad u_2 = e_3,$$

one has

$$t_{10} = u_1^{\dagger} T^1 z_0 = \frac{1}{2}, \quad t_{20} = u_2^{\dagger} T^1 z_0 = 0.$$

Therefore the shell-normal reduced J -split is

$$U_{\parallel}(T^1) = \begin{pmatrix} \Re(\kappa_J) & 0 \\ 0 & 0 \end{pmatrix},$$

and hence

$$u_{\parallel}(T^1) = \frac{1}{2} [(U_{\parallel})_{11} - (U_{\parallel})_{22}] = \frac{1}{2} \Re(\kappa_J).$$

Thus

$$c_{\parallel} = \alpha_X m_{\parallel} u_{\parallel} = \frac{1}{2} \alpha_X m_{\parallel} \Re(\kappa_J).$$

Using

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3}, \quad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3,$$

we get

$$\lambda_{\parallel}(T^1) = \frac{1}{4} \alpha_X m_{\parallel} \Re(\kappa_J) I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{1024} \alpha_X m_{\parallel} \Re(\kappa_J) \theta_0^3 \kappa^3.$$

CXCI. 2. FIX THE TRANSVERSE SEED

Now choose the pure transverse E -direction in the explicit canonical split:

$$\boxed{\xi_E v_E = 1, \quad \xi_O v_O = 0.}$$

Then

$$c_E = \beta_X, \quad c_O = 0.$$

Using

$$\alpha = c_E I_\theta, \quad \beta = c_O I_\theta, \quad I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa,$$

we obtain

$$\boxed{\alpha = \beta_X I_\theta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa, \quad \beta = 0.}$$

Thus the reduced first-order symbol is

$$\boxed{A_1(p) = \lambda_{\parallel} (T^1) p_{\parallel} \sigma_3 - \frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa (p_1 \sigma_1 + p_2 \sigma_2).}$$

CXCII. 3. EXACT DETERMINANT

Since

$$\det M = \lambda_{\parallel} (\alpha^2 + \beta^2),$$

for this branch we get

$$\det M = \lambda_{\parallel} (T^1) \alpha^2.$$

Now

$$\alpha^2 = \left(\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa \right)^2 = \frac{3\pi^2}{256} \beta_X^2 \theta_0^2 \kappa^2.$$

Hence

$$\boxed{\det M = -\frac{9\sqrt{3}\pi^3}{262144} \alpha_X \beta_X^2 m_{\parallel} \Re(\kappa_J) \theta_0^5 \kappa^5.}$$

Therefore the explicit sufficient condition for a nondegenerate Weyl/Dirac witness in this branch is

$$\boxed{\alpha_X \neq 0, \quad \beta_X \neq 0, \quad m_{\parallel} \neq 0, \quad \Re(\kappa_J) \neq 0, \quad \theta_0 \kappa \neq 0.}$$

Under these conditions,

$$\boxed{\lambda_{\parallel} \neq 0, \quad \alpha \neq 0, \quad \beta = 0, \quad \det M \neq 0.}$$

CXCIII. 4. PURE O -BRANCH VARIANT

If instead we choose

$$\xi_{Ev_E} = 0, \quad \xi_{Ov_O} = 1,$$

then

$$\alpha = 0, \quad \beta = -\frac{\sqrt{3}\pi}{16}\beta_X\theta_0\kappa,$$

while the same $\lambda_{\parallel}(T^1)$ survives. Hence

$$\det M = \lambda_{\parallel}(T^1)\beta^2$$

and the same nondegeneracy condition follows.

CXCIV. 5. HONEST INTERPRETATION

This does *not* prove that the full complete microscopic TECT action has already been executed at the final transverse derivative-bookkeeping level.

What it does prove is:

1. inside the explicit canonical split, there exists a fully concrete branch with

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0), \quad \det M \neq 0;$$

2. the longitudinal part is realized by the actual internal generator choice $T_{\star} = T^1$;
3. the transverse part is realized by the explicit E - or O -seed direction of the already fixed seed space.

CXCV. EXPLICIT COMPLETION OF THE CANONICAL WITNESS BRANCH $T_{\star} = T^1$

CXCVI. 1. LONGITUDINAL PART

In the canonical locked basis

$$z_0 = e_1, \quad u_1 = e_2, \quad u_2 = e_3,$$

the active generator $T_{\star} = T^1$ has

$$t_{10} = u_1^{\dagger} T^1 z_0 = \frac{1}{2}, \quad t_{20} = u_2^{\dagger} T^1 z_0 = 0.$$

Hence the shell-normal reduced J -split is

$$u_{\parallel}(T^1) = \frac{1}{2} \Re(\kappa_J).$$

Therefore

$$L_{\star}(T^1) = m_{\parallel} u_{\parallel}(T^1) = \frac{m_{\parallel}}{2} \Re(\kappa_J).$$

Using the exact reduced longitudinal formula

$$\lambda_{\parallel} = \frac{1}{2} \alpha_X L_{\star} I_{\theta'^3},$$

together with

$$I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3,$$

we get

$$\boxed{\lambda_{\parallel}(T^1) = -\frac{3\sqrt{3}\pi}{1024} \alpha_X m_{\parallel} \Re(\kappa_J) \theta_0^3 \kappa^3.} \quad (397)$$

CXCVII. 2. TRANSVERSE PART

Now choose the pure E -seed direction

$$R_{\perp} = 1, \quad \varphi = 0.$$

Then the transverse part of the master symbol reduces to

$$\beta_X I_{\theta}(p_1 \sigma_1 + p_2 \sigma_2),$$

so

$$\boxed{\alpha = \beta_X I_{\theta} = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa, \quad \beta = 0.} \quad (398)$$

Therefore the full reduced first-order symbol is

$$\boxed{A_1(p) = -\frac{3\sqrt{3}\pi}{1024} \alpha_X m_{\parallel} \Re(\kappa_J) \theta_0^3 \kappa^3 p_{\parallel} \sigma_3 - \frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa (p_1 \sigma_1 + p_2 \sigma_2).} \quad (399)$$

CXCVIII. 3. DETERMINANT AND NONDEGENERACY

The reduced Weyl determinant is

$$\det M = \lambda_{\parallel} (\alpha^2 + \beta^2).$$

Since $\beta = 0$, this becomes

$$\det M = \lambda_{\parallel} \alpha^2.$$

Now

$$\alpha^2 = \left(\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa \right)^2 = \frac{3\pi^2}{256} \beta_X^2 \theta_0^2 \kappa^2.$$

Substituting (397),

$$\boxed{\det M = -\frac{9\sqrt{3}\pi^3}{262144} \alpha_X \beta_X^2 m_{\parallel} \Re(\kappa_J) \theta_0^5 \kappa^5.} \quad (400)$$

Hence an explicit sufficient condition for nondegeneracy is

$$\boxed{\alpha_X \neq 0, \quad \beta_X \neq 0, \quad m_{\parallel} \neq 0, \quad \Re(\kappa_J) \neq 0, \quad \theta_0 \kappa \neq 0.} \quad (401)$$

Under these conditions,

$$\lambda_{\parallel} \neq 0, \quad \alpha \neq 0, \quad \det M \neq 0.$$

CXCIX. 4. INFRARED VELOCITIES

The reduced velocities are

$$v_{\parallel}^{\text{IR}} = |\lambda_{\parallel}|, \quad v_{\perp}^{\text{IR}} = \sqrt{\alpha^2 + \beta^2} = |\alpha|.$$

Thus

$$\boxed{v_{\parallel}^{\text{IR}}(T^1) = \frac{3\sqrt{3}\pi}{1024} |\alpha_X m_{\parallel} \Re(\kappa_J)| \theta_0^3 \kappa^3,} \quad (402)$$

$$v_{\perp}^{\text{IR}}(T^1) = \frac{\sqrt{3}\pi}{16} |\beta_X| \theta_0 \kappa. \quad (403)$$

Therefore the anisotropy ratio is

$$\rho(T^1) = \frac{v_{\parallel}^{\text{IR}}}{v_{\perp}^{\text{IR}}} = \frac{3}{64} \frac{|\alpha_X|}{|\beta_X|} |m_{\parallel} \Re(\kappa_J)| \theta_0^2 \kappa^2. \quad (404)$$

up to the overall sign convention.

CC. 5. PURE *O*-VARIANT

If instead one chooses

$$R_{\perp} = 1, \quad \varphi = \frac{\pi}{2},$$

then

$$\alpha = 0, \quad \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa,$$

while the same $\lambda_{\parallel}(T^1)$ survives. Hence the determinant becomes

$$\det M = \lambda_{\parallel} \beta^2,$$

with exactly the same nondegeneracy condition (401).

CCI. 6. FINAL EXPLICIT VERDICT FOR THIS BRANCH

The branch

$$T_{\star} = T^1, \quad R_{\perp} = 1, \quad \varphi = 0$$

is a fully explicit finite witness branch.

Its reduced coefficients are

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{1024} \alpha_X m_{\parallel} \Re(\kappa_J) \theta_0^3 \kappa^3,$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa, \quad \beta = 0,$$

and its determinant is

$$\det M = -\frac{9\sqrt{3}\pi^3}{262144} \alpha_X \beta_X^2 m_{\parallel} \Re(\kappa_J) \theta_0^5 \kappa^5.$$

Therefore this branch is nondegenerate whenever

$$\alpha_X \neq 0, \quad \beta_X \neq 0, \quad m_{\parallel} \neq 0, \quad \Re(\kappa_J) \neq 0, \quad \theta_0 \kappa \neq 0.$$

CCII. FINAL SUMMARY

A fully explicit finite nondegenerate witness branch is obtained by choosing

$$T_\star = T^1, \quad R_\perp = 1, \quad \varphi = 0.$$

Then

$$u_\parallel(T^1) = \frac{1}{2}\Re(\kappa_J), \quad L_\star(T^1) = \frac{m_\parallel}{2}\Re(\kappa_J),$$

and therefore

$$\lambda_\parallel(T^1) = -\frac{3\sqrt{3}\pi}{1024}\alpha_X m_\parallel \Re(\kappa_J) \theta_0^3 \kappa^3.$$

For the pure E -transverse seed,

$$\alpha = -\frac{\sqrt{3}\pi}{16}\beta_X \theta_0 \kappa, \quad \beta = 0.$$

Hence

$$\det M = -\frac{9\sqrt{3}\pi^3}{262144}\alpha_X \beta_X^2 m_\parallel \Re(\kappa_J) \theta_0^5 \kappa^5.$$

So the branch is nondegenerate whenever

$$\alpha_X \neq 0, \quad \beta_X \neq 0, \quad m_\parallel \neq 0, \quad \Re(\kappa_J) \neq 0, \quad \theta_0 \kappa \neq 0.$$

Thus one explicit witness branch is already completely finite and concrete. The only remaining unresolved microscopic input is the actual refined locked Bloch value of

$$\Re(\kappa_J).$$

CCIII. ELIMINATION OF THE AUXILIARY κ_J AND EXACT CLOSURE OF THE $T_\star = T^1$ WITNESS BRANCH

CCIV. 1. SCALAR REDUCTION OF THE PURE- JJ LONGITUDINAL FACTOR

The current refined locked + Class II notes imply that the principal pure- JJ axial tensor has the isotropic shell form

$$C_{jm}^{(JJ)} = \Xi_{JJ} \delta_{jm}. \quad (405)$$

Hence

$$\hat{n} \cdot C^{(JJ)} G_\star = \Xi_{JJ} |G_\star|. \quad (406)$$

In the canonical locked basis and minimal explicit Class II split,

$$\Xi_{JJ} = \mathcal{N}_{JJ} L_\star, \quad L_\star = m_\parallel u_\parallel. \quad (407)$$

Therefore the first-shell Fourier coefficient becomes

$$\hat{A}_{JJ}(G_\star) = i \mathcal{N}_{JJ} L_\star |G_\star| A_{G_\star}. \quad (408)$$

This shows that the previously introduced shell-normal prefactor κ_J is not an independent microscopic datum. Its shell-relevant role is completely absorbed into the scalar reduction factor

$$\mathcal{N}_{JJ} |G_\star| A_{G_\star}.$$

CCV. 2. CANONICAL $T_\star = T^1$ BRANCH

Fix the canonical locked basis

$$z_0 = e_1, \quad u_1 = e_2, \quad u_2 = e_3.$$

Take the active mediator direction

$$\boxed{T_\star = T^1.}$$

Then

$$t_{10} = u_1^\dagger T^1 z_0 = \frac{1}{2}, \quad t_{20} = u_2^\dagger T^1 z_0 = 0.$$

Therefore the shell-normal J -singlet bias is

$$\boxed{u_\parallel(T^1) \neq 0} \tag{409}$$

whenever the canonical J -kernel is not tuned to equalize the two tangent channels.

Equivalently, the remaining longitudinal scalar is simply

$$\boxed{L_\star(T^1) = m_\parallel u_\parallel(T^1).} \tag{410}$$

CCVI. 3. PURE LONGITUDINAL FOURIER COEFFICIENT

Substituting (410) into (408),

$$\boxed{\hat{\mathcal{A}}_{JJ}(G_\star; T^1) = i \mathcal{N}_{JJ} m_\parallel u_\parallel(T^1) |G_\star| A_{G_\star}.} \tag{411}$$

Hence, if

$$\boxed{m_\parallel \neq 0, \quad u_\parallel(T^1) \neq 0, \quad A_{G_\star} \neq 0, \quad \alpha_X \neq 0,} \tag{412}$$

then

$$\hat{\mathcal{A}}_{JJ}(G_\star) \neq 0 \implies \hat{V}_\parallel^{(JJ)}(G_\star) \neq 0 \implies \lambda_\parallel \neq 0.$$

CCVII. 4. EXACT LONGITUDINAL COEFFICIENT

Inside the exact Branch-B reduced dictionary,

$$\boxed{\lambda_\parallel = \frac{1}{2} I_{\theta^3} \lambda_c^{(JJ)}, \quad I_{\theta^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3.} \tag{413}$$

The pure- JJ carrier-side scalar is already reduced to $L_\star = m_\parallel u_\parallel$ up to the fixed microscopic normalization. So introduce the explicit branch-normalization constant

$$\boxed{\lambda_c^{(JJ)} = \Gamma_{JJ} m_\parallel u_\parallel(T^1),} \tag{414}$$

where

$$\Gamma_{JJ}$$

is the fixed shell/Bloch normalization determined by the projected refined locked basis. Then

$$\boxed{\lambda_\parallel(T^1) = -\frac{3\sqrt{3}\pi}{512} \Gamma_{JJ} m_\parallel u_\parallel(T^1) \theta_0^3 \kappa^3.} \tag{415}$$

Equivalently, using (411), one may regard

$$\Gamma_{JJ} \propto \alpha_X \mathcal{N}_{JJ} |G_\star| A_{G_\star}$$

in the first-shell dominant microscopic reduction.

CCVIII. 5. EXPLICIT TRANSVERSE COMPLETION

Choose the pure transverse E -seed:

$$R_{\perp} = 1, \quad \varphi = 0.$$

Then

$$\alpha = I_{\theta} \beta_X = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa, \quad \beta = 0, \quad (416)$$

where

$$I_{\theta} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa.$$

Therefore the reduced first-order symbol is

$$A_1(p) = \lambda_{\parallel}(T^1) p_{\parallel} \sigma_3 - \frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa (p_1 \sigma_1 + p_2 \sigma_2). \quad (417)$$

CCIX. 6. EXACT DETERMINANT

Since

$$\det M = \lambda_{\parallel}(\alpha^2 + \beta^2),$$

and $\beta = 0$, we obtain

$$\det M = \lambda_{\parallel}(T^1) \left(\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa \right)^2. \quad (418)$$

Substituting (415),

$$\det M = -\frac{9\sqrt{3}\pi^3}{131072} \Gamma_{JJ} \beta_X^2 m_{\parallel} u_{\parallel}(T^1) \theta_0^5 \kappa^5. \quad (419)$$

Thus the exact explicit sufficient condition for this branch is

$$\Gamma_{JJ} \neq 0, \quad \beta_X \neq 0, \quad m_{\parallel} \neq 0, \quad u_{\parallel}(T^1) \neq 0, \quad \theta_0 \kappa \neq 0. \quad (420)$$

CCX. 7. FINAL PRACTICAL MEANING

The previous notation $\Re(\kappa_J)$ is not needed anymore. The true microscopic longitudinal input is

$$L_* = m_{\parallel} u_{\parallel},$$

and its shell/Fourier promotion is controlled by

$$\mathcal{N}_{JJ} |G_*| A_{G_*},$$

equivalently by the one fixed branch normalization Γ_{JJ} .

So the explicit canonical branch

$$T_* = T^1, \quad R_{\perp} = 1, \quad \varphi = 0$$

is completely finite once the following scalars are inserted:

$$m_{\parallel}, \quad u_{\parallel}(T^1), \quad A_{G_*}, \quad \mathcal{N}_{JJ} \text{ (equiv. } \Gamma_{JJ}), \quad \beta_X.$$

CCXI. FINAL SUMMARY

The auxiliary notation κ_J is not needed as an independent microscopic input.

In the current refined locked + Class II working branch, the pure- JJ longitudinal tensor collapses to

$$C_{jm}^{(JJ)} = \Xi_{JJ} \delta_{jm},$$

and in the canonical locked basis this scalar is identified as

$$\Xi_{JJ} = \mathcal{N}_{JJ} L_*, \quad L_* = m_{\parallel} u_{\parallel}.$$

Hence the first-shell Fourier coefficient is

$$\hat{\mathcal{A}}_{JJ}(G_*) = i \mathcal{N}_{JJ} m_{\parallel} u_{\parallel} |G_*| A_{G_*}.$$

So the true microscopic longitudinal inputs are not

$$\Re(\kappa_J)$$

by itself, but rather the finite scalar data

$$m_{\parallel}, \quad u_{\parallel}, \quad A_{G_*}, \quad \mathcal{N}_{JJ}.$$

For the explicit canonical witness branch

$$T_{\star} = T^1, \quad R_{\perp} = 1, \quad \varphi = 0,$$

the reduced coefficients are

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \Gamma_{JJ} m_{\parallel} u_{\parallel} (T^1) \theta_0^3 \kappa^3,$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa, \quad \beta = 0,$$

with $\Gamma_{JJ} \propto \alpha_X \mathcal{N}_{JJ} |G_*| A_{G_*}$ in the first-shell dominant reduction.

Therefore an explicit sufficient condition for a nondegenerate witness branch is

$$\Gamma_{JJ} \neq 0, \quad \beta_X \neq 0, \quad m_{\parallel} \neq 0, \quad u_{\parallel}(T^1) \neq 0, \quad \theta_0 \kappa \neq 0.$$

Under these conditions,

$$\lambda_{\parallel} \neq 0, \quad \alpha \neq 0, \quad \beta = 0, \quad \det M \neq 0.$$

So the true remaining longitudinal microscopic frontier is the explicit scalar evaluation of

$$\mathcal{N}_{JJ},$$

together with the already isolated finite inputs

$$m_{\parallel}, \quad u_{\parallel}, \quad A_{G_*}.$$

CCXII. DIRECT COMPLETION OF THE EXPLICIT BRANCH IN THE LATEST ACCEPTANCE-STAGE NORMALIZATION

Choose

$$T_{\star} = T^1, \quad R_{\perp} = 1, \quad \varphi = 0.$$

Then the longitudinal coefficient is exactly

$$\lambda_{\parallel} = -\frac{\alpha_X \beta_X}{2M_X^2} C_{\parallel} I_3. \quad (421)$$

Using

$$I_3 = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3,$$

this becomes

$$\lambda_{\parallel} = \frac{3\sqrt{3}\pi}{512} \frac{\alpha_X \beta_X}{M_X^2} C_{\parallel} \theta_0^3 \kappa^3. \quad (422)$$

The transverse coefficients are

$$\alpha = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa, \quad \beta = 0. \quad (423)$$

Hence the reduced first-order symbol is

$$A_1(p) = \frac{3\sqrt{3}\pi}{512} \frac{\alpha_X \beta_X}{M_X^2} C_{\parallel} \theta_0^3 \kappa^3 p_{\parallel} \sigma_3 - \frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa (p_1 \sigma_1 + p_2 \sigma_2). \quad (424)$$

Since

$$\det M = \lambda_{\parallel} (\alpha^2 + \beta^2),$$

and $\beta = 0$, one gets

$$\alpha^2 = \frac{3\pi^2}{256} \beta_X^2 \theta_0^2 \kappa^2.$$

Therefore

$$\det M = \frac{9\sqrt{3}\pi^3}{131072} \frac{\alpha_X \beta_X^3}{M_X^2} C_{\parallel} \theta_0^5 \kappa^5. \quad (425)$$

So the exact sufficient condition for this branch is

$$\alpha_X \neq 0, \quad \beta_X \neq 0, \quad C_{\parallel} \neq 0, \quad \theta_0 \kappa \neq 0. \quad (426)$$

Equivalently, since

$$C_{\parallel} \neq 0 \iff (\hat{n} \cdot \vec{W}_+)_{11} \neq (\hat{n} \cdot \vec{W}_+)_{22},$$

the condition can be written as

$$\alpha_X \neq 0, \quad \beta_X \neq 0, \quad (\hat{n} \cdot \vec{W}_+)_{11} \neq (\hat{n} \cdot \vec{W}_+)_{22}, \quad \theta_0 \kappa \neq 0. \quad (427)$$

CCXIII. DIRECT DIAGONAL-DIFFERENCE COMPLETION OF THE EXPLICIT WITNESS BRANCH

CCXIV. 1. DEFINE THE ACTUAL REMAINING LONGITUDINAL SCALAR

Introduce the exact projected diagonal difference

$$\Delta W_{\parallel} := (\hat{n} \cdot \vec{W}_+)_{11} - (\hat{n} \cdot \vec{W}_+)_{22}. \quad (428)$$

Then the reduced longitudinal coefficient is exactly

$$\lambda_{\parallel} = \frac{1}{2} \Delta W_{\parallel}. \quad (429)$$

Equivalently, in the Branch-B one-mediator normalization,

$$\lambda_{\parallel} = -\frac{\alpha_X \beta_X}{2M_X^2} C_{\parallel} I_3, \quad I_3 = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3, \quad (430)$$

so

$$C_{\parallel} = -\frac{M_X^2}{\alpha_X \beta_X I_3} \Delta W_{\parallel}. \quad (431)$$

CCXV. 2. FIX THE TRANSVERSE BRANCH

Choose the explicit pure- E transverse seed

$$R_{\perp} = 1, \quad \varphi = 0.$$

Then

$$\alpha = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa, \quad \beta = 0. \quad (432)$$

Hence the reduced first-order symbol is

$$A_1(p) = \frac{1}{2} \Delta W_{\parallel} p_{\parallel} \sigma_3 - \frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa (p_1 \sigma_1 + p_2 \sigma_2). \quad (433)$$

CCXVI. 3. EXACT DETERMINANT

Since

$$\det M = \lambda_{\parallel} (\alpha^2 + \beta^2),$$

and $\beta = 0$, we get

$$\det M = \lambda_{\parallel} \alpha^2.$$

Now

$$\alpha^2 = \left(\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa \right)^2 = \frac{3\pi^2}{256} \beta_X^2 \theta_0^2 \kappa^2.$$

Substituting (429),

$$\det M = \frac{3\pi^2}{512} \beta_X^2 \theta_0^2 \kappa^2 \Delta W_{\parallel}. \quad (434)$$

Using (430), the same determinant may also be written as

$$\det M = \frac{9\sqrt{3}\pi^3}{131072} \frac{\alpha_X \beta_X^3}{M_X^2} C_{\parallel} \theta_0^5 \kappa^5. \quad (435)$$

CCXVII. 4. EXACT NONDEGENERACY CONDITION FOR THIS EXPLICIT BRANCH

Therefore the branch

$$\boxed{T_\star = T^1, \quad R_\perp = 1, \quad \varphi = 0}$$

is nondegenerate iff

$$\boxed{\Delta W_\parallel \neq 0, \quad \beta_X \neq 0, \quad \theta_0 \kappa \neq 0.} \quad (436)$$

Equivalently, in the one-mediator calibration language,

$$\boxed{\alpha_X \neq 0, \quad \beta_X \neq 0, \quad C_\parallel \neq 0, \quad \theta_0 \kappa \neq 0.} \quad (437)$$

CCXVIII. 5. IF THE SCALAR CORE REMAINS DIAGONAL-SYMMETRIC

The exact longitudinal split formula is

$$\boxed{2\lambda_\parallel = (D_1^{(0)} - D_2^{(0)}) - (D_1^{(\text{med})} - D_2^{(\text{med})}).} \quad (438)$$

Hence

$$\boxed{\Delta W_\parallel = (D_1^{(0)} - D_2^{(0)}) - (D_1^{(\text{med})} - D_2^{(\text{med})}).} \quad (439)$$

So, if the explicit safe scalar core/refined tree-level block remains diagonal-symmetric,

$$D_1^{(0)} - D_2^{(0)} = 0,$$

then the entire longitudinal decision reduces to

$$\boxed{\Delta W_\parallel = -(D_1^{(\text{med})} - D_2^{(\text{med})}).} \quad (440)$$

In that case, the working-branch Gate-2 question is literally the sign/nonvanishing of the mediated diagonal asymmetry.

CCXIX. 6. FINAL EXPLICIT VERDICT

Thus the actual explicit branch calculation is already compressed to one scalar:

$$\boxed{\Delta W_\parallel = (\hat{n} \cdot \vec{W}_+)_{11} - (\hat{n} \cdot \vec{W}_+)_{22}.} \quad (441)$$

Once this is computed from the projected refined locked Bloch operator, everything else follows immediately:

$$\lambda_\parallel = \frac{1}{2} \Delta W_\parallel, \quad \alpha = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa, \quad \beta = 0,$$

and

$$\det M = \frac{3\pi^2}{512} \beta_X^2 \theta_0^2 \kappa^2 \Delta W_\parallel.$$

CCXX. FINAL SUMMARY

For the explicit witness branch

$$T_\star = T^1, \quad R_\perp = 1, \quad \varphi = 0,$$

the remaining longitudinal microscopic input is the single projected diagonal difference

$$\Delta W_\parallel = (\hat{n} \cdot \vec{W}_+)_{11} - (\hat{n} \cdot \vec{W}_+)_{22}.$$

The reduced coefficients are then exactly

$$\lambda_\parallel = \frac{1}{2} \Delta W_\parallel,$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa, \quad \beta = 0.$$

Hence the determinant is

$$\det M = \frac{3\pi^2}{512} \beta_X^2 \theta_0^2 \kappa^2 \Delta W_\parallel.$$

Equivalently, using the one-mediator Branch-B calibration,

$$\lambda_\parallel = -\frac{\alpha_X \beta_X}{2M_X^2} C_\parallel I_3, \quad C_\parallel = -\frac{M_X^2}{\alpha_X \beta_X I_3} \Delta W_\parallel.$$

Therefore the exact nondegeneracy condition for this explicit branch is

$$\Delta W_\parallel \neq 0, \quad \beta_X \neq 0, \quad \theta_0 \kappa \neq 0,$$

or equivalently

$$\alpha_X \neq 0, \quad \beta_X \neq 0, \quad C_\parallel \neq 0, \quad \theta_0 \kappa \neq 0.$$

So the entire remaining Gate-2 bottleneck is now the direct projected matrix-element computation of

$$(\hat{n} \cdot \vec{W}_+)_{11} - (\hat{n} \cdot \vec{W}_+)_{22}.$$

CCXXI. ACTUAL DIAGONAL-DIFFERENCE COMPUTATION FOR THE EXPLICIT BRANCH $T_\star = T^1$

CCXXII. 1. EXACT SPLIT

The exact longitudinal split is

$$2\lambda_\parallel = \left(D_1^{(0)} - D_2^{(0)} \right) - \left(D_1^{(\text{med})} - D_2^{(\text{med})} \right). \quad (442)$$

Equivalently,

$$\Delta W_\parallel := (\hat{n} \cdot \vec{W}_+)_{11} - (\hat{n} \cdot \vec{W}_+)_{22} = \left(D_1^{(0)} - D_2^{(0)} \right) - \left(D_1^{(\text{med})} - D_2^{(\text{med})} \right), \quad (443)$$

and

$$\lambda_\parallel = \frac{1}{2} \Delta W_\parallel. \quad (444)$$

CCXXIII. 2. TREE-LEVEL CORE CONTRIBUTION

In the present safe scalar/refined-tree branch, the direct amplitude/scalar sector contributes only to the identity channel, and the direct projector-stiffness tree level contributes zero to the traceless transport kernel.

Therefore, in the current explicit tree-level branch,

$$\boxed{D_1^{(0)} - D_2^{(0)} = 0.} \quad (445)$$

Hence the exact longitudinal coefficient reduces to

$$\boxed{2\lambda_{\parallel} = -\left(D_1^{(\text{med})} - D_2^{(\text{med})}\right).} \quad (446)$$

CCXXIV. 3. MEDIATED RESPONSE MATRIX

The mediated quadratic response in the adapted frame is

$$\boxed{r_{mn}^{\text{med}} = -\frac{1}{M_X^2} \sum_{i,a} \overline{\mathcal{C}_{ia;m}} \mathcal{C}_{ia;n},} \quad (447)$$

so the longitudinal Pauli component is

$$\boxed{s_3^{\text{med}} = \frac{1}{2} (r_{11}^{\text{med}} - r_{22}^{\text{med}}) = -\frac{1}{2M_X^2} \left[\sum_{i,a} |\mathcal{C}_{ia;1}|^2 - \sum_{i,a} |\mathcal{C}_{ia;2}|^2 \right].} \quad (448)$$

Thus a sufficient condition for a nonzero mediated longitudinal channel is

$$\boxed{\sum_{i,a} |\mathcal{C}_{ia;1}|^2 \neq \sum_{i,a} |\mathcal{C}_{ia;2}|^2.} \quad (449)$$

CCXXV. 4. EXPLICIT CANONICAL BRANCH $T_{\star} = T^1$

Take the canonical locked basis

$$z_0 = e_1, \quad u_1 = e_2, \quad u_2 = e_3,$$

and choose the active mediator direction

$$\boxed{T_{\star} = T^1.}$$

Then

$$t_{10} = u_1^{\dagger} T^1 z_0 = \frac{1}{2}, \quad t_{20} = u_2^{\dagger} T^1 z_0 = 0.$$

In the minimal single-generator shell truncation, this means that the mediated current amplitudes take the form

$$\boxed{\mathcal{C}_{ia;1} = \frac{1}{2} \mathcal{A}_{ia}, \quad \mathcal{C}_{ia;2} = 0,} \quad (450)$$

for some shell-normal amplitude family $\mathcal{A}_{ia}$ determined by the actual background-linear cross term.

Therefore

$$\sum_{i,a} |\mathcal{C}_{ia;1}|^2 = \frac{1}{4} \sum_{i,a} |\mathcal{A}_{ia}|^2, \quad \sum_{i,a} |\mathcal{C}_{ia;2}|^2 = 0. \quad (451)$$

Define

$$\Sigma_J^{(T^1)} := \sum_{i,a} |\mathcal{A}_{ia}|^2. \quad (452)$$

Then

$$s_3^{\text{med}}(T^1) = -\frac{1}{8M_X^2} \Sigma_J^{(T^1)}. \quad (453)$$

As long as the shell/background-linear cross term is not identically zero,

$$\Sigma_J^{(T^1)} > 0,$$

hence

$$s_3^{\text{med}}(T^1) \neq 0. \quad (454)$$

CCXXVI. 5. DIAGONAL-DIFFERENCE AND LONGITUDINAL COEFFICIENT

Since the current explicit branch has $D_1^{(0)} - D_2^{(0)} = 0$, the whole diagonal difference is mediated:

$$\Delta W_{\parallel} = -\left(D_1^{(\text{med})} - D_2^{(\text{med})}\right). \quad (455)$$

Introduce the positive shell-normal mediated diagonal asymmetry

$$\Delta D_{\text{med}}^{(T^1)} := D_1^{(\text{med})} - D_2^{(\text{med})} = \frac{1}{4M_X^2} \Sigma_J^{(T^1)}, \quad (456)$$

up to the fixed transport-normalization convention relating $D^{(\text{med})}$ to the adapted response matrix.

Then

$$\Delta W_{\parallel}^{(T^1)} = -\Delta D_{\text{med}}^{(T^1)}, \quad \lambda_{\parallel}^{(T^1)} = -\frac{1}{2} \Delta D_{\text{med}}^{(T^1)}. \quad (457)$$

Therefore, provided the shell/background-linear cross term is active,

$$\Delta W_{\parallel}^{(T^1)} \neq 0, \quad \lambda_{\parallel}^{(T^1)} \neq 0. \quad (458)$$

CCXXVII. 6. TRANSVERSE COMPLETION OF THE SAME BRANCH

Now choose the pure E -transverse seed

$$R_{\perp} = 1, \quad \varphi = 0.$$

Then

$$\alpha = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa, \quad \beta = 0. \quad (459)$$

Hence

$$\det M = \lambda_{\parallel}^{(T^1)} \alpha^2. \quad (460)$$

So if

$$\beta_X \neq 0, \quad \theta_0 \kappa \neq 0, \quad \Sigma_J^{(T^1)} > 0,$$

then

$$\det M \neq 0. \quad (461)$$

CCXXVIII. 7. FINAL EXPLICIT VERDICT

Thus, in the explicit branch

$$\boxed{T_\star = T^1, \quad R_\perp = 1, \quad \varphi = 0,}$$

the tree-level core contribution to the diagonal difference vanishes, while the mediated block gives a nonzero diagonal asymmetry as soon as the actual shell/background-linear current amplitude is nontrivial.

Equivalently,

$$\boxed{\Delta W_\parallel^{(T^1)} = -\left(D_1^{(\text{med})} - D_2^{(\text{med})}\right) \neq 0} \quad (462)$$

under the finite condition

$$\Sigma_J^{(T^1)} > 0.$$

Hence

$$\lambda_\parallel \neq 0, \quad \alpha \neq 0, \quad \beta = 0, \quad \det M \neq 0$$

for this branch.

CCXXIX. ACTUAL COMPLETION OF THE EXPLICIT $T_\star = T^1$ BRANCH

The exact longitudinal split is

$$2\lambda_\parallel = (D_1^{(0)} - D_2^{(0)}) - (D_1^{(\text{med})} - D_2^{(\text{med})}).$$

In the present safe scalar tree branch,

$$D_1^{(0)} - D_2^{(0)} = 0,$$

so

$$\lambda_\parallel = -\frac{1}{2}(D_1^{(\text{med})} - D_2^{(\text{med})}).$$

Take the canonical locked basis

$$z_0 = e_1, \quad u_1 = e_2, \quad u_2 = e_3,$$

and choose the active generator

$$T_\star = T^1.$$

Then

$$t_{10} = u_1^\dagger T^1 z_0 = \frac{1}{2}, \quad t_{20} = u_2^\dagger T^1 z_0 = 0.$$

Hence, in the minimal single-generator shell truncation,

$$\mathcal{C}_{ia;1} = \frac{1}{2} \mathcal{A}_{ia}, \quad \mathcal{C}_{ia;2} = 0.$$

Using

$$r_{mn}^{\text{med}} = -\frac{1}{M_X^2} \sum_{i,a} \overline{\mathcal{C}_{ia;m}} \mathcal{C}_{ia;n},$$

we get

$$r_{11}^{\text{med}} = -\frac{1}{4M_X^2} \sum_{i,a} |\mathcal{A}_{ia}|^2, \quad r_{22}^{\text{med}} = 0, \quad r_{12}^{\text{med}} = 0.$$

Define

$$\Sigma_J^{(T^1)} := \sum_{i,a} |\mathcal{A}_{ia}|^2.$$

Then

$$s_3^{\text{med}} = \frac{1}{2}(r_{11}^{\text{med}} - r_{22}^{\text{med}}) = -\frac{1}{8M_X^2} \Sigma_J^{(T^1)}.$$

Let $\nu_{\parallel} > 0$ be the fixed transport normalization relating the mediated response to the diagonal derivative split. Then

$$D_1^{(\text{med})} - D_2^{(\text{med})} = -\frac{\nu_{\parallel}}{4M_X^2} \Sigma_J^{(T^1)},$$

and therefore

$$\Delta W_{\parallel}^{(T^1)} = \frac{\nu_{\parallel}}{4M_X^2} \Sigma_J^{(T^1)}, \quad \lambda_{\parallel}^{(T^1)} = \frac{\nu_{\parallel}}{8M_X^2} \Sigma_J^{(T^1)}.$$

Now choose the pure even transverse seed

$$R_{\perp} = 1, \quad \varphi = 0.$$

Then

$$\alpha = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa, \quad \beta = 0.$$

Hence

$$\det M = \frac{3\pi^2}{2048} \frac{\nu_{\parallel} \beta_X^2}{M_X^2} \Sigma_J^{(T^1)} \theta_0^2 \kappa^2.$$

Therefore, as long as

$$\beta_X \neq 0, \quad \theta_0 \kappa \neq 0, \quad \Sigma_J^{(T^1)} > 0,$$

the explicit branch

$$T_{\star} = T^1, \quad R_{\perp} = 1, \quad \varphi = 0$$

satisfies

$$\lambda_{\parallel} \neq 0, \quad \alpha \neq 0, \quad \beta = 0, \quad \det M \neq 0.$$

CCXXX. EXPLICIT CANONICAL BRANCH AND CONTINUED COEFFICIENT EXTRACTION

CCXXXI. SHORT EXPLANATION

We adopt the canonical explicit one-mediator normalization

$$C_{\parallel} = C_E = C_O = 1,$$

and use the one-mediator transport-side coefficient point

$$\lambda_c^{(1)} = -\frac{\alpha_X \beta_X}{M_X^2}, \quad \tilde{\Lambda}_E^{(1)} = -\frac{\beta_X \gamma_E}{M_X^2}, \quad \tilde{\Lambda}_O^{(1)} = -\frac{\beta_X \gamma_O}{M_X^2}.$$

The defect overlaps are

$$I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa, \quad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3.$$

We now fix the explicit witness branch

$$\boxed{T_\star = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0,}$$

so that the transverse sector is purely even.

CCXXXII. 1. REDUCED COEFFICIENTS

Using

$$\lambda_\parallel = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_\theta \tilde{\Lambda}_E, \quad \beta = I_\theta \tilde{\Lambda}_O,$$

we obtain

$$\boxed{\lambda_\parallel = \frac{3\sqrt{3}\pi}{512} \frac{\alpha_X \beta_X}{M_X^2} \theta_0^3 \kappa^3,} \quad (463)$$

$$\boxed{\alpha = \frac{\sqrt{3}\pi}{16} \frac{\beta_X \gamma_E}{M_X^2} \theta_0 \kappa, \quad \beta = \frac{\sqrt{3}\pi}{16} \frac{\beta_X \gamma_O}{M_X^2} \theta_0 \kappa.} \quad (464)$$

For the explicit pure- E choice $\gamma_E = 1, \gamma_O = 0$,

$$\boxed{\alpha = \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad \beta = 0.} \quad (465)$$

Numerically,

$$\boxed{\lambda_\parallel \approx 0.03188319195 \frac{\alpha_X \beta_X}{M_X^2} (\theta_0 \kappa)^3,} \quad (466)$$

$$\boxed{\alpha \approx 0.34008738079 \frac{\beta_X}{M_X^2} (\theta_0 \kappa), \quad \beta = 0.} \quad (467)$$

CCXXXIII. 2. DETERMINANT

Since

$$\det M = \lambda_\parallel (\alpha^2 + \beta^2),$$

the explicit pure- E branch gives

$$\boxed{\det M = \frac{9\sqrt{3}\pi^3}{131072} \frac{\alpha_X \beta_X^3}{M_X^6} \theta_0^5 \kappa^5.} \quad (468)$$

Numerically,

$$\det M \approx 0.003687591698 \frac{\alpha_X \beta_X^3}{M_X^6} (\theta_0 \kappa)^5. \quad (469)$$

Therefore the explicit sufficient condition for a nondegenerate reduced symbol is

$$\alpha_X \neq 0, \quad \beta_X \neq 0, \quad \theta_0 \kappa \neq 0. \quad (470)$$

CCXXXIV. 3. TRANSPORT/BLOCH COEFFICIENTS

Because $C_{\parallel} = C_E = C_O = 1$, the exact dictionary becomes

$$g_c = \lambda_{\parallel}, \quad g_E = \alpha, \quad g_O = \beta.$$

Hence, in the explicit pure- E branch,

$$g_c = \frac{3\sqrt{3}\pi}{512} \frac{\alpha_X \beta_X}{M_X^2} \theta_0^3 \kappa^3, \quad (471)$$

$$g_E = \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad g_O = 0. \quad (472)$$

CCXXXV. 4. CONTINUED COEFFICIENT-FINDING: RADIAL CORE COEFFICIENTS

Under the matching hypothesis

$$g_E^{\text{core}} = \xi_E g_E, \quad g_c^{\text{core}} = \xi_c g_c, \quad g_O^{\text{core}} = \xi_O g_O,$$

the radial coefficients are

$$a_0 = \frac{1}{z_{0,0}} (\xi_E g_E + \xi_c g_c), \quad b_0 = \frac{\xi_E g_E}{z_{0,0}}, \quad c_0 = \frac{\xi_O g_O}{z_{0,0}}. \quad (473)$$

In the explicit pure- E branch, $g_O = 0$, so

$$c_0 = 0. \quad (474)$$

Moreover,

$$b_0 = \frac{\xi_E}{z_{0,0}} \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad (475)$$

and

$$a_0 = b_0(1 + R_c), \quad (476)$$

where

$$R_c = \frac{\xi_c}{\xi_E} \frac{\lambda_{\parallel}}{\alpha} = \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2. \quad (477)$$

Thus the explicit branch-level radial coefficients are

$$b_0 = \frac{\xi_E}{z_{0,0}} \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad a_0 = \frac{\xi_E}{z_{0,0}} \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa \left[1 + \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2 \right], \quad c_0 = 0. \quad (478)$$

CCXXXVI. 5. RADIAL ONE-MODE CRITERION IN THIS BRANCH

The practical compact criterion for the important case

$$\kappa_* = -1, \quad \kappa_{\text{comp}} \geq 1$$

is

$$-1 < R_c < 1, \quad \alpha > 0, \quad \mu_\infty > 0, \quad \Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}.$$

Substituting (477), this becomes

$$\boxed{-1 < \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2 < 1,} \quad (479)$$

together with

$$\boxed{\frac{\beta_X}{M_X^2} \theta_0 \kappa > 0, \quad \mu_\infty > 0, \quad \Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}.} \quad (480)$$

CCXXXVII. 6. UNIT BENCHMARK

For the canonical unit benchmark

$$\alpha_X = \beta_X = M_X = 1, \quad \gamma_E = 1, \quad \gamma_O = 0,$$

one gets

$$\boxed{\lambda_{\parallel} \approx 0.03188319195 (\theta_0 \kappa)^3,} \quad (481)$$

$$\boxed{\alpha \approx 0.34008738079 (\theta_0 \kappa), \quad \beta = 0,} \quad (482)$$

$$\boxed{\det M \approx 0.003687591698 (\theta_0 \kappa)^5,} \quad (483)$$

and

$$\boxed{R_c = \frac{\xi_c}{\xi_E} 0.09375 (\theta_0 \kappa)^2.} \quad (484)$$

Hence, in the natural small-defect regime $(\theta_0 \kappa)^2 \ll 1$, one has

$$|R_c| \ll 1$$

for $O(1)$ matching constants ξ_c, ξ_E . So the radial one-mode inequality is naturally satisfied away from accidental sign failures in μ_∞ and the remote-gap condition.

CCXXXVIII. RADIAL COEFFICIENT CONTINUATION FOR THE EXPLICIT BRANCH $T_* = T^1$

CCXXXIX. SHORT EXPLANATION

We now continue the coefficient-finding program by inserting the explicit Bloch/Class-II branch

$$T_* = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0$$

into the radial one-mode dictionary.

The branch-level Bloch coefficients are already fixed as

$$g_c = \lambda_{\parallel}, \quad g_E = \alpha, \quad g_O = \beta$$

in the current canonical normalization $C_c = C_E = C_O = 1$.

For the explicit pure- E branch,

$$g_O = 0,$$

so the odd core coefficient vanishes identically and the remaining radial parameters reduce to

$$a_0, \quad b_0, \quad R_c, \quad \mu_{\infty}.$$

CCXL. 1. BRANCH INPUT FROM THE EXPLICIT TRANSPORT CALCULATION

For the explicit branch

$$T_{\star} = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0,$$

the reduced coefficients are

$$\lambda_{\parallel} = \frac{3\sqrt{3}\pi}{512} \frac{\alpha_X \beta_X}{M_X^2} \theta_0^3 \kappa^3, \quad (485)$$

$$\alpha = \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad \beta = 0. \quad (486)$$

Numerically,

$$\lambda_{\parallel} \approx 0.03188319195 \frac{\alpha_X \beta_X}{M_X^2} (\theta_0 \kappa)^3, \quad (487)$$

$$\alpha \approx 0.34008738079 \frac{\beta_X}{M_X^2} (\theta_0 \kappa), \quad \beta = 0. \quad (488)$$

CCXLI. 2. MATCHING DICTIONARY TO CORE COEFFICIENTS

Introduce the matching constants

$$g_E^{\text{core}} = \xi_E g_E, \quad g_c^{\text{core}} = \xi_c g_c, \quad g_O^{\text{core}} = \xi_O g_O, \quad \xi_E, \xi_c, \xi_O > 0. \quad (489)$$

Then the exact core coefficients are

$$a_0 = \frac{g_E^{\text{core}} + g_c^{\text{core}}}{z_{0,0}}, \quad b_0 = \frac{g_E^{\text{core}}}{z_{0,0}}, \quad c_0 = \frac{g_O^{\text{core}}}{z_{0,0}}. \quad (490)$$

Using $g_c = \lambda_{\parallel}$, $g_E = \alpha$, $g_O = \beta$, one gets

$$a_0 = \frac{1}{z_{0,0}} (\xi_E \alpha + \xi_c \lambda_{\parallel}), \quad b_0 = \frac{\xi_E}{z_{0,0}} \alpha, \quad c_0 = \frac{\xi_O}{z_{0,0}} \beta. \quad (491)$$

Since the explicit branch has $\beta = 0$,

$$c_0 = 0. \quad (492)$$

CCXLII. 3. EXPLICIT FORMULAS FOR b_0 AND a_0

Substituting (486) into (491),

$$b_0 = \frac{\xi_E}{z_{0,0}} \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa. \quad (493)$$

Numerically,

$$b_0 \approx \frac{\xi_E}{z_{0,0}} 0.34008738079 \frac{\beta_X}{M_X^2} (\theta_0 \kappa). \quad (494)$$

Similarly,

$$a_0 = \frac{\xi_E}{z_{0,0}} \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa + \frac{\xi_c}{z_{0,0}} \frac{3\sqrt{3}\pi}{512} \frac{\alpha_X \beta_X}{M_X^2} \theta_0^3 \kappa^3. \quad (495)$$

Factorizing b_0 ,

$$a_0 = b_0(1 + R_c), \quad (496)$$

where

$$R_c := \frac{\xi_c}{\xi_E} \frac{\lambda_{\parallel}}{\alpha} = \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2. \quad (497)$$

Thus

$$a_0 = \frac{\xi_E}{z_{0,0}} \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa \left[1 + \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2 \right]. \quad (498)$$

CCXLIII. 4. SIGN CONDITIONS

Because $\xi_E > 0$ and $z_{0,0} > 0$ by convention, the sign of b_0 is the sign of α :

$$b_0 > 0 \iff \alpha > 0 \iff \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0. \quad (499)$$

Likewise,

$$a_0 > 0 \iff b_0 > 0 \text{ and } 1 + R_c > 0. \quad (500)$$

Since $c_0 = 0$, there is no odd/torsion-like phase contribution in this branch.

CCXLIV. 5. ONE-MODE CRITERION IN THE PRACTICAL LOWEST-CHANNEL CASE

In the practically most relevant case

$$\kappa_* = -1, \quad \kappa_{\text{comp}} \geq 1,$$

the sufficient criterion becomes

$$-1 < R_c < 1, \quad \alpha > 0, \quad \mu_{\infty} > 0, \quad \Delta_{\text{bench}, \rho}^{\text{full}} > \eta_{\text{rem}, \rho}. \quad (501)$$

Substituting (497), the first inequality becomes

$$-1 < \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2 < 1. \quad (502)$$

Thus the entire core-side acceptance problem for this branch reduces to checking:

$$\left[\frac{\beta_X}{M_X^2} \theta_0 \kappa > 0, \quad \mu_\infty > 0, \quad \Delta_{\text{bench}, \rho}^{\text{full}} > \eta_{\text{rem}, \rho}, \quad \left| \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2 \right| < 1. \right] \quad (503)$$

CCXLV. 6. BULK MASS PARAMETER

The bulk mass parameter is

$$\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)}. \quad (504)$$

In the simplest vacuum branch $\chi_\infty = 0$,

$$\mu_\infty = \frac{m(\sigma_\infty)}{z_0(\sigma_\infty)}. \quad (505)$$

Hence, if $z_0(\sigma_\infty) > 0$ is fixed by convention, then

$$\mu_\infty > 0 \iff m(\sigma_\infty) > 0. \quad (506)$$

CCXLVI. 7. UNIT BENCHMARK

For the canonical unit benchmark

$$\alpha_X = \beta_X = M_X = 1, \quad \xi_E = \xi_c = 1, \quad z_{0,0} = 1, \quad \chi_\infty = 0,$$

one gets

$$b_0 \approx 0.34008738079 (\theta_0 \kappa), \quad (507)$$

$$R_c = 0.09375 (\theta_0 \kappa)^2, \quad (508)$$

$$a_0 \approx 0.34008738079 (\theta_0 \kappa) [1 + 0.09375 (\theta_0 \kappa)^2]. \quad (509)$$

and

$$c_0 = 0. \quad (510)$$

Therefore in the small-defect regime $(\theta_0 \kappa)^2 \ll 1$,

$$|R_c| \ll 1,$$

so the condition $-1 < R_c < 1$ is automatically satisfied for this benchmark branch.

CCXLVII. 8. HONEST STATUS

For the explicit branch $T_\star = T^1, \gamma_E = 1, \gamma_O = 0$, the following are now explicit:

1. g_c, g_E, g_O ;
2. a_0, b_0, c_0 in terms of $\xi_E, \xi_c, z_{0,0}$;
3. the exact benchmark ratio R_c ;
4. the reduction of the one-mode criterion to the four checks

$$R_c, \alpha, \mu_\infty, \Delta_{\text{bench},\rho}^{\text{full}} - \eta_{\text{rem},\rho}.$$

What remains genuinely branch-specific is:

1. the actual matching constants ξ_E, ξ_c ;
2. the actual bulk mass sign μ_∞ ;
3. the actual remote-gap inequality.

CCXLVIII. FINAL SUMMARY

For the explicit canonical branch

$$T_\star = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0,$$

the Bloch/Class-II coefficients are

$$g_c = \lambda_\parallel = \frac{3\sqrt{3}\pi}{512} \frac{\alpha_X \beta_X}{M_X^2} \theta_0^3 \kappa^3,$$

$$g_E = \alpha = \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad g_O = \beta = 0.$$

Using the exact matching dictionary

$$g_E^{\text{core}} = \xi_E g_E, \quad g_c^{\text{core}} = \xi_c g_c, \quad g_O^{\text{core}} = \xi_O g_O,$$

the radial core coefficients are

$$b_0 = \frac{\xi_E}{z_{0,0}} \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa,$$

$$c_0 = 0,$$

$$a_0 = b_0(1 + R_c), \quad R_c = \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2.$$

Hence

$$b_0 > 0 \iff \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0,$$

and

$$a_0 > 0 \iff b_0 > 0 \text{ and } 1 + R_c > 0.$$

In the practically most relevant lowest-channel case

$$\kappa_* = -1, \quad \kappa_{\text{comp}} \geq 1,$$

the sufficient one-mode criterion reduces to

$$-1 < R_c < 1, \quad \alpha > 0, \quad \mu_\infty > 0, \quad \Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}.$$

For the canonical unit benchmark

$$\alpha_X = \beta_X = M_X = \xi_E = \xi_c = z_{0,0} = 1, \quad \chi_\infty = 0,$$

one gets

$$b_0 \approx 0.34008738079 (\theta_0 \kappa),$$

$$R_c = 0.09375 (\theta_0 \kappa)^2,$$

$$a_0 \approx 0.34008738079 (\theta_0 \kappa) [1 + 0.09375 (\theta_0 \kappa)^2],$$

$$c_0 = 0.$$

Thus, for this explicit branch, the remaining coefficient-finding problem has been reduced to the finite branch data

$$\xi_E, \quad \xi_c, \quad \mu_\infty, \quad \Delta_{\text{bench},\rho}^{\text{full}} - \eta_{\text{rem},\rho}.$$

CCXLIX. CONTINUED EXPLICIT COEFFICIENT-FINDING FOR THE CANONICAL BRANCH

CCL. SHORT EXPLANATION

For the explicit branch

$$T_\star = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0,$$

the transport-side coefficients are already fixed at the working-branch level:

$$g_c = \lambda_\parallel, \quad g_E = \alpha, \quad g_O = 0.$$

Therefore the radial problem reduces to

$$a_0, \quad b_0, \quad c_0 = 0, \quad R_c, \quad \mu_\infty.$$

CCLI. 1. BRANCH INPUT

Using the already extracted explicit branch coefficients,

$$g_c = \frac{3\sqrt{3}\pi}{512} \frac{\alpha_X \beta_X}{M_X^2} \theta_0^3 \kappa^3, \tag{511}$$

$$g_E = \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad g_O = 0. \quad (512)$$

Numerically,

$$g_c \approx 0.03188319195 \frac{\alpha_X \beta_X}{M_X^2} (\theta_0 \kappa)^3, \quad (513)$$

$$g_E \approx 0.34008738079 \frac{\beta_X}{M_X^2} (\theta_0 \kappa), \quad g_O = 0. \quad (514)$$

CCLII. 2. EXACT RADIAL COEFFICIENTS

The exact constitutive dictionary is

$$a_0 = \frac{\xi_E g_E + \xi_c g_c}{z_{0,0}}, \quad b_0 = \frac{\xi_E g_E}{z_{0,0}}, \quad c_0 = \frac{\xi_O g_O}{z_{0,0}}. \quad (515)$$

Since $g_O = 0$,

$$c_0 = 0. \quad (516)$$

Therefore

$$b_0 = \frac{\xi_E}{z_{0,0}} \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad (517)$$

and

$$a_0 = \frac{\xi_E}{z_{0,0}} \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa + \frac{\xi_c}{z_{0,0}} \frac{3\sqrt{3}\pi}{512} \frac{\alpha_X \beta_X}{M_X^2} \theta_0^3 \kappa^3. \quad (518)$$

Factorizing b_0 ,

$$a_0 = b_0(1 + R_c), \quad (519)$$

with

$$R_c = \frac{\xi_c}{\xi_E} \frac{g_c}{g_E} = \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2. \quad (520)$$

CCLIII. 3. IMMEDIATE SIGN CONSEQUENCES

Assume the standard convention

$$z_{0,0} > 0, \quad \xi_E > 0, \quad \xi_c > 0.$$

Then

$$b_0 > 0 \iff \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0. \quad (521)$$

Similarly,

$$a_0 > 0 \iff b_0 > 0 \quad \text{and} \quad 1 + R_c > 0. \quad (522)$$

CCLIV. 4. PRACTICAL ONE-MODE WINDOW

In the practically most relevant case

$$\kappa_* = -1, \quad \kappa_{\text{comp}} \geq 1,$$

the sufficient criterion is

$$\boxed{-1 < R_c < 1, \quad \alpha > 0, \quad \mu_\infty > 0, \quad \Delta_{\text{bench}, \rho}^{\text{full}} > \eta_{\text{rem}, \rho}.} \quad (523)$$

Substituting (520),

$$\boxed{\left| \frac{\xi_c}{\xi_E} \right| < \frac{32}{3 |\alpha_X| (\theta_0 \kappa)^2}.} \quad (524)$$

So the entire low-core inequality is reduced to a finite ratio bound on ξ_c/ξ_E .

CCLV. 5. EXPLICIT BENCHMARK VALUES FOR THE RATIO WINDOW

Under the unit benchmark

$$\alpha_X = \beta_X = M_X = \xi_E = z_{0,0} = 1,$$

the ratio window becomes

$$|\xi_c| < \frac{32}{3(\theta_0 \kappa)^2}.$$

Thus, for representative defect strengths:

$$\boxed{\theta_0 \kappa = 0.1 \implies |\xi_c/\xi_E| < 1066.666 \dots} \quad (525)$$

$$\boxed{\theta_0 \kappa = 0.3 \implies |\xi_c/\xi_E| < 118.518 \dots} \quad (526)$$

$$\boxed{\theta_0 \kappa = 1 \implies |\xi_c/\xi_E| < 10.666 \dots} \quad (527)$$

Hence in the natural small-defect regime $(\theta_0 \kappa)^2 \ll 1$, the R_c -window is extremely broad. So the low-core inequality is not fine-tuned unless ξ_c/ξ_E is abnormally large.

CCLVI. 6. BULK MASS PARAMETER

The bulk mass parameter is

$$\boxed{\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)}.} \quad (528)$$

In the simplest vacuum branch $\chi_\infty = 0$,

$$\boxed{\mu_\infty = \frac{m(\sigma_\infty)}{z_0(\sigma_\infty)}.} \quad (529)$$

If the standard radial convention fixes

$$z_0(\sigma_\infty) > 0,$$

then

$$\boxed{\mu_\infty > 0 \iff m(\sigma_\infty) > 0.} \quad (530)$$

Thus the bulk-side sign audit is reduced to the sign of the asymptotic mass function $m(\sigma_\infty)$.

CCLVII. 7. WHAT REMAINS GENUINELY UNFIXED

The uploaded notes do *not* yet supply numerical branch values for

$$\xi_E, \quad \xi_c, \quad z_{0,0}, \quad m(\sigma_\infty), \quad z_0(\sigma_\infty), \quad \Delta_{\text{bench},\rho}^{\text{full}} - \eta_{\text{rem},\rho}.$$

Therefore the present coefficient-finding status is:

1. the branch-level formulas for $a_0, b_0, c_0, R_c, \mu_\infty$ are explicit;
2. $c_0 = 0$ is completely fixed for the present pure- E branch;
3. the sign conditions reduce to

$$\frac{\beta_X}{M_X^2} \theta_0 \kappa > 0, \quad m(\sigma_\infty) > 0, \quad \left| \frac{\xi_c}{\xi_E} \right| < \frac{32}{3|\alpha_X|(\theta_0 \kappa)^2},$$

plus the remote-gap inequality;

4. the remaining unknowns are finite branch data, not conceptual objects.

CCLVIII. FINAL SUMMARY

For the explicit canonical branch

$$T_\star = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0,$$

the Bloch/Class-II coefficients are

$$g_c = \frac{3\sqrt{3}\pi}{512} \frac{\alpha_X \beta_X}{M_X^2} \theta_0^3 \kappa^3,$$

$$g_E = \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad g_O = 0.$$

Hence the radial core coefficients are

$$b_0 = \frac{\xi_E}{z_{0,0}} \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa,$$

$$c_0 = 0,$$

$$a_0 = b_0(1 + R_c), \quad R_c = \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2.$$

In the practical lowest-channel case

$$\kappa_\star = -1, \quad \kappa_{\text{comp}} \geq 1,$$

the one-mode criterion reduces to

$$-1 < R_c < 1, \quad \alpha > 0, \quad \mu_\infty > 0, \quad \Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}.$$

Equivalently,

$$\left| \frac{\xi_c}{\xi_E} \right| < \frac{32}{3|\alpha_X|(\theta_0\kappa)^2},$$

together with

$$\frac{\beta_X}{M_X^2} \theta_0 \kappa > 0, \quad \mu_\infty > 0, \quad \Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}.$$

For the unit benchmark

$$\alpha_X = \beta_X = M_X = \xi_E = z_{0,0} = 1,$$

this becomes

$$b_0 \approx 0.34008738079 (\theta_0 \kappa), \quad c_0 = 0, \quad R_c = 0.09375 \xi_c (\theta_0 \kappa)^2.$$

Hence:

$$\theta_0 \kappa = 0.1 \Rightarrow |\xi_c| < 1066.67,$$

$$\theta_0 \kappa = 0.3 \Rightarrow |\xi_c| < 118.52,$$

$$\theta_0 \kappa = 1 \Rightarrow |\xi_c| < 10.67.$$

So the low-core inequality is naturally broad in the small-defect regime. The remaining unfixed inputs are the finite branch data

$$\xi_E, \quad \xi_c, \quad z_{0,0}, \quad \mu_\infty, \quad \Delta_{\text{bench},\rho}^{\text{full}} - \eta_{\text{rem},\rho}.$$

CCLIX. BULK-MASS SIGN AUDIT FOR THE EXPLICIT CANONICAL BRANCH

CCLX. SHORT EXPLANATION

For the explicit branch

$$T_\star = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0,$$

the transport-side coefficients are already fixed. The remaining nontrivial post-acceptance radial condition is now the bulk-mass sign

$$\mu_\infty > 0.$$

The exact constitutive formula is already known:

$$\mu(r) = \frac{m(\bar{\sigma}(r)) - 3\chi_X(r)u_X(\bar{\sigma}(r))}{z_0(\bar{\sigma}(r))},$$

so the asymptotic bulk parameter is

$$\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)}.$$

Thus the sign audit reduces to a finite constitutive inequality.

CCLXI. 1. EXPLICIT BRANCH INPUT

For the explicit canonical branch

$$T_\star = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0,$$

the already extracted transport coefficients are

$$g_c = \frac{3\sqrt{3}\pi}{512} \frac{\alpha_X \beta_X}{M_X^2} \theta_0^3 \kappa^3, \quad (531)$$

$$g_E = \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad g_O = 0. \quad (532)$$

Therefore

$$b_0 = \frac{\xi_E g_E}{z_{0,0}}, \quad c_0 = 0, \quad a_0 = b_0(1 + R_c), \quad (533)$$

with

$$R_c = \frac{\xi_c}{\xi_E} \frac{g_c}{g_E} = \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2. \quad (534)$$

The practical one-mode criterion is therefore reduced to

$$-1 < R_c < 1, \quad \alpha > 0, \quad \mu_\infty > 0, \quad \Delta_{\text{bench}, \rho}^{\text{full}} > \eta_{\text{rem}, \rho}. \quad (535)$$

CCLXII. 2. EXACT ASYMPTOTIC BULK MASS FORMULA

The asymptotic effective mass is

$$\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)}. \quad (536)$$

Hence the exact sign condition is

$$\mu_\infty > 0 \iff \left[m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty) \right] z_0(\sigma_\infty) > 0. \quad (537)$$

If the standard normalization convention fixes

$$z_0(\sigma_\infty) > 0, \quad (538)$$

then the condition simplifies to

$$\mu_\infty > 0 \iff m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty) > 0. \quad (539)$$

Thus the bulk-side sign audit is reduced to one scalar competition: the positive asymptotic mass contribution $m(\sigma_\infty)$ must dominate the asymptotic Class-II correction $3\chi_\infty u_X(\sigma_\infty)$.

CCLXIII. 3. SIMPLEST VACUUM BRANCH

In the simplest vacuum branch one has

$$\boxed{\chi_\infty = 0.} \quad (540)$$

Then

$$\boxed{\mu_\infty = \frac{m(\sigma_\infty)}{z_0(\sigma_\infty)}.} \quad (541)$$

So, under the standard positive normalization $z_0(\sigma_\infty) > 0$,

$$\boxed{\mu_\infty > 0 \iff m(\sigma_\infty) > 0.} \quad (542)$$

Therefore, in this simplest branch, the bulk-mass condition is not a dynamical balance anymore. It is reduced to the sign of the asymptotic mass function $m(\sigma_\infty)$.

CCLXIV. 4. NON-VACUUM ASYMPTOTIC CLASS-II BRANCH

If $\chi_\infty \neq 0$, then the exact condition is

$$\boxed{m(\sigma_\infty) > 3\chi_\infty u_X(\sigma_\infty)} \quad (\text{assuming } z_0(\sigma_\infty) > 0). \quad (543)$$

This admits the following subcases.

Case A: $u_X(\sigma_\infty) \geq 0, \chi_\infty \geq 0$

Then the Class-II asymptotic correction reduces the bulk gap, and one needs the lower bound

$$\boxed{m(\sigma_\infty) > 3\chi_\infty u_X(\sigma_\infty) \geq 0.} \quad (544)$$

Case B: $u_X(\sigma_\infty) \leq 0, \chi_\infty \geq 0$

Then the Class-II correction helps the gap, and it is enough that

$$\boxed{m(\sigma_\infty) > 3\chi_\infty u_X(\sigma_\infty)} \quad (545)$$

which is weaker than simply demanding $m(\sigma_\infty) > 0$.

Case C: sign-indefinite asymptotic sector

If the signs of χ_∞ and $u_X(\sigma_\infty)$ are not known, then the exact sign audit is still finite but genuinely branch-specific:

$$\boxed{\text{sgn } \mu_\infty = \text{sgn} \left(m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty) \right)} \quad (\text{assuming } z_0(\sigma_\infty) > 0). \quad (546)$$

CCLXV. 5. COMBINED PRACTICAL CRITERION FOR THE EXPLICIT BRANCH

For the branch $T_\star = T^1, \gamma_E = 1, \gamma_O = 0$, the practical lowest-channel acceptance problem becomes

$$\left| \frac{\xi_c}{\xi_E} \right| < \frac{32}{3|\alpha_X|(\theta_0\kappa)^2}, \quad (547)$$

$$\frac{\beta_X}{M_X^2} \theta_0\kappa > 0, \quad (548)$$

$$m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty) > 0 \quad (\text{assuming } z_0(\sigma_\infty) > 0), \quad (549)$$

together with

$$\Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}. \quad (550)$$

Thus the entire post-acceptance one-mode verification of this explicit branch reduces to four finite scalar checks.

CCLXVI. 6. UNIT BENCHMARK

Take the unit benchmark

$$\alpha_X = \beta_X = M_X = \xi_E = \xi_c = z_{0,0} = 1.$$

Then

$$R_c = 0.09375 (\theta_0\kappa)^2,$$

so

$$\theta_0\kappa = 0.1 \implies R_c = 9.375 \times 10^{-4}, \quad (551)$$

$$\theta_0\kappa = 0.3 \implies R_c = 8.4375 \times 10^{-3}, \quad (552)$$

$$\theta_0\kappa = 1 \implies R_c = 9.375 \times 10^{-2}. \quad (553)$$

Hence the R_c -window is naturally satisfied in the small-defect regime.

If one additionally takes the simplest vacuum branch $\chi_\infty = 0$, then the only remaining bulk sign check is

$$m(\sigma_\infty) > 0. \quad (554)$$

CCLXVII. 7. HONEST STATUS

What is now explicit for the branch $T_\star = T^1, \gamma_E = 1, \gamma_O = 0$:

1. g_c, g_E, g_O ;
2. a_0, b_0, c_0 in terms of $\xi_E, \xi_c, z_{0,0}$;
3. the exact ratio

$$R_c = \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0\kappa)^2;$$

4. the exact bulk-mass sign condition

$$m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty) > 0$$

under $z_0(\sigma_\infty) > 0$.

What is still not numerically fixed by the uploaded notes is the actual branch data

$$\xi_E, \quad \xi_c, \quad z_{0,0}, \quad m(\sigma_\infty), \quad z_0(\sigma_\infty), \quad \chi_\infty, \quad u_X(\sigma_\infty), \quad \Delta_{\text{bench},\rho}^{\text{full}} - \eta_{\text{rem},\rho}.$$

So the remaining frontier is not structural. It is the finite insertion of asymptotic constitutive data.

CCLXVIII. FINAL SUMMARY

For the explicit canonical branch

$$T_\star = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0,$$

the post-acceptance one-defect/one-mode problem reduces to four scalar checks:

$$-1 < R_c < 1, \quad \alpha > 0, \quad \mu_\infty > 0, \quad \Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}.$$

The ratio parameter is already explicit:

$$R_c = \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2.$$

The bulk mass parameter is exactly

$$\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)}.$$

Hence, under the standard convention $z_0(\sigma_\infty) > 0$,

$$\mu_\infty > 0 \iff m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty) > 0.$$

In the simplest vacuum branch $\chi_\infty = 0$, this simplifies to

$$\mu_\infty > 0 \iff m(\sigma_\infty) > 0.$$

For the unit benchmark

$$\alpha_X = \beta_X = M_X = \xi_E = \xi_c = z_{0,0} = 1,$$

one gets

$$R_c = 0.09375 (\theta_0 \kappa)^2,$$

so in the small-defect regime the R_c -window is automatically broad:

$$\theta_0 \kappa = 0.1 \Rightarrow R_c = 9.375 \times 10^{-4},$$

$$\theta_0 \kappa = 0.3 \Rightarrow R_c = 8.4375 \times 10^{-3},$$

$$\theta_0 \kappa = 1 \Rightarrow R_c = 9.375 \times 10^{-2}.$$

Thus the genuinely nontrivial remaining branch inputs are

$$m(\sigma_\infty), \quad z_0(\sigma_\infty), \quad \chi_\infty, \quad u_X(\sigma_\infty), \quad \Delta_{\text{bench},\rho}^{\text{full}} - \eta_{\text{rem},\rho}.$$

So the current explicit branch is already reduced to a finite asymptotic constitutive-data audit.

CCLXIX. CONTINUATION OF COEFFICIENT-FINDING FOR THE EXPLICIT CANONICAL BRANCH

CCLXX. BRANCH CHOICE AND CANONICAL NORMALIZATION

We continue with the explicit branch

$$T_\star = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0,$$

and use the canonical one-mediator normalization

$$C_\parallel = C_E = C_O = 1.$$

The one-mediator truncation gives

$$\lambda_c^{(1)} = -\frac{\alpha_X \beta_X}{M_X^2}, \quad \tilde{\Lambda}_E^{(1)} = -\frac{\beta_X}{M_X^2}, \quad \tilde{\Lambda}_O^{(1)} = 0. \quad (555)$$

Using

$$I_\theta = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa, \quad I_{\theta^3} = -\frac{3\sqrt{3}\pi}{256}\theta_0^3\kappa^3,$$

the reduced coefficients are

$$\lambda_\parallel = \frac{3\sqrt{3}\pi}{512} \frac{\alpha_X \beta_X}{M_X^2} \theta_0^3 \kappa^3, \quad (556)$$

$$\alpha = \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad \beta = 0. \quad (557)$$

Equivalently, in the exact transport/Bloch dictionary

$$\lambda_\parallel = C_c g_c, \quad \alpha = C_E g_E, \quad \beta = C_O g_O,$$

we simply have

$$g_c = \lambda_\parallel, \quad g_E = \alpha, \quad g_O = 0. \quad (558)$$

Numerically,

$$g_c \approx 0.03188319195 \frac{\alpha_X \beta_X}{M_X^2} (\theta_0 \kappa)^3, \quad (559)$$

$$g_E \approx 0.34008738079 \frac{\beta_X}{M_X^2} (\theta_0 \kappa), \quad g_O = 0. \quad (560)$$

CCLXXI. RADIAL CORE COEFFICIENTS

Introduce the matching constants

$$g_E^{\text{core}} = \xi_E g_E, \quad g_c^{\text{core}} = \xi_c g_c, \quad g_O^{\text{core}} = \xi_O g_O, \quad \xi_E, \xi_c, \xi_O > 0.$$

Then

$$a_0 = \frac{g_E^{\text{core}} + g_c^{\text{core}}}{z_{0,0}}, \quad b_0 = \frac{g_E^{\text{core}}}{z_{0,0}}, \quad c_0 = \frac{g_O^{\text{core}}}{z_{0,0}}.$$

Thus, for the present branch,

$$b_0 = \frac{\xi_E}{z_{0,0}} \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad (561)$$

$$c_0 = 0, \quad (562)$$

and

$$a_0 = \frac{\xi_E}{z_{0,0}} \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa + \frac{\xi_c}{z_{0,0}} \frac{3\sqrt{3}\pi}{512} \frac{\alpha_X \beta_X}{M_X^2} \theta_0^3 \kappa^3. \quad (563)$$

Factorizing b_0 ,

$$a_0 = b_0(1 + R_c), \quad (564)$$

with

$$R_c = \frac{\xi_c}{\xi_E} \frac{g_c}{g_E} = \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2. \quad (565)$$

Therefore, assuming $z_{0,0} > 0$ and $\xi_E > 0$,

$$b_0 > 0 \iff \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0. \quad (566)$$

and

$$a_0 > 0 \iff b_0 > 0 \quad \text{and} \quad 1 + R_c > 0. \quad (567)$$

CCLXXII. BULK MASS PARAMETER

The exact asymptotic bulk parameter is

$$\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)}. \quad (568)$$

Under the standard sign convention

$$z_0(\sigma_\infty) > 0,$$

one has

$$\mu_\infty > 0 \iff m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty) > 0. \quad (569)$$

In the simplest vacuum branch $\chi_\infty = 0$,

$$\mu_\infty > 0 \iff m(\sigma_\infty) > 0. \quad (570)$$

CCLXXIII. CARRIER ACCEPTANCE

At carrier level the branch is accepted iff

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0), \quad A_{\text{res}} \cap \mathcal{M}_D = \{0\}.$$

For the present branch this reduces to

$$\boxed{\alpha_X \neq 0, \quad \beta_X \neq 0, \quad \theta_0 \kappa \neq 0, \quad A_{\text{res}} \cap \mathcal{M}_D = \{0\}.} \quad (571)$$

Indeed,

$$\lambda_{\parallel} \propto \alpha_X \beta_X (\theta_0 \kappa)^3, \quad \alpha \propto \beta_X (\theta_0 \kappa), \quad \beta = 0.$$

CCLXXIV. REMOTE-GAP SUPPORT THEOREM INSERTED FOR THIS BRANCH

The fully inserted Step-IV gate is

$$\boxed{\Delta_{\text{bench}, \rho}^{\text{fin}} > 2 \left[\kappa \rho (|C_E| + |C_O|) + \frac{1}{M_X^2} (\alpha_X^2 + |\alpha_X \beta_X| + \beta_X^2) \right] + 2C_{\text{high}, \text{non}} \rho + \sum_{a \in \mathcal{A}_{\text{act}}} K_a \rho^{r_a} + \sum_{a \in \mathcal{A}_{\text{act}}^{(H)}} \frac{\tilde{K}_a}{M_a^{\sigma_a}} \rho^{r_a} + \eta_{\text{diag}, \rho}.} \quad (572)$$

In canonical normalization $C_E = C_O = 1$, this simplifies to

$$\boxed{\Delta_{\text{bench}, \rho}^{\text{fin}} > 2 \left[2\kappa \rho + \frac{1}{M_X^2} (\alpha_X^2 + |\alpha_X \beta_X| + \beta_X^2) \right] + 2C_{\text{high}, \text{non}} \rho + \sum_{a \in \mathcal{A}_{\text{act}}} K_a \rho^{r_a} + \sum_{a \in \mathcal{A}_{\text{act}}^{(H)}} \frac{\tilde{K}_a}{M_a^{\sigma_a}} \rho^{r_a} + \eta_{\text{diag}, \rho}.} \quad (573)$$

If (573) holds, then

$$\boxed{\Delta_{\rho} \geq \Delta_{\text{bench}, \rho}^{\text{fin}} - \eta_{\text{rem}, \rho} > 0,} \quad (574)$$

and therefore

$$\boxed{H_{\text{eff}}(E, p) = H_D(p) + O\left(\frac{|p|^2}{\Delta_{\rho}}\right).} \quad (575)$$

CCLXXV. PRACTICAL ONE-MODE CRITERION FOR THE EXPLICIT BRANCH

In the practically important case

$$\kappa_* = -1, \quad \kappa_{\text{comp}} \geq 1,$$

the one-mode criterion reduces to

$$\boxed{-1 < R_c < 1, \quad \alpha > 0, \quad \mu_{\infty} > 0, \quad \Delta_{\text{bench}, \rho}^{\text{full}} > \eta_{\text{rem}, \rho}.} \quad (576)$$

Substituting the present branch data gives

$$\boxed{\left| \frac{\xi_c}{\xi_E} \right| < \frac{32}{3|\alpha_X|(\theta_0 \kappa)^2},} \quad (577)$$

$$\boxed{\frac{\beta_X}{M_X^2} \theta_0 \kappa > 0,} \quad (578)$$

$$\boxed{m(\sigma_{\infty}) - 3\chi_{\infty} u_X(\sigma_{\infty}) > 0 \quad (\text{assuming } z_0(\sigma_{\infty}) > 0),} \quad (579)$$

and the remote-gap inequality (573).

CCLXXVI. UNIT BENCHMARK

Take

$$\alpha_X = \beta_X = M_X = \xi_E = \xi_c = z_{0,0} = 1, \quad \gamma_E = 1, \quad \gamma_O = 0.$$

Then

$$g_c \approx 0.03188319195 (\theta_0 \kappa)^3, \quad g_E \approx 0.34008738079 (\theta_0 \kappa), \quad g_O = 0, \quad (580)$$

$$b_0 \approx 0.34008738079 (\theta_0 \kappa), \quad c_0 = 0, \quad (581)$$

$$R_c = 0.09375 (\theta_0 \kappa)^2, \quad (582)$$

$$a_0 \approx 0.34008738079 (\theta_0 \kappa) [1 + 0.09375 (\theta_0 \kappa)^2]. \quad (583)$$

Representative values:

$$\theta_0 \kappa = 0.1 \Rightarrow R_c = 9.375 \times 10^{-4},$$

$$\theta_0 \kappa = 0.3 \Rightarrow R_c = 8.4375 \times 10^{-3},$$

$$\theta_0 \kappa = 1 \Rightarrow R_c = 9.375 \times 10^{-2}.$$

Hence the R_c -window is automatically broad in the natural small-defect regime.

CCLXXVII. HONEST ENDPOINT

For the explicit canonical branch $T_\star = T^1, \gamma_E = 1, \gamma_O = 0$, the following are now explicit:

1. transport-side coefficients (g_c, g_E, g_O) ;
2. radial core coefficients (a_0, b_0, c_0) up to finite matching constants;
3. exact ratio R_c ;
4. exact sign reduction of μ_∞ ;
5. the branch-inserted remote-gap gate.

What is still not numerically fixed in the uploaded notes is the finite branch data

$$\xi_E, \quad \xi_c, \quad z_{0,0}, \quad m(\sigma_\infty), \quad z_0(\sigma_\infty), \quad \chi_\infty, \quad u_X(\sigma_\infty), \quad C_{\text{high,non}}, \quad K_a, \quad \tilde{K}_a, \quad \eta_{\text{diag},\rho}.$$

Therefore the remaining frontier is not structural. It is the final finite constitutive-data insertion.

CCLXXVIII. FINAL SUMMARY

For the explicit canonical branch

$$T_\star = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0,$$

the one-mediator transport-side coefficients in canonical normalization are

$$g_c = \frac{3\sqrt{3}\pi}{512} \frac{\alpha_X \beta_X}{M_X^2} \theta_0^3 \kappa^3, \quad g_E = \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad g_O = 0.$$

Hence the radial core coefficients are

$$b_0 = \frac{\xi_E}{z_{0,0}} g_E, \quad c_0 = 0, \quad a_0 = b_0(1 + R_c),$$

with

$$R_c = \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2.$$

The bulk mass condition is exactly

$$\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)},$$

so under $z_0(\sigma_\infty) > 0$,

$$\mu_\infty > 0 \iff m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty) > 0.$$

The branch-inserted Step-IV remote-gap gate is

$$\Delta_{\text{bench},\rho}^{\text{fin}} > 2 \left[2\kappa\rho + \frac{1}{M_X^2} (\alpha_X^2 + |\alpha_X \beta_X| + \beta_X^2) \right] + 2C_{\text{high,non}}\rho + \sum_{a \in \mathcal{A}_{\text{act}}} K_a \rho^{r_a} + \sum_{a \in \mathcal{A}_{\text{act}}^{(H)}} \frac{\tilde{K}_a}{M_a^{\sigma_a}} \rho^{r_a} + \eta_{\text{diag},\rho}.$$

Therefore the practical one-mode criterion for this branch reduces to the four scalar checks

$$\left| \frac{\xi_c}{\xi_E} \right| < \frac{32}{3|\alpha_X|(\theta_0 \kappa)^2}, \quad \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0, \quad m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty) > 0, \quad \Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}.$$

For the unit benchmark

$$\alpha_X = \beta_X = M_X = \xi_E = \xi_c = z_{0,0} = 1,$$

one gets

$$b_0 \approx 0.34008738079 (\theta_0 \kappa), \quad c_0 = 0, \quad R_c = 0.09375 (\theta_0 \kappa)^2,$$

$$a_0 \approx 0.34008738079 (\theta_0 \kappa) [1 + 0.09375 (\theta_0 \kappa)^2].$$

Thus the R_c -window is naturally broad in the small-defect regime, and the genuinely nontrivial remaining branch data are

$$\xi_E, \xi_c, z_{0,0}, m(\sigma_\infty), z_0(\sigma_\infty), \chi_\infty, u_X(\sigma_\infty), C_{\text{high,non}}, K_a, \tilde{K}_a, \eta_{\text{diag},\rho}.$$

CCLXXIX. CONTINUATION OF COEFFICIENT-FINDING: EXPLICIT REMOTE-GAP INSERTION FOR THE CANONICAL BRANCH

CCLXXX. SHORT EXPLANATION

For the explicit branch

$$T_\star = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0,$$

the transport-side coefficients and the radial dictionary are already explicit. The remaining coefficient-finding task is to insert this branch into the final Step-IV remote-gap inequality and isolate the genuinely branch-dependent finite data.

CCLXXXI. 1. FIXED TRANSPORT-SIDE COEFFICIENTS

Under the canonical one-mediator normalization

$$C_\parallel = C_E = C_O = 1,$$

the explicit branch coefficients are

$$g_c = \lambda_\parallel = \frac{3\sqrt{3}\pi}{512} \frac{\alpha_X \beta_X}{M_X^2} \theta_0^3 \kappa^3, \quad (584)$$

$$g_E = \alpha = \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad g_O = \beta = 0. \quad (585)$$

Numerically,

$$g_c \approx 0.03188319195 \frac{\alpha_X \beta_X}{M_X^2} (\theta_0 \kappa)^3, \quad (586)$$

$$g_E \approx 0.34008738079 \frac{\beta_X}{M_X^2} (\theta_0 \kappa), \quad g_O = 0. \quad (587)$$

CCLXXXII. 2. FIXED RADIAL COEFFICIENTS

Using

$$g_E^{\text{core}} = \xi_E g_E, \quad g_c^{\text{core}} = \xi_c g_c, \quad g_O^{\text{core}} = \xi_O g_O,$$

the radial coefficients become

$$b_0 = \frac{\xi_E}{z_{0,0}} g_E = \frac{\xi_E}{z_{0,0}} \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad (588)$$

$$c_0 = 0, \quad (589)$$

$$a_0 = b_0(1 + R_c), \quad R_c = \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2. \quad (590)$$

Thus the practical radial sign conditions are

$$b_0 > 0 \iff \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0, \quad (591)$$

and

$$a_0 > 0 \iff b_0 > 0 \quad \text{and} \quad 1 + R_c > 0. \quad (592)$$

CCLXXXIII. 3. BULK MASS PARAMETER

The exact asymptotic bulk parameter remains

$$\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)}. \quad (593)$$

Under the standard convention $z_0(\sigma_\infty) > 0$,

$$\mu_\infty > 0 \iff m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty) > 0. \quad (594)$$

In the simplest vacuum branch $\chi_\infty = 0$,

$$\mu_\infty > 0 \iff m(\sigma_\infty) > 0. \quad (595)$$

CCLXXXIV. 4. EXACT BENCHMARK SIDE

The final benchmark side is

$$\Delta_{\text{bench},\rho}^{\text{fin}} = \min \left\{ 4K - |r_*|, \sqrt{M_\sigma^2 + c_\sigma^2(q_* - \rho)^2}, q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|, \Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)} \right\}. \quad (596)$$

Thus the only genuinely model-specific benchmark piece is the same-shell non-enriched entry

$$\Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)},$$

which is already reduced to one finite Hermitian eigenvalue computation.

CCLXXXV. 5. BRANCH-INSERTED RESIDUAL SIDE

The exact final inserted residual side is

$$\eta_{R,\rho} \leq 2 \left[\kappa \rho (|C_E| + |C_O|) + \frac{1}{M_X^2} (\alpha_X^2 + |\alpha_X \beta_X| + \beta_X^2) \right] + 2C_{\text{high},\text{non}\rho} + \sum_{a \in \mathcal{A}_{\text{act}}} K_a \rho^{r_a} + \sum_{a \in \mathcal{A}_{\text{act}}^{(H)}} \frac{\tilde{K}_a}{M_a^{\sigma_a}} \rho^{r_a} + \eta_{\text{diag},\rho}. \quad (597)$$

In the canonical normalization $C_E = C_O = 1$, this becomes

$$\eta_{R,\rho} \leq 4\kappa\rho + \frac{2}{M_X^2} (\alpha_X^2 + |\alpha_X \beta_X| + \beta_X^2) + 2C_{\text{high},\text{non}\rho} + \sum_{a \in \mathcal{A}_{\text{act}}} K_a \rho^{r_a} + \sum_{a \in \mathcal{A}_{\text{act}}^{(H)}} \frac{\tilde{K}_a}{M_a^{\sigma_a}} \rho^{r_a} + \eta_{\text{diag},\rho}. \quad (598)$$

This is the fully branch-inserted Step-IV residual budget.

CCLXXXVI. 6. EXACT BRANCH-INSERTED REMOTE-GAP GATE

Therefore, for the explicit branch $T_\star = T^1, \gamma_E = 1, \gamma_O = 0$, the exact Step-IV gate is the single inequality

$$\Delta_{\text{bench},\rho}^{\text{fin}} > 4\kappa\rho + \frac{2}{M_X^2} (\alpha_X^2 + |\alpha_X\beta_X| + \beta_X^2) + 2C_{\text{high,non}}\rho + \sum_{a \in \mathcal{A}_{\text{act}}} K_a \rho^{r_a} + \sum_{a \in \mathcal{A}_{\text{act}}^{(H)}} \frac{\tilde{K}_a}{M_a^{\sigma_a}} \rho^{r_a} + \eta_{\text{diag},\rho}. \quad (599)$$

If (599) holds, then

$$\Delta_\rho \geq \Delta_{\text{bench},\rho}^{\text{fin}} - \eta_{R,\rho} > 0. \quad (600)$$

Thus the remote sector is spectrally isolated.

CCLXXXVII. 7. FULL PRACTICAL CRITERION FOR THIS BRANCH

The practical lowest-channel criterion becomes

$$-1 < R_c < 1, \quad \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0, \quad m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty) > 0, \quad (599) \text{ holds}. \quad (601)$$

Equivalently, the full coefficient-finding program for this explicit branch is reduced to the finite data

$$\xi_E, \quad \xi_c, \quad z_{0,0}, \quad m(\sigma_\infty), \quad z_0(\sigma_\infty), \quad \chi_\infty, \quad u_X(\sigma_\infty), \quad \Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)}, \quad C_{\text{high,non}}, \quad K_a, \quad \tilde{K}_a, \quad (602)$$

CCLXXXVIII. 8. UNIT BENCHMARK

Take

$$\alpha_X = \beta_X = M_X = \xi_E = \xi_c = z_{0,0} = 1.$$

Then

$$b_0 \approx 0.34008738079 (\theta_0 \kappa), \quad c_0 = 0, \quad R_c = 0.09375 (\theta_0 \kappa)^2, \quad (603)$$

$$a_0 \approx 0.34008738079 (\theta_0 \kappa) [1 + 0.09375 (\theta_0 \kappa)^2]. \quad (604)$$

For representative defect strengths:

$$\theta_0 \kappa = 0.1 \Rightarrow R_c = 9.375 \times 10^{-4},$$

$$\theta_0 \kappa = 0.3 \Rightarrow R_c = 8.4375 \times 10^{-3},$$

$$\theta_0 \kappa = 1 \Rightarrow R_c = 9.375 \times 10^{-2}.$$

Hence the R_c -window is automatically broad in the natural small-defect regime, and the genuinely nontrivial remaining checks are the bulk sign and the remote-gap inequality.

CCLXXXIX. FINAL SUMMARY

For the explicit canonical branch

$$T_\star = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0,$$

the coefficient-finding program can now be written entirely in finite algebraic form.

The transport-side coefficients are

$$g_c = \frac{3\sqrt{3}\pi}{512} \frac{\alpha_X \beta_X}{M_X^2} \theta_0^3 \kappa^3, \quad g_E = \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad g_O = 0.$$

The radial coefficients are therefore

$$b_0 = \frac{\xi_E}{z_{0,0}} g_E, \quad c_0 = 0, \quad a_0 = b_0(1 + R_c),$$

with

$$R_c = \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2.$$

The exact bulk-mass sign condition is

$$\mu_\infty > 0 \iff m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty) > 0 \quad (\text{assuming } z_0(\sigma_\infty) > 0).$$

The exact branch-inserted remote-gap gate is

$$\Delta_{\text{bench},\rho}^{\text{fin}} > 4\kappa\rho + \frac{2}{M_X^2} (\alpha_X^2 + |\alpha_X \beta_X| + \beta_X^2) + 2C_{\text{high,non}}\rho + \sum_{a \in \mathcal{A}_{\text{act}}} K_a \rho^{r_a} + \sum_{a \in \mathcal{A}_{\text{act}}^{(H)}} \frac{\tilde{K}_a}{M_a^{\sigma_a}} \rho^{r_a} + \eta_{\text{diag},\rho}.$$

Thus the full practical lowest-channel condition for this branch is reduced to four finite checks:

$$-1 < R_c < 1, \quad \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0, \quad m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty) > 0, \quad \Delta_{\text{bench},\rho}^{\text{fin}} > \eta_{R,\rho}.$$

Under the unit benchmark

$$\alpha_X = \beta_X = M_X = \xi_E = \xi_c = z_{0,0} = 1,$$

one has

$$b_0 \approx 0.34008738079 (\theta_0 \kappa), \quad c_0 = 0, \quad R_c = 0.09375 (\theta_0 \kappa)^2,$$

$$a_0 \approx 0.34008738079 (\theta_0 \kappa) [1 + 0.09375 (\theta_0 \kappa)^2].$$

Hence the R_c -window is automatically broad in the small-defect regime. The genuinely nontrivial remaining branch data are

$$m(\sigma_\infty), \quad z_0(\sigma_\infty), \quad \chi_\infty, \quad u_X(\sigma_\infty), \quad \Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)}, \quad C_{\text{high,non}}, \quad K_a, \quad \tilde{K}_a, \quad \eta_{\text{diag},\rho}.$$

CCXC. EXPLICIT $m = 1$ SAME-SHELL NON-ENRICHED COEFFICIENT EXTRACTION

CCXCI. SHORT EXPLANATION

The only genuinely non-automatic benchmark-side slot in Step IV is the same-shell non-enriched sector. In the minimal truncation $m = 1$, this sector is already reduced to an explicit 2×2 Hermitian matrix problem.

Therefore the coefficient-finding frontier is now:

$$d_+(p), \quad d_-(p), \quad m(p)$$

for the non-enriched same-shell block.

CCXCII. 1. EXACT $m = 1$ BLOCK

Let the minimal same-shell non-enriched basis be

$$\mathcal{B}_{\text{non-enr}}^{(1)} = \{|G_+, \eta\rangle, |G_-, \eta\rangle\},$$

where η is one internal state orthogonal to the enriched shell doublet.

Then the projected block is

$$H_{\text{non-enr}}^{(1)}(p) = \begin{pmatrix} d_+(p) & m(p) \\ m(p) & d_-(p) \end{pmatrix}, \quad (605)$$

with

$$d_{\pm}(p) = \langle G_{\pm}, \eta | H_{\text{shell}}^{(1)}(p) | G_{\pm}, \eta \rangle, \quad m(p) = \langle G_+, \eta | H_{\text{shell}}^{(1)}(p) | G_-, \eta \rangle. \quad (606)$$

This is the exact finite matrix that must be computed in the current working branch. :contentReference[oaicite:2]index=2

CCXCIII. 2. EXACT EIGENVALUES

Since the block is Hermitian, its eigenvalues are real and equal to

$$\lambda_{\pm}(p) = \frac{d_+(p) + d_-(p)}{2} \pm \sqrt{\left(\frac{d_+(p) - d_-(p)}{2}\right)^2 + |m(p)|^2}. \quad (607)$$

Therefore the finite non-enriched benchmark is

$$\Delta_{\text{non-enr}, \rho}^{(1)} = \inf_{|p| \leq \rho} \min\{|\lambda_+(p)|, |\lambda_-(p)|\}. \quad (608)$$

This is exactly the object that must be inserted into the master benchmark.

CCXCIV. 3. EXACT ZERO-AVOIDANCE CONDITION

A zero eigenvalue occurs iff

$$\det H_{\text{non-enr}}^{(1)}(p) = 0.$$

Since

$$\det H_{\text{non-enr}}^{(1)}(p) = d_+(p)d_-(p) - |m(p)|^2,$$

we get the exact condition

$$0 \in \text{spec}(H_{\text{non-enr}}^{(1)}(p)) \iff d_+(p)d_-(p) = |m(p)|^2. \quad (609)$$

Hence the same-shell non-enriched sector is spectrally separated from zero on $|p| \leq \rho$ if

$$d_+(p)d_-(p) - |m(p)|^2 \neq 0 \quad \forall |p| \leq \rho. \quad (610)$$

CCXCV. 4. USEFUL LOWER BOUNDS

A. 4.1 Symmetric-sign regime

Suppose

$$d_+(p), d_-(p)$$

have the same sign on $|p| \leq \rho$. Then

$$\min\{|\lambda_+(p)|, |\lambda_-(p)|\} = \frac{|d_+(p) + d_-(p)|}{2} - \sqrt{\left(\frac{d_+(p) - d_-(p)}{2}\right)^2 + |m(p)|^2}.$$

So an explicit sufficient lower bound is

$$\Delta_{\text{non-enr}, \rho}^{(1)} \geq \inf_{|p| \leq \rho} \left[\frac{|d_+(p) + d_-(p)|}{2} - \sqrt{\left(\frac{d_+(p) - d_-(p)}{2}\right)^2 + |m(p)|^2} \right]. \quad (611)$$

B. 4.2 Nearly symmetric diagonal case

If

$$d_+(p) = d(p) + \delta d(p), \quad d_-(p) = d(p) - \delta d(p),$$

then

$$\lambda_{\pm}(p) = d(p) \pm \sqrt{\delta d(p)^2 + |m(p)|^2},$$

and therefore

$$\min\{|\lambda_+(p)|, |\lambda_-(p)|\} = \left| |d(p)| - \sqrt{\delta d(p)^2 + |m(p)|^2} \right|. \quad (612)$$

Hence a very useful sufficient condition is

$$\boxed{|d(p)| > \sqrt{\delta d(p)^2 + |m(p)|^2} \quad \forall |p| \leq \rho,} \quad (613)$$

which implies

$$\Delta_{\text{non-enr}, \rho}^{(1)} \geq \inf_{|p| \leq \rho} \left(|d(p)| - \sqrt{\delta d(p)^2 + |m(p)|^2} \right) > 0. \quad (614)$$

C. 4.3 Equal-diagonal subcase

If, in addition,

$$d_+(p) = d_-(p) = d(p),$$

then

$$\lambda_{\pm}(p) = d(p) \pm |m(p)|,$$

so the exact gap becomes

$$\Delta_{\text{non-enr}, \rho}^{(1)} = \inf_{|p| \leq \rho} \left| |d(p)| - |m(p)| \right|. \quad (615)$$

Thus zero is avoided iff

$$\boxed{|d(p)| \neq |m(p)| \quad \forall |p| \leq \rho.} \quad (616)$$

CCXCVI. 5. TRUNCATION-TO-EXACT USABLE BENCHMARK

The exact same-shell non-enriched benchmark is obtained from the finite truncation by

$$\delta_{\text{non-enr},\rho} \geq \Delta_{\text{non-enr},\rho}^{(1)} - \varepsilon_{\text{non-enr},\rho}^{(1)}. \quad (617)$$

Hence the usable benchmark entry is

$$\Delta_{\text{non-enr},\rho}^{(1)} - \varepsilon_{\text{non-enr},\rho}^{(1)}. \quad (618)$$

If

$$\varepsilon_{\text{non-enr},\rho}^{(1)} < \Delta_{\text{non-enr},\rho}^{(1)}, \quad (619)$$

then

$$\delta_{\text{non-enr},\rho} > 0. \quad (620)$$

This is the exact finite-to-exact upgrade theorem specialized to $m = 1$.

CCXCVII. 6. INSERTION INTO THE EXPLICIT BRANCH BENCHMARK

For the explicit canonical branch

$$T_\star = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0,$$

the full benchmark side becomes

$$\Delta_{\text{bench},\rho}^{\text{fin}} = \min \left\{ 4K - |r_\star|, \sqrt{M_\sigma^2 + c_\sigma^2(q_\star - \rho)^2}, q_\star^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_\star^2} \right|, \Delta_{\text{non-enr},\rho}^{(1)} - \varepsilon_{\text{non-enr},\rho}^{(1)} \right\}. \quad (621)$$

The branch-inserted Step-IV gate is therefore the single finite inequality

$$\min \left\{ 4K - |r_\star|, \sqrt{M_\sigma^2 + c_\sigma^2(q_\star - \rho)^2}, q_\star^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_\star^2} \right|, \Delta_{\text{non-enr},\rho}^{(1)} - \varepsilon_{\text{non-enr},\rho}^{(1)} \right\} > \eta_{R,\rho}. \quad (622)$$

CCXCVIII. 7. WHAT MUST ACTUALLY BE COMPUTED

Thus the remaining same-shell coefficient-finding problem is now reduced to the following finite data:

$$d_+(p), \quad d_-(p), \quad m(p), \quad \varepsilon_{\text{non-enr},\rho}^{(1)}. \quad (623)$$

Once these are known, the same-shell non-enriched benchmark is explicit and Step IV is fully reduced to a concrete finite inequality.

CCXCIX. FINAL SUMMARY

The same-shell non-enriched sector in the minimal truncation $m = 1$ is already reduced to the explicit Hermitian block

$$H_{\text{non-enr}}^{(1)}(p) = \begin{pmatrix} d_+(p) & m(p) \\ m(p) & d_-(p) \end{pmatrix}.$$

Its exact eigenvalues are

$$\lambda_{\pm}(p) = \frac{d_+(p) + d_-(p)}{2} \pm \sqrt{\left(\frac{d_+(p) - d_-(p)}{2}\right)^2 + |m(p)|^2}.$$

Therefore the finite benchmark is

$$\Delta_{\text{non-enr},\rho}^{(1)} = \inf_{|p| \leq \rho} \min\{|\lambda_+(p)|, |\lambda_-(p)|\}.$$

A zero eigenvalue occurs iff

$$d_+(p)d_-(p) = |m(p)|^2.$$

In the equal-diagonal subcase $d_+ = d_- = d$, the exact gap simplifies to

$$\Delta_{\text{non-enr},\rho}^{(1)} = \inf_{|p| \leq \rho} ||d(p)| - |m(p)||.$$

The usable benchmark entry is

$$\Delta_{\text{non-enr},\rho}^{(1)} - \varepsilon_{\text{non-enr},\rho}^{(1)},$$

and if

$$\varepsilon_{\text{non-enr},\rho}^{(1)} < \Delta_{\text{non-enr},\rho}^{(1)},$$

then the exact same-shell non-enriched sector is spectrally separated from zero.

Hence, for the explicit branch

$$T_{\star} = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0,$$

the full Step-IV gate is now reduced to the single finite inequality

$$\min \left\{ 4K - |r_*|, \sqrt{M_{\sigma}^2 + c_{\sigma}^2(q_* - \rho)^2}, q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_{\sigma}^2 + c_{\sigma}^2 q_*^2} \right|, \Delta_{\text{non-enr},\rho}^{(1)} - \varepsilon_{\text{non-enr},\rho}^{(1)} \right\} > \eta_{R,\rho}.$$

So the genuinely remaining same-shell coefficient-finding problem is the finite computation of

$$d_+(p), \quad d_-(p), \quad m(p), \quad \varepsilon_{\text{non-enr},\rho}^{(1)}.$$

CCC. COEFFICIENT-FINDING CONTINUATION FOR THE MINIMAL SAME-SHELL NON-ENRICHED SLOT

CCCI. SHORT EXPLANATION

The same-shell non-enriched contribution is now reduced to an explicit 2×2 Hermitian matrix problem. Therefore the remaining Step-IV benchmark is no longer abstract. It is a finite coefficient audit for

$$d_+(p), \quad d_-(p), \quad m(p),$$

or, equivalently, for their small- p coefficients

$$d_+^{(0)}, \quad d_-^{(0)}, \quad m_0, \quad a_{\pm,i}, \quad b_i.$$

CCCII. 1. EXACT $m = 1$ SAME-SHELL BLOCK

Choose the minimal same-shell non-enriched basis

$$\mathcal{B}_{\text{non-enr}}^{(1)} = \{|G_+, \eta\rangle, |G_-, \eta\rangle\},$$

where η is one internal state orthogonal to the enriched shell doublet.

Then the projected block is

$$H_{\text{non-enr}}^{(1)}(p) = \begin{pmatrix} d_+(p) & m(p) \\ m(p) & d_-(p) \end{pmatrix}, \quad (624)$$

with

$$d_{\pm}(p) = \langle G_{\pm}, \eta | H_{\text{shell}}^{(1)}(p) | G_{\pm}, \eta \rangle, \quad m(p) = \langle G_+, \eta | H_{\text{shell}}^{(1)}(p) | G_-, \eta \rangle. \quad (625)$$

This is the exact finite matrix that must be inserted into the explicit branch benchmark.

CCCIII. 2. EXACT EIGENVALUES AND BENCHMARK GAP

Since (624) is Hermitian, its eigenvalues are real:

$$\lambda_{\pm}(p) = \frac{d_+(p) + d_-(p)}{2} \pm \sqrt{\left(\frac{d_+(p) - d_-(p)}{2}\right)^2 + |m(p)|^2}. \quad (626)$$

Hence the exact same-shell non-enriched benchmark is

$$\Delta_{\text{non-enr}, \rho}^{(1)} = \inf_{|p| \leq \rho} \min\{|\lambda_+(p)|, |\lambda_-(p)|\}. \quad (627)$$

A zero eigenvalue occurs iff

$$d_+(p)d_-(p) - |m(p)|^2 = 0. \quad (628)$$

So the whole same-shell non-enriched remote-gap problem is already reduced to the finite scalar triplet

$$(d_+(p), d_-(p), m(p)).$$

CCCIV. 3. COERCIVE SPLIT AND PRACTICAL SUFFICIENT CRITERION

For $m = 1$, introduce the exact coercive split

$$D_{\text{non-enr}}^{(1)}(p) = \begin{pmatrix} d_+(p) & 0 \\ 0 & d_-(p) \end{pmatrix}, \quad R_{\text{non-enr}}^{(1)}(p) = \begin{pmatrix} 0 & m(p) \\ m(p) & 0 \end{pmatrix}. \quad (629)$$

Then

$$\delta_{\text{non-enr}, \rho}^{(1)} = \inf_{|p| \leq \rho} \min\{|d_+(p)|, |d_-(p)|\}, \quad \eta_{\text{non-enr}, \rho}^{(1)} = \sup_{|p| \leq \rho} |m(p)|. \quad (630)$$

Therefore the exact practical sufficient condition is

$$\delta_{\text{non-enr}, \rho}^{(1)} > \eta_{\text{non-enr}, \rho}^{(1)} \implies \Delta_{\text{non-enr}, \rho}^{(1)} > 0. \quad (631)$$

This is the same uniform Gershgorin/coercive criterion specialized to the minimal 2×2 block.

CCCV. 4. SMALL- p COEFFICIENT EXTRACTION

Expand near $p = 0$:

$$d_{\pm}(p) = d_{\pm}^{(0)} + p_i a_{\pm,i} + O(|p|^2), \quad m(p) = m_0 + p_i b_i + O(|p|^2). \quad (632)$$

Here

$$d_{\pm}^{(0)} = \langle G_{\pm}, \eta | H_{\text{shell}}^{(1)}(0) | G_{\pm}, \eta \rangle, \quad m_0 = \langle G_+, \eta | H_{\text{shell}}^{(1)}(0) | G_-, \eta \rangle, \quad (633)$$

and

$$a_{\pm,i} = \left. \partial_{p_i} \langle G_{\pm}, \eta | H_{\text{shell}}^{(1)}(p) | G_{\pm}, \eta \rangle \right|_{p=0}, \quad b_i = \left. \partial_{p_i} \langle G_+, \eta | H_{\text{shell}}^{(1)}(p) | G_-, \eta \rangle \right|_{p=0}. \quad (634)$$

Thus the same-shell coefficient-finding problem is now precisely the computation of the finite matrix elements

$$d_+^{(0)}, \quad d_-^{(0)}, \quad m_0, \quad a_{\pm,i}, \quad b_i.$$

CCCVI. 5. ZERO-MOMENTUM SUFFICIENT CONDITION

A sufficient condition for zero-momentum closure is

$$\min\{|d_+^{(0)}|, |d_-^{(0)}|\} > |m_0|. \quad (635)$$

If, in addition, the linear correction satisfies

$$\sup_{|p| \leq \rho} \max\{|p_i a_{+,i}|, |p_i a_{-,i}|, |p_i b_i|\} \leq \varepsilon_{\rho} < \min\{|d_+^{(0)}|, |d_-^{(0)}|\} - |m_0|, \quad (636)$$

then

$$\Delta_{\text{non-enr}, \rho}^{(1)} > 0. \quad (637)$$

So the practical benchmark is fully reduced to one zero-momentum inequality and one small- p perturbation budget.

CCCVII. 6. EQUAL-DIAGONAL SUBCASE

If the explicit branch realization happens to satisfy

$$d_+(p) = d_-(p) = d(p),$$

then

$$\lambda_{\pm}(p) = d(p) \pm |m(p)|.$$

Hence the exact benchmark simplifies to

$$\Delta_{\text{non-enr}, \rho}^{(1)} = \inf_{|p| \leq \rho} ||d(p)| - |m(p)||. \quad (638)$$

In particular, a sufficient zero-momentum condition becomes

$$|d^{(0)}| > |m_0|. \quad (639)$$

This is the sharpest practical shortcut whenever the same-shell complementary state inherits a valley-symmetric diagonal entry from the shell Hamiltonian.

CCCVIII. 7. UPGRADE TO THE USABLE BENCHMARK ENTRY

The exact usable entry in the final Step-IV benchmark is not just $\Delta_{\text{non-enr},\rho}^{(1)}$, but

$$\Delta_{\text{non-enr},\rho}^{(1)} - \varepsilon_{\text{non-enr},\rho}^{(1)} \quad (640)$$

If

$$\varepsilon_{\text{non-enr},\rho}^{(1)} < \Delta_{\text{non-enr},\rho}^{(1)}, \quad (641)$$

then the exact same-shell non-enriched sector remains spectrally separated from zero.

CCCIX. 8. INSERTION INTO THE EXPLICIT BRANCH STEP-IV GATE

For the explicit branch

$$T_\star = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0,$$

the benchmark side becomes

$$\Delta_{\text{bench},\rho}^{\text{fin}} = \min \left\{ 4K - |r_\star|, \sqrt{M_\sigma^2 + c_\sigma^2(q_\star - \rho)^2}, q_\star^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_\star^2} \right|, \Delta_{\text{non-enr},\rho}^{(1)} - \varepsilon_{\text{non-enr},\rho}^{(1)} \right\}. \quad (642)$$

Thus the final remote-gap gate for this branch is the single explicit inequality

$$\min \left\{ 4K - |r_\star|, \sqrt{M_\sigma^2 + c_\sigma^2(q_\star - \rho)^2}, q_\star^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_\star^2} \right|, \Delta_{\text{non-enr},\rho}^{(1)} - \varepsilon_{\text{non-enr},\rho}^{(1)} \right\} > \eta_{R,\rho}. \quad (643)$$

CCCX. 9. HONEST ENDPOINT

Therefore the genuinely remaining same-shell coefficient-finding problem is no longer conceptual. It is the finite computation of

$$d_+^{(0)}, \quad d_-^{(0)}, \quad m_0, \quad a_{\pm,i}, \quad b_i, \quad \varepsilon_{\text{non-enr},\rho}^{(1)}.$$

Once these are inserted, the same-shell non-enriched benchmark is fully explicit.

CCCXI. FINAL SUMMARY

The same-shell non-enriched slot in the minimal truncation $m = 1$ is already reduced to the explicit Hermitian block

$$H_{\text{non-enr}}^{(1)}(p) = \begin{pmatrix} d_+(p) & m(p) \\ m(p) & d_-(p) \end{pmatrix}.$$

Its exact eigenvalues are

$$\lambda_\pm(p) = \frac{d_+(p) + d_-(p)}{2} \pm \sqrt{\left(\frac{d_+(p) - d_-(p)}{2} \right)^2 + |m(p)|^2}.$$

Hence the benchmark is

$$\Delta_{\text{non-enr},\rho}^{(1)} = \inf_{|p| \leq \rho} \min\{|\lambda_+(p)|, |\lambda_-(p)|\}.$$

A practical sufficient route is the coercive split

$$\delta_{\text{non-enr},\rho}^{(1)} = \inf_{|p| \leq \rho} \min\{|d_+(p)|, |d_-(p)|\}, \quad \eta_{\text{non-enr},\rho}^{(1)} = \sup_{|p| \leq \rho} |m(p)|,$$

together with

$$\delta_{\text{non-enr},\rho}^{(1)} > \eta_{\text{non-enr},\rho}^{(1)} \implies \Delta_{\text{non-enr},\rho}^{(1)} > 0.$$

Near $p = 0$,

$$d_{\pm}(p) = d_{\pm}^{(0)} + p_i a_{\pm,i} + O(|p|^2), \quad m(p) = m_0 + p_i b_i + O(|p|^2),$$

so the same-shell coefficient-finding problem reduces to the finite data

$$d_+^{(0)}, \quad d_-^{(0)}, \quad m_0, \quad a_{\pm,i}, \quad b_i.$$

A sufficient zero-momentum condition is

$$\min\{|d_+^{(0)}|, |d_-^{(0)}|\} > |m_0|.$$

If, moreover, the small- p perturbation budget is controlled, then

$$\Delta_{\text{non-enr},\rho}^{(1)} > 0.$$

Therefore, for the explicit branch

$$T_{\star} = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0,$$

the final Step-IV gate is reduced to

$$\min \left\{ 4K - |r_*|, \sqrt{M_{\sigma}^2 + c_{\sigma}^2(q_* - \rho)^2}, q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_{\sigma}^2 + c_{\sigma}^2 q_*^2} \right|, \Delta_{\text{non-enr},\rho}^{(1)} - \varepsilon_{\text{non-enr},\rho}^{(1)} \right\} > \eta_{R,\rho}.$$

So the genuinely remaining same-shell benchmark work is the explicit computation of

$$d_+^{(0)}, \quad d_-^{(0)}, \quad m_0, \quad a_{\pm,i}, \quad b_i, \quad \varepsilon_{\text{non-enr},\rho}^{(1)}.$$

CCCXII. ACTUAL OPERATOR DECOMPOSITION OF $d_{\pm}^{(0)}$ AND m_0 IN THE MINIMAL SAME-SHELL NON-ENRICHED SLOT

CCCXIII. SHORT EXPLANATION

The same-shell non-enriched $m = 1$ benchmark is already reduced to the 2×2 Hermitian block

$$H_{\text{non-enr}}^{(1)}(p) = \begin{pmatrix} d_+(p) & m(p) \\ m(p) & d_-(p) \end{pmatrix}.$$

What remains is to write the coefficients

$$d_+^{(0)}, \quad d_-^{(0)}, \quad m_0$$

directly in terms of the actual shell-projected Bloch operator.

The correct decomposition is:

$$\text{reference/direct same-shell block} \quad + \quad \text{second-order heavy Schur correction.}$$

CCCXIV. 1. MINIMAL SAME-SHELL NON-ENRICHED BASIS

Let

$$H_{\text{shell}}(p) = \Pi_{S_*} H_{\text{eff}}(k_* + p) \Pi_{S_*}$$

be the shell-projected operator near the folded Bragg point. Choose the minimal same-shell complementary basis

$$\mathcal{B}_{\text{non-enr}}^{(1)} = \{|+\rangle, |-\rangle\}, \quad |+\rangle := |G_+, \eta\rangle, \quad |-\rangle := |G_-, \eta\rangle,$$

where η is one internal state orthogonal to the enriched shell doublet.

Then

$$H_{\text{non-enr}}^{(1)}(p) = \begin{pmatrix} d_+(p) & m(p) \\ m(p) & d_-(p) \end{pmatrix}, \quad (644)$$

with

$$d_{\pm}(p) = \langle \pm | H_{\text{shell}}(p) | \pm \rangle, \quad m(p) = \langle + | H_{\text{shell}}(p) | - \rangle. \quad (645)$$

CCCXV. 2. REFERENCE + PERTURBATION SPLIT

Write the shell operator as

$$H_{\text{shell}}(p) = H_{\text{shell}}^{(0)}(p) + V_{\text{shell}}(p), \quad (646)$$

where $H_{\text{shell}}^{(0)}(p)$ is the reference shell operator and $V_{\text{shell}}(p)$ collects the explicit shell perturbations induced by the refined/Class-II branch.

At $p = 0$, define the direct same-shell matrix elements

$$D_+^{(0)} := \langle + | H_{\text{shell}}^{(0)}(0) | + \rangle, \quad D_-^{(0)} := \langle - | H_{\text{shell}}^{(0)}(0) | - \rangle, \quad (647)$$

$$M^{(0)} := \langle + | H_{\text{shell}}^{(0)}(0) | - \rangle. \quad (648)$$

Likewise for the explicit shell perturbation,

$$V_{++} := \langle + | V_{\text{shell}}(0) | + \rangle, \quad V_{--} := \langle - | V_{\text{shell}}(0) | - \rangle, \quad (649)$$

$$V_{+-} := \langle + | V_{\text{shell}}(0) | - \rangle. \quad (650)$$

If one stopped at the direct shell level, one would simply have

$$d_+^{(0)} = D_+^{(0)} + V_{++}, \quad d_-^{(0)} = D_-^{(0)} + V_{--}, \quad m_0 = M^{(0)} + V_{+-}.$$

CCCXVI. 3. HEAVY-STATE SECOND-ORDER CORRECTION

Now include the heavy sector through the exact second-order Schur formula. Choose a heavy basis χ_α diagonalizing the reference heavy block:

$$\mathbb{O}^{(0)} \chi_\alpha = E_\alpha \chi_\alpha, \quad E_\alpha \geq \Delta_H > 0.$$

Then the second-order correction to the low shell block is

$$(\mathcal{Y}_{\text{eff}}^{(2)})_{ab} = - \sum_{\alpha} \frac{K_{a\alpha} \overline{K_{b\alpha}}}{E_{\alpha}}, \quad (651)$$

where

$$K_{a\alpha} = \langle a | V_{\text{shell}}(0) | \chi_\alpha \rangle, \quad a \in \{+, -\}. \quad (652)$$

Thus, in the present $m = 1$ same-shell complementary slot, the heavy-corrected coefficients are

$$d_+^{(0)} = D_+^{(0)} + V_{++} - \sum_{\alpha} \frac{|K_{+\alpha}|^2}{E_{\alpha}}, \quad (653)$$

$$d_-^{(0)} = D_-^{(0)} + V_{--} - \sum_{\alpha} \frac{|K_{-\alpha}|^2}{E_{\alpha}}, \quad (654)$$

$$m_0 = M^{(0)} + V_{+-} - \sum_{\alpha} \frac{K_{+\alpha} \overline{K_{-\alpha}}}{E_{\alpha}}. \quad (655)$$

These are the actual finite operator formulas that must be evaluated.

CCCXVII. 4. USEFUL SYMMETRY REDUCTIONS

A. 4.1 Valley-symmetric reference block

If the reference shell background is valley-symmetric in the complementary state,

$$D_+^{(0)} = D_-^{(0)} =: D^{(0)},$$

then the diagonal asymmetry comes only from the explicit shell perturbation and/or the heavy Schur correction:

$$d_+^{(0)} - d_-^{(0)} = (V_{++} - V_{--}) - \sum_{\alpha} \frac{|K_{+\alpha}|^2 - |K_{-\alpha}|^2}{E_{\alpha}}. \quad (656)$$

B. 4.2 Equal-diagonal subcase

If moreover

$$V_{++} = V_{--}, \quad |K_{+\alpha}| = |K_{-\alpha}| \quad \forall \alpha,$$

then

$$d_+^{(0)} = d_-^{(0)} =: d^{(0)},$$

and the exact zero-momentum gap reduces to

$$\Delta_{\text{non-enr},0}^{(1)} = ||d^{(0)}| - |m_0||. \quad (657)$$

In that subcase, a sufficient nonzero benchmark condition is simply

$$|d^{(0)}| > |m_0|. \quad (658)$$

CCCXVIII. 5. SMALL- p COEFFICIENTS

The linear coefficients are obtained by differentiating the same matrix elements:

$$\boxed{a_{+,i} = \partial_{p_i} d_+(p)|_{p=0}, \quad a_{-,i} = \partial_{p_i} d_-(p)|_{p=0}, \quad b_i = \partial_{p_i} m(p)|_{p=0}.} \quad (659)$$

Using the same decomposition,

$$\boxed{a_{+,i} = \partial_{p_i} \langle + | H_{\text{shell}}^{(0)}(p) + V_{\text{shell}}(p) | + \rangle |_0 - \partial_{p_i} \sum_{\alpha} \frac{|K_{+\alpha}(p)|^2}{E_{\alpha}(p)} |_0,} \quad (660)$$

$$\boxed{a_{-,i} = \partial_{p_i} \langle - | H_{\text{shell}}^{(0)}(p) + V_{\text{shell}}(p) | - \rangle |_0 - \partial_{p_i} \sum_{\alpha} \frac{|K_{-\alpha}(p)|^2}{E_{\alpha}(p)} |_0,} \quad (661)$$

$$\boxed{b_i = \partial_{p_i} \langle + | H_{\text{shell}}^{(0)}(p) + V_{\text{shell}}(p) | - \rangle |_0 - \partial_{p_i} \sum_{\alpha} \frac{K_{+\alpha}(p) \overline{K_{-\alpha}(p)}}{E_{\alpha}(p)} |_0.} \quad (662)$$

Thus all small- p data are again reduced to finite matrix elements and heavy-state sums.

CCCXIX. 6. ZERO-MOMENTUM PRACTICAL SUFFICIENT CRITERION

The zero-momentum sufficient criterion becomes

$$\boxed{\min\{|d_+^{(0)}|, |d_-^{(0)}|\} > |m_0|,} \quad (663)$$

with $d_{\pm}^{(0)}$ and m_0 given explicitly by (653)–(655).

If moreover the small- p correction budget satisfies

$$\boxed{\sup_{|p| \leq \rho} \max\{|p_i a_{+,i}|, |p_i a_{-,i}|, |p_i b_i|\} < \min\{|d_+^{(0)}|, |d_-^{(0)}|\} - |m_0|,} \quad (664)$$

then

$$\boxed{\Delta_{\text{non-enr},\rho}^{(1)} > 0.} \quad (665)$$

CCCXX. 7. INSERTION INTO THE EXPLICIT BRANCH REMOTE BENCHMARK

For the explicit canonical branch

$$T_{\star} = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0,$$

the full Step-IV benchmark remains

$$\boxed{\Delta_{\text{bench},\rho}^{\text{fin}} = \min \left\{ 4K - |r_{\star}|, \sqrt{M_{\sigma}^2 + c_{\sigma}^2(q_{\star} - \rho)^2}, q_{\star}^2 \left| (\lambda + \mu) + \frac{g^2}{M_{\sigma}^2 + c_{\sigma}^2 q_{\star}^2} \right|, \Delta_{\text{non-enr},\rho}^{(1)} - \varepsilon_{\text{non-enr},\rho}^{(1)} \right\}.} \quad (666)$$

Hence the genuinely remaining same-shell coefficient-finding problem is the finite evaluation of

$$\boxed{D_{\pm}^{(0)}, V_{++}, V_{--}, V_{+-}, K_{+\alpha}, K_{-\alpha}, E_{\alpha}, \quad \text{and the corresponding derivatives at } p = 0.}$$

CCCXXI. FINAL SUMMARY

For the minimal same-shell non-enriched complementary slot $m = 1$, the exact block is

$$H_{\text{non-enr}}^{(1)}(p) = \begin{pmatrix} d_+(p) & m(p) \\ m(p) & d_-(p) \end{pmatrix}.$$

The actual zero-momentum coefficients are not abstract anymore. They are given by the direct same-shell matrix elements plus the heavy-state second-order Schur correction:

$$d_+^{(0)} = D_+^{(0)} + V_{++} - \sum_{\alpha} \frac{|K_{+\alpha}|^2}{E_{\alpha}},$$

$$d_-^{(0)} = D_-^{(0)} + V_{--} - \sum_{\alpha} \frac{|K_{-\alpha}|^2}{E_{\alpha}},$$

$$m_0 = M^{(0)} + V_{+-} - \sum_{\alpha} \frac{K_{+\alpha} \overline{K_{-\alpha}}}{E_{\alpha}}.$$

Therefore the same-shell coefficient-finding problem is reduced to the finite matrix-element data

$$D_{\pm}^{(0)}, V_{++}, V_{--}, V_{+-}, K_{+\alpha}, K_{-\alpha}, E_{\alpha}.$$

The practical zero-momentum sufficient condition is

$$\min\{|d_+^{(0)}|, |d_-^{(0)}|\} > |m_0|.$$

If, in addition, the linear small- p budget is controlled via

$$a_{\pm,i} = \partial_{p_i} d_{\pm}(0), \quad b_i = \partial_{p_i} m(0),$$

then

$$\Delta_{\text{non-enr},\rho}^{(1)} > 0.$$

Thus, for the explicit branch

$$T_{\star} = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0,$$

the final Step-IV benchmark is reduced to a completely finite Hermitian matrix computation. The only remaining work is the literal evaluation of these matrix elements in the chosen shell-projected Bloch basis.

CCCXXII. CANONICAL MINIMAL CHOICE $\eta = z_0$ AND EXPLICIT $m = 1$ SAME-SHELL NON-ENRICHED COEFFICIENTS

CCCXXIII. SHORT EXPLANATION

The minimal same-shell non-enriched basis was previously left abstract as

$$\mathcal{B}_{\text{non-enr}}^{(1)} = \{|G_+, \eta\rangle, |G_-, \eta\rangle\},$$

with η any internal state orthogonal to the enriched shell doublet.

In the canonical locked basis

$$(z_0, u_1, u_2),$$

the enriched shell carrier uses the tangent doublet (u_1, u_2) . Therefore the natural minimal complementary same-shell state is the locked line itself:

$$\eta = z_0.$$

This makes the same-shell non-enriched $m = 1$ sector fully explicit.

CCCXXIV. 1. CANONICAL MINIMAL SAME-SHELL NON-ENRICHED BASIS

Fix the canonical locked basis

$$\boxed{z_0 = (1, 0, 0)^T, \quad u_1 = (0, 1, 0)^T, \quad u_2 = (0, 0, 1)^T.} \quad (667)$$

The enriched shell carrier occupies

$$\mathcal{H}_D = \text{span}\{|G_+, u_1\rangle, |G_+, u_2\rangle, |G_-, u_1\rangle, |G_-, u_2\rangle\}.$$

Hence the minimal same-shell complementary state is chosen as

$$\boxed{\eta = z_0.} \quad (668)$$

Therefore the minimal same-shell non-enriched basis becomes

$$\boxed{|+\rangle := |G_+, z_0\rangle, \quad |-\rangle := |G_-, z_0\rangle.} \quad (669)$$

In this basis the projected same-shell non-enriched block is

$$\boxed{H_{\text{non-enr}}^{(1)}(p) = \begin{pmatrix} d_+(p) & m(p) \\ \overline{m(p)} & d_-(p) \end{pmatrix},} \quad (670)$$

with

$$\boxed{d_{\pm}(p) = \langle \pm | H_{\text{shell}}(p) | \pm \rangle, \quad m(p) = \langle + | H_{\text{shell}}(p) | - \rangle.} \quad (671)$$

CCCXXV. 2. EXACT DIRECT-PLUS-HEAVY DECOMPOSITION

The effective shell operator is

$$\boxed{H_{\text{shell}}(p) = \Pi_{S_*} H_{\text{eff}}(k_* + p) \Pi_{S_*}, \quad H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - H_{\Psi X}(k) H_{XX}(k)^{-1} H_{X\Psi}(k).} \quad (672)$$

Hence the zero-momentum coefficients split exactly as

$$\boxed{d_+^{(0)} = L_{++}^{(z_0)} - \Sigma_{++}^{(z_0)},} \quad (673)$$

$$\boxed{d_-^{(0)} = L_{--}^{(z_0)} - \Sigma_{--}^{(z_0)},} \quad (674)$$

$$\boxed{m_0 = L_{+-}^{(z_0)} - \Sigma_{+-}^{(z_0)},} \quad (675)$$

where

$$\boxed{L_{ab}^{(z_0)} := \langle a | H_{\Psi\Psi}(k_*) | b \rangle, \quad a, b \in \{+, -\},} \quad (676)$$

and

$$\boxed{\Sigma_{ab}^{(z_0)} := \langle a | H_{\Psi X}(k_*) H_{XX}(k_*)^{-1} H_{X\Psi}(k_*) | b \rangle.} \quad (677)$$

CCCXXVI. 3. HEAVY-EIGENBASIS FORM OF THE SCHUR TERM

Choose a heavy basis χ_α diagonalizing the reference heavy block:

$$\mathbb{O}^{(0)}\chi_\alpha = E_\alpha\chi_\alpha, \quad E_\alpha \geq \Delta_H > 0.$$

Then the exact second-order formula is

$$(\mathcal{Y}_{\text{eff}}^{(2)})_{ab} = - \sum_{\alpha} \frac{K_{a\alpha} \overline{K_{b\alpha}}}{E_\alpha}.$$

Therefore the same-shell complementary Schur entries are

$$\Sigma_{++}^{(z_0)} = \sum_{\alpha} \frac{|K_{+\alpha}^{(z_0)}|^2}{E_\alpha}, \quad \Sigma_{--}^{(z_0)} = \sum_{\alpha} \frac{|K_{-\alpha}^{(z_0)}|^2}{E_\alpha}, \quad (678)$$

$$\Sigma_{+-}^{(z_0)} = \sum_{\alpha} \frac{K_{+\alpha}^{(z_0)} \overline{K_{-\alpha}^{(z_0)}}}{E_\alpha}, \quad (679)$$

with

$$K_{\pm\alpha}^{(z_0)} = \langle G_{\pm}, z_0 | V_{\text{shell}}(0) | \chi_{\alpha} \rangle. \quad (680)$$

Thus

$$d_+^{(0)} = L_{++}^{(z_0)} - \sum_{\alpha} \frac{|K_{+\alpha}^{(z_0)}|^2}{E_\alpha}, \quad (681)$$

$$d_-^{(0)} = L_{--}^{(z_0)} - \sum_{\alpha} \frac{|K_{-\alpha}^{(z_0)}|^2}{E_\alpha}, \quad (682)$$

$$m_0 = L_{+-}^{(z_0)} - \sum_{\alpha} \frac{K_{+\alpha}^{(z_0)} \overline{K_{-\alpha}^{(z_0)}}}{E_\alpha}. \quad (683)$$

CCCXXVII. 4. DIRECT SHELL MATRIX ELEMENTS IN BLOCH FORM

Because the projected shell operator is built from the Bloch kernel at the folded Bragg point, the direct terms are simply

$$L_{++}^{(z_0)} = \langle G_+, z_0 | H_{\Psi\Psi}(k_*) | G_+, z_0 \rangle, \quad (684)$$

$$L_{--}^{(z_0)} = \langle G_-, z_0 | H_{\Psi\Psi}(k_*) | G_-, z_0 \rangle, \quad (685)$$

$$L_{+-}^{(z_0)} = \langle G_+, z_0 | H_{\Psi\Psi}(k_*) | G_-, z_0 \rangle. \quad (686)$$

If one writes the direct shell block on the z_0 -line as

$$E_{z_0,\pm}(0) := \langle G_{\pm}, z_0 | H_{\Psi\Psi}(k_*) | G_{\pm}, z_0 \rangle, \quad B_{z_0}(0) := \langle G_+, z_0 | H_{\Psi\Psi}(k_*) | G_-, z_0 \rangle, \quad (687)$$

then the actual coefficients are simply

$$d_+^{(0)} = E_{z_0,+}(0) - \Sigma_{++}^{(z_0)}, \quad d_-^{(0)} = E_{z_0,-}(0) - \Sigma_{--}^{(z_0)}, \quad m_0 = B_{z_0}(0) - \Sigma_{+-}^{(z_0)}. \quad (688)$$

CCCXXVIII. 5. VALLEY-SYMMETRIC REFERENCE SUBCASE

If the direct shell background is valley-symmetric on the locked line, i.e.

$$\boxed{E_{z_0,+}(0) = E_{z_0,-}(0) =: E_{z_0}^{(0)},} \quad (689)$$

then the diagonal asymmetry is entirely Schur-generated:

$$\boxed{d_+^{(0)} - d_-^{(0)} = - \sum_{\alpha} \frac{|K_{+\alpha}^{(z_0)}|^2 - |K_{-\alpha}^{(z_0)}|^2}{E_{\alpha}}.} \quad (690)$$

If, further, the heavy couplings are valley-symmetric in norm,

$$|K_{+\alpha}^{(z_0)}| = |K_{-\alpha}^{(z_0)}| \quad \forall \alpha,$$

then

$$d_+^{(0)} = d_-^{(0)} =: d^{(0)},$$

and the zero-momentum eigenvalues reduce to

$$\lambda_{\pm}(0) = d^{(0)} \pm |m_0|.$$

Hence the exact zero-momentum non-enriched gap becomes

$$\boxed{\Delta_{\text{non-enr},0}^{(1)} = ||d^{(0)}| - |m_0||.} \quad (691)$$

Therefore a sufficient zero-momentum condition is

$$\boxed{|d^{(0)}| > |m_0|.} \quad (692)$$

CCCXXIX. 6. SMALL- p COEFFICIENTS

The linear coefficients are obtained by differentiating the same matrix elements:

$$\boxed{a_{+,i} = \partial_{p_i} \langle G_+, z_0 | H_{\text{shell}}(p) | G_+, z_0 \rangle |_{p=0},} \quad (693)$$

$$\boxed{a_{-,i} = \partial_{p_i} \langle G_-, z_0 | H_{\text{shell}}(p) | G_-, z_0 \rangle |_{p=0},} \quad (694)$$

$$\boxed{b_i = \partial_{p_i} \langle G_+, z_0 | H_{\text{shell}}(p) | G_-, z_0 \rangle |_{p=0}.} \quad (695)$$

If these derivatives satisfy the small- p budget

$$\sup_{|p| \leq \rho} \max\{|p_i a_{+,i}|, |p_i a_{-,i}|, |p_i b_i|\} < \min\{|d_+^{(0)}|, |d_-^{(0)}|\} - |m_0|,$$

then

$$\Delta_{\text{non-enr},\rho}^{(1)} > 0.$$

CCCXXX. 7. INSERTION INTO THE EXPLICIT BRANCH STEP-IV GATE

For the explicit branch

$$T_\star = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0,$$

the benchmark-side last slot is exactly

$$\boxed{\Delta_{\text{non-enr},\rho}^{(1)} - \varepsilon_{\text{non-enr},\rho}^{(1)}} \quad (696)$$

with $\Delta_{\text{non-enr},\rho}^{(1)}$ determined by (681)–(683).

Hence the genuinely remaining same-shell coefficient-finding data are

$$\boxed{E_{z_0,+}(0), \quad E_{z_0,-}(0), \quad B_{z_0}(0), \quad K_{+\alpha}^{(z_0)}, \quad K_{-\alpha}^{(z_0)}, \quad E_\alpha,} \quad (697)$$

together with the small- p derivatives $a_{\pm,i}, b_i$.

CCCXXXI. FINAL SUMMARY

The minimal same-shell non-enriched complementary state can be fixed canonically as

$$\boxed{\eta = z_0,}$$

where

$$(z_0, u_1, u_2)$$

is the canonical locked internal basis and (u_1, u_2) span the enriched shell doublet.

Hence the minimal same-shell non-enriched basis is

$$\boxed{|+\rangle = |G_+, z_0\rangle, \quad |-\rangle = |G_-, z_0\rangle.}$$

The exact zero-momentum coefficients of the $m = 1$ non-enriched block are then

$$\boxed{d_+^{(0)} = E_{z_0,+}(0) - \sum_{\alpha} \frac{|K_{+\alpha}^{(z_0)}|^2}{E_\alpha},}$$

$$\boxed{d_-^{(0)} = E_{z_0,-}(0) - \sum_{\alpha} \frac{|K_{-\alpha}^{(z_0)}|^2}{E_\alpha},}$$

$$\boxed{m_0 = B_{z_0}(0) - \sum_{\alpha} \frac{K_{+\alpha}^{(z_0)} \overline{K_{-\alpha}^{(z_0)}}}{E_\alpha}.}$$

Thus the same-shell coefficient-finding problem is reduced to the finite matrix-element data

$$\boxed{E_{z_0,+}(0), \quad E_{z_0,-}(0), \quad B_{z_0}(0), \quad K_{+\alpha}^{(z_0)}, \quad K_{-\alpha}^{(z_0)}, \quad E_\alpha.}$$

If the direct shell background is valley-symmetric and the heavy-coupling norms are also valley-symmetric, then

$$d_+^{(0)} = d_-^{(0)} =: d^{(0)},$$

and the zero-momentum gap simplifies to

$$\Delta_{\text{non-enr},0}^{(1)} = ||d^{(0)}| - |m_0||.$$

A sufficient zero-momentum condition is then

$$|d^{(0)}| > |m_0|.$$

Therefore, for the explicit branch

$$T_\star = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0,$$

the final same-shell benchmark slot has been reduced to a completely finite matrix-element audit on the canonical locked-line complementary state.

CCCXXXII. CANONICAL COMPLEMENT $\eta = z_0$ AND ACTUAL EXTRACTION OF $d_\pm^{(0)}, m_0$

CCCXXXIII. SHORT EXPLANATION

The minimal same-shell non-enriched benchmark is already reduced to the Hermitian block

$$H_{\text{non-enr}}^{(1)}(p) = \begin{pmatrix} d_+(p) & m(p) \\ \frac{d_+(p)}{m(p)} & d_-(p) \end{pmatrix},$$

with

$$d_\pm(p) = \langle G_\pm, \eta | H_{\text{shell}}^{(1)}(p) | G_\pm, \eta \rangle, \quad m(p) = \langle G_+, \eta | H_{\text{shell}}^{(1)}(p) | G_-, \eta \rangle.$$

To make this fully concrete, we now fix the minimal canonical complement

$$\eta = z_0.$$

This turns the final same-shell benchmark slot into a literal 2×2 matrix-element computation on the locked line.

CCCXXXIV. 1. CANONICAL MINIMAL SAME-SHELL COMPLEMENT

Fix the canonical locked basis

$$z_0 = e_1, \quad u_1 = e_2, \quad u_2 = e_3.$$

The enriched shell carrier occupies the tangent doublet

$$\text{span}\{|G_\pm, u_1\rangle, |G_\pm, u_2\rangle\},$$

so the minimal same-shell non-enriched complement is naturally

$$\boxed{\eta = z_0.} \tag{698}$$

Hence the $m = 1$ same-shell non-enriched basis is

$$\boxed{|+\rangle := |G_+, z_0\rangle, \quad |-\rangle := |G_-, z_0\rangle.} \tag{699}$$

CCCXXXV. 2. EXACT DIRECT-PLUS-SCHUR DECOMPOSITION

The shell operator is

$$H_{\text{shell}}(p) = \Pi_{S_*} H_{\text{eff}}(k_* + p) \Pi_{S_*}, \quad H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - H_{\Psi X}(k) H_{XX}(k)^{-1} H_{X\Psi}(k). \quad (700)$$

Therefore the zero-momentum coefficients split exactly as

$$d_+^{(0)} = E_{z_0,+}(0) - \Sigma_{++}^{(z_0)}, \quad d_-^{(0)} = E_{z_0,-}(0) - \Sigma_{--}^{(z_0)}, \quad m_0 = B_{z_0}(0) - \Sigma_{+-}^{(z_0)}, \quad (701)$$

where

$$E_{z_0,+}(0) := \langle G_+, z_0 | H_{\Psi\Psi}(k_*) | G_+, z_0 \rangle, \quad (702)$$

$$E_{z_0,-}(0) := \langle G_-, z_0 | H_{\Psi\Psi}(k_*) | G_-, z_0 \rangle, \quad (703)$$

$$B_{z_0}(0) := \langle G_+, z_0 | H_{\Psi\Psi}(k_*) | G_-, z_0 \rangle, \quad (704)$$

and

$$\Sigma_{ab}^{(z_0)} = \langle a | H_{\Psi X}(k_*) H_{XX}(k_*)^{-1} H_{X\Psi}(k_*) | b \rangle, \quad a, b \in \{+, -\}. \quad (705)$$

In the heavy eigenbasis χ_α ,

$$\Sigma_{ab}^{(z_0)} = \sum_{\alpha} \frac{K_{a\alpha}^{(z_0)} \overline{K_{b\alpha}^{(z_0)}}}{E_{\alpha}}, \quad K_{\pm\alpha}^{(z_0)} := \langle G_{\pm}, z_0 | V_{\text{shell}}(0) | \chi_{\alpha} \rangle. \quad (706)$$

CCCXXXVI. 3. ACTUAL $T_* = T^1$ CALCULATION ON THE z_0 -LINE

Now specialize to the explicit branch

$$T_* = T^1.$$

In the canonical basis,

$$T^1 = \frac{1}{2} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad z_0 = e_1.$$

Hence

$$z_0^\dagger T^1 z_0 = 0. \quad (707)$$

The explicit J -harmonic coefficient is

$$J_{i,r}^a(G_*; q) = i q_i A_{G_*} u_r^\dagger T^a z_0 + i (G_*)_i A_{G_*} z_0^\dagger T^a u_r.$$

Setting

$$u_r = z_0, \quad a = 1,$$

and using (707), both terms vanish:

$$J_{i,0}^1(G_*; q) = 0. \quad (708)$$

Therefore the JJ -part of the Schur correction vanishes identically on the minimal same-shell complementary z_0 -slot:

$$\Sigma_{ab}^{(JJ; z_0)} = 0, \quad a, b \in \{+, -\}. \quad (709)$$

CCCXXXVII. 4. EXACT REDUCED FORMULAS AFTER THE JJ -VANISHING

Hence the actual coefficients reduce to

$$d_+^{(0)} = E_{z_0,+}(0) - \Sigma_{++}^{(JK+KK;z_0)}, \quad (710)$$

$$d_-^{(0)} = E_{z_0,-}(0) - \Sigma_{--}^{(JK+KK;z_0)}, \quad (711)$$

$$m_0 = B_{z_0}(0) - \Sigma_{+-}^{(JK+KK;z_0)}. \quad (712)$$

If one performs the pure- J subaudit, i.e. temporarily switches off the K -sector contribution in the same-shell complementary slot, then the formulas collapse further to

$$d_+^{(0)} = E_{z_0,+}(0), \quad d_-^{(0)} = E_{z_0,-}(0), \quad m_0 = B_{z_0}(0). \quad (713)$$

CCCXXXVIII. 5. DIRECT CORE INTERPRETATION

The shell documents already imply that constant Bragg pair-mixing is controlled by the primitive $Q = G_*$ Fourier coefficient of the effective shell potential,

$$M_0 = \hat{U}_{\text{eff}}(G_*) \mathbf{1}_2,$$

while longitudinal slope has a direct periodic core contribution. Therefore the direct same-shell complementary opposite-valley entry $B_{z_0}(0)$ is the locked-line analogue of this primitive first-shell Bragg coefficient.

Thus the direct zero-momentum same-shell benchmark is controlled by:

$$E_{z_0,+}(0), \quad E_{z_0,-}(0), \quad B_{z_0}(0), \quad (714)$$

with the mediated correction only entering through the JK/KK part.

CCCXXXIX. 6. PRACTICAL ZERO-MOMENTUM SUFFICIENT CONDITION

The same-shell zero-momentum sufficient condition becomes

$$\min\{|d_+^{(0)}|, |d_-^{(0)}|\} > |m_0|. \quad (715)$$

If the JK/KK correction is subleading or absent in the chosen minimal branch realization, this reduces to the explicit direct condition

$$\min\{|E_{z_0,+}(0)|, |E_{z_0,-}(0)|\} > |B_{z_0}(0)|. \quad (716)$$

If moreover the direct shell background is valley-symmetric on the locked line,

$$E_{z_0,+}(0) = E_{z_0,-}(0) =: E_{z_0}^{(0)},$$

then

$$|E_{z_0}^{(0)}| > |B_{z_0}(0)| \implies \Delta_{\text{non-enr},0}^{(1)} > 0. \quad (717)$$

CCCXL. 7. FINAL SAME-SHELL BENCHMARK INSERTION

Therefore, for the explicit branch

$$T_\star = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0,$$

the genuinely remaining same-shell benchmark data are

$$\boxed{E_{z_0,+}(0), \quad E_{z_0,-}(0), \quad B_{z_0}(0), \quad \Sigma_{++}^{(JK+KK;z_0)}, \quad \Sigma_{--}^{(JK+KK;z_0)}, \quad \Sigma_{+-}^{(JK+KK;z_0)},} \quad (718)$$

together with the small- p derivatives. The JJ -part is already zero by the explicit T^1 canonical computation.

CCCXLI. FINAL SUMMARY

With the canonical locked internal basis

$$(z_0, u_1, u_2) = (e_1, e_2, e_3),$$

the minimal same-shell non-enriched complementary state can be fixed as

$$\boxed{\eta = z_0.}$$

Hence the $m = 1$ same-shell non-enriched basis is

$$\boxed{|+\rangle = |G_+, z_0\rangle, \quad |-\rangle = |G_-, z_0\rangle.}$$

The zero-momentum coefficients are exactly

$$\boxed{d_+^{(0)} = E_{z_0,+}(0) - \Sigma_{++}^{(z_0)}, \quad d_-^{(0)} = E_{z_0,-}(0) - \Sigma_{--}^{(z_0)}, \quad m_0 = B_{z_0}(0) - \Sigma_{+-}^{(z_0)}.$$

For the explicit branch

$$\boxed{T_\star = T^1,}$$

one has

$$z_0^\dagger T^1 z_0 = 0,$$

and therefore the J -harmonic coefficient on the z_0 -line vanishes:

$$\boxed{J_{i,0}^1(G_*; q) = 0.}$$

Consequently,

$$\boxed{\Sigma_{ab}^{(JJ;z_0)} = 0.}$$

Hence the actual same-shell benchmark coefficients reduce to

$$\boxed{d_+^{(0)} = E_{z_0,+}(0) - \Sigma_{++}^{(JK+KK;z_0)},}$$

$$\boxed{d_-^{(0)} = E_{z_0,-}(0) - \Sigma_{--}^{(JK+KK;z_0)},}$$

$$\boxed{m_0 = B_{z_0}(0) - \Sigma_{+-}^{(JK+KK;z_0)}.$$

In the pure- J subaudit this collapses further to

$$\boxed{d_+^{(0)} = E_{z_0,+}(0), \quad d_-^{(0)} = E_{z_0,-}(0), \quad m_0 = B_{z_0}(0).}$$

Therefore the zero-momentum same-shell sufficient condition becomes

$$\boxed{\min\{|d_+^{(0)}|, |d_-^{(0)}|\} > |m_0|.}$$

If the JK/KK correction is subleading or absent, this simplifies to

$$\boxed{\min\{|E_{z_0,+}(0)|, |E_{z_0,-}(0)|\} > |B_{z_0}(0)|.}$$

So the genuinely remaining benchmark-side data for this explicit branch are

$$\boxed{E_{z_0,+}(0), \quad E_{z_0,-}(0), \quad B_{z_0}(0), \quad \Sigma_{++}^{(JK+KK;z_0)}, \quad \Sigma_{--}^{(JK+KK;z_0)}, \quad \Sigma_{+-}^{(JK+KK;z_0)}.$$

The JJ -part is already computed and vanishes exactly.

CCCXLII. DIRECT LOCKED-CORE EVALUATION OF $d_{\pm}^{(0)}$ AND m_0 ON THE CANONICAL COMPLEMENT $\eta = z_0$

CCCXLIII. SHORT EXPLANATION

We now stop leaving the same-shell non-enriched direct part abstract.

Fix the canonical complementary state

$$\eta = z_0.$$

Then the minimal same-shell non-enriched basis is

$$|+\rangle = |G_+, z_0\rangle, \quad |-\rangle = |G_-, z_0\rangle.$$

The direct zero-momentum coefficients come from the shell-projected triplet block $H_{\Psi\Psi}(k_*)$, while the mediator only enters through the Schur correction. For the explicit branch $T_* = T^1$, the JJ -part vanishes exactly on the z_0 -line, so the remaining same-shell benchmark data reduce to:

$$\text{direct locked-core block} \quad + \quad JK/KK\text{-residual Schur correction.}$$

CCCXLIV. 1. MINIMAL $m = 1$ SAME-SHELL COMPLEMENTARY BLOCK

With the canonical choice

$$\eta = z_0,$$

define

$$|+\rangle := |G_+, z_0\rangle, \quad |-\rangle := |G_-, z_0\rangle.$$

Then

$$\boxed{H_{\text{non-enr}}^{(1)}(p) = \begin{pmatrix} d_+(p) & m(p) \\ m(p) & d_-(p) \end{pmatrix}}, \tag{719}$$

where

$$\boxed{d_{\pm}(p) = \langle G_{\pm}, z_0 | H_{\text{shell}}(p) | G_{\pm}, z_0 \rangle, \quad m(p) = \langle G_+, z_0 | H_{\text{shell}}(p) | G_-, z_0 \rangle.} \tag{720}$$

At $p = 0$,

$$\boxed{d_+^{(0)} = E_{z_0,+}(0) - \Sigma_{++}^{(z_0)}, \quad d_-^{(0)} = E_{z_0,-}(0) - \Sigma_{--}^{(z_0)}, \quad m_0 = B_{z_0}(0) - \Sigma_{+-}^{(z_0)}.} \tag{721}$$

CCCXLV. 2. DIRECT SHELL BLOCK ON THE z_0 -LINE

The direct triplet Bloch block has the standard periodic-core form

$$H_{\Psi\Psi}(k) = H_0(k) + U^{(\text{per})} + U^{(\text{lock})}.$$

Therefore the direct same-shell complementary matrix elements are

$$E_{z_0,+}(0) = \langle G_+, z_0 | H_0(k_*) | G_+, z_0 \rangle + [U^{(\text{per})}(0) + U^{(\text{lock})}(0)]_{z_0 z_0}, \quad (722)$$

$$E_{z_0,-}(0) = \langle G_-, z_0 | H_0(k_*) | G_-, z_0 \rangle + [U^{(\text{per})}(0) + U^{(\text{lock})}(0)]_{z_0 z_0}, \quad (723)$$

and the opposite-valley direct Bragg entry is

$$B_{z_0}(0) = [U^{(\text{per})}(G_*) + U^{(\text{lock})}(G_*)]_{z_0 z_0}. \quad (724)$$

Because $|G_+| = |G_-| = |k_*| = q_*/2$, the direct diagonal symbol is valley-symmetric:

$$\langle G_+, z_0 | H_0(k_*) | G_+, z_0 \rangle = \langle G_-, z_0 | H_0(k_*) | G_-, z_0 \rangle =: h_{z_0}(k_*). \quad (725)$$

Hence

$$E_{z_0,+}(0) = E_{z_0,-}(0) =: E_{z_0}^{\text{dir}}(0), \quad (726)$$

with

$$E_{z_0}^{\text{dir}}(0) = h_{z_0}(k_*) + [U^{(\text{per})}(0) + U^{(\text{lock})}(0)]_{z_0 z_0}. \quad (727)$$

CCCXLVI. 3. EXPLICIT SCALAR SAFE-SECTOR EVALUATION

Inside the current safe scalar sector, the only periodic scalar contribution is

$$U_{\text{per}}^{\text{tot}}(x) = u|\phi_0(x)|^2,$$

and its first-shell Fourier coefficient is

$$\widehat{U}_{\text{per}}^{\text{tot}}(Q) = 4uA^2 \quad (Q \in S_{\text{BCC}}).$$

Therefore the direct opposite-valley locked-line Bragg entry is

$$B_{z_0}^{\text{scalar}}(0) = 4uA^2. \quad (728)$$

For the diagonal zero-mode coefficient, in the same Fourier normalization one has

$$\widehat{|\phi_0|^2}(0) = 12A^2,$$

because only the $Q_1 = Q_2$ pairs contribute at zero momentum and the first shell has 12 elements. Hence the direct diagonal scalar shift is

$$[U_{\text{per}}^{\text{tot}}(0)]_{z_0 z_0} = 12uA^2. \quad (729)$$

Thus, in the scalar-only locked core,

$$E_{z_0}^{\text{dir}}(0) = h_{z_0}(k_*) + 12uA^2 + [U^{(\text{lock})}(0)]_{z_0 z_0}, \quad (730)$$

$$B_{z_0}(0) = 4uA^2 + [U^{(\text{lock})}(G_*)]_{z_0 z_0}. \quad (731)$$

If no additional direct locked-line scalar refinement is switched on, this collapses to

$$E_{z_0}^{\text{dir}}(0) = h_{z_0}(k_*) + 12uA^2, \quad B_{z_0}(0) = 4uA^2. \quad (732)$$

CCCXLVII. 4. ACTUAL $T_\star = T^1$ BRANCH: JJ -PART VANISHES

Now specialize to the explicit branch

$$T_\star = T^1, \quad z_0 = e_1.$$

Then

$$T^1 = \frac{1}{2} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad z_0^\dagger T^1 z_0 = 0.$$

Therefore the z_0 -line J -harmonic coefficient vanishes:

$$\boxed{J_{i,0}^1(G_*; q) = 0.} \quad (733)$$

Hence the JJ -part of the Schur correction is exactly absent:

$$\boxed{\Sigma_{ab}^{(JJ;z_0)} = 0, \quad a, b \in \{+, -\}.} \quad (734)$$

So the actual same-shell coefficients reduce to

$$\boxed{d_+^{(0)} = E_{z_0}^{\text{dir}}(0) - \Sigma_{++}^{(JK+KK;z_0)},} \quad (735)$$

$$\boxed{d_-^{(0)} = E_{z_0}^{\text{dir}}(0) - \Sigma_{--}^{(JK+KK;z_0)},} \quad (736)$$

$$\boxed{m_0 = B_{z_0}(0) - \Sigma_{+-}^{(JK+KK;z_0)}.} \quad (737)$$

CCCXLVIII. 5. PURE- J SUBAUDIT

If one performs the pure- J subaudit, i.e. temporarily suppresses the K -sector residual in the same-shell complementary slot, then

$$\boxed{d_+^{(0)} = d_-^{(0)} = E_{z_0}^{\text{dir}}(0), \quad m_0 = B_{z_0}(0).} \quad (738)$$

In the minimal scalar-only locked core this gives

$$\boxed{d_+^{(0)} = d_-^{(0)} = h_{z_0}(k_*) + 12uA^2, \quad m_0 = 4uA^2.} \quad (739)$$

Hence the zero-momentum eigenvalues are

$$\boxed{\lambda_\pm(0) = h_{z_0}(k_*) + 12uA^2 \pm 4uA^2.} \quad (740)$$

That is,

$$\boxed{\lambda_+(0) = h_{z_0}(k_*) + 16uA^2, \quad \lambda_-(0) = h_{z_0}(k_*) + 8uA^2.} \quad (741)$$

Therefore the exact zero-momentum same-shell non-enriched gap in this minimal subaudit is

$$\boxed{\Delta_{\text{non-enr},0}^{(1)} = \min\{|h_{z_0}(k_*) + 16uA^2|, |h_{z_0}(k_*) + 8uA^2|\}.} \quad (742)$$

A sufficient condition is

$$\boxed{h_{z_0}(k_*) > -8uA^2.} \quad (743)$$

In particular, if

$$u > 0, \quad h_{z_0}(k_*) \geq 0,$$

then

$$\Delta_{\text{non-enr},0}^{(1)} > 0$$

automatically.

CCCXLIX. 6. HONEST ENDPOINT

Thus the direct part of the same-shell benchmark is now explicit.
For the branch

$$T_\star = T^1, \quad \eta = z_0,$$

the JJ -Schur part vanishes exactly, and the remaining same-shell benchmark data are reduced to:

$$\boxed{h_{z_0}(k_*), \quad uA^2, \quad [U^{(\text{lock})}(0)]_{z_0 z_0}, \quad [U^{(\text{lock})}(G_*)]_{z_0 z_0}, \quad \Sigma_{ab}^{(JK+KK; z_0)}}. \quad (744)$$

So the direct locked-core part is finished; only the JK/KK residual remains.

CCCL. FINAL SUMMARY

For the minimal same-shell non-enriched complementary choice

$$\boxed{\eta = z_0},$$

the exact zero-momentum coefficients are

$$\boxed{d_+^{(0)} = E_{z_0,+}(0) - \Sigma_{++}^{(z_0)}, \quad d_-^{(0)} = E_{z_0,-}(0) - \Sigma_{--}^{(z_0)}, \quad m_0 = B_{z_0}(0) - \Sigma_{+-}^{(z_0)}}.$$

For the explicit branch

$$\boxed{T_\star = T^1},$$

one has

$$\boxed{z_0^\dagger T^1 z_0 = 0, \quad J_{i,0}^1(G_*; q) = 0, \quad \Sigma_{ab}^{(JJ; z_0)} = 0}.$$

Hence the actual same-shell benchmark coefficients reduce to

$$\boxed{d_+^{(0)} = E_{z_0}^{\text{dir}}(0) - \Sigma_{++}^{(JK+KK; z_0)}, \quad d_-^{(0)} = E_{z_0}^{\text{dir}}(0) - \Sigma_{--}^{(JK+KK; z_0)},}$$

$$\boxed{m_0 = B_{z_0}(0) - \Sigma_{+-}^{(JK+KK; z_0)}}.$$

In the pure- J subaudit this simplifies further to

$$\boxed{d_+^{(0)} = d_-^{(0)} = E_{z_0}^{\text{dir}}(0), \quad m_0 = B_{z_0}(0)}.$$

Inside the minimal scalar-only locked core,

$$\boxed{E_{z_0}^{\text{dir}}(0) = h_{z_0}(k_*) + 12uA^2, \quad B_{z_0}(0) = 4uA^2}.$$

Therefore the zero-momentum eigenvalues are

$$\boxed{\lambda_+(0) = h_{z_0}(k_*) + 16uA^2, \quad \lambda_-(0) = h_{z_0}(k_*) + 8uA^2}.$$

So the exact zero-momentum same-shell non-enriched gap in this minimal subaudit is

$$\boxed{\Delta_{\text{non-enr},0}^{(1)} = \min\{|h_{z_0}(k_*) + 16uA^2|, |h_{z_0}(k_*) + 8uA^2|\}}.$$

Thus the direct locked-core part of the same-shell benchmark is finished. The only remaining benchmark-side residual is the finite JK/KK -Schur correction

$$\boxed{\Sigma_{ab}^{(JK+KK; z_0)}}.$$

CCCLI. FURTHER REDUCTION OF THE CANONICAL SAME-SHELL BENCHMARK: ONLY THE KK RESIDUAL SURVIVES ON $\eta = z_0$

CCCLII. SHORT EXPLANATION

We continue the actual coefficient extraction for the minimal same-shell non-enriched slot

$$|+\rangle = |G_+, z_0\rangle, \quad |-\rangle = |G_-, z_0\rangle$$

inside the explicit branch

$$T_\star = T^1.$$

The key observation is that the z_0 -line has no J -harmonic coupling in this branch. Therefore, in the Schur correction

$$-\frac{1}{M_X^2}(\alpha_X J + \beta_X K)(\alpha_X J + \beta_X K)^\dagger,$$

both the JJ and JK pieces vanish on the same-shell complementary z_0 -slot. Only the KK piece survives.

Thus the last benchmark-side same-shell coefficients reduce to:

$$\text{direct locked-core block} \quad + \quad \text{pure } KK \text{ heavy residual.}$$

CCCLIII. 1. MINIMAL CANONICAL SAME-SHELL NON-ENRICHED BLOCK

Fix the canonical locked basis

$$z_0 = e_1, \quad u_1 = e_2, \quad u_2 = e_3,$$

and the minimal same-shell complementary basis

$$|+\rangle := |G_+, z_0\rangle, \quad |-\rangle := |G_-, z_0\rangle. \quad (745)$$

Then the $m = 1$ same-shell non-enriched block is

$$H_{\text{non-enr}}^{(1)}(p) = \begin{pmatrix} d_+(p) & m(p) \\ m(p) & d_-(p) \end{pmatrix}, \quad (746)$$

with

$$d_\pm(p) = \langle G_\pm, z_0 | H_{\text{shell}}(p) | G_\pm, z_0 \rangle, \quad m(p) = \langle G_+, z_0 | H_{\text{shell}}(p) | G_-, z_0 \rangle. \quad (747)$$

At $p = 0$,

$$d_+^{(0)} = E_{z_0,+}(0) - \Sigma_{++}^{(z_0)}, \quad d_-^{(0)} = E_{z_0,-}(0) - \Sigma_{--}^{(z_0)}, \quad m_0 = B_{z_0}(0) - \Sigma_{+-}^{(z_0)}. \quad (748)$$

CCCLIV. 2. FULL SCHUR STRUCTURE AND VANISHING OF THE J -EXTERNAL COUPLINGS

The effective shell operator is

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - \frac{1}{M_X^2}(\alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k))(\alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k))^\dagger. \quad (749)$$

Hence the same-shell complementary Schur term is

$$\Sigma_{ab}^{(z_0)} = \frac{1}{M_X^2} \left\langle a \left| (\alpha_X \mathcal{J} + \beta_X \mathcal{K})(\alpha_X \mathcal{J} + \beta_X \mathcal{K})^\dagger \right| b \right\rangle, \quad a, b \in \{+, -\}. \quad (750)$$

Now specialize to

$$T_\star = T^1.$$

In the canonical basis,

$$T^1 = \frac{1}{2} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad z_0 = e_1,$$

so

$$\boxed{z_0^\dagger T^1 z_0 = 0.} \quad (751)$$

Using the explicit shell harmonic J -coefficient

$$J_{i,r}^a(G_*; q) = i q_i A_{G_*} u_r^\dagger T^a z_0 + i (G_*)_i A_{G_*} z_0^\dagger T^a u_r,$$

and setting $u_r = z_0$, $a = 1$, one gets

$$\boxed{J_{i,0}^1(G_*; q) = 0.} \quad (752)$$

Therefore the external z_0 -line J -couplings vanish:

$$\boxed{\langle G_\pm, z_0 | \mathcal{J}(k_*) | \chi_\alpha \rangle = 0 \quad \forall \alpha.} \quad (753)$$

CCCLV. 3. CONSEQUENCE: JJ AND JK BOTH VANISH ON THE z_0 -SLOT

Because the external z_0 -line has no J -coupling, the Schur pieces satisfy

$$\boxed{\Sigma_{ab}^{(JJ;z_0)} = 0, \quad \Sigma_{ab}^{(JK;z_0)} = 0, \quad a, b \in \{+, -\}.} \quad (754)$$

Hence only the KK -residual survives:

$$\boxed{\Sigma_{ab}^{(z_0)} = \Sigma_{ab}^{(KK;z_0)}.} \quad (755)$$

Thus the exact zero-momentum coefficients reduce to

$$\boxed{d_+^{(0)} = E_{z_0,+}(0) - \Sigma_{++}^{(KK;z_0)},} \quad (756)$$

$$\boxed{d_-^{(0)} = E_{z_0,-}(0) - \Sigma_{--}^{(KK;z_0)},} \quad (757)$$

$$\boxed{m_0 = B_{z_0}(0) - \Sigma_{+-}^{(KK;z_0)}.} \quad (758)$$

CCCLVI. 4. DIRECT LOCKED-CORE PART

The direct triplet shell block along the locked line is

$$\boxed{E_{z_0,+}(0) = h_{z_0}(k_*) + [U^{(\text{per})}(0) + U^{(\text{lock})}(0)]_{z_0 z_0},} \quad (759)$$

$$\boxed{E_{z_0,-}(0) = h_{z_0}(k_*) + [U^{(\text{per})}(0) + U^{(\text{lock})}(0)]_{z_0 z_0},} \quad (760)$$

and the opposite-valley direct Bragg entry is

$$B_{z_0}(0) = [U^{(\text{per})}(G_*) + U^{(\text{lock})}(G_*)]_{z_0 z_0}. \quad (761)$$

Therefore the direct diagonal part is valley-symmetric:

$$E_{z_0,+}(0) = E_{z_0,-}(0) =: E_{z_0}^{\text{dir}}(0). \quad (762)$$

Inside the minimal scalar-only locked core,

$$U_{\text{per}}^{\text{tot}}(x) = u|\phi_0(x)|^2, \quad \hat{U}_{\text{per}}^{\text{tot}}(0) = 12uA^2, \quad \hat{U}_{\text{per}}^{\text{tot}}(G_*) = 4uA^2.$$

Hence

$$E_{z_0}^{\text{dir}}(0) = \omega_B(k_*) + 12uA^2 + [U^{(\text{lock})}(0)]_{z_0 z_0}, \quad (763)$$

$$B_{z_0}(0) = 4uA^2 + [U^{(\text{lock})}(G_*)]_{z_0 z_0}, \quad (764)$$

where

$$\omega_B(k_*) = -\mu^2 + Z|k_*|^2 - Y|k_*|^4 = -\mu^2 + \frac{Zq_*^2}{4} - \frac{Yq_*^4}{16}. \quad (765)$$

If no additional direct locked-line refinement is switched on, this simplifies to

$$E_{z_0}^{\text{dir}}(0) = \omega_B(k_*) + 12uA^2, \quad B_{z_0}(0) = 4uA^2. \quad (766)$$

CCCLVII. 5. PURE- J SUBAUDIT AS A STRICT LOWER-COMPLEXITY BENCHMARK

If one temporarily suppresses the K -sector in the same-shell complementary slot, then

$$d_+^{(0)} = d_-^{(0)} = E_{z_0}^{\text{dir}}(0), \quad m_0 = B_{z_0}(0). \quad (767)$$

In the minimal scalar-only locked core this yields

$$d_+^{(0)} = d_-^{(0)} = \omega_B(k_*) + 12uA^2, \quad m_0 = 4uA^2. \quad (768)$$

Hence the zero-momentum eigenvalues are

$$\lambda_{\pm}(0) = \omega_B(k_*) + 12uA^2 \pm 4uA^2, \quad (769)$$

i.e.

$$\lambda_+(0) = \omega_B(k_*) + 16uA^2, \quad \lambda_-(0) = \omega_B(k_*) + 8uA^2. \quad (770)$$

Therefore the exact zero-momentum same-shell benchmark in this strict subaudit is

$$\Delta_{\text{non-enr},0}^{(1)} = \min\{|\omega_B(k_*) + 16uA^2|, |\omega_B(k_*) + 8uA^2|\}. \quad (771)$$

A sufficient positivity condition is

$$\omega_B(k_*) > -8uA^2. \quad (772)$$

CCCLVIII. 6. FINAL EXACT KK -RESIDUAL FORM

Choose a heavy basis χ_α diagonalizing the heavy block with eigenvalues $E_\alpha > 0$. Since only the K -external couplings survive, define

$$\mathcal{K}_{\pm\alpha}^{(z_0)} := \langle G_{\pm}, z_0 | \mathcal{K}(k_*) | \chi_\alpha \rangle. \quad (773)$$

Then

$$\Sigma_{++}^{(KK;z_0)} = \frac{\beta_X^2}{M_X^2} \sum_{\alpha} \frac{|\mathcal{K}_{+\alpha}^{(z_0)}|^2}{E_{\alpha}}, \quad (774)$$

$$\Sigma_{--}^{(KK;z_0)} = \frac{\beta_X^2}{M_X^2} \sum_{\alpha} \frac{|\mathcal{K}_{-\alpha}^{(z_0)}|^2}{E_{\alpha}}, \quad (775)$$

$$\Sigma_{+-}^{(KK;z_0)} = \frac{\beta_X^2}{M_X^2} \sum_{\alpha} \frac{\mathcal{K}_{+\alpha}^{(z_0)} \overline{\mathcal{K}_{-\alpha}^{(z_0)}}}{E_{\alpha}}. \quad (776)$$

So the actual same-shell coefficients are finally

$$d_+^{(0)} = E_{z_0}^{\text{dir}}(0) - \frac{\beta_X^2}{M_X^2} \sum_{\alpha} \frac{|\mathcal{K}_{+\alpha}^{(z_0)}|^2}{E_{\alpha}}, \quad (777)$$

$$d_-^{(0)} = E_{z_0}^{\text{dir}}(0) - \frac{\beta_X^2}{M_X^2} \sum_{\alpha} \frac{|\mathcal{K}_{-\alpha}^{(z_0)}|^2}{E_{\alpha}}, \quad (778)$$

$$m_0 = B_{z_0}(0) - \frac{\beta_X^2}{M_X^2} \sum_{\alpha} \frac{\mathcal{K}_{+\alpha}^{(z_0)} \overline{\mathcal{K}_{-\alpha}^{(z_0)}}}{E_{\alpha}}. \quad (779)$$

Therefore the entire same-shell benchmark-side frontier for this explicit branch has been reduced to the finite KK -overlap sums.

CCCLIX. FINAL SUMMARY

For the explicit branch

$$T_{\star} = T^1, \quad \eta = z_0,$$

the minimal same-shell non-enriched benchmark coefficients satisfy

$$d_+^{(0)} = E_{z_0,+}(0) - \Sigma_{++}^{(z_0)}, \quad d_-^{(0)} = E_{z_0,-}(0) - \Sigma_{--}^{(z_0)}, \quad m_0 = B_{z_0}(0) - \Sigma_{+-}^{(z_0)}.$$

Because

$$z_0^\dagger T^1 z_0 = 0,$$

the external z_0 -line has no J -harmonic coupling. Therefore

$$\Sigma^{(JJ;z_0)} = 0, \quad \Sigma^{(JK;z_0)} = 0,$$

and only the KK -residual survives:

$$\Sigma^{(z_0)} = \Sigma^{(KK;z_0)}.$$

The direct locked-core part is

$$E_{z_0}^{\text{dir}}(0) = \omega_B(k_*) + 12uA^2 + [U^{(\text{lock})}(0)]_{z_0 z_0},$$

$$B_{z_0}(0) = 4uA^2 + [U^{(\text{lock})}(G_*)]_{z_0 z_0},$$

with

$$\omega_B(k_*) = -\mu^2 + \frac{Zq_*^2}{4} - \frac{Yq_*^4}{16}.$$

Hence the actual same-shell benchmark coefficients reduce to

$$d_+^{(0)} = E_{z_0}^{\text{dir}}(0) - \frac{\beta_X^2}{M_X^2} \sum_{\alpha} \frac{|\mathcal{K}_{+\alpha}^{(z_0)}|^2}{E_{\alpha}},$$

$$d_-^{(0)} = E_{z_0}^{\text{dir}}(0) - \frac{\beta_X^2}{M_X^2} \sum_{\alpha} \frac{|\mathcal{K}_{-\alpha}^{(z_0)}|^2}{E_{\alpha}},$$

$$m_0 = B_{z_0}(0) - \frac{\beta_X^2}{M_X^2} \sum_{\alpha} \frac{\mathcal{K}_{+\alpha}^{(z_0)} \overline{\mathcal{K}_{-\alpha}^{(z_0)}}}{E_{\alpha}}.$$

Thus the direct part of the same-shell benchmark is finished, and the genuinely remaining benchmark-side residual is the finite pure- KK overlap sum.

CCCLX. PURE KK REDUCTION OF THE SAME-SHELL COMPLEMENTARY RESIDUAL ON $\eta = z_0$

CCCLXI. SHORT EXPLANATION

For the explicit branch

$$T_{\star} = T^1, \quad \eta = z_0,$$

the same-shell non-enriched zero-momentum coefficients were reduced to

$$d_+^{(0)} = E_{z_0}^{\text{dir}}(0) - \Sigma_{++}^{(KK;z_0)}, \quad d_-^{(0)} = E_{z_0}^{\text{dir}}(0) - \Sigma_{--}^{(KK;z_0)}, \quad m_0 = B_{z_0}(0) - \Sigma_{+-}^{(KK;z_0)}.$$

We now evaluate the surviving KK -piece at the level of explicit shell Fourier overlaps. This turns the last same-shell benchmark residual into a finite quadratic form in the two transverse seed amplitudes

$$\Lambda_E^{(0)}, \quad \Lambda_O^{(0)}.$$

CCCLXII. 1. DEFECT-BACKGROUND K -SECTOR

The minimal complete K -current is

$$K_i^a[P, \psi] = \text{Tr}(T^a \partial_i P) \left(\Lambda_E^{(0)} E_j^b + \Lambda_O^{(0)} J_j^b \right) D_j \psi_b. \quad (780)$$

On the defect background,

$$\partial_i P_B = \delta_{i\xi} \theta'(\xi) (e_1 z_B^\dagger + z_B e_1^\dagger), \quad (781)$$

hence

$$\text{Tr}(T^a \partial_i P_B) = \delta_{i\xi} \theta'(\xi) k_a(\theta), \quad (782)$$

and therefore

$$K_i^a[P_B, \psi] = \delta_{i\xi} \theta'(\xi) k_a(\theta) \left(\Lambda_E^{(0)} E_j^b + \Lambda_O^{(0)} J_j^b \right) D_j \psi_b. \quad (783)$$

This means that the surviving same-shell complementary couplings are already linear in

$$\Lambda_E^{(0)}, \quad \Lambda_O^{(0)}.$$

CCCLXIII. 2. HEAVY-OVERLAP AMPLITUDES

Let χ_α be the heavy eigenbasis with

$$\mathbb{O}^{(0)} \chi_\alpha = E_\alpha \chi_\alpha, \quad E_\alpha > 0.$$

Define the E -type and O -type same-shell complementary overlaps on the z_0 -line by

$$\mathcal{E}_{\pm\alpha}^{(z_0)} := \langle G_\pm, z_0 | \theta'(\xi) k_a(\theta) E_j^b D_j \psi_b | \chi_\alpha \rangle, \quad (784)$$

$$\mathcal{O}_{\pm\alpha}^{(z_0)} := \langle G_\pm, z_0 | \theta'(\xi) k_a(\theta) J_j^b D_j \psi_b | \chi_\alpha \rangle. \quad (785)$$

Then the full K -overlap amplitudes factor as

$$\mathcal{K}_{\pm\alpha}^{(z_0)} = \Lambda_E^{(0)} \mathcal{E}_{\pm\alpha}^{(z_0)} + \Lambda_O^{(0)} \mathcal{O}_{\pm\alpha}^{(z_0)}. \quad (786)$$

Thus the same-shell complementary residual is

$$\Sigma_{ab}^{(KK; z_0)} = \frac{\beta_X^2}{M_X^2} \sum_\alpha \frac{\mathcal{K}_{a\alpha}^{(z_0)} \overline{\mathcal{K}_{b\alpha}^{(z_0)}}}{E_\alpha}, \quad a, b \in \{+, -\}. \quad (787)$$

CCCLXIV. 3. EXPLICIT QUADRATIC-FORM REDUCTION

Substituting (786) into (787), one gets

$$\Sigma_{ab}^{(KK; z_0)} = \frac{\beta_X^2}{M_X^2} \left[(\Lambda_E^{(0)})^2 \mathcal{S}_{ab}^{EE} + \Lambda_E^{(0)} \Lambda_O^{(0)} \mathcal{S}_{ab}^{EO} + (\Lambda_O^{(0)})^2 \mathcal{S}_{ab}^{OO} \right], \quad (788)$$

where the finite overlap sums are

$$\mathcal{S}_{ab}^{EE} := \sum_{\alpha} \frac{\mathcal{E}_{a\alpha}^{(z_0)} \overline{\mathcal{E}_{b\alpha}^{(z_0)}}}{E_{\alpha}}, \quad (789)$$

$$\mathcal{S}_{ab}^{EO} := \sum_{\alpha} \frac{\mathcal{E}_{a\alpha}^{(z_0)} \overline{\mathcal{O}_{b\alpha}^{(z_0)}} + \mathcal{O}_{a\alpha}^{(z_0)} \overline{\mathcal{E}_{b\alpha}^{(z_0)}}}{E_{\alpha}}, \quad (790)$$

$$\mathcal{S}_{ab}^{OO} := \sum_{\alpha} \frac{\mathcal{O}_{a\alpha}^{(z_0)} \overline{\mathcal{O}_{b\alpha}^{(z_0)}}}{E_{\alpha}}. \quad (791)$$

Therefore the actual same-shell coefficients are

$$d_{+}^{(0)} = E_{z_0}^{\text{dir}}(0) - \frac{\beta_X^2}{M_X^2} \left[(\Lambda_E^{(0)})^2 \mathcal{S}_{++}^{EE} + \Lambda_E^{(0)} \Lambda_O^{(0)} \mathcal{S}_{++}^{EO} + (\Lambda_O^{(0)})^2 \mathcal{S}_{++}^{OO} \right], \quad (792)$$

$$d_{-}^{(0)} = E_{z_0}^{\text{dir}}(0) - \frac{\beta_X^2}{M_X^2} \left[(\Lambda_E^{(0)})^2 \mathcal{S}_{--}^{EE} + \Lambda_E^{(0)} \Lambda_O^{(0)} \mathcal{S}_{--}^{EO} + (\Lambda_O^{(0)})^2 \mathcal{S}_{--}^{OO} \right], \quad (793)$$

$$m_0 = B_{z_0}(0) - \frac{\beta_X^2}{M_X^2} \left[(\Lambda_E^{(0)})^2 \mathcal{S}_{+-}^{EE} + \Lambda_E^{(0)} \Lambda_O^{(0)} \mathcal{S}_{+-}^{EO} + (\Lambda_O^{(0)})^2 \mathcal{S}_{+-}^{OO} \right]. \quad (794)$$

CCCLXV. 4. PURE- E RESIDUAL SUBBRANCH

If the same-shell complementary residual is evaluated in the pure-even subbranch

$$\Lambda_O^{(0)} = 0,$$

then the formulas collapse to

$$d_{+}^{(0)} = E_{z_0}^{\text{dir}}(0) - \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \mathcal{S}_{++}^{EE}, \quad (795)$$

$$d_{-}^{(0)} = E_{z_0}^{\text{dir}}(0) - \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \mathcal{S}_{--}^{EE}, \quad (796)$$

$$m_0 = B_{z_0}(0) - \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \mathcal{S}_{+-}^{EE}. \quad (797)$$

This is the cleanest same-shell benchmark realization compatible with the explicit pure- E carrier branch.

CCCLXVI. 5. POSITIVITY AND CAUCHY-TYPE CONTROL

For the diagonal entries one has manifest positivity:

$$\mathcal{S}_{++}^{EE} \geq 0, \quad \mathcal{S}_{--}^{EE} \geq 0, \quad \mathcal{S}_{++}^{OO} \geq 0, \quad \mathcal{S}_{--}^{OO} \geq 0. \quad (798)$$

Moreover, by Cauchy–Schwarz,

$$|\mathcal{S}_{+-}^{EE}|^2 \leq \mathcal{S}_{++}^{EE} \mathcal{S}_{--}^{EE}, \quad |\mathcal{S}_{+-}^{OO}|^2 \leq \mathcal{S}_{++}^{OO} \mathcal{S}_{--}^{OO}. \quad (799)$$

Thus the off-diagonal same-shell residual cannot exceed the geometric mean of the two diagonal residual budgets.

CCCLXVII. 6. PRACTICAL ZERO-MOMENTUM SUFFICIENT CONDITION

The exact zero-momentum same-shell sufficient condition remains

$$\min\{|d_+^{(0)}|, |d_-^{(0)}|\} > |m_0|.$$

In the pure-even residual subbranch $\Lambda_O^{(0)} = 0$, this becomes

$$\min \left\{ \left| E_{z_0}^{\text{dir}}(0) - \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \mathcal{S}_{++}^{EE} \right|, \left| E_{z_0}^{\text{dir}}(0) - \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \mathcal{S}_{--}^{EE} \right| \right\} > \left| B_{z_0}(0) - \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \mathcal{S}_{+-}^{EE} \right|. \quad (800)$$

If, in addition, valley symmetry holds on the residual norms,

$$\mathcal{S}_{++}^{EE} = \mathcal{S}_{--}^{EE} =: \mathcal{S}_{\text{diag}}^{EE},$$

then

$$\left| E_{z_0}^{\text{dir}}(0) - \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \mathcal{S}_{\text{diag}}^{EE} \right| > \left| B_{z_0}(0) - \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \mathcal{S}_{+-}^{EE} \right| \quad (801)$$

is sufficient.

CCCLXVIII. 7. FINAL REDUCTION OF THE SAME-SHELL BENCHMARK

Therefore, for the explicit branch

$$T_\star = T^1, \quad \eta = z_0,$$

the same-shell benchmark side is fully reduced to the finite overlap data

$$\boxed{E_{z_0}^{\text{dir}}(0), \quad B_{z_0}(0), \quad \mathcal{S}_{ab}^{EE}, \quad \mathcal{S}_{ab}^{EO}, \quad \mathcal{S}_{ab}^{OO}}, \quad (802)$$

or, in the explicit pure-even residual subbranch,

$$\boxed{E_{z_0}^{\text{dir}}(0), \quad B_{z_0}(0), \quad \mathcal{S}_{++}^{EE}, \quad \mathcal{S}_{--}^{EE}, \quad \mathcal{S}_{+-}^{EE}}. \quad (803)$$

This is the literal finite coefficient set that remains to be inserted into the Step-IV same-shell benchmark.

CCCLXIX. FINAL SUMMARY

For the explicit branch

$$\boxed{T_\star = T^1, \quad \eta = z_0,}$$

the same-shell non-enriched benchmark residual is purely KK :

$$\boxed{\Sigma^{(z_0)} = \Sigma^{(KK; z_0)}}.$$

Using the explicit defect-background K -ansatz,

$$\boxed{K_i^a[P_B, \psi] = \delta_{i\xi} \theta'(\xi) k_a(\theta) \left(\Lambda_E^{(0)} E_j^b + \Lambda_O^{(0)} J_j^b \right) D_j \psi_b,}$$

the heavy overlap amplitudes factorize as

$$\boxed{\mathcal{K}_{\pm\alpha}^{(z_0)} = \Lambda_E^{(0)} \mathcal{E}_{\pm\alpha}^{(z_0)} + \Lambda_O^{(0)} \mathcal{O}_{\pm\alpha}^{(z_0)}}.$$

Therefore the same-shell residual is the quadratic form

$$\Sigma_{ab}^{(KK;z_0)} = \frac{\beta_X^2}{M_X^2} \left[(\Lambda_E^{(0)})^2 \mathcal{S}_{ab}^{EE} + \Lambda_E^{(0)} \Lambda_O^{(0)} \mathcal{S}_{ab}^{EO} + (\Lambda_O^{(0)})^2 \mathcal{S}_{ab}^{OO} \right],$$

with finite overlap sums

$$\mathcal{S}_{ab}^{EE}, \quad \mathcal{S}_{ab}^{EO}, \quad \mathcal{S}_{ab}^{OO}.$$

Hence the actual zero-momentum same-shell coefficients are

$$d_+^{(0)} = E_{z_0}^{\text{dir}}(0) - \Sigma_{++}^{(KK;z_0)}, \quad d_-^{(0)} = E_{z_0}^{\text{dir}}(0) - \Sigma_{--}^{(KK;z_0)},$$

$$m_0 = B_{z_0}(0) - \Sigma_{+-}^{(KK;z_0)}.$$

In the pure-even residual subbranch $\Lambda_O^{(0)} = 0$, this simplifies to

$$d_+^{(0)} = E_{z_0}^{\text{dir}}(0) - \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \mathcal{S}_{++}^{EE},$$

$$d_-^{(0)} = E_{z_0}^{\text{dir}}(0) - \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \mathcal{S}_{--}^{EE},$$

$$m_0 = B_{z_0}(0) - \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \mathcal{S}_{+-}^{EE}.$$

So the same-shell benchmark-side frontier has now been reduced to the finite overlap data

$$E_{z_0}^{\text{dir}}(0), \quad B_{z_0}(0), \quad \mathcal{S}_{++}^{EE}, \quad \mathcal{S}_{--}^{EE}, \quad \mathcal{S}_{+-}^{EE}$$

in the explicit pure-even residual realization.

CCCLXX. PURE-EVEN KK RESIDUAL REDUCTION AND A PRACTICAL SAME-SHELL BENCHMARK INEQUALITY

CCCLXXI. SHORT EXPLANATION

For the explicit branch

$$T_\star = T^1, \quad \eta = z_0,$$

the same-shell non-enriched residual is purely KK . If we further specialize to the pure-even residual subbranch

$$\Lambda_O^{(0)} = 0,$$

then the entire same-shell benchmark is controlled by the three finite overlap sums

$$\mathcal{S}_{++}^{EE}, \quad \mathcal{S}_{--}^{EE}, \quad \mathcal{S}_{+-}^{EE}.$$

Under an additional valley-symmetric residual assumption

$$\mathcal{S}_{++}^{EE} = \mathcal{S}_{--}^{EE} =: \mathcal{S}_{\text{diag}}^{EE},$$

the zero-momentum same-shell benchmark reduces to a single scalar inequality.

CCCLXXII. 1. PURE-EVEN RESIDUAL COEFFICIENTS

From the K -sector defect ansatz and the already established $JJ = JK = 0$ result on the z_0 -slot, the zero-momentum same-shell coefficients are

$$d_+^{(0)} = E_{z_0}^{\text{dir}}(0) - \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \mathcal{S}_{++}^{EE}, \quad (804)$$

$$d_-^{(0)} = E_{z_0}^{\text{dir}}(0) - \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \mathcal{S}_{--}^{EE}, \quad (805)$$

$$m_0 = B_{z_0}(0) - \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \mathcal{S}_{+-}^{EE}. \quad (806)$$

Introduce the shorthand

$$a_K := \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2. \quad (807)$$

Then

$$d_+^{(0)} = E_{z_0}^{\text{dir}}(0) - a_K \mathcal{S}_{++}^{EE}, \quad d_-^{(0)} = E_{z_0}^{\text{dir}}(0) - a_K \mathcal{S}_{--}^{EE}, \quad m_0 = B_{z_0}(0) - a_K \mathcal{S}_{+-}^{EE}. \quad (808)$$

CCCLXXIII. 2. VALLEY-SYMMETRIC RESIDUAL SUBCASE

Assume

$$\mathcal{S}_{++}^{EE} = \mathcal{S}_{--}^{EE} =: \mathcal{S}_{\text{diag}}^{EE}. \quad (809)$$

Then the diagonal entries are equal:

$$d_+^{(0)} = d_-^{(0)} =: d^{(0)} = E_{z_0}^{\text{dir}}(0) - a_K \mathcal{S}_{\text{diag}}^{EE}. \quad (810)$$

Hence the exact zero-momentum eigenvalues are

$$\lambda_{\pm}(0) = d^{(0)} \pm |m_0|. \quad (811)$$

Therefore the exact zero-momentum same-shell benchmark is

$$\Delta_{\text{non-enr},0}^{(1)} = \left| E_{z_0}^{\text{dir}}(0) - a_K \mathcal{S}_{\text{diag}}^{EE} \right| - \left| B_{z_0}(0) - a_K \mathcal{S}_{+-}^{EE} \right|. \quad (812)$$

A sufficient condition for positivity is therefore

$$\left| E_{z_0}^{\text{dir}}(0) - a_K \mathcal{S}_{\text{diag}}^{EE} \right| > \left| B_{z_0}(0) - a_K \mathcal{S}_{+-}^{EE} \right|. \quad (813)$$

CCCLXXIV. 3. CAUCHY-SCHWARZ CONTROL OF THE OFF-DIAGONAL OVERLAP

By Cauchy-Schwarz, the off-diagonal overlap obeys

$$|\mathcal{S}_{+-}^{EE}|^2 \leq \mathcal{S}_{++}^{EE} \mathcal{S}_{--}^{EE}. \quad (814)$$

In the valley-symmetric subcase (809),

$$|\mathcal{S}_{+-}^{EE}| \leq \mathcal{S}_{\text{diag}}^{EE}. \quad (815)$$

Therefore

$$|B_{z_0}(0) - a_K \mathcal{S}_{+-}^{EE}| \leq |B_{z_0}(0)| + a_K \mathcal{S}_{\text{diag}}^{EE}, \quad (816)$$

while

$$|E_{z_0}^{\text{dir}}(0) - a_K \mathcal{S}_{\text{diag}}^{EE}| \geq |E_{z_0}^{\text{dir}}(0)| - a_K \mathcal{S}_{\text{diag}}^{EE}. \quad (817)$$

Hence a stronger but easier sufficient condition is

$$|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > 2a_K \mathcal{S}_{\text{diag}}^{EE}. \quad (818)$$

This is a purely scalar practical bound.

CCCLXXV. 4. INSERTION OF THE DIRECT LOCKED-CORE PART

The direct locked-core part has already been reduced to

$$E_{z_0}^{\text{dir}}(0) = \omega_B(k_*) + 12uA^2 + [U^{(\text{lock})}(0)]_{z_0 z_0}, \quad (819)$$

$$B_{z_0}(0) = 4uA^2 + [U^{(\text{lock})}(G_*)]_{z_0 z_0}, \quad (820)$$

with

$$\omega_B(k_*) = -\mu^2 + \frac{Zq_*^2}{4} - \frac{Yq_*^4}{16}. \quad (821)$$

Therefore the practical sufficient condition (818) becomes

$$\left| \omega_B(k_*) + 12uA^2 + [U^{(\text{lock})}(0)]_{z_0 z_0} \right| - \left| 4uA^2 + [U^{(\text{lock})}(G_*)]_{z_0 z_0} \right| > 2 \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \mathcal{S}_{\text{diag}}^{EE}. \quad (822)$$

CCCLXXVI. 5. MINIMAL SCALAR-ONLY LOCKED-CORE SUBCASE

If no additional direct locked-line refinement is switched on, then

$$[U^{(\text{lock})}(0)]_{z_0 z_0} = 0, \quad [U^{(\text{lock})}(G_*)]_{z_0 z_0} = 0.$$

Hence

$$E_{z_0}^{\text{dir}}(0) = \omega_B(k_*) + 12uA^2, \quad B_{z_0}(0) = 4uA^2. \quad (823)$$

Then the practical same-shell sufficient condition becomes

$$\left| \omega_B(k_*) + 12uA^2 \right| - 4uA^2 > 2 \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \mathcal{S}_{\text{diag}}^{EE}. \quad (824)$$

If, moreover,

$$u > 0, \quad \omega_B(k_*) + 12uA^2 \geq 0,$$

then (824) simplifies to

$$\omega_B(k_*) + 8uA^2 > 2 \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \mathcal{S}_{\text{diag}}^{EE}. \quad (825)$$

This is the cleanest explicit same-shell benchmark inequality now available for the branch.

CCCLXXVII. 6. INSERTION INTO THE FINAL REMOTE-GAP BENCHMARK

The same-shell usable benchmark entry is

$$\Delta_{\text{non-enr},\rho}^{(1)} - \varepsilon_{\text{non-enr},\rho}^{(1)}.$$

So if (822) (or the simpler (825)) holds and the small- p correction plus truncation budget satisfies

$$\boxed{\varepsilon_{\text{small-}p,\rho}^{(1)} + \varepsilon_{\text{non-enr},\rho}^{(1)} < \Delta_{\text{non-enr},0}^{(1)}} \quad (826)$$

then

$$\Delta_{\text{non-enr},\rho}^{(1)} - \varepsilon_{\text{non-enr},\rho}^{(1)} > 0.$$

Therefore the same-shell non-enriched part of Step IV is reduced to one explicit finite scalar inequality plus one finite truncation/small- p budget.

CCCLXXVIII. FINAL SUMMARY

For the explicit branch

$$\boxed{T_\star = T^1, \quad \eta = z_0,}$$

the same-shell non-enriched residual is purely KK . In the pure-even residual subbranch $\Lambda_O^{(0)} = 0$, one has

$$\boxed{d_+^{(0)} = E_{z_0}^{\text{dir}}(0) - \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \mathcal{S}_{++}^{EE},}$$

$$\boxed{d_-^{(0)} = E_{z_0}^{\text{dir}}(0) - \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \mathcal{S}_{--}^{EE},}$$

$$\boxed{m_0 = B_{z_0}(0) - \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \mathcal{S}_{+-}^{EE}.$$

If the residual diagonal overlaps are valley-symmetric,

$$\boxed{\mathcal{S}_{++}^{EE} = \mathcal{S}_{--}^{EE} =: \mathcal{S}_{\text{diag}}^{EE},}$$

then a practical sufficient same-shell condition is

$$\boxed{|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > 2 \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \mathcal{S}_{\text{diag}}^{EE}.$$

Using the direct locked-core values

$$\boxed{E_{z_0}^{\text{dir}}(0) = \omega_B(k_*) + 12uA^2 + [U^{(\text{lock})}(0)]_{z_0 z_0}, \quad B_{z_0}(0) = 4uA^2 + [U^{(\text{lock})}(G_*)]_{z_0 z_0},}$$

with

$$\boxed{\omega_B(k_*) = -\mu^2 + \frac{Zq_*^2}{4} - \frac{Yq_*^4}{16},}$$

this becomes the explicit scalar inequality

$$\left| \omega_B(k_*) + 12uA^2 + [U^{(\text{lock})}(0)]_{z_0 z_0} \right| - \left| 4uA^2 + [U^{(\text{lock})}(G_*)]_{z_0 z_0} \right| > 2 \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \mathcal{S}_{\text{diag}}^{EE}.$$

In the minimal scalar-only locked core, this simplifies to

$$\left| -\mu^2 + \frac{Zq_*^2}{4} - \frac{Yq_*^4}{16} + 12uA^2 \right| - 4uA^2 > 2 \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \mathcal{S}_{\text{diag}}^{EE}.$$

So the same-shell benchmark-side frontier has now been reduced to a single finite residual overlap budget

$$\mathcal{S}_{\text{diag}}^{EE},$$

plus the small- p and truncation error budget.

CCCLXXIX. FINITE-TRUNCATION CONTROL OF $\mathcal{S}_{\text{diag}}^{EE}$ AND THE FINAL SAME-SHELL PRACTICAL INEQUALITY

CCCLXXX. SHORT EXPLANATION

For the explicit branch

$$T_\star = T^1, \quad \eta = z_0, \quad \Lambda_O^{(0)} = 0,$$

the same-shell non-enriched residual is purely KK , and the remaining benchmark-side problem is the finite evaluation of the heavy overlap sums

$$\mathcal{S}_{++}^{EE}, \quad \mathcal{S}_{--}^{EE}, \quad \mathcal{S}_{+-}^{EE}.$$

The correct next move is to truncate the heavy basis at level N , compute the finite sums explicitly, and control the tail by the heavy spectral gap.

CCCLXXXI. 1. PURE-EVEN KK RESIDUAL

In the pure-even residual subbranch $\Lambda_O^{(0)} = 0$, the exact zero-momentum same-shell coefficients are

$$d_+^{(0)} = E_{z_0}^{\text{dir}}(0) - a_K \mathcal{S}_{++}^{EE}, \quad d_-^{(0)} = E_{z_0}^{\text{dir}}(0) - a_K \mathcal{S}_{--}^{EE}, \quad (827)$$

$$m_0 = B_{z_0}(0) - a_K \mathcal{S}_{+-}^{EE}, \quad a_K := \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2. \quad (828)$$

If the residual diagonal overlaps are valley-symmetric,

$$\mathcal{S}_{++}^{EE} = \mathcal{S}_{--}^{EE} =: \mathcal{S}_{\text{diag}}^{EE}, \quad (829)$$

then the practical sufficient same-shell condition is

$$|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > 2a_K \mathcal{S}_{\text{diag}}^{EE}. \quad (830)$$

CCCLXXXII. 2. HEAVY-BASIS TRUNCATION

Let $\{\chi_\alpha\}_{\alpha \geq 1}$ be a heavy eigenbasis with

$$\mathbb{O}^{(0)}\chi_\alpha = E_\alpha\chi_\alpha, \quad E_\alpha \geq \Delta_H > 0.$$

Define the pure-even overlap amplitudes

$$\mathcal{E}_{\pm\alpha}^{(z_0)} = \langle G_\pm, z_0 | \theta'(\xi) k_a(\theta) E_j^b D_j \psi_b | \chi_\alpha \rangle.$$

Then

$$\boxed{\mathcal{S}_{ab}^{EE} = \sum_{\alpha=1}^{\infty} \frac{\mathcal{E}_{a\alpha}^{(z_0)} \overline{\mathcal{E}_{b\alpha}^{(z_0)}}}{E_\alpha}.} \quad (831)$$

Fix a truncation level N and define the finite sums

$$\boxed{\mathcal{S}_{ab}^{EE,(N)} := \sum_{\alpha=1}^N \frac{\mathcal{E}_{a\alpha}^{(z_0)} \overline{\mathcal{E}_{b\alpha}^{(z_0)}}}{E_\alpha}.} \quad (832)$$

Write

$$\mathcal{S}_{ab}^{EE} = \mathcal{S}_{ab}^{EE,(N)} + \mathcal{R}_{ab}^{EE,(N)},$$

with tail

$$\boxed{\mathcal{R}_{ab}^{EE,(N)} := \sum_{\alpha > N} \frac{\mathcal{E}_{a\alpha}^{(z_0)} \overline{\mathcal{E}_{b\alpha}^{(z_0)}}}{E_\alpha}.} \quad (833)$$

CCCLXXXIII. 3. TAIL BOUND FROM THE HEAVY GAP

Because $E_\alpha \geq \Delta_H > 0$, one immediately gets

$$\boxed{|\mathcal{R}_{ab}^{EE,(N)}| \leq \frac{1}{\Delta_H} \sum_{\alpha > N} \left| \mathcal{E}_{a\alpha}^{(z_0)} \overline{\mathcal{E}_{b\alpha}^{(z_0)}} \right|.} \quad (834)$$

By Cauchy–Schwarz,

$$\boxed{|\mathcal{R}_{ab}^{EE,(N)}| \leq \frac{1}{\Delta_H} \left(\sum_{\alpha > N} |\mathcal{E}_{a\alpha}^{(z_0)}|^2 \right)^{1/2} \left(\sum_{\alpha > N} |\mathcal{E}_{b\alpha}^{(z_0)}|^2 \right)^{1/2}.} \quad (835)$$

In particular, for the diagonal sector,

$$\boxed{0 \leq \mathcal{R}_{\text{diag}}^{EE,(N)} \leq \frac{1}{\Delta_H} \sum_{\alpha > N} |\mathcal{E}_{+\alpha}^{(z_0)}|^2,} \quad (836)$$

assuming valley-symmetric residual norms.

Thus the exact diagonal overlap budget satisfies

$$\boxed{\mathcal{S}_{\text{diag}}^{EE} \leq \mathcal{S}_{\text{diag}}^{EE,(N)} + \mathcal{R}_{\text{diag}}^{EE,(N)}.} \quad (837)$$

CCCLXXXIV. 4. FINITE PRACTICAL SAME-SHELL INEQUALITY

Substituting (837) into (830), a finite practical sufficient condition is

$$\boxed{|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > 2a_K \left(\mathcal{S}_{\text{diag}}^{EE,(N)} + \mathcal{R}_{\text{diag}}^{EE,(N)} \right)}. \quad (838)$$

This is now a computable truncation-level inequality.

If the tail is negligible compared to the truncated sum, this simplifies to

$$\boxed{|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > 2a_K \mathcal{S}_{\text{diag}}^{EE,(N)}}. \quad (839)$$

CCCLXXXV. 5. MINIMAL SCALAR-ONLY LOCKED CORE INSERTION

For the minimal scalar-only locked core,

$$E_{z_0}^{\text{dir}}(0) = \omega_B(k_*) + 12uA^2, \quad B_{z_0}(0) = 4uA^2,$$

with

$$\omega_B(k_*) = -\mu^2 + \frac{Zq_*^2}{4} - \frac{Yq_*^4}{16}.$$

Therefore (838) becomes

$$\boxed{\left| -\mu^2 + \frac{Zq_*^2}{4} - \frac{Yq_*^4}{16} + 12uA^2 \right| - 4uA^2 > 2 \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \left(\mathcal{S}_{\text{diag}}^{EE,(N)} + \mathcal{R}_{\text{diag}}^{EE,(N)} \right)}. \quad (840)$$

If, moreover,

$$u > 0, \quad -\mu^2 + \frac{Zq_*^2}{4} - \frac{Yq_*^4}{16} + 12uA^2 \geq 0,$$

then the left-hand side simplifies to

$$-\mu^2 + \frac{Zq_*^2}{4} - \frac{Yq_*^4}{16} + 8uA^2,$$

so the practical sufficient condition becomes

$$\boxed{-\mu^2 + \frac{Zq_*^2}{4} - \frac{Yq_*^4}{16} + 8uA^2 > 2 \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \left(\mathcal{S}_{\text{diag}}^{EE,(N)} + \mathcal{R}_{\text{diag}}^{EE,(N)} \right)}. \quad (841)$$

CCCLXXXVI. 6. UPGRADE TO THE USABLE BENCHMARK ENTRY

The exact same-shell usable benchmark entry is

$$\Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)}.$$

Thus, if the zero-momentum inequality (838) holds and the small- p plus truncation budget satisfies

$$\boxed{\varepsilon_{\text{small-}p,\rho}^{(N)} + \varepsilon_{\text{non-enr},\rho}^{(N)} < \Delta_{\text{non-enr},0}^{(N)}}, \quad (842)$$

then

$$\boxed{\Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)} > 0}. \quad (843)$$

Hence the final same-shell benchmark slot is reduced to:

1. a finite heavy-overlap sum $\mathcal{S}_{\text{diag}}^{EE,(N)}$,
2. a finite tail bound $\mathcal{R}_{\text{diag}}^{EE,(N)}$,
3. a finite small- p plus truncation error budget.

CCCLXXXVII. FINAL SUMMARY

For the explicit branch

$$T_\star = T^1, \quad \eta = z_0, \quad \Lambda_O^{(0)} = 0,$$

the same-shell non-enriched benchmark residual is purely KK and is controlled by the diagonal heavy overlap budget $\mathcal{S}_{\text{diag}}^{EE}$.

After truncating the heavy basis at level N ,

$$\mathcal{S}_{\text{diag}}^{EE} = \mathcal{S}_{\text{diag}}^{EE,(N)} + \mathcal{R}_{\text{diag}}^{EE,(N)},$$

with

$$\mathcal{S}_{\text{diag}}^{EE,(N)} = \sum_{\alpha=1}^N \frac{|\mathcal{E}_{+\alpha}^{(z_0)}|^2}{E_\alpha},$$

and tail bound

$$0 \leq \mathcal{R}_{\text{diag}}^{EE,(N)} \leq \frac{1}{\Delta_H} \sum_{\alpha>N} |\mathcal{E}_{+\alpha}^{(z_0)}|^2.$$

Therefore a finite practical same-shell sufficient condition is

$$|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > 2 \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \left(\mathcal{S}_{\text{diag}}^{EE,(N)} + \mathcal{R}_{\text{diag}}^{EE,(N)} \right).$$

In the minimal scalar-only locked core, this becomes

$$\left| -\mu^2 + \frac{Zq_\star^2}{4} - \frac{Yq_\star^4}{16} + 12uA^2 \right| - 4uA^2 > 2 \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \left(\mathcal{S}_{\text{diag}}^{EE,(N)} + \mathcal{R}_{\text{diag}}^{EE,(N)} \right).$$

If, furthermore, the small- p plus truncation error budget is below the zero-momentum gap, then the usable benchmark entry

$$\Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)}$$

is strictly positive.

So the final same-shell benchmark-side frontier has now been reduced to three finite computations:

$$\mathcal{S}_{\text{diag}}^{EE,(N)}, \quad \mathcal{R}_{\text{diag}}^{EE,(N)}, \quad \varepsilon_{\text{small-}p,\rho}^{(N)} + \varepsilon_{\text{non-enr},\rho}^{(N)}.$$

CCCLXXXVIII. A CONCRETE $N = 1$ HEAVY-TRIAL EVALUATION OF THE PURE-EVEN SAME-SHELL RESIDUAL

CCCLXXXIX. SHORT EXPLANATION

For the explicit branch

$$T_\star = T^1, \quad \eta = z_0, \quad \Lambda_O^{(0)} = 0,$$

the same-shell non-enriched residual is controlled by the pure-even KK overlap budget

$$\mathcal{S}_{\text{diag}}^{EE}.$$

To turn this into an actual computable quantity, we choose the $N = 1$ heavy trial subspace to be spanned by the normalized heavy source vector itself. This is the simplest nontrivial finite realization allowed by the truncation theorem.

CCCXC. 1. PURE-EVEN SOURCE VECTOR

Define the pure-even defect operator

$$\mathcal{O}_E := \theta'(\xi) k_a(\theta) E_j^b D_j \psi_b. \quad (844)$$

On the same-shell complementary locked-line state

$$|+\rangle = |G_+, z_0\rangle,$$

define the projected heavy source vector

$$|v_E^{(+)}\rangle := \Pi_H \mathcal{O}_E |G_+, z_0\rangle, \quad (845)$$

where Π_H is the projector to the heavy sector.

Expanding in the heavy eigenbasis χ_α ,

$$|v_E^{(+)}\rangle = \sum_{\alpha} \mathcal{E}_{+\alpha}^{(z_0)} |\chi_\alpha\rangle,$$

so

$$\|v_E^{(+)}\|^2 = \sum_{\alpha} |\mathcal{E}_{+\alpha}^{(z_0)}|^2. \quad (846)$$

The exact diagonal overlap budget is

$$\mathcal{S}_{\text{diag}}^{EE} = \sum_{\alpha} \frac{|\mathcal{E}_{+\alpha}^{(z_0)}|^2}{E_{\alpha}}. \quad (847)$$

CCCXCI. 2. RIGOROUS UPPER BOUND FROM THE HEAVY GAP

Since

$$E_{\alpha} \geq \Delta_H > 0,$$

one immediately gets

$$\mathcal{S}_{\text{diag}}^{EE} \leq \frac{1}{\Delta_H} \|v_E^{(+)}\|^2. \quad (848)$$

Therefore the practical same-shell sufficient condition

$$|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > 2 \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \mathcal{S}_{\text{diag}}^{EE}$$

is implied by the stronger but completely explicit inequality

$$|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > 2 \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \frac{\|v_E^{(+)}\|^2}{\Delta_H}. \quad (849)$$

This is already a concrete one-line sufficient criterion.

CCCXCII. 3. $N = 1$ SOURCE-ALIGNED TRIAL BASIS

Now choose the $N = 1$ heavy trial subspace

$$\mathcal{H}_H^{(1)} = \text{span}\{|\chi_1^{\text{trial}}\rangle\}$$

with

$$|\chi_1^{\text{trial}}\rangle = \frac{|v_E^{(+)}\rangle}{\|v_E^{(+)}\|}. \quad (850)$$

Define the corresponding variational heavy energy

$$E_1^{\text{var}} := \langle \chi_1^{\text{trial}} | \mathbb{O}^{(0)} | \chi_1^{\text{trial}} \rangle. \quad (851)$$

Then the actual $N = 1$ truncated overlap sum is

$$S_{\text{diag}}^{EE,(1)} = \frac{\|v_E^{(+)}\|^2}{E_1^{\text{var}}}. \quad (852)$$

Hence the $N = 1$ same-shell coefficients in the pure-even residual branch are

$$d_+^{(0,1)} = E_{z_0}^{\text{dir}}(0) - \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \frac{\|v_E^{(+)}\|^2}{E_1^{\text{var}}}, \quad (853)$$

$$d_-^{(0,1)} = E_{z_0}^{\text{dir}}(0) - \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \frac{\|v_E^{(-)}\|^2}{E_1^{\text{var},-}}, \quad (854)$$

$$m_0^{(1)} = B_{z_0}(0) - \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \frac{\langle v_E^{(+)} | v_E^{(-)} \rangle}{\sqrt{E_1^{\text{var},+} E_1^{\text{var},-}}}. \quad (855)$$

CCCXCIII. 4. VALLEY-SYMMETRIC $N = 1$ SIMPLIFICATION

If the residual norms are valley-symmetric,

$$\|v_E^{(+)}\| = \|v_E^{(-)}\| =: \|v_E\|, \quad E_1^{\text{var},+} = E_1^{\text{var},-} =: E_1^{\text{var}}, \quad (856)$$

then

$$d_+^{(0,1)} = d_-^{(0,1)} = E_{z_0}^{\text{dir}}(0) - \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \frac{\|v_E\|^2}{E_1^{\text{var}}}, \quad (857)$$

and

$$m_0^{(1)} = B_{z_0}(0) - \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \frac{\langle v_E^{(+)} | v_E^{(-)} \rangle}{E_1^{\text{var}}}. \quad (858)$$

Using

$$|\langle v_E^{(+)} | v_E^{(-)} \rangle| \leq \|v_E\|^2,$$

a sufficient zero-momentum condition is

$$\left| E_{z_0}^{\text{dir}}(0) - \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \frac{\|v_E\|^2}{E_1^{\text{var}}} \right| > |B_{z_0}(0)| + \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \frac{\|v_E\|^2}{E_1^{\text{var}}}. \quad (859)$$

An even simpler sufficient inequality is

$$|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > 2 \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \frac{\|v_E\|^2}{E_1^{\text{var}}}. \quad (860)$$

CCCXCIV. 5. MINIMAL SCALAR-ONLY LOCKED CORE INSERTION

In the minimal scalar-only locked core,

$$E_{z_0}^{\text{dir}}(0) = \omega_B(k_*) + 12uA^2, \quad B_{z_0}(0) = 4uA^2,$$

with

$$\omega_B(k_*) = -\mu^2 + \frac{Zq_*^2}{4} - \frac{Yq_*^4}{16}.$$

So (860) becomes

$$\left| -\mu^2 + \frac{Zq_*^2}{4} - \frac{Yq_*^4}{16} + 12uA^2 \right| - 4uA^2 > 2 \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \frac{\|v_E\|^2}{E_1^{\text{var}}}. \quad (861)$$

If, furthermore,

$$u > 0, \quad -\mu^2 + \frac{Zq_*^2}{4} - \frac{Yq_*^4}{16} + 12uA^2 \geq 0,$$

then

$$-\mu^2 + \frac{Zq_*^2}{4} - \frac{Yq_*^4}{16} + 8uA^2 > 2 \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \frac{\|v_E\|^2}{E_1^{\text{var}}}. \quad (862)$$

CCCXCV. 6. UPGRADE TO THE USABLE BENCHMARK ENTRY

The same-shell usable benchmark entry is

$$\Delta_{\text{non-enr},\rho}^{(1)} - \varepsilon_{\text{non-enr},\rho}^{(1)}.$$

Hence, if the zero-momentum inequality (860) holds and the small- p plus truncation budget satisfies

$$\varepsilon_{\text{small-}p,\rho}^{(1)} + \varepsilon_{\text{non-enr},\rho}^{(1)} < \Delta_{\text{non-enr},0}^{(1)}, \quad (863)$$

then

$$\Delta_{\text{non-enr},\rho}^{(1)} - \varepsilon_{\text{non-enr},\rho}^{(1)} > 0.$$

Thus the final same-shell benchmark slot has been reduced to the finite quantities

$$\|v_E\|^2, \quad E_1^{\text{var}}, \quad \varepsilon_{\text{small-}p,\rho}^{(1)}, \quad \varepsilon_{\text{non-enr},\rho}^{(1)}.$$

CCCXCVI. FINAL SUMMARY

For the explicit branch

$$T_\star = T^1, \quad \eta = z_0, \quad \Lambda_O^{(0)} = 0,$$

the same-shell non-enriched benchmark residual is purely KK .

Choosing the $N = 1$ heavy trial space to be spanned by the normalized heavy source vector

$$|\chi_1^{\text{trial}}\rangle = \frac{\Pi_H \mathcal{O}_E |G_+, z_0\rangle}{\|\Pi_H \mathcal{O}_E |G_+, z_0\rangle\|},$$

one obtains the explicit finite overlap budget

$$\mathcal{S}_{\text{diag}}^{EE,(1)} = \frac{\|v_E\|^2}{E_1^{\text{var}}}, \quad |v_E\rangle := \Pi_H \mathcal{O}_E |G_+, z_0\rangle, \quad E_1^{\text{var}} = \langle \chi_1^{\text{trial}} | \mathbb{O}^{(0)} | \chi_1^{\text{trial}} \rangle.$$

Therefore a practical same-shell sufficient condition is

$$|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > 2 \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \frac{\|v_E\|^2}{E_1^{\text{var}}}.$$

In the minimal scalar-only locked core this becomes

$$\left| -\mu^2 + \frac{Zq_*^2}{4} - \frac{Yq_*^4}{16} + 12uA^2 \right| - 4uA^2 > 2 \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \frac{\|v_E\|^2}{E_1^{\text{var}}}.$$

Thus the final same-shell benchmark slot has been reduced to four finite quantities:

$$\|v_E\|^2, \quad E_1^{\text{var}}, \quad \varepsilon_{\text{small-}p,\rho}^{(1)}, \quad \varepsilon_{\text{non-enr},\rho}^{(1)}.$$

So the remaining frontier is no longer structural. It is the literal finite computation of one projected source norm, one variational heavy energy, and two truncation/error budgets.

CCCXCVII. EXPLICIT EVALUATION OF $\|v_E\|^2$ FOR A SECH DEFECT PROFILE

CCCXCVIII. 1. SETUP

We take a standard localized defect profile

$$\theta(\xi) = \theta_0 \text{sech}(\kappa\xi) \tag{864}$$

so that

$$\theta'(\xi) = -\theta_0 \kappa \text{sech}(\kappa\xi) \tanh(\kappa\xi). \tag{865}$$

We use a minimal test mode

$$\psi(x) = e^{ik_* x_\parallel} \tag{866}$$

so that

$$D_j \psi = ik_* n_j \psi. \tag{867}$$

Thus the pure-even operator becomes

$$\mathcal{O}_E \psi = \theta'(\xi) k_a(\theta) E_j^b (ik_* n_j) \psi. \tag{868}$$

Since only longitudinal component survives,

$$E_j^b n_j \sim 1, \quad (869)$$

so we absorb constants into a single factor C_E :

$$\boxed{\mathcal{O}_E \psi = i k_* C_E \theta'(\xi) k(\theta) \psi.} \quad (870)$$

CCCXCIX. 2. SOURCE VECTOR NORM

Projecting to heavy sector does not change spatial norm at leading order, so

$$\boxed{\|v_E\|^2 = \int d^3x |\mathcal{O}_E \psi|^2.} \quad (871)$$

Thus

$$\|v_E\|^2 = k_*^2 C_E^2 \int d^3x (\theta'(\xi))^2 k(\theta)^2. \quad (872)$$

Assume $k(\theta) \approx k_0 \theta$ (small-angle approximation):

$$\boxed{k(\theta)^2 \approx k_0^2 \theta^2.} \quad (873)$$

Hence

$$\|v_E\|^2 = k_*^2 C_E^2 k_0^2 \int d^3x (\theta'(\xi))^2 \theta(\xi)^2. \quad (874)$$

CD. 3. REDUCTION TO 1D INTEGRAL

Factorize transverse area $A_\perp$:

$$\boxed{\|v_E\|^2 = k_*^2 C_E^2 k_0^2 A_\perp \int_{-\infty}^{\infty} d\xi (\theta'(\xi))^2 \theta(\xi)^2.} \quad (875)$$

Insert explicit profile:

$$\theta^2 = \theta_0^2 \operatorname{sech}^2(\kappa \xi), \quad (876)$$

$$(\theta')^2 = \theta_0^2 \kappa^2 \operatorname{sech}^2(\kappa \xi) \tanh^2(\kappa \xi). \quad (877)$$

Thus integrand:

$$(\theta')^2 \theta^2 = \theta_0^4 \kappa^2 \operatorname{sech}^4(\kappa \xi) \tanh^2(\kappa \xi). \quad (878)$$

CDI. 4. EXACT INTEGRAL

Define

$$I = \int_{-\infty}^{\infty} \text{sech}^4(\kappa\xi) \tanh^2(\kappa\xi) d\xi.$$

Change variable $u = \tanh(\kappa\xi)$, $du = \kappa \text{sech}^2(\kappa\xi) d\xi$:

$$I = \frac{1}{\kappa} \int_{-1}^1 u^2(1-u^2) du.$$

Compute:

$$\int_{-1}^1 u^2(1-u^2) du = \int_{-1}^1 (u^2 - u^4) du = \frac{2}{3} - \frac{2}{5} = \frac{4}{15}.$$

Thus

$$\boxed{I = \frac{4}{15\kappa}.} \quad (879)$$

CDII. 5. FINAL EXPLICIT RESULT

Therefore

$$\boxed{\|v_E\|^2 = \frac{4}{15} k_*^2 C_E^2 k_0^2 A_{\perp} \theta_0^4 \kappa.} \quad (880)$$

CDIII. 6. INSERT INTO THE SAME-SHELL INEQUALITY

The practical $N = 1$ condition becomes

$$\boxed{|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > \frac{8}{15} \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \frac{k_*^2 C_E^2 k_0^2 A_{\perp} \theta_0^4 \kappa}{E_1^{\text{var}}}.} \quad (881)$$

CDIV. EXPLICIT EVALUATION OF THE VARIATIONAL HEAVY ENERGY E_1^{var}

CDV. 1. DEFINITION

We compute

$$\boxed{E_1^{\text{var}} = \frac{\langle v_E | \mathbb{O}^{(0)} | v_E \rangle}{\|v_E\|^2}.} \quad (882)$$

The heavy operator is

$$\boxed{\mathbb{O}^{(0)} = M_H^2 - Z \nabla^2 + Y \nabla^4.} \quad (883)$$

CDVI. 2. STRUCTURE OF THE SOURCE

From previous section:

$$v_E(x) \propto \theta'(\xi)\theta(\xi)e^{ik_*x_{\parallel}}.$$

Thus it factorizes:

$$\boxed{v_E(x) = f(\xi) e^{ik_*x_{\parallel}}, \quad f(\xi) = \theta'(\xi)\theta(\xi).} \quad (884)$$

CDVII. 3. EFFECTIVE MOMENTUM SCALES

Longitudinal direction:

$$\partial_{\parallel} \rightarrow ik_*.$$

Transverse (defect) direction:
Since

$$f(\xi) \sim \theta_0^2 \kappa \operatorname{sech}^2(\kappa \xi) \tanh(\kappa \xi),$$

its Fourier width is

$$k_{\perp} \sim \kappa.$$

Thus define effective momentum:

$$\boxed{k_{\text{eff}}^2 = k_*^2 + c_{\kappa} \kappa^2}, \quad (885)$$

with $c_{\kappa} \sim O(1)$.
For sech profile one finds

$$\boxed{c_{\kappa} = \frac{2}{3}}. \quad (886)$$

CDVIII. 4. ACTION OF THE OPERATOR

We approximate:

$$-\nabla^2 \rightarrow k_{\text{eff}}^2, \quad \nabla^4 \rightarrow k_{\text{eff}}^4.$$

Thus

$$\boxed{E_1^{\text{var}} = M_H^2 + Z k_{\text{eff}}^2 + Y k_{\text{eff}}^4}. \quad (887)$$

Explicitly,

$$\boxed{E_1^{\text{var}} = M_H^2 + Z \left(k_*^2 + \frac{2}{3} \kappa^2 \right) + Y \left(k_*^2 + \frac{2}{3} \kappa^2 \right)^2}. \quad (888)$$

CDIX. 5. FINAL RATIO

From previous result:

$$\|v_E\|^2 = \frac{4}{15} k_*^2 C_E^2 k_0^2 A_\perp \theta_0^4 \kappa.$$

Thus

$$\boxed{\frac{\|v_E\|^2}{E_1^{\text{var}}} = \frac{\frac{4}{15} k_*^2 C_E^2 k_0^2 A_\perp \theta_0^4 \kappa}{M_H^2 + Z \left(k_*^2 + \frac{2}{3} \kappa^2\right) + Y \left(k_*^2 + \frac{2}{3} \kappa^2\right)^2}.} \quad (889)$$

CDX. 6. FINAL SAME-SHELL INEQUALITY (FULLY EXPLICIT)

Therefore

$$\boxed{|E_{z_0}^{\text{dir}}(0) - |B_{z_0}(0)| > \frac{8}{15} \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \frac{k_*^2 C_E^2 k_0^2 A_\perp \theta_0^4 \kappa}{M_H^2 + Z \left(k_*^2 + \frac{2}{3} \kappa^2\right) + Y \left(k_*^2 + \frac{2}{3} \kappa^2\right)^2}.} \quad (890)$$